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MASS-TRANSFER OPERATIONS

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MASS - TRANSFER OPERATIONS

Third Edition

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University of Rhode Island*

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MASS-TRANSFER OPERATIONS

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This book is dedicated
to the memory of my dear wife, Gertrude,
whose help with this edition was sorely missed.

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PREFACE

My purpose in presenting the third edition of this book continues to be that of the previous edition: "to provide a vehicle for teaching, either through a formal course or through self-study, the techniques of, and principles of equipment design for, the mass-transfer operations of chemical engineering." As before, these operations are largely the responsibility of the chemical engineer, but increasingly practitioners of other engineering disciplines are finding them necessary for their work. This is especially true for those engaged in pollution control and environment protection, where separation processes predominate, and in, for example, extractive metallurgy, where more sophisticated and diverse methods of separation are increasingly relied upon.

I have taken this opportunity to improve and modernize many of the explanations, to modernize the design data, and to lighten the writing as best I could. There are now included discussions of such topics as the surface-stretch theory of mass-transfer, transpiration cooling, new types of tray towers, heatless adsorbers, and the like. Complete design methods are presented for mixer-settler and sieve-tray extractors, sparged vessels, and mechanically agitated vessels for gas-liquid, liquid-liquid, and solid-liquid contact, adiabatic packed-tower absorbers, and evaporative coolers. There are new worked examples and problems for student practice. In order to keep the length of the book within reasonable limits, the brief discussion of the so-called less conventional operations in the last chapter of the previous edition has been omitted.

One change will be immediately evident to those familiar with previous editions; the new edition is written principally in the SI system of units. In order to ease the transition to this system, an important change was made: of the more than 1000 numbered equations, all but 25 can now be used with any system of consistent units, SI, English engineering, Metric engineering, cgs, or whatever. The few equations which remain dimensionally inconsistent are given in SI, and also by footnote or other means in English engineering units. All tables of engineering data, important dimensions in the text, and most student problems

are treated similarly. An extensive list of conversion factors from other systems to SI is included in Chapter 1; these will cover all quantities needed for the use of this book. I hope this book will stimulate the transition to SI, the advantages of which become increasingly clear as one becomes familiar with it.

I remain as before greatly indebted to many firms and publications for permission to use their material, and most of all to the many engineers and scientists whose works provide the basis for a book of this sort. I am also indebted to Edward C. Hohmann and William R. Schowalter as well as to several anonymous reviewers who provided useful suggestions. Thanks are due to the editorial staff of the publisher, all of whom have been most helpful.

Robert E. Treybal

Robert E. Treybal passed away while this book was in production. We are grateful to Mark M. Friedman who, in handling the proofs of this book, has contributed significantly to the usability of this text.

THE MASS-TRANSFER OPERATIONS

A substantial number of the unit operations of chemical engineering are concerned with the problem of changing the compositions of solutions and mixtures through methods not necessarily involving chemical reactions. Usually these operations are directed toward separating a substance into its component parts. For mixtures, such separations may be entirely mechanical, e.g., the filtration of a solid from a suspension in a liquid, the classification of a solid into fractions of different particle size by screening, or the separation of particles of a ground solid according to their density. On the other hand, if the operations involve changes in composition of solutions, they are known as the mass-transfer operations and it is these which concern us here.

The importance of these operations is profound. There is scarcely any chemical process which does not require a preliminary purification of raw materials or final separation of products from by-products, and for these the mass-transfer operations are usually used. One can perhaps most readily develop an immediate appreciation of the part these separations play in a processing plant by observing the large number of towers which bristle from a modern petroleum refinery, in each of which a mass-transfer separation operation takes place. Frequently the major part of the cost of a process is that for the separations. These separation or purification costs depend directly upon the ratio of final to initial concentration of the separated substances, and if this ratio is large, the product costs are large. Thus, sulfuric acid is a relatively low-priced product in part because sulfur is found naturally in a relatively pure state, whereas pure uranium is expensive because of the low concentration in which it is found in nature.

The mass-transfer operations are characterized by transfer of a substance through another on a molecular scale. For example, when water evaporates from

a pool into an airstream flowing over the water surface, molecules of water vapor diffuse through those of the air at the surface into the main portion of the airstream, whence they are carried away. It is not bulk movement as a result of a pressure difference, as in pumping a liquid through a pipe, with which we are primarily concerned. In the problems at hand, the mass transfer is a result of a concentration difference, or gradient, the diffusing substance moving from a place of high to one of low concentration.

CLASSIFICATION OF THE MASS-TRANSFER OPERATIONS

It is useful to classify the operations and to cite examples of each, in order to indicate the scope of this book and to provide a vehicle for some definitions of terms which are commonly used.

Direct Contact of Two Immiscible Phases

This category is by far the most important of all and includes the bulk of the mass-transfer operations. Here we take advantage of the fact that in a two-phase system of several components at equilibrium, with few exceptions the compositions of the phases are different. The various components, in other words, are differently distributed between the phases.

In some instances, the separation thus afforded leads immediately to a pure substance because one of the phases at equilibrium contains only one constituent. For example, the equilibrium vapor in contact with a liquid aqueous salt solution contains no salt regardless of the concentration of the liquid. Similarly the equilibrium solid in contact with such a liquid salt solution is either pure water or pure salt, depending upon which side of the eutectic composition the liquid happens to be. Starting with the liquid solution, one can then obtain a complete separation by boiling off the water. Alternatively, pure salt or pure water can be produced by partly freezing the solution; or, in principle at least, both can be obtained pure by complete solidification followed by mechanical separation of the eutectic mixture of crystals. In cases like these, when the two phases are first formed, they are immediately at their final equilibrium compositions and the establishment of equilibrium is not a time-dependent process. Such separations, with one exception, are not normally considered to be among the mass-transfer operations.

In the mass-transfer operations, neither equilibrium phase consists of only one component. Consequently when the two phases are initially contacted, they will not (except fortuitously) be of equilibrium compositions. The system then attempts to reach equilibrium by a relatively slow diffusive movement of the constituents, which transfer in part between the phases in the process. Separations are therefore never complete, although, as will be shown, they can be brought as near completion as desired (but not totally) by appropriate manipulations.

The three states of aggregation, gas, liquid, and solid, permit six possibilities of phase contact.

Gas-gas Since with very few exceptions all gases are completely soluble in each other, this category is not practically realized.

Gas-liquid If all components of the system distribute between the phases at equilibrium, the operation is known as *fractional distillation* (or frequently just *distillation*). In this instance the gas phase is created from the liquid by application of heat; or conversely, the liquid is created from the gas by removal of heat. For example, if a liquid solution of acetic acid and water is partially vaporized by heating, it is found that the newly created vapor phase and the residual liquid both contain acetic acid and water but in proportions at equilibrium which are different for the two phases and different from those in the original solution. If the vapor and liquid are separated mechanically from each other and the vapor condensed, two solutions, one richer in acetic acid and the other richer in water, are obtained. In this way a certain degree of separation of the original components has been accomplished.

Both phases may be solutions, each containing, however, only one common component (or group of components) which distributes between the phases. For example, if a mixture of ammonia and air is contacted with liquid water, a large portion of the ammonia, but essentially no air, will dissolve in the liquid and in this way the air-ammonia mixture can be separated. The operation is known as *gas absorption*. On the other hand, if air is brought into contact with an ammonia-water solution, some of the ammonia leaves the liquid and enters the gas phase, an operation known as *desorption* or *stripping*. The difference is purely in the direction of solute transfer.

If the liquid phase is a pure liquid containing but one component while the gas contains two or more, the operation is *humidification* or *dehumidification*, depending upon the direction of transfer (this is the exception mentioned earlier). For example, contact of dry air with liquid water results in evaporation of some water into the air (humidification of the air). Conversely, contact of very moist air with pure liquid water may result in condensation of part of the moisture in the air (dehumidification). In both cases, diffusion of water vapor through air is involved, and we include these among the mass-transfer operations.

Gas-solid Classification of the operations in this category according to the number of components which appear in the two phases is again convenient.

If a solid solution is partially vaporized without the appearance of a liquid phase, the newly formed vapor phase and the residual solid each contains all the original components, but in different proportions, and the operation is *fractional sublimation*. As in distillation, the final compositions are established by interdiffusion of the components between the phases. While such an operation is theoretically possible, practically it is not generally done because of the inconvenience of dealing with solid phases in this manner.

Not all components may be present in both phases, however. If a solid which is moistened with a volatile liquid is exposed to a relatively dry gas, the liquid leaves the solid and diffuses into the gas, an operation generally known as *drying*, sometimes as *desorption*. A homely example is drying laundry by exposure to air, and there are many industrial counterparts such as drying lumber or the removal of moisture from a wet filter cake by exposure to dry gas. In this case, the diffusion is, of course, from the solid to the gas phase. If the diffusion takes place in the opposite direction, the operation is known as *adsorption*. For example, if a mixture of water vapor and air is brought into contact with activated silica gel, the water vapor diffuses to the solid, which retains it strongly, and the air is thus dried. In other instances, a gas mixture may contain several components each of which is adsorbed on a solid but to different extents (*fractional adsorption*). For example, if a mixture of propane and propylene gases is brought into contact with activated carbon, the two hydrocarbons are both adsorbed, but to different extents, thus leading to a separation of the gas mixture.

When the gas phase is a pure vapor, as in the sublimation of a volatile solid from a mixture with one which is nonvolatile, the operation depends more on the rate of application of heat than on concentration difference and is essentially nondiffusional. The same is true of the condensation of a vapor to the condition of a pure solid, where the rate depends on the rate of heat removal.

Liquid-liquid Separations involving the contact of two insoluble liquid phases are known as *liquid-extraction operations*. A simple example is the familiar laboratory procedure: if an acetone-water solution is shaken in a separatory funnel with carbon tetrachloride and the liquids allowed to settle, a large portion of the acetone will be found in the carbon tetrachloride-rich phase and will thus have been separated from the water. A small amount of the water will also have been dissolved by the carbon tetrachloride, and a small amount of the latter will have entered the water layer, but these effects are relatively minor. As another possibility, a solution of acetic acid and acetone can be separated by adding it to the insoluble mixture of water and carbon tetrachloride. After shaking and settling, both acetone and acetic acid will be found in both liquid phases, but in different proportions. Such an operation is known as *fractional extraction*. Another form of fractional extraction can be effected by producing two liquid phases from a single-phase solution by cooling the latter below its critical solution temperature. The two phases which form will be of different composition.

Liquid-solid When all the constituents are present in both phases at equilibrium, we have the operation of *fractional crystallization*. Perhaps the most interesting examples of this are the special techniques of *zone refining*, used to obtain ultrapure metals and semiconductors, and *adductive crystallization*, where a substance, such as urea, has a crystal lattice which will selectively entrap long straight-chain molecules like the paraffin hydrocarbons but will exclude branched molecules.

Cases where the phases are solutions (or mixtures) containing but one common component occur more frequently. Selective solution of a component from a solid mixture by a liquid solvent is known as *leaching* (sometimes also as solvent extraction), and as examples we cite the leaching of gold from its ores by cyanide solutions and of cottonseed oil from the seeds by hexane. The diffusion is, of course, from the solid to the liquid phase. If the diffusion is in the opposite direction, the operation is known as *adsorption*. Thus, the colored material which contaminates impure cane sugar solutions can be removed by contacting the liquid solutions with activated carbon, whereupon the colored substances are retained on the surface of the solid carbon.

Solid-solid Because of the extraordinarily slow rates of diffusion within solid phases, there is no industrial separation operation in this category.

Phases Separated by a Membrane

These operations are used relatively infrequently, although they are rapidly increasing in importance. The membranes operate in different ways, depending upon the nature of the separation to be made. In general, however, they serve to prevent intermingling of two miscible phases. They also prevent ordinary hydrodynamic flow, and movement of substances through them is by diffusion. And they permit a component separation by selectively controlling passage of the components from one side to the other.

Gas-gas In *gaseous diffusion* or *effusion*, the membrane is microporous. If a gas mixture whose components are of different molecular weights is brought into contact with such a diaphragm, the various components of the gas pass through the pores at rates dependent upon the molecular weights. This leads to different compositions on opposite sides of the membrane and consequent separation of the mixture. In this manner large-scale separation of the isotopes of uranium, in the form of gaseous uranium hexafluoride, is carried out. In *permeation*, the membrane is not porous, and the gas transmitted through the membrane first dissolves in it and then diffuses through. Separation in this case is brought about principally by difference in solubility of the components. Thus, helium can be separated from natural gas by selective permeation through fluorocarbon-polymer membranes.

Gas-liquid These are *permeation* separations where, for example, a liquid solution of alcohol and water is brought into contact with a suitable nonporous membrane, in which the alcohol preferentially dissolves. After passage through the membrane the alcohol is vaporized on the far side.

Liquid-liquid The separation of a crystalline substance from a colloid, by contact of their solution with a liquid solvent with an intervening membrane permeable only to the solvent and the dissolved crystalline substance, is known as *dialysis*. For example, aqueous beet-sugar solutions containing undesired

colloidal material are freed of the latter by contact with water with an intervening semipermeable membrane. Sugar and water diffuse through the membrane, but the larger colloidal particles cannot. *Fractional dialysis* for separating two crystalline substances in solution makes use of the difference in membrane permeability for the substances. If an electromotive force is applied across the membrane to assist in the diffusion of charged particles, the operation is *electrodialysis*. If a solution is separated from the pure solvent by a membrane which is permeable only to the solvent, the solvent diffuses into the solution, an operation known as *osmosis*. This is not a separation operation, of course, but by superimposing a pressure to oppose the osmotic pressure the flow of solvent is reversed, and the solvent and solute of a solution can be separated by *reverse osmosis*. This is one of the processes which may become important in the desalination of seawater.

Direct Contact of Miscible Phases

Because of the difficulty in maintaining concentration gradients without mixing the fluid, the operations in this category are not generally considered practical industrially except in unusual circumstances.

Thermal diffusion involves the formation of a concentration difference within a single liquid or gaseous phase by imposition of a temperature gradient upon the fluid, thus making a separation of the components of the solution possible. In this way, ^3He is separated from its mixture with ^4He .

If a condensable vapor, such as steam, is allowed to diffuse through a gas mixture, it will preferentially carry one of the components along with it, thus making a separation by the operation known as *sweep diffusion*. If the two zones within the gas phase where the concentrations are different are separated by a screen containing relatively large openings, the operation is called *atmolysis*.

If a gas mixture is subjected to a very rapid *centrifugation*, the components will be separated because of the slightly different forces acting on the various molecules owing to their different masses. The heavier molecules thus tend to accumulate at the periphery of the centrifuge. This method is also used for separation of uranium isotopes.

Use of Surface Phenomena

Substances which when dissolved in a liquid produce a solution of lowered surface tension (in contact with a gas) are known to concentrate in solution at the liquid surface. By forming a foam of large surface, as by bubbling air through the solution, and collecting the foam, the solute can be concentrated. In this manner, detergents have been separated from water, for example. The operation is known as *foam separation*. It is not to be confused with the flotation processes of the ore-dressing industries, where insoluble solid particles are removed from slurries by collection into froths.

This classification is not exhaustive but it does categorize all the major mass-transfer operations. Indeed, new operations continue to be developed,

some of which can be fit into more than one category. This book includes gas-liquid, liquid-liquid, and solid-fluid operations, all of which involve direct contact of two immiscible phases and make up the great bulk of the applications of the transfer operations.

Direct and indirect operations The operations depending upon contact of two immiscible phases particularly can be further subclassified into two types. The *direct* operations produce the two phases from a single-phase solution by addition or removal of heat. Fractional distillation, fractional crystallization, and one form of fractional extraction are of this type. The *indirect* operations involve addition of a foreign substance and include gas absorption and stripping, adsorption, drying, leaching, liquid extraction, and certain types of fractional crystallization.

It is characteristic of the direct operations that the products are obtained directly, free of added substance; these operations are therefore sometimes favored over the indirect if they can be used.

If the separated products are required relatively pure, the disadvantages of the indirect operations incurred by addition of a foreign substance are several. The removed substance is obtained as a solution, which in this case must in turn be separated, either to obtain the pure substance or the added substance for reuse, and this represents an expense. The separation of added substance and product can rarely be complete, which may lead to difficulty in meeting product specifications. In any case, addition of a foreign substance may add to the problems of building corrosion-resistant equipment, and the cost of inevitable losses must be borne. Obviously the indirect methods are used only because they are, in the net, less costly than the direct methods if there is a choice. Frequently there is no choice.

When the separated substance need not be obtained pure, many of these disadvantages may disappear. For example, in ordinary drying, the water-vapor-air mixture is discarded since neither constituent need be recovered. In the production of hydrochloric acid by washing a hydrogen chloride-containing gas with water, the acid-water solution is sold directly without separation.

CHOICE OF SEPARATION METHOD

The chemical engineer faced with the problem of separating the components of a solution must ordinarily choose from several possible methods. While the choice is usually limited by the peculiar physical characteristics of the materials to be handled, the necessity for making a decision nevertheless almost always exists. Until the fundamentals of the various operations have been clearly understood, of course, no basis for such a decision is available, but it is well at least to establish the nature of the alternatives at the beginning.

One can sometimes choose between using a mass-transfer operation of the sort discussed in this book and a purely mechanical separation method. For example, in the separation of a desired mineral from its ore, it may be possible

to use either the mass-transfer operation of leaching with a solvent or the purely mechanical methods of flotation. Vegetable oils can be separated from the seeds in which they occur by expression or by leaching with a solvent. A vapor can be removed from a mixture with a permanent gas by the mechanical operation of compression or by the mass-transfer operations of gas absorption or adsorption. Sometimes both mechanical and mass-transfer operations are used, especially where the former are incomplete, as in processes for recovering vegetable oils wherein expression is followed by leaching. A more commonplace example is wringing water from wet laundry followed by air drying. It is characteristic that at the end of the operation the substance removed by mechanical methods is pure, while if removed by diffusional methods it is associated with another substance.

One can also frequently choose between a purely mass-transfer operation and a chemical reaction or a combination of both. Water can be removed from an ethanol-water solution either by causing it to react with unslaked lime or by special methods of distillation, for example. Hydrogen sulfide can be separated from other gases either by absorption in a liquid solvent with or without simultaneous chemical reaction or by chemical reaction with ferric oxide. Chemical methods ordinarily destroy the substance removed, while mass-transfer methods usually permit its eventual recovery in unaltered form without great difficulty.

There are also choices to be made within the mass-transfer operations. For example, a gaseous mixture of oxygen and nitrogen may be separated by preferential adsorption of the oxygen on activated carbon, by adsorption, by distillation, or by gaseous effusion. A liquid solution of acetic acid may be separated by distillation, by liquid extraction with a suitable solvent, or by adsorption with a suitable adsorbent.

The principal basis for choice in any case is cost: that method which costs the least is usually the one to be used. Occasionally other factors also influence the decision, however. The simplest operation, while it may not be the least costly, is sometimes desired because it will be trouble-free. Sometimes a method will be discarded because of imperfect knowledge of design methods or unavailability of data for design, so that results cannot be guaranteed. Favorable previous experience with one method may be given strong consideration.

METHODS OF CONDUCTING THE MASS-TRANSFER OPERATIONS

Several characteristics of these operations influence our method of dealing with them and are described in terms which require definition at the start.

Solute Recovery and Fractionation

If the components of a solution fall into two distinct groups of quite different properties, so that one can imagine that one group of components constitutes the

solvent and the other group the solute, separation according to these groups is usually relatively easy and amounts to a *solute-recovery* or *solute-removal* operation. For example, a gas consisting of methane, pentane, and hexane can be imagined to consist of methane as solvent with pentane plus hexane as solute, the solvent and solute in this case differing considerably in at least one property, vapor pressure. A simple gas-absorption operation, washing the mixture with a nonvolatile hydrocarbon oil, will easily provide a new solution of pentane plus hexane in the oil, essentially methane-free; and the residual methane will be essentially free of pentane and hexane. On the other hand, a solution consisting of pentane and hexane alone cannot be classified so readily. While the component properties differ, the differences are small, and to separate them into relatively pure components requires a different technique. Such separations are termed *fractionations*, and in this case we might use fractional distillation as a method.

Whether a solute-recovery or fractionation procedure is used may depend upon the property chosen to be exploited. For example, to separate a mixture of propanol and butanol from water by a gas-liquid contacting method, which depends on vapor pressures, requires fractionation (fractional distillation) because the vapor pressures of the components are not greatly different. But nearly complete separation of the combined alcohols from water can be obtained by liquid extraction of the solution with a hydrocarbon, using solute-recovery methods because the solubility of the alcohols as a group and water in hydrocarbons is greatly different. The separation of propanol from butanol, however, requires a fractionation technique (fractional extraction or fractional distillation, for example), because all their properties are very similar.

Unsteady-State Operation

It is characteristic of unsteady-state operation that concentrations at any point in the apparatus change with time. This may result from changes in concentrations of feed materials, flow rates, or conditions of temperature or pressure. In any case, *batch* operations are always of the unsteady-state type. In purely batch operations, all the phases are stationary from a point of view outside the apparatus, i.e., no flow in or out, even though there may be relative motion within. The familiar laboratory extraction procedure of shaking a solution with an immiscible solvent is an example. In *semibatch* operations, one phase is stationary while the other flows continuously in and out of the apparatus. As an example, we may cite the case of a drier where a quantity of wet solid is contacted continuously with fresh air, which carries away the vaporized moisture until the solid is dry.

Steady-State Operation

It is characteristic of steady-state operation that concentrations at any position in the apparatus remain constant with passage of time. This requires continuous, invariable flow of all phases into and out of the apparatus, a persistence of the

flow regime within the apparatus, constant concentrations of the feed streams, and unchanging conditions of temperature and pressure.

Stagewise Operation

If two insoluble phases are first allowed to come into contact so that the various diffusing substances can distribute themselves between the phases, and if the phases are then mechanically separated, the entire operation and the equipment required to carry it out are said to constitute one *stage*, e.g., laboratory batch extraction in a separatory funnel. The operation can be carried on in continuous fashion (steady-state) or batchwise fashion, however. For separations requiring greater concentration changes, a series of stages can be arranged so that the phases flow through the assembled stages from one to the other, e.g., in countercurrent flow. Such an assemblage is called a *cascade*. In order to establish a standard for the measurement of performance, the *equilibrium, ideal, or theoretical*, stage is defined as one where the effluent phases are in equilibrium, so that a longer time of contact will bring about no additional change of composition. The approach to equilibrium realized in any stage is then defined as the *stage of efficiency*.

Continuous-Contact (Differential-Contact) Operation

In this case the phases flow through the equipment in continuous, intimate contact throughout, without repeated physical separation and recontacting. The nature of the method requires the operation to be either semibatch or steady-state, and the resulting change in compositions may be equivalent to that given by a fraction of an ideal stage or by many stages. Equilibrium between two phases at any position in the equipment is never established; indeed, should equilibrium occur anywhere in the system, the result would be equivalent to the effect of an infinite number of stages.

The essential difference between stagewise and continuous-contact operation can be summarized. In the case of the stagewise operation the diffusional flow of matter between the phases is allowed to reduce the concentration difference which causes the flow. If allowed to continue long enough, an equilibrium is established, after which no further diffusional flow occurs. The rate of diffusion and the time then determine the stage efficiency realized in any particular situation. On the other hand, in continuous-contact operation the departure from equilibrium is deliberately maintained, and the diffusional flow between the phases may continue without interruption. Which method will be used depends to some extent on the stage efficiency that can be practically realized. A high stage efficiency can mean a relatively inexpensive plant and one whose performance can be reliably predicted. A low stage efficiency, on the other hand, may make the continuous-contact methods more desirable for reasons of cost and certainty.

DESIGN PRINCIPLES

There are four major factors to be established in the design of any plant involving the diffusional operations: the number of equilibrium stages or their equivalent, the time of phase contact required, the permissible rate of flow, and the energy requirements.

Number of Equilibrium Stages

In order to determine the number of equilibrium stages required in a cascade to bring about a specified degree of separation, or the equivalent quantity for a continuous-contact device, the equilibrium characteristics of the system and material-balance calculations are required.

Time Requirement

In stagewise operations the time of contact is intimately connected with stage efficiency, whereas for continuous-contact equipment the time leads ultimately to the volume or length of the required device. The factors which help establish the time are several. Material balances permit calculation of the relative quantities required of the various phases. The equilibrium characteristics of the system establish the ultimate concentrations possible, and the rate of transfer of material between phases depends upon the departure from equilibrium which is maintained. The rate of transfer additionally depends upon the physical properties of the phases as well as the flow regime within the equipment.

It is important to recognize that, for a given degree of intimacy of contact of the phases, the time of contact required is independent of the total quantity of the phases to be processed.

Permissible Flow Rate

This factor enters into consideration of semibatch and steady-state operations, where it leads to the determination of the cross-sectional area of the equipment. Considerations of fluid dynamics establish the permissible flow rate, and material balances determine the absolute quantity of each of the streams required.

Energy Requirements

Heat and mechanical energies are ordinarily required to carry out the diffusional operations. Heat is necessary for the production of any temperature changes, for the creation of new phases (such as vaporization of a liquid), and for overcoming heat-of-solution effects. Mechanical energy is required for fluid and solid transport, for dispersing liquids and gases, and for operating moving parts of machinery.

The ultimate design, consequently, requires us to deal with the equilibrium characteristics of the system, material balances, diffusional rates, fluid dynamics, and energy requirements. In what follows, basic considerations of diffusion rates are discussed first (Part One) and these are later applied to specific operations. The principal operations, in turn, are subdivided into three categories, depending upon the nature of the insoluble phases contacted, gas-liquid (Part Two), liquid-liquid (Part Three), and solid-fluid (Part Four), since the equilibrium and fluid dynamics of the systems are most readily studied in such a grouping.

UNIT SYSTEMS

The principal unit system of this book is the SI (*Système International d'Unités*), but to accommodate other systems, practically all (992 of a total of 1017) numbered equations are written so that they can be used with *any* consistent set of units. In order to permit this, it is necessary to include in all expressions involving dimensions of both force and mass the conversion factor g_c , defined through Newton's second law of motion,

$$F = \frac{MA}{g_c}$$

where F = force

M = mass

A = acceleration

For the SI and cgs (centimeter-gram-second) system of units, g_c is unnecessary, but it can be assigned a numerical value of unity for purposes of calculation. There are four unit systems commonly used in chemical engineering, and values of g_c corresponding to these are listed in Table 1.1.

For engineering work, SI and English engineering units are probably most important. Consequently the coefficients of the 25 dimensionally inconsistent equations which cannot be used directly with any unit system are given first for SI, then by footnote or other means for English engineering units. Tables and graphs of engineering data are similarly treated.†

Table 1.2 lists the basic quantities as expressed in SI together with the unit abbreviations, Table 1.3 lists the unit prefixes needed for this book, and Table 1.4 lists some of the constants needed in several systems. Finally, Table 1.5 lists the conversion factors into SI for all quantities needed for this book. The boldface letters for each quantity represent the fundamental dimensions: **F** = force, **L** = length, **M** = mass, **mole** = mole, **T** = temperature, **Θ** = time. The list of notation at the end of each chapter gives the symbols used, their meaning, and dimensions.

† In practice, some departure from the standard systems is fairly common. Thus, for example, pressures are still frequently quoted in standard atmospheres, millimeters of mercury, bars, or kilograms force per square meter, depending upon the local common practice in the past.

Table 1.1 Conversion factors g_c for the common unit systems

Fundamental quantity	System			
	SI	English engineering†	cgs	Metric engineering‡
Mass M	Kilogram, kg	Pound mass, lb	Gram, g	Kilogram mass, kg
Length L	Meter, m	Foot, ft	Centimeter, cm	Meter, m
Time Θ	Second, s	Second, s, or hour, h	Second, s	Second, s
Force F	Newton, N	Pound force, lb _f	Dyne, dyn	Kilogram force, kg _f
g_c	$1 \text{ kg} \cdot \text{m}/\text{N} \cdot \text{s}^2$	$32.174 \text{ lb} \cdot \text{ft}/\text{lb}_f \cdot \text{s}^2$ or $4.1698 \times 10^8 \text{ lb} \cdot \text{ft}/\text{lb}_f \cdot \text{h}^2$	$1 \text{ g} \cdot \text{cm}/\text{dyn} \cdot \text{s}^2$	$9.80665 \text{ kg} \cdot \text{m}/\text{kg}_f \cdot \text{s}^2$

† Note that throughout this book lb alone is used as the abbreviation pound mass and lb_f is used for pound force.

‡ Note that throughout this book kg alone is used as the abbreviation for kilogram mass and kg_f is used for kilogram force.

Table 1.2 Basic SI units

Force = newton, N

Length = meter, m

Mass = kilogram, kg

Mole = kilogram mole, kmol

Temperature = kelvin = K

Time = second, s

Pressure = newton/meter², N/m² = pascal, Pa

Energy = newton-meter, N · m = joule, J

Power = newton-meter/second, N · m/s = watt, W

Frequency = 1/second, s⁻¹ = hertz, Hz

Table 1.3 Prefixes for SI units

Amount	Multiple	Prefix	Symbol
1 000 000	10 ⁶	mega	M
1 000	10 ³	kilo	k
100	10 ²	hecto	h
10	10	deka	da
0.1	10 ⁻¹	deci	d
0.01	10 ⁻²	centi	c
0.001	10 ⁻³	milli	m
0.000 001	10 ⁻⁶	micro	μ
0.000 000 001	10 ⁻⁹	nano	n

Table 1.4 Constants

Acceleration of gravity†	Molar volume of ideal gases at standard conditions (STP) (0°C, 1 std atm)‡
9.807 m/s ²	
980.7 cm/s ²	22.41 m ³ /kmol
32.2 ft/s ²	22.41 l/g mol
4.17 × 10 ⁸ ft/h ²	359 ft ³ /lb mol
Gas constant R	Conversion factor g _c
8314 N · m/kmol · K	1 kg · m/N · s ²
1.987 cal/g mol · K	1 g · cm/dyn · s ²
82.06 atm · cm ³ /g mol · K	9.80665 kg · m/kg _f · s ²
0.7302 atm · ft ³ /lb mol · °R	32.174 lb · ft/lb _f · s ²
1545 lb _f · ft/lb mol · °R	4.1698 × 10 ⁸ lb · ft/lb _f · h ²
1.987 Btu/lb mol · °R	
847.8 kg _f · m/kmol · K	

† Approximate, depends on location.

‡ Standard conditions are abbreviated STP, for standard temperature and pressure.

Table 1.5 Conversion factors to SI units

Length

Length, L

ft(0.3048) = m

in(0.0254) = m

in(25.4) = mm

cm(0.01) = m

Å(10⁻¹⁰) = m

μm(10⁻⁶) = m

Area, L²

ft²(0.0929) = m²

in²(6.452 × 10⁻⁴) = m²

in²(645.2) = mm²

cm²(10⁻⁴) = m²

Volume, L³

ft³(0.02832) = m³

cm³(10⁻⁶) = m³

l(10⁻³) = m³

U.S. gal(3.285 × 10⁻³) = m³

U.K. gal(4.546 × 10⁻³) = m³

Specific area, L²/L³

(ft²/ft³)(3.2804) = m²/m³

(cm²/cm³)(100) = m²/m³

Velocity, L/Θ

(ft/s)(0.3048) = m/s

(ft/min)(5.08 × 10⁻³) = m/s

(ft/h)(8.467 × 10⁻⁵) = m/s

Table 1.5 (Continued)

Acceleration, L/Θ^2

$$(ft/s^2)(0.3048) = m/s^2$$

$$(ft/h^2)(2.352 \times 10^{-8}) = m/s^2$$

$$(cm/s^2)(0.01) = m/s^2$$

Diffusivity, kinematic viscosity, L^2/Θ

$$(ft^2/h)(2.581 \times 10^{-5}) = m^2/s$$

$$(cm^2/s)(10^{-4}) = m^2/s$$

$$St(10^{-4}) = m^2/s^2$$

$$cSt(10^{-6}) = m^2/s^2$$

Volume rate, L^3/Θ

$$(ft^3/s)(0.02832) = m^3/s$$

$$(ft^3/min)(4.72 \times 10^{-4}) = m^3/s$$

$$(ft^3/h)(7.867 \times 10^{-6}) = m^3/s$$

$$(U.S. \text{ gal/min})(6.308 \times 10^{-5}) = m^3/s$$

$$(U.K. \text{ gal/min})(7.577 \times 10^{-5}) = m^3/s$$

Mass**Mass, M**

$$lb(0.4536) = kg$$

$$ton(907.2) = kg$$

$$t(1000) = kg^b$$

Density, concentration, M/L^3

$$(lb/ft^3)(16.019) = kg/m^3$$

$$(lb/U.S. \text{ gal})(119.8) = kg/m^3$$

$$(lb/U.K. \text{ gal})(99.78) = kg/m^3$$

$$(g/cm^3)(1000) = kg/m^3 = g/l$$

Specific volume, L^3/M

$$(ft^3/lb)(0.0624) = m^3/kg$$

$$(cm^3/g)(0.001) = m^3/kg$$

Mass rate, M/Θ

$$(lb/s)(0.4536) = kg/s$$

$$(lb/min)(7.56 \times 10^{-3}) = kg/s$$

$$(lb/h)(1.26 \times 10^{-4}) = kg/s$$

Mass rate/length, $M/L\Theta$

$$(lb/ft \cdot h)(4.134 \times 10^{-4}) = kg/m \cdot s$$

Viscosity, $M/L\Theta$

$$(lb/ft \cdot s)(1.488) = kg/m \cdot s$$

$$(lb/ft \cdot h)(4.134 \times 10^{-4}) = kg/m \cdot s$$

$$P(0.1) = kg/m \cdot s^c$$

$$cP(0.001) = kg/m \cdot s^c$$

$$N \cdot s/m^2 = kg/m \cdot s$$

Mass flux, mass velocity, $M/L^2\Theta$

$$(lb/ft^2 \cdot h)(1.356 \times 10^{-3}) = kg/m^2 \cdot s$$

$$(g/cm^2 \cdot s)(10) = kg/m^2 \cdot s$$

Molar flux, molar mass velocity, $\text{mole}/L^2\Theta$

$$(lb \text{ mol}/ft^2 \cdot h)(1.356 \times 10^{-3}) = kmol/m^2 \cdot s$$

$$(g \text{ mol}/cm^2 \cdot s)(10) = kmol/m^2 \cdot s$$

Table 1.5 (Continued)

 Mass-transfer coefficient, $\text{mole/L}^2\Theta(\text{F/L}^2)$ and others

$$K_g, k_g(\text{lb mol/ft}^2 \cdot \text{h} \cdot \text{atm})(1.338 \times 10^{-8}) = \text{kmol/m}^2 \cdot \text{s} \cdot (\text{N/m}^2)$$

$$K_L, K_c, k_L, k_c(\text{lb mol/ft}^2 \cdot \text{h} \cdot (\text{lb mol/ft}^3))(8.465 \times 10^{-5}) = \text{kmol/m}^2 \cdot \text{s} \cdot (\text{kmol/m}^3)$$

$$K_{gx}, k_{gx}, K_y, k_y(\text{lb mol/ft}^2 \cdot \text{h} \cdot \text{mole fraction})(1.356 \times 10^{-3}) = \text{kmol/m}^2 \cdot \text{s} \cdot \text{mole fraction}$$

$$K_Y, k_Y(\text{lb/ft}^2 \cdot \text{h} \cdot (\text{lb A/lb B}))(1.356 \times 10^{-3}) = \text{kg/m}^2 \cdot \text{s} \cdot (\text{kg A/kg B})$$

$$F_G, F_L(\text{lb mol/ft}^2 \cdot \text{h})(1.356 \times 10^{-3}) = \text{kmol/m}^2 \cdot \text{s}$$

 Volumetric mass-transfer coefficient, $\text{mole/L}^3\Theta(\text{F/L}^3)$ and others

 For volumetric coefficients of the type Ka , ka , Fa , and the like, multiply the conversion factors for the coefficient by that for a .

 Force

 Force, F

$$\text{lb}_f(4.448) = \text{N}$$

$$\text{kg}_f(9.807) = \text{N}$$

$$\text{kp}(9.807) = \text{N}^d$$

$$\text{dyn}(10^{-5}) = \text{N}$$

 Interfacial tension, surface tension, F/L

$$(\text{lb}_f/\text{ft})(14.59) = \text{N/m}$$

$$(\text{dyn/cm})(10^{-3}) = \text{N/m}$$

$$(\text{erg/cm}^2)(10^{-3}) = \text{N/m}$$

$$\text{kg/s}^2 = \text{N/m}$$

 Pressure, F/L^2

$$(\text{lb}_f/\text{ft}^2)(47.88) = \text{N/m}^2 = \text{Pa}$$

$$(\text{lb}_f/\text{in}^2)(6895) = \text{N/m}^2 = \text{Pa}$$

$$\text{std atm}(1.0133 \times 10^5) = \text{N/m}^2 = \text{Pa}$$

$$\text{inHg}(3386) = \text{N/m}^2 = \text{Pa}$$

$$\text{inH}_2\text{O}(249.1) = \text{N/m}^2 = \text{Pa}$$

$$(\text{dyn/cm}^2)(10^{-1}) = \text{N/m}^2 = \text{Pa}$$

$$\text{cmH}_2\text{O}(98.07) = \text{N/m}^2 = \text{Pa}$$

$$\text{mmHg}(133.3) = \text{N/m}^2 = \text{Pa}$$

$$\text{torr}(133.3) = \text{N/m}^2 = \text{Pa}$$

$$(\text{kp/m}^2)(9.807) = \text{N/m}^2 = \text{Pa}$$

$$\text{bar}(10^5) = \text{N/m}^2 = \text{Pa}$$

$$(\text{kg}_f/\text{cm}^2)(9.807 \times 10^4) = \text{N/m}^2 = \text{Pa}$$

 Pressure drop/length, $(F/L^2)/L$

$$[(\text{lb}_f/\text{ft}^2)/\text{ft}](157.0) = (\text{N/m}^2)/\text{m} = \text{Pa/m}$$

$$(\text{inH}_2\text{O}/\text{ft})(817) = (\text{N/m}^2)/\text{m} = \text{Pa/m}$$

 Energy, work, heat, FL

$$(\text{ft} \cdot \text{lb}_f)(1.356) = \text{N} \cdot \text{m} = \text{J}$$

$$\text{Btu}(1055) = \text{N} \cdot \text{m} = \text{J}$$

$$\text{Chu}(1900) = \text{N} \cdot \text{m} = \text{J}^e$$

$$\text{erg}(10^{-7}) = \text{N} \cdot \text{m} = \text{J}$$

$$\text{cal}(4.187) = \text{N} \cdot \text{m} = \text{J}$$

$$\text{kcal}(4187) = \text{N} \cdot \text{m} = \text{J}$$

$$(\text{kW} \cdot \text{h})(3.6 \times 10^6) = \text{N} \cdot \text{m} = \text{J}$$

 Enthalpy, FL/M

$$(\text{Btu/lb})(2326) = \text{N} \cdot \text{m/kg} = \text{J/kg}$$

$$(\text{cal/g})(4187) = \text{N} \cdot \text{m/kg} = \text{J/kg}$$

Table 1.5 (Continued)

Molar enthalpy, FL/mole

$$(\text{Btu/lb mol})(2326) = N \cdot \text{m/kmol} = \text{J/kmol}$$

$$(\text{cal/g mol})(4187) = N \cdot \text{m/kmol} = \text{J/kmol}$$

Heat capacity, specific heat, FL/MT

$$(\text{Btu/lb} \cdot ^\circ\text{F})(4187) = N \cdot \text{m/kg} \cdot \text{K} = \text{J/kg} \cdot \text{K}$$

$$(\text{cal/g} \cdot ^\circ\text{C})(4187) = N \cdot \text{m/kg} \cdot \text{K} = \text{J/kg} \cdot \text{K}$$

Molar heat capacity, FL/mole T

$$(\text{Btu/lb mol} \cdot ^\circ\text{F})(4187) = N \cdot \text{m/kmol} \cdot \text{K} = \text{J/kmol} \cdot \text{K}$$

$$(\text{cal/g} \cdot ^\circ\text{C})(4187) = N \cdot \text{m/kmol} \cdot \text{K} = \text{J/kmol} \cdot \text{K}$$

Energy flux, FL/L²Θ

$$(\text{Btu/ft}^2 \cdot \text{h})(3.155) = N \cdot \text{m/m}^2 \cdot \text{s} = \text{W/m}^2$$

$$(\text{cal/cm}^2 \cdot \text{s})(4.187 \times 10^4) = N \cdot \text{m/m}^2 \cdot \text{s} = \text{W/m}^2$$

Thermal conductivity, FL²/L²ΘT = FL/L²Θ(T/L)

$$(\text{Btu} \cdot \text{ft/ft}^2 \cdot \text{h} \cdot ^\circ\text{F})(1.7307) = N \cdot \text{m/m} \cdot \text{s} \cdot \text{K} = \text{W/m} \cdot \text{K}$$

$$(\text{kcal} \cdot \text{m/m}^2 \cdot \text{h} \cdot ^\circ\text{C})(1.163) = N \cdot \text{m/m} \cdot \text{s} \cdot \text{K} = \text{W/m} \cdot \text{K}$$

$$(\text{cal} \cdot \text{cm/cm}^2 \cdot \text{s} \cdot ^\circ\text{C})(418.7) = N \cdot \text{m/m} \cdot \text{s} \cdot \text{K} = \text{W/m} \cdot \text{K}$$

Heat-transfer coefficient, FL/L²ΘT

$$(\text{Btu/ft}^2 \cdot \text{h} \cdot ^\circ\text{F})(5.679) = N \cdot \text{m/m}^2 \cdot \text{s} \cdot \text{K} = \text{W/m}^2 \cdot \text{K}$$

$$(\text{cal/cm}^2 \cdot \text{s} \cdot ^\circ\text{C})(4.187 \times 10^4) = N \cdot \text{m/m}^2 \cdot \text{s} \cdot \text{K} = \text{W/m}^2 \cdot \text{K}$$

Power, FL/Θ

$$(\text{ft} \cdot \text{lb}_f/\text{s})(1.356) = N \cdot \text{m/s} = \text{W}$$

$$\text{hp}(745.7) = N \cdot \text{m/s} = \text{W}$$

$$(\text{Btu/min})(4.885 \times 10^{-3}) = N \cdot \text{m/s} = \text{W}$$

$$(\text{Btu/h})(0.2931) = N \cdot \text{m/s} = \text{W}$$

Power/volume, FL/L³Θ

$$(\text{ft} \cdot \text{lb}_f/\text{ft}^3 \cdot \text{s})(47.88) = N \cdot \text{m/m}^3 \cdot \text{s} = \text{W/m}^3$$

$$(\text{hp}/1000 \text{ U.S. gal})(197) = N \cdot \text{m/m}^3 \cdot \text{s} = \text{W/m}^3$$

Power/mass, FL/M

$$(\text{ft} \cdot \text{lb}_f/\text{lb} \cdot \text{s})(2.988) = N \cdot \text{m/kg} \cdot \text{s} = \text{W/kg}$$

^a St is the abbreviation for stokes.

^b t is the abbreviation for metric ton (= 1000 kg).

^c P is the abbreviation for poise.

^d kp is the abbreviation for kilopond = kg force, kg_f.

^e Chu is the abbreviation for centigrade heat unit.

In reading elsewhere, the student may come upon an empirical equation which it would be useful to convert into SI units. The procedure for this is shown in the following example.

Illustration 1.1 The minimum speed for a four-bladed paddle in an unbaffled, agitated vessel to mix two immiscible liquids is reported to be [S. Nagata et al., *Trans. Soc. Chem. Eng. Jap.*, 8, 43 (1950)]

$$N' = \frac{30\,600}{T'^{3/2}} \left(\frac{\mu'}{\rho'} \right)^{0.111} \left(\frac{\Delta \rho'}{\rho'} \right)^{0.26}$$

where N' = impeller speed $r/h \equiv h^{-1}$

T' = vessel diameter, ft

μ' = continuous-liquid viscosity, lb/ft · h

ρ' = continuous-liquid density, lb/ft³

$\Delta\rho'$ = difference in density of liquids, lb/ft³

Substitution of the units or dimensions of the various quantities shows that neither the dimensions nor the units on the left of the equals sign are the same, respectively, as those on the right. In other words, the equation is dimensionally inconsistent, and the coefficient 30 600 can be used only for the units listed above. Compute the coefficient required for the equation to be used with SI units: $N = s^{-1}$, $T = m$, $\mu = kg/m \cdot s$, $\rho = kg/m^3$, and $\Delta\rho = kg/m^3$.

SOLUTION The conversion factors are taken from Table 1.5. The procedure involves substitution of each SI unit together with necessary conversion factor to convert it into the unit of the equation as reported, the inverse of the usual use of Table 1.5. Thus,

$$\frac{N s^{-1}}{2.778 \times 10^{-4}} = N' h^{-1}$$

Then

$$\frac{N}{2.778 \times 10^{-4}} = \frac{30\,600}{(T/0.3048)^{3/2}} \left[\frac{\mu/(4.134 \times 10^{-4})}{16.019} \right]^{0.111} \left(\frac{\Delta\rho/16.019}{\rho/16.019} \right)^{0.26}$$

or

$$N = \frac{4.621}{T^{3/2}} \left(\frac{\mu}{\rho} \right)^{0.111} \left(\frac{\Delta\rho}{\rho} \right)^{0.26}$$

which is suitable for SI units.

PART ONE

DIFFUSION AND MASS TRANSFER

We have seen that most of the mass-transfer operations used for separating the components of a solution achieve this result by bringing the solution to be separated into contact with another insoluble phase. As will be developed, the rate at which a component is then transferred from one phase to the other depends upon a so-called *mass-transfer*, or rate, *coefficient* and upon the degree of departure of the system from equilibrium. The transfer stops when equilibrium is attained.

The rate coefficients for the various components in a given phase will differ from each other to the greatest extent under conditions where molecular diffusion prevails, but even then the difference is not really large. For example, gases and vapors diffusing through air will show transfer coefficients whose ratio at most may be 3 or 4 to 1. The same is true when various substances diffuse through a liquid such as water. Under conditions of turbulence, where molecular diffusion is relatively unimportant, the transfer coefficients become much more nearly alike for all components. Consequently, while in principle some separation of the components could be achieved by taking advantage of their different transfer coefficients, the degree of separation attainable in this manner is small. This is especially significant when it is considered that we frequently wish to obtain products which are nearly pure substances, where the ratio of components may be of the order of 1000 or 10 000 to 1, or even larger.

Therefore we depend almost entirely upon the differences in concentration which exist at equilibrium, and not upon the difference in transfer coefficients, for making separations. Nevertheless, the mass-transfer coefficients are of great importance, since, as they regulate the rate at which equilibrium is approached, they control the time required for separation and therefore the size and cost of the equipment used. The transfer coefficients are also important in governing the size of equipment used for entirely different purposes, such as carrying out

chemical reactions. For example, the rate at which a reaction between two gases occurs on a solid catalyst is frequently governed by the rate of transfer of the gases to the catalyst surface and the rate of transfer of the product away from the catalyst.

The mass-transfer coefficients, their relationship to the phenomenon of diffusion, fluid motion, and to related rate coefficients, such as those describing heat transfer, are treated in Part One.

MOLECULAR DIFFUSION IN FLUIDS

Molecular diffusion is concerned with the movement of individual molecules through a substance by virtue of their thermal energy. The kinetic theory of gases provides a means of visualizing what occurs, and indeed it was the success of this theory in quantitatively describing the diffusional phenomena which led to its rapid acceptance. In the case of a simplified kinetic theory, a molecule is imagined to travel in a straight line at a uniform velocity until it collides with another molecule, whereupon its velocity changes both in magnitude and direction. The average distance the molecule travels between collisions is its mean free path, and the average velocity is dependent upon the temperature. The molecule thus travels a highly zigzag path, the net distance in one direction which it moves in a given time, the rate of diffusion, being only a small fraction of the length of its actual path. For this reason the diffusion rate is very slow, although we can expect it to increase with decreasing pressure, which reduces the number of collisions, and with increased temperature, which increases the molecular velocity.

The importance of the barrier molecular collision presents to diffusive movement is profound. Thus, for example, it can be computed through the kinetic theory that the rate of evaporation of water at 25°C into a complete vacuum is roughly 3.3 kg/s per square meter of water surface. But placing a layer of stagnant air at 1 std atm pressure and only 0.1 mm thick above the water surface reduces the rate by a factor of about 600. The same general mechanism prevails also for the liquid state, but because of the considerably higher molecular concentration, we find even slower diffusion rates than in gases.

The phenomenon of molecular diffusion ultimately leads to a completely uniform concentration of substances throughout a solution which may initially

have been nonuniform. Thus, for example, if a drop of blue copper sulfate solution is placed in a beaker of water, the copper sulfate eventually permeates the entire liquid. The blue color in time becomes everywhere uniform, and no subsequent change occurs.

We must distinguish at the start, however, between molecular diffusion, which is a slow process, and the more rapid mixing which can be brought about by mechanical stirring and convective movement of the fluid. Visualize a tank 1.5 m in diameter into which has been placed a salt solution to a depth of 0.75 m. Imagine that a 0.75-m-deep layer of pure water has been carefully placed over the brine without disturbing the brine in any way. If the contents of the tank are left completely undisturbed, the salt will, by molecular diffusion, completely permeate the liquid, ultimately coming everywhere to one-half its concentration in the original brine. But the process is very slow, and it can be calculated that the salt concentration at the top surface will still be only 87.5 percent of its final value after 10 years and will reach 99 percent of its final value only after 28 years. On the other hand, it has been demonstrated that a simple paddle agitator rotating in the tank at 22 r/min will bring about complete uniformity in about 60 s [27]. The mechanical agitation has produced rapid movement of relatively large chunks or eddies of fluid characteristic of turbulent motion, which have carried the salt with them. This method of solute transfer is known as eddy or turbulent diffusion, as opposed to molecular diffusion. Of course, within each eddy, no matter how small, uniformity is achieved only by molecular diffusion, which is the ultimate process. We see then that molecular diffusion is the mechanism of mass transfer in stagnant fluids or in fluids which are moving only in laminar flow, although it is nevertheless always present even in highly developed turbulent flow.

In a two-phase system not at equilibrium, such as a layer of ammonia and air as a gas solution in contact with a layer of liquid water, spontaneous alteration through molecular diffusion also occurs, ultimately bringing the entire system to a state of equilibrium, whereupon alteration stops. At the end, we observe that the concentration of any constituent is the same throughout a phase, but it will not necessarily be the same in both phases. Thus the ammonia concentration will be uniform throughout the liquid and uniform at a different value throughout the gas. On the other hand, the chemical potential of the ammonia (or its activity if the same reference state is used), which is differently dependent upon concentration in the two phases, will be uniform everywhere throughout the system at equilibrium, and it is this uniformity which has brought the diffusive process to a halt. Evidently, then, the true driving force for diffusion is activity or chemical potential, and not concentration. In multiphase systems, however, we customarily deal with diffusional processes in each phase separately, and within one phase it is usually described in terms of that which is most readily observed, namely, concentration changes.

Molecular Diffusion

We have noted that if a solution is everywhere uniform in concentration of its constituents, no alteration occurs but that as long as it is not uniform, the

solution is spontaneously brought to uniformity by diffusion, the substances moving from a place of high concentration to one of low. The rate at which a solute moves at any point in any direction must therefore depend on the concentration gradient at that point and in that direction. In describing this quantitatively, we need an appropriate measure of rate.

Rates will be most conveniently described in terms of a molar flux, or mol/(area)(time), the area being measured in a direction normal to the diffusion. In a nonuniform solution even containing only two constituents, however, both constituents must diffuse if uniformity is the ultimate result, and this leads to the use of two fluxes to describe the motion of one constituent: N , the flux relative to a fixed location in space; and J , the flux of a constituent relative to the average molar velocity of all constituents. The first of these is of importance in the applications to design of equipment, but the second is more characteristic of the nature of the constituent. For example, a fisherman is most interested in the rate at which a fish swims upstream against the flowing current to reach his baited hook (analogous to N), but the velocity of the fish relative to the stream (analogous to J) is more characteristic of the swimming ability of the fish.

The diffusivity, or diffusion coefficient, D_{AB} of a constituent A in solution in B, which is a measure of its diffusive mobility, is then defined as the ratio of its flux J_A to its concentration gradient

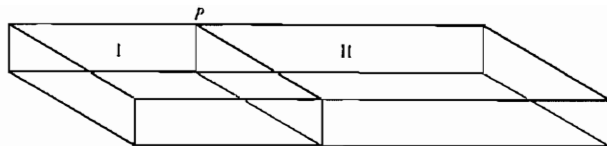
$$\text{Fick's 1st Law} \quad J_A = -D_{AB} \frac{\partial c_A}{\partial z} = -c D_{AB} \frac{\partial x_A}{\partial z} \quad (2.1)$$

which is Fick's first law written for the z direction. The negative sign emphasizes that diffusion occurs in the direction of a drop in concentration. The diffusivity is a characteristic of a constituent and its environment (temperature, pressure, concentration, whether in liquid, gas, or solid solution, and the nature of the other constituents).

Consider the box of Fig. 2.1, which is separated into two parts by the partition P . Into section I, 1 kg water (A) is placed and into section II 1 kg ethanol (B) (the densities of the liquids are different, and the partition is so located that the depths of the liquids in each section are the same). Imagine the partition to be carefully removed, thus allowing diffusion of both liquids to occur. When diffusion stops, the concentration will be uniform throughout at 50 mass percent of each constituent, and the masses and moles of each constituent in the two regions will be as indicated in the figure. It is clear that while the water has diffused to the right and the ethanol to the left, there has been a net mass movement to the right, so that if the box had originally been balanced on a knife-edge, at the end of the process it would have tipped downward to the right. If the direction to the right is taken as positive, the flux N_A of A relative to the fixed position P has been positive and the flux N_B of B has been negative. For a condition of steady state, the net flux is

$$N_A + N_B = N \quad (2.2)$$

The movement of A is made up of two parts, namely, that resulting from the bulk motion N and the fraction x_A of N which is A and that resulting from



	I			II	
	kg	kmol		kg	kmol
Initially:					
H ₂ O	100	5.55	EtOH	100	2.17
Finally:					
H ₂ O	44.08	2.45		55.92	3.10
EtOH	44.08	0.96		55.92	1.21
Total	88.16	3.41	Total:	111.84	4.31

Figure 2.1 Diffusion in a binary solution.

diffusion J_A :

$$N_A = N x_A + J_A \quad (2.3)$$

$$N_A = (N_A + N_B) \frac{c_A}{c} - D_{AB} \frac{\partial c_A}{\partial z} \quad (2.4)$$

The counterpart of Eq. (2.4) for B is

$$N_B = (N_A + N_B) \frac{c_B}{c} - D_{BA} \frac{\partial c_B}{\partial z} \quad (2.5)$$

Adding these gives

$$-D_{AB} \frac{\partial c_A}{\partial z} = D_{BA} \frac{\partial c_B}{\partial z} \quad (2.6)$$

or $J_A = -J_B$. If $c_A + c_B = \text{const}$, it follows that $D_{AB} = D_{BA}$ at the prevailing concentration and temperature.

All the above has considered diffusion in only one direction, but in general concentration gradients, velocities, and diffusional fluxes exist in all directions, so that counterparts of Eqs. (2.1) to (2.6) for all three directions in the cartesian coordinate system exist. In certain solids, the diffusivity D_{AB} may also be direction-sensitive, although in fluids which are true solutions it is not.

The Equation of Continuity

Consider the volume element of fluid of Fig. 2.2, where a fluid is flowing through the element. We shall need a material balance for a component of the fluid applicable to a differential fluid volume of this type.

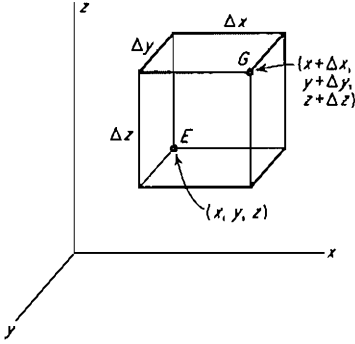


Figure 2.2 An elemental fluid volume.

The mass rate of flow of component A into the three faces with a common corner at E is

$$M_A [(N_{A,x})_x \Delta y \Delta z + (N_{A,y})_y \Delta x \Delta z + (N_{A,z})_z \Delta x \Delta y]$$

where $N_{A,x}$ signifies the x-directed flux and $(N_{A,x})_x$ its value at location x. Similarly the mass rate of flow out of the three faces with a common corner at G is

$$M_A [(N_{A,x})_{x+\Delta x} \Delta y \Delta z + (N_{A,y})_{y+\Delta y} \Delta x \Delta z + (N_{A,z})_{z+\Delta z} \Delta x \Delta y]$$

The total component A in the element is $\Delta x \Delta y \Delta z \rho_A$, and its rate of accumulation is therefore $\Delta x \Delta y \Delta z \partial \rho_A / \partial \theta$. If, in addition, A is generated by chemical reaction at the rate R_A mol/(volume)(time), its production rate is $M_A R_A \Delta x \Delta y \Delta z$, mass/time. Since, in general,

Rate out - rate in + rate of accumulation = rate of generation

then

$$\begin{aligned} M_A \{ [(N_{A,x})_{x+\Delta x} - (N_{A,x})_x] \Delta y \Delta z + [(N_{A,y})_{y+\Delta y} - (N_{A,y})_y] \Delta x \Delta z \\ + [(N_{A,z})_{z+\Delta z} - (N_{A,z})_z] \Delta x \Delta y \} + \Delta x \Delta y \Delta z \frac{\partial \rho_A}{\partial \theta} = M_A R_A \Delta x \Delta y \Delta z \end{aligned} \quad (2.7)$$

Dividing by $\Delta x \Delta y \Delta z$ and taking the limit as the three distances become zero gives

$$M_A \left(\frac{\partial N_{A,x}}{\partial x} + \frac{\partial N_{A,y}}{\partial y} + \frac{\partial N_{A,z}}{\partial z} \right) + \frac{\partial \rho_A}{\partial \theta} = M_A R_A \quad (2.8)$$

Similarly, for component B

$$M_B \left(\frac{\partial N_{B,x}}{\partial x} + \frac{\partial N_{B,y}}{\partial y} + \frac{\partial N_{B,z}}{\partial z} \right) + \frac{\partial \rho_B}{\partial \theta} = M_B R_B \quad (2.9)$$

The total material balance is obtained by adding those for A and B

$$\frac{\partial (M_A N_A + M_B N_B)_x}{\partial x} + \frac{\partial (M_A N_A + M_B N_B)_y}{\partial y} + \frac{\partial (M_A N_A + M_B N_B)_z}{\partial z} + \frac{\partial \rho}{\partial \theta} = 0 \quad (2.10)$$

where $\rho = \rho_A + \rho_B$ = the solution density, since the mass rate of generation of A and B must equal zero.

Now the counterpart of Eq. (2.3) in terms of masses and in the x direction is

$$M_A N_{A,x} = u_x \rho_A + M_A J_{A,x} \quad (2.11)$$

where u_x is the mass-average velocity such that

$$\rho u_x = u_{A,x} \rho_A + u_{B,x} \rho_B = M_A N_{A,x} + M_B N_{B,x} \quad (2.12)$$

Therefore,
$$\frac{\partial(M_A N_A + M_B N_B)_x}{\partial x} = \rho \frac{\partial u_x}{\partial x} + u_x \frac{\partial \rho}{\partial x}$$

Equation (2.10) therefore becomes

$$\rho \left(\frac{\partial u_x}{\partial x} + \frac{\partial u_y}{\partial y} + \frac{\partial u_z}{\partial z} \right) + u_x \frac{\partial \rho}{\partial x} + u_y \frac{\partial \rho}{\partial y} + u_z \frac{\partial \rho}{\partial z} + \frac{\partial \rho}{\partial \theta} = 0 \quad (2.13)$$

which is the equation of continuity, or a mass balance, for total substance. If the solution density is constant, it becomes

$$\frac{\partial u_x}{\partial x} + \frac{\partial u_y}{\partial y} + \frac{\partial u_z}{\partial z} = 0 \quad (2.14)$$

Returning to the balance for component A, we see from Eq. (2.11) that

$$M_A \frac{\partial N_{A,x}}{\partial x} = u_x \frac{\partial \rho_A}{\partial x} + \rho_A \frac{\partial u_x}{\partial x} + M_A \frac{\partial J_{A,x}}{\partial x} = u_x \frac{\partial \rho_A}{\partial x} + \rho_A \frac{\partial u_x}{\partial x} - M_A D_{AB} \frac{\partial^2 c_A}{\partial x^2} \quad (2.15)$$

Equation (2.8) then becomes

$$u_x \frac{\partial \rho_A}{\partial x} + u_y \frac{\partial \rho_A}{\partial y} + u_z \frac{\partial \rho_A}{\partial z} + \rho_A \left(\frac{\partial u_x}{\partial x} + \frac{\partial u_y}{\partial y} + \frac{\partial u_z}{\partial z} \right) - M_A D_{AB} \left(\frac{\partial^2 c_A}{\partial x^2} + \frac{\partial^2 c_A}{\partial y^2} + \frac{\partial^2 c_A}{\partial z^2} \right) + \frac{\partial \rho_A}{\partial \theta} = M_A R_A \quad (2.16)$$

which is the equation of continuity for substance A. For a solution of constant density, we can apply Eq. (2.14) to the terms multiplying ρ_A . Dividing by M_A , we then have

$$u_x \frac{\partial c_A}{\partial x} + u_y \frac{\partial c_A}{\partial y} + u_z \frac{\partial c_A}{\partial z} + \frac{\partial c_A}{\partial \theta} = D_{AB} \left(\frac{\partial^2 c_A}{\partial x^2} + \frac{\partial^2 c_A}{\partial y^2} + \frac{\partial^2 c_A}{\partial z^2} \right) + R_A \quad (2.17)$$

In the special case where the velocity equals zero and there is no chemical reaction, it reduces to Fick's second law

$$\frac{\partial c_A}{\partial \theta} = D_{AB} \left(\frac{\partial^2 c_A}{\partial x^2} + \frac{\partial^2 c_A}{\partial y^2} + \frac{\partial^2 c_A}{\partial z^2} \right) \quad (2.18)$$

This is frequently applicable to diffusion in solids and to limited situations in fluids.

In similar fashion, it is possible to derive the equations for a differential energy balance. For a fluid of constant density, the result is

$$u_x \frac{\partial t}{\partial x} + u_y \frac{\partial t}{\partial y} + u_z \frac{\partial t}{\partial z} + \frac{\partial t}{\partial \theta} = \alpha \left(\frac{\partial^2 t}{\partial x^2} + \frac{\partial^2 t}{\partial y^2} + \frac{\partial^2 t}{\partial z^2} \right) + \frac{Q}{\rho C_p} \quad (2.19)$$

where $\alpha = k/\rho C_p$ and Q is the rate of heat generation within the fluid per unit volume from a chemical reaction. The significance of the similarities between Eqs. (2.17) and (2.19) will be developed in Chap. 3.

STEADY-STATE MOLECULAR DIFFUSION IN FLUIDS AT REST AND IN LAMINAR FLOW

Applying Eq. (2.4) to the case of diffusion only in the z -direction, with N_A and N_B both constant (steady state), we can readily separate the variables, and if D_{AB} is constant, it can be integrated

$$\int_{c_{A1}}^{c_{A2}} \frac{-dc_A}{N_{AC} - c_A(N_A + N_B)} = \frac{1}{cD_{AB}} \int_{z_1}^{z_2} dz \quad (2.20)$$

where 1 indicates the beginning of the diffusion path (c_A high) and 2 the end of the diffusion path (c_A low). Letting $z_2 - z_1 = z$, we get

$$\frac{1}{N_A + N_B} \ln \frac{N_A c - c_{A2}(N_A + N_B)}{N_A c - c_{A1}(N_A + N_B)} = \frac{z}{c D_{AB}} \quad (2.21)$$

or

$$N_A = \frac{N_A}{N_A + N_B} \frac{D_{AB} c}{z} \ln \frac{N_A / (N_A + N_B) - c_{A2}/c}{N_A / (N_A + N_B) - c_{A1}/c} \quad (2.22)$$

Integration under steady-state conditions where the flux N_A is not constant is also possible. Consider radial diffusion from the surface of a solid sphere into a fluid, for example. Equation (2.20) can be applied, but the flux is a function of distance owing to the geometry. Most practical problems which deal with such matters, however, are concerned with diffusion under turbulent conditions, and the transfer coefficients which are then used are based upon a flux expressed in terms of some arbitrarily chosen area, such as the surface of the sphere. These matters are considered in Chap. 3.

Molecular Diffusion in Gases

When the ideal-gas law can be applied, Eq. (2.21) can be written in a form more convenient for use with gases. Thus,

$$\frac{c_A}{c} = \frac{\bar{p}_A}{p_t} = y_A \quad (2.23)$$

where $\bar{p}_A$ = partial pressure of component A

p_t = total pressure

y_A = mole fraction concentration†

Further,

$$c = \frac{n}{V} = \frac{p_t}{RT} \quad (2.24)$$

so that Eq. (2.22) becomes $\frac{N_A}{N_A + N_B} \frac{D_{AB} p_t}{RTz} \ln \frac{[N_A / (N_A + N_B)] p_t - \bar{p}_{A2}}{[N_A / (N_A + N_B)] p_t - \bar{p}_{A1}}$

$$N_A = \frac{N_A}{N_A + N_B} \frac{D_{AB} p_t}{RTz} \ln \frac{[N_A / (N_A + N_B)] p_t - \bar{p}_{A2}}{[N_A / (N_A + N_B)] p_t - \bar{p}_{A1}} \quad (2.25)$$

or

$$N_A = \frac{N_A}{N_A + N_B} \frac{D_{AB} p_t}{RTz} \ln \frac{N_A / (N_A + N_B) - y_{A2}}{N_A / (N_A + N_B) - y_{A1}} \quad (2.26)$$

In order to use these equations the relation between N_A and N_B must be known. This is usually fixed by other considerations. For example, if methane is being cracked on a catalyst,



under circumstances such that CH_4 (A) diffuses to the cracking surface and H_2 (B) diffuses back, the reaction stoichiometry fixes the relationship $N_B = -2N_A$,

† The component subscript A on y_A will differentiate mole fraction from the y meaning distance in the y direction.

and

$$\frac{N_A}{N_A + N_B} = \frac{N_A}{N_A - 2N_A} = -1$$

On other occasions, in the absence of chemical reaction, the ratio can be fixed by enthalpy considerations. In the case of the purely separational operations, there are two situations which frequently arise.

Steady-state diffusion of A through nondiffusing B This might occur, for example, if ammonia (A) were being absorbed from air (B) into water. In the gas phase, since air does not dissolve appreciably in water, and if we neglect the evaporation of water, only the ammonia diffuses. Thus, $N_B = 0$, $N_A = \text{const}$,

$$\frac{N_A}{N_A + N_B} = 1$$

and Eq. (2.25) becomes

$$N_A = \frac{D_{AB} p_t}{RTz} \ln \frac{p_t - \bar{p}_{A2}}{p_t - \bar{p}_{A1}} \quad (2.27)$$

Since $p_t - \bar{p}_{A2} = \bar{p}_{B2}$, $p_t - \bar{p}_{A1} = \bar{p}_{B1}$, $\bar{p}_{B2} - \bar{p}_{B1} = \bar{p}_{A1} - \bar{p}_{A2}$, then

$$N_A = \frac{D_{AB} p_t}{RTz} \frac{\bar{p}_{A1} - \bar{p}_{A2}}{\bar{p}_{B2} - \bar{p}_{B1}} \ln \frac{\bar{p}_{B2}}{\bar{p}_{B1}} \quad (2.28)$$

If we let

$$\frac{\bar{p}_{B2} - \bar{p}_{B1}}{\ln(\bar{p}_{B2}/\bar{p}_{B1})} = \bar{p}_{B,M} \quad (\text{log mean}) \quad (2.29)$$

then

$$N_A = \frac{D_{AB} p_t}{RTz \bar{p}_{B,M}} (\bar{p}_{A1} - \bar{p}_{A2}) \quad (2.30)$$

This equation is shown graphically in Fig. 2.3. Substance A diffuses by virtue of its concentration gradient, $-\bar{d}p_A/dz$. Substance B is also diffusing relative to

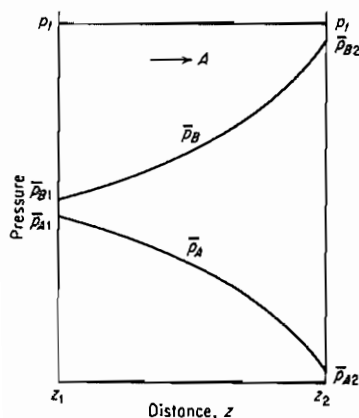


Figure 2.3 Diffusion of A through stagnant B.

the average molar velocity at a flux J_B , which depends upon $-d\bar{p}_B/dz$, but like a fish which swims upstream at the same velocity as the water flows downstream, $N_B = 0$ relative to a fixed place in space.

Steady-state equimolar counterdiffusion This is a situation which frequently pertains in distillation operations. $N_A = -N_B = \text{const.}$ Equation (2.25) becomes indeterminate, but we can go back to Eq. (2.4), which, for gases, becomes

$$N_A = (N_A + N_B) \frac{\bar{p}_A}{p_i} - \frac{D_{AB}}{RT} \frac{d\bar{p}_A}{dz} \quad (2.31)$$

or, for this case,

$$N_A = - \frac{D_{AB}}{RT} \frac{d\bar{p}_A}{dz} \quad (2.32)$$

$$\int_{z_1}^{z_2} dz = - \frac{D_{AB}}{RTN_A} \int_{\bar{p}_{A1}}^{\bar{p}_{A2}} d\bar{p}_A \quad (2.33)$$

$$N_A = \frac{D_{AB}}{RTz} (\bar{p}_{A1} - \bar{p}_{A2}) \quad (2.34)$$

This is shown graphically in Fig. 2.4.

Steady-state diffusion in multicomponent mixtures The expressions for diffusion in multicomponent systems become very complicated, but they can frequently be handled by using an effective diffusivity in Eq. (2.25), where the effective diffusivity of a component can be synthesized from its binary diffusivities with each of the other constituents [1].† Thus, in Eq. (2.25), $N_A + N_B$ is replaced by $\sum_{i=A}^n N_i$, where N_i is positive if diffusion is in the same direction as that of A and negative if in the opposite direction and D_{AB} is replaced by the effective $D_{A,m}$

$$D_{A,m} = \frac{N_A - y_A \sum_{i=A}^n N_i}{\sum_{i=A}^n \frac{1}{D_{Ai}} (y_i N_A - y_A N_i)} \quad (2.35)$$

The D_{Ai} are the binary diffusivities. This indicates that $D_{A,m}$ may vary considerably from one end of the diffusion path to the other, but a linear variation with distance can usually be assumed for practical calculations [1]. A common situation is when all the N 's except N_A are zero; i.e., all but one component is stagnant. Equation (2.35) then becomes [23]

$$D_{A,m} = \frac{1 - y_A}{\sum_{i=B}^n \frac{y_i}{D_{Ai}}} = \frac{1}{\sum_{i=B}^n \frac{y'_i}{D_{Ai}}} \quad (2.36)$$

Pg 655, 656
for gas & liq.

† Numbered references appear at the end of the chapter.

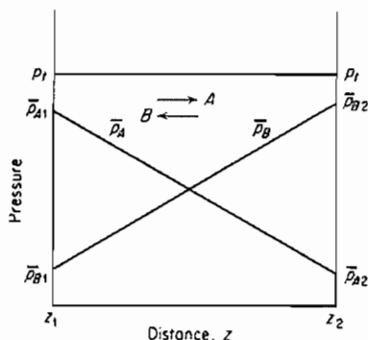


Figure 2.4 Equimolar counterdiffusion.

where y'_i is the mole fraction of component i on an A-free basis. The limitations of Eq. (2.35) and some suggestions for dealing with them have been considered [21].

Illustration 2.1 Oxygen (A) is diffusing through carbon monoxide (B) under steady-state conditions, with the carbon monoxide nondiffusing. The total pressure is $1 \times 10^5 \text{ N/m}^2$, and the temperature 0°C . The partial pressure of oxygen at two planes 2.0 mm apart is, respectively, 13 000 and 6500 N/m^2 . The diffusivity for the mixture is $1.87 \times 10^{-5} \text{ m}^2/\text{s}$. Calculate the rate of diffusion of oxygen in kmol/s through each square meter of the two planes.

SOLUTION Equation (2.30) applies. $D_{AB} = 1.87 \times 10^{-5} \text{ m}^2/\text{s}$, $p_t = 10^5 \text{ N/m}^2$, $z = 0.002 \text{ m}$, $R = 8314 \text{ N} \cdot \text{m}/\text{kmol} \cdot \text{K}$, $T = 273 \text{ K}$; $\bar{p}_{A,1} = 13 \times 10^3$, $\bar{p}_{B,1} = 10^5 - 13 \times 10^3 = 87 \times 10^3$, $\bar{p}_{A,2} = 6500$, $\bar{p}_{B,2} = 10^5 - 6500 = 93.5 \times 10^3$, all in N/m^2 .

$$\bar{p}_{B,M} = \frac{\bar{p}_{B,1} - \bar{p}_{B,2}}{\ln(\bar{p}_{B,1}/\bar{p}_{B,2})} = \frac{(87 - 93.5)(10^3)}{\ln(87/93.5)} = 90\,200 \text{ N/m}^2$$

$$N_A = \frac{D_{AB} p_t}{RTz \bar{p}_{B,M}} (\bar{p}_{A,1} - \bar{p}_{A,2}) = \frac{(1.87 \times 10^{-5})(10^5)(13 - 6.5)(10^3)}{8314(273)(0.002)(90.2 \times 10^3)} \\ = 2.97 \times 10^{-5} \text{ kmol/m}^2 \cdot \text{s} \quad \text{Ans.}$$

Illustration 2.2 Recalculate the rate of diffusion of oxygen (A) in Illustration 2.1, assuming that the nondiffusing gas is a mixture of methane (B) and hydrogen (C) in the volume ratio 2 : 1. The diffusivities are estimated to be $D_{\text{O}_2-\text{H}_2} = 6.99 \times 10^{-5}$, $D_{\text{O}_2-\text{CH}_4} = 1.86 \times 10^{-5} \text{ m}^2/\text{s}$.

SOLUTION Equation (2.25) will become Eq. (2.30) for this case. $p_t = 10^5 \text{ N/m}^2$, $T = 273 \text{ K}$, $\bar{p}_{A,1} = 13 \times 10^3$, $\bar{p}_{A,2} = 6500$, $\bar{p}_{B,M} = 90.2 \times 10^3$, all in N/m^2 ; $z = 0.002 \text{ m}$, $R = 8314 \text{ N} \cdot \text{m}/\text{kmol} \cdot \text{K}$, as in Illustration 2.1. In Eq. (2.36), $y'_B = 2/(2 + 1) = 0.667$, $y'_C = 1 - 0.667 = 0.333$, whence

$$D_{A,M} = \frac{1}{y'_B/D_{AB} + y'_C/D_{A,C}} = \frac{1}{0.667/(1.86 \times 10^{-5}) + 0.333/(6.99 \times 10^{-5})} \\ = 2.46 \times 10^{-5} \text{ m}^2/\text{s}$$

Therefore Eq. (2.30) becomes

$$N_A = \frac{(2.46 \times 10^{-5})(13\,000 - 6500)}{8314(273)(0.002)(90\,200)} = 3.91 \times 10^{-5} \text{ kmol/m}^2 \cdot \text{s} \quad \text{Ans.}$$

Table 2.1 Diffusivities of gases at standard atmospheric pressure, 101.3 kN/m²

System	Temp, °C	Diffusivity, m ² /s × 10 ⁵	Ref.
H ₂ -CH ₄	0	6.25†	3
O ₂ -N ₂	0	1.81	3
CO-O ₂	0	1.85	3
CO ₂ -O ₂	0	1.39	3
Air-NH ₃	0	1.98	26
Air-H ₂ O	25.9	2.58	7
	59.0	3.05	7
Air-ethanol	0	1.02	14
Air- <i>n</i> -butanol	25.9	0.87	7
	59.0	1.04	7
Air-ethyl acetate	25.9	0.87	7
	59.0	1.06	7
Air-aniline	25.9	0.74	7
	59.0	0.90	7
Air-chlorobenzene	25.9	0.74	7
	59.0	0.90	7
Air-toluene	25.9	0.86	7
	59.0	0.92	7

† For example, $D_{H_2-CH_4} = 6.25 \times 10^{-5} \text{ m}^2/\text{s}$.

Diffusivity of Gases

The diffusivity, or diffusion coefficient, D is a property of the system dependent upon temperature, pressure, and nature of the components. An advanced kinetic theory [12] predicts that in binary mixtures there will be only a small effect of composition. The dimensions of diffusivity can be established from its definition, Eq. (2.1), and are length²/time. Most of the values for D reported in the literature are expressed as cm²/s; the SI dimensions are m²/s. Conversion factors are listed in Table 1.5. A few typical data are listed in Table 2.1; a longer list is available in "The Chemical Engineers' Handbook" [18]. For a complete review, see Ref. 17.

Expressions for estimating D in the absence of experimental data are based on considerations of the kinetic theory of gases. The Wilke-Lee modification [25] of the Hirschfelder-Bird-Spotz method [11] is recommended for mixtures of nonpolar gases or of a polar with a nonpolar gas†

$$D_{AB} = \frac{10^{-4} (1.084 - 0.249 \sqrt{1/M_A + 1/M_B}) T^{3/2} \sqrt{1/M_A + 1/M_B}}{p_r (r_{AB})^2 f(kT/\epsilon_{AB})} \quad (2.37)$$

$\frac{\text{N/m}^2}{\text{atm}}$ $\frac{\text{cm}}{\text{m}}$

† The listed units must be used in Eq. (2.37). For D_{AB} , p_r , and T in units of feet, pounds force, hours, and degrees Rankine and all other quantities as listed above, multiply the right-hand side of Eq. (2.37) by 334.7.

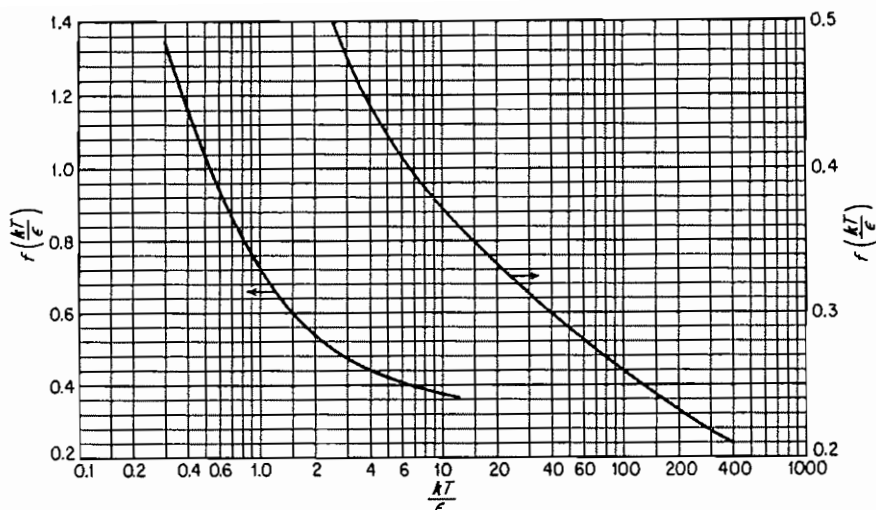


Figure 2.5 Collision function for diffusion.

where D_{AB} = diffusivity, m^2/s

T = absolute temperature, K

M_A, M_B = molecular weight of A and B, respectively, kg/kmol

p_i = abs pressure, N/m^2

r_{AB} = molecular separation at collision, $\text{nm} = (r_A + r_B)/2$

ϵ_{AB} = energy of molecular attraction $= \sqrt{\epsilon_A \epsilon_B}$

k = Boltzmann's constant

$f(kT/\epsilon_{AB})$ = collision function given by Fig. 2.5

The values of r and ϵ , such as those listed in Table 2.2, can be calculated from other properties of gases, such as viscosity. If necessary, they can be estimated for each component empirically [25]

$$r = 1.18 v^{1/3} \quad (2.38)$$

$$\frac{\epsilon}{k} = 1.21 T_b \quad (2.39)$$

where v is the molal volume of liquid at normal boiling point, m^3/kmol (estimate from Table 2.3), and T_b is the normal boiling point in Kelvins. In using Table 2.3, the contributions for the constituent atoms are added together. Thus, for toluene, C_7H_8 , $v = 7(0.0148) + 8(0.0037) - 0.015 = 0.1182$. Diffusion through air, when the constituents of the air remain in fixed proportions, is handled as if the air were a single substance.

Illustration 2.3 Estimate the diffusivity of ethanol vapor, $\text{C}_2\text{H}_5\text{OH}$, (A), through air (B) at 1 std atm pressure, 0°C .

SOLUTION $T = 273 \text{ K}$, $p_i = 101.3 \text{ kN}/\text{m}^2$, $M_A = 46.07$, $M_B = 29$. From Table 2.2 for air, $\epsilon_B/k = 78.6$, $r_B = 0.3711 \text{ nm}$. Values for ethanol will be estimated through Eqs. (2.38) and

Table 2.2 Force constants of gases as determined from viscosity data

Gas	ϵ/k , K	r , nm	Gas	ϵ/k , K	r , nm
Air	78.6	0.3711	HCl	344.7	0.3339
CCl ₄	322.7	0.5947	He	10.22	0.2551
CH ₃ OH	481.8	0.3626	H ₂	59.7	0.2827
CH ₄	148.6	0.3758	H ₂ O	809.1	0.2641
CO	91.7	0.3690	H ₂ S	301.1	0.3623
CO ₂	195.2	0.3941	NH ₃	558.3	0.2900
CS ₂	467	0.4483	NO	116.7	0.3492
C ₂ H ₆	215.7	0.4443	N ₂	71.4	0.3798
C ₃ H ₈	237.1	0.5118	N ₂ O	232.4	0.3828
C ₆ H ₆	412.3	0.5349	O ₂	106.7	0.3467
Cl ₂	316	0.4217	SO ₂	335.4	0.4112

† From R. A. Svehla, *NASA Tech. Rept. R-132*, Lewis Research Center, Cleveland, Ohio, 1962.

(2.39). From Table 2.3, $\epsilon_A = 2(0.0148) + 6(0.0037) + 0.0074 = 0.0592$, whence $r_A = 1.18(0.0592)^{1/3} = 0.46$ nm. The normal boiling point is $T_{b,A} = 351.4$ K and $\epsilon_A/k = 1.21(351.4) = 425$.

$$r_{AB} = \frac{0.46 + 0.3711}{2} = 0.416 \quad \frac{\epsilon_{AB}}{k} = \sqrt{425(78.6)} = 170.7$$

$$\frac{kT}{\epsilon_{AB}} = \frac{273}{170.7} = 1.599$$

Fig. 2.5:

$$f\left(\frac{kT}{\epsilon_{AB}}\right) = 0.595 \quad \sqrt{\frac{1}{M_A} + \frac{1}{M_B}} = 0.237$$

Eq. (2.37):

$$D_{AB} = \frac{10^{-4}[1.084 - 0.249(0.237)](273^{3/2})(0.237)}{(101.3 \times 10^3)(0.416)^2(0.595)}$$

$$= 1.05 \times 10^{-5} \text{ m}^2/\text{s}$$

The observed value (Table 2.1) is $1.02 \times 10^{-5} \text{ m}^2/\text{s}$.

Table 2.3 Atomic and molecular volumes

Atomic volume, $\text{m}^3/1000 \text{ atoms} \times 10^3$		Molecular volume, $\text{m}^3/\text{kmol} \times 10^3$		Atomic volume, $\text{m}^3/1000 \text{ atoms} \times 10^3$		Molecular volume, $\text{m}^3/\text{kmol} \times 10^3$	
Carbon	14.8	H ₂	14.3	Oxygen	7.4	NH ₃	25.8
Hydrogen	3.7	O ₂	25.6	In methyl esters	9.1	H ₂ O	18.9
Chlorine	24.6	N ₂	31.2	In higher esters	11.0	H ₂ S	32.9
Bromine	27.0	Air	29.9	In acids	12.0	COS	51.5
Iodine	37.0	CO	30.7	In methyl ethers	9.9	Cl ₂	48.4
Sulfur	25.6	CO ₂	34.0	In higher ethers	11.0	Br ₂	53.2
Nitrogen	15.6	SO ₂	44.8	Benzene ring: subtract	15	I ₂	71.5
In primary amines	10.5	NO	23.6	Naphthalene ring: subtract	30		
In secondary amines	12.0	N ₂ O	36.4				

Equation (2.37) shows D varying almost as $T^{3/2}$ (although a more correct temperature variation is given by considering also the collision function of Fig. 2.5) and inversely as the pressure, which will serve for pressures up to about 1500 kN/m^2 (15 atm) [19].

The coefficient of self-diffusion, or D for a gas diffusing through itself, can be determined experimentally only by very special techniques involving, for example, the use of radioactive tracers. It can be estimated from Eq. (2.37) by setting $A = B$.

Molecular Diffusion in Liquids

The integration of Eq. (2.4) to put it in the form of Eq. (2.22) requires the assumption that D_{AB} and c are constant. This is satisfactory for binary gas mixtures but not in the case of liquids, where both may vary considerably with concentration. Nevertheless, in view of our very meager knowledge of the D 's, it is customary to use Eq. (2.22), together with an average c and the best average D_{AB} available. Equation (2.22) is also conveniently written†

$$N_A = \frac{N_A}{N_A + N_B} \frac{D_{AB}}{z} \left(\frac{\rho}{M} \right)_{av} \ln \frac{N_A / (N_A + N_B) - x_{A2}}{N_A / (N_A + N_B) - x_{A1}} \quad (2.40)$$

where ρ and M are the solution density and molecular weight, respectively. As for gases, the value of $N_A / (N_A + N_B)$ must be established by the circumstances prevailing. For the most commonly occurring cases, we have, as for gases:

1. *Steady-state diffusion of A through nondiffusing B.* $N_A = \text{const}$, $N_B = 0$, whence

$$N_A = \frac{D_{AB}}{z x_{BM}} \left(\frac{\rho}{M} \right)_{av} (x_{A1} - x_{A2}) \quad (2.41)$$

where

$$x_{BM} = \frac{x_{B2} - x_{B1}}{\ln (x_{B2} / x_{B1})} \quad (2.42)$$

2. *Steady-state equimolar counterdiffusion.* $N_A = -N_B = \text{const}$.

$$N_A = \frac{D_{AB}}{z} (c_{A1} - c_{A2}) = \frac{D_{AB}}{z} \left(\frac{\rho}{M} \right)_{av} (x_{A1} - x_{A2}) \quad (2.43)$$

Illustration 2.4 Calculate the rate of diffusion of acetic acid (A) across a film of nondiffusing water (B) solution 1 mm thick at 17°C when the concentrations on opposite sides of the film are, respectively, 9 and 3 wt % acid. The diffusivity of acetic acid in the solution is $0.95 \times 10^{-9} \text{ m}^2/\text{s}$.

SOLUTION Equation (2.41) applies. $z = 0.001 \text{ m}$, $M_A = 60.03$, $M_B = 18.02$. At 17°C , the

† The component subscript on x_A indicates mole fraction A, to distinguish it from x meaning distance in the x direction.

density of the 9% solution is 1012 kg/m³. Therefore,

$$x_{A1} = \frac{0.09/60.03}{0.09/60.03 + 0.91/18.02} = \frac{0.0015}{0.0520} = 0.0288 \text{ mole fraction acetic acid}$$

$$x_{B1} = 1 - 0.0288 = 0.9712 \text{ mole fraction water}$$

$$M = \frac{1}{0.0520} = 19.21 \text{ kg/kmol} \quad \frac{\rho}{M} = \frac{1012}{19.21} = 52.7 \text{ kmol/m}^3$$

Similarly the density of the 3% solution is 1003.2 kg/m³, $x_{A2} = 0.0092$, $x_{B2} = 0.9908$, $M = 18.40$, and $\rho/M = 54.5$.

$$\left(\frac{\rho}{M}\right)_{av} = \frac{52.7 + 54.5}{2} = 53.6 \text{ kmol/m}^3 \quad x_{BM} = \frac{0.9908 - 0.9712}{\ln(0.9908/0.9712)} = 0.980$$

Eq. (2.41):

$$N_A = \frac{0.95 \times 10^{-9}}{0.001(0.980)} 53.6(0.0288 - 0.0092) = 1.018 \times 10^{-6} \text{ kmol/m}^2 \cdot \text{s} \quad \text{Ans.}$$

Diffusivity of Liquids

The dimensions for diffusivity in liquids are the same as those for gases, length²/time. Unlike the case for gases, however, the diffusivity varies appreciably with concentration. A few typical data are listed in Table 2.4, and larger lists are available [6, 8, 10, 15, 17].

Estimates of the diffusivity in the absence of data cannot be made with anything like the accuracy with which they can be made for gases because no sound theory of the structure of liquids has been developed. For dilute solutions of nonelectrolytes, the empirical correlation of Wilke and Chang [23, 24] is recommended.†

$$D_{AB}^0 = \frac{(117.3 \times 10^{-18})(\varphi M_B)^{0.5} T}{\mu v_A^{0.6}} \quad (2.44)$$

where D_{AB}^0 = diffusivity of A in very dilute solution in solvent B, m²/s

M_B = molecular weight of solvent, kg/kmol

T = temperature, K

μ = solution viscosity, kg/m · s

v_A = solute molal volume at normal boiling point, m³/kmol

= 0.0756 for water as solute

φ = association factor for solvent

= 2.26 for water as solvent [9]

= 1.9 for methanol as solvent

= 1.5 for ethanol as solvent

= 1.0 for unassociated solvents, e.g., benzene and ethyl ether

The value of v_A may be the true value [9] or, if necessary, estimated from the data of Table 2.3, except when water is the diffusing solute, as noted above. The

† The listed units must be used for Eq. (2.44). For D , μ , and T in units of feet, hours, pounds mass, and degrees Rankine, with v_A as listed above, multiply the right-hand side of Eq. (2.44) by 5.20×10^7 .

Table 2.4 Liquid diffusivities [8]

Solute	Solvent	Temp, °C	Solute concentration, kmol/m ³	Diffusivity,† m ² /s × 10 ⁹
Cl ₂	Water	16	0.12	1.26
HCl	Water	0	9	2.7
			2	1.8
		10	9	3.3
			2.5	2.5
		16	0.5	2.44
NH ₃	Water	5	3.5	1.24
		15	1.0	1.77
CO ₂	Water	10	0	1.46
		20	0	1.77
NaCl	Water	18	0.05	1.26
			0.2	1.21
			1.0	1.24
			3.0	1.36
			5.4	1.54
Methanol	Water	15	0	1.28
Acetic acid	Water	12.5	1.0	0.82
			0.01	0.91
		18.0	1.0	0.96
Ethanol	Water	10	3.75	0.50
			0.05	0.83
		16	2.0	0.90
n-Butanol	Water	15	0	0.77
CO ₂	Ethanol	17	0	3.2
Chloroform	Ethanol	20	2.0	1.25

† For example, D for Cl₂ in water is 1.26×10^{-9} m²/s.

association factor for a solvent can be estimated only when diffusivities in that solvent have been experimentally measured. If a value of ϕ is in doubt, the empirical correlation of Scheibel [20] can be used to estimate D . There is also some doubt about the ability of Eq. (2.44) to handle solvents of very high viscosity, say 0.1 kg/m · s (100 cP) or more. Excellent reviews of all these matters are available [4, 19].

The diffusivity in concentrated solutions differs from that in dilute solutions because of changes in viscosity with concentration and also because of changes in the degree of nonideality of the solution [16]

$$D_A \mu = (D_{BA}^0 \mu_A)^{x_A} (D_{AB}^0 \mu_B)^{x_B} \left(1 + \frac{d \log \gamma_A}{d \log x_A} \right) \quad (2.45)$$

where D_{AB}^0 is the diffusivity of A at infinite dilution in B and D_{BA}^0 the diffusivity of B at infinite dilution in A. The activity coefficient γ_A can typically be obtained from vapor-liquid equilibrium data as the ratio (at ordinary pressures)

of the real to ideal partial pressures of A in the vapor in equilibrium with a liquid of concentration x_A :

$$\gamma_A = \frac{\bar{p}_A}{x_A p_A} = \frac{y_A p_t}{x_A p_A} \quad (2.46)$$

and the derivative $(d \log \gamma_A)/(d \log x_A)$ can be obtained graphically as the slope of a graph of $\log \gamma_A$ vs. $\log x_A$.

For strong electrolytes dissolved in water, the diffusion rates are those of the individual ions, which move more rapidly than the large, undissociated molecules, although the positively and negatively charged ions must move at the same rate in order to maintain electrical neutrality of the solution. Estimates of these effects have been thoroughly reviewed [8, 19] but are beyond the scope of this book.

Illustration 2.5 Estimate the diffusivity of mannitol, $\text{CH}_2\text{OH}(\text{CHOH})_4\text{CH}_2\text{OH}$, $\text{C}_6\text{H}_{14}\text{O}_6$, in dilute solution in water at 20°C. Compare with the observed value, $0.56 \times 10^{-9} \text{ m}^2/\text{s}$.

SOLUTION From the data of Table 2.3

$$v_A = 0.0148(6) + 0.0037(14) + 0.0074(6) = 0.185$$

For water as solvent, $\phi = 2.26$, $M_B = 18.02$, $T = 293 \text{ K}$. For dilute solutions, the viscosity μ may be taken as that for water, $0.001005 \text{ kg/m} \cdot \text{s}$. Eq. (2.44):

$$D_{AB} = \frac{(117.3 \times 10^{-18})[2.26(18.02)]^{0.5}(293)}{0.001005(0.185)^{0.6}} = 0.601 \times 10^{-9} \text{ m}^2/\text{s} \quad \text{Ans.}$$

Illustration 2.6 Estimate the diffusivity of mannitol in dilute water solution at 70°C and compare with the observed value, $1.56 \times 10^{-9} \text{ m}^2/\text{s}$.

SOLUTION At 20°C, the observed $D_{AB} = 0.56 \times 10^{-9} \text{ m}^2/\text{s}$, and $\mu = 1.005 \times 10^{-3} \text{ kg/m} \cdot \text{s}$ (Illustration 2.5). At 70°C, the viscosity of water is $0.4061 \times 10^{-3} \text{ kg/m} \cdot \text{s}$. Equation (2.44) indicates that $D_{AB}\mu/T$ should be constant:

$$\frac{D_{AB}(0.4061 \times 10^{-3})}{70 + 273} = \frac{(0.56 \times 10^{-9})(1.005 \times 10^{-3})}{20 + 273}$$

$$D_{AB} = 1.62 \times 10^{-9} \text{ m}^2/\text{s} \text{ at } 70^\circ\text{C} \quad \text{Ans.}$$

Applications of Molecular Diffusion

While the flux relative to the average molar velocity J always means transfer *down* a concentration gradient, the flux N need not. For example, consider the dissolution of a hydrated salt crystal such as $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$ into pure water at 20°C. The solution in contact with the crystal surface contains Na_2CO_3 and H_2O at a concentration corresponding to the solubility of Na_2CO_3 in H_2O , or 0.0353 mole fraction Na_2CO_3 and 0.9647 mole fraction H_2O . For the Na_2CO_3 , transfer is from the crystal surface at a concentration 0.0353 outward to 0 mole fraction Na_2CO_3 in the bulk liquid. But the water of crystallization which dissolves must transfer outward in the ratio 10 mol H_2O to 1 mol Na_2CO_3 from a concentration

at the crystal surface of 0.9647 to 1.0 mole fraction in the bulk liquid, or transfer up a concentration gradient. Application of Eq. (2.40) confirms this. The expressions developed for the rate of mass transfer under conditions where molecular diffusion defines the mechanism of mass transfer (fluids which are stagnant or in laminar flow) are of course directly applicable to the experimental measurement of the diffusivities, and they are extensively used for this.

In the practical applications of the mass-transfer operations, the fluids are always in motion, even in batch processes, so that we do not have stagnant fluids. While occasionally the moving fluids are entirely in laminar flow, more frequently the motion is turbulent. If the fluid is in contact with a solid surface, where the fluid velocity is zero, there will be a region in predominantly laminar flow adjacent to the surface. Mass transfer must then usually take place through the laminar region, and molecular diffusion predominates there. When two immiscible fluids in motion are in contact and mass transfer occurs between them, there may be no laminar region, even at the interface between the fluids.

In practical situations like these, it has become customary to describe the mass-transfer flux in terms of mass-transfer coefficients. The relationships of this chapter are then rarely used directly to determine mass-transfer rates, but they are particularly useful in establishing the form of the mass-transfer coefficient-rate equations and in computing the mass-transfer coefficients for laminar flow.

MOMENTUM AND HEAT TRANSFER IN LAMINAR FLOW

In the flow of a fluid past a phase boundary, such as that through which mass transfer occurs, there will be a velocity gradient within the fluid, which results in a transfer of momentum through the fluid. In some cases there is also a transfer of heat by virtue of a temperature gradient. The processes of mass, momentum, and heat transfer under these conditions are intimately related, and it is useful to consider this point briefly.

Momentum Transfer

Consider the velocity profile for the case of a gas flowing past a flat plate, as in Fig. 2.6. Since the velocity at the solid surface is zero, there must necessarily be a layer (the laminar sublayer) adjacent to the surface where the flow is predominantly laminar. Within this region, the fluid can be imagined as being made up of thin layers sliding over each other at increasing velocities at increasing distances from the plate. The force per unit area parallel to the surface, or shearing stress τ , required to maintain their velocities is proportional to the velocity gradient,

$$\tau_{gc} = -\mu \frac{du}{dz} \quad (2.47)$$

where μ is the viscosity and z is measured as increasing in the direction toward

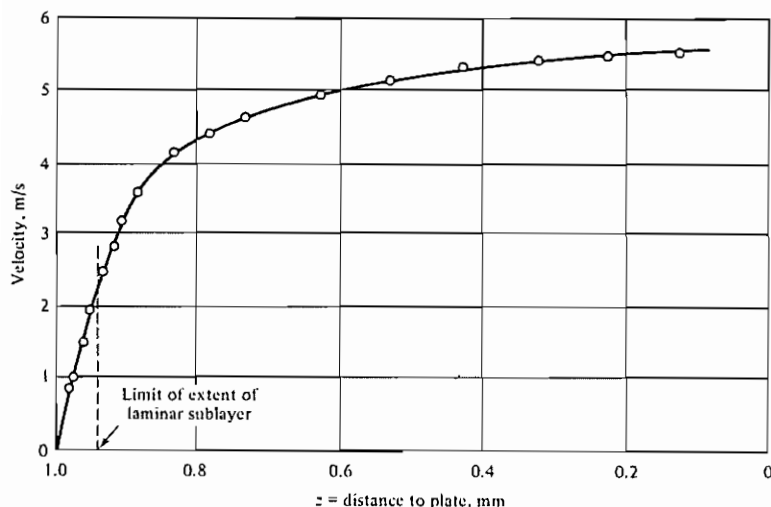


Figure 2.6 Velocity profile, flow of air along a flat plate. [Page, *et al.*, *Ind. Eng. Chem.*, **44**, 424 (1952).]

the surface. This can be written

$$\tau_{gc} = -\frac{\mu}{\rho} \frac{d(u\rho)}{dz} = -\nu \frac{d(u\rho)}{dz} \quad (2.48)$$

where ν is the kinematic viscosity, μ/ρ .

The kinematic viscosity has the same dimensions as diffusivity, length²/time, while the quantity $u\rho$ can be looked upon as a volumetric momentum concentration. The quantity τ_{gc} is the rate of momentum transfer per unit area, or flux of momentum. Equation (2.48) is therefore a rate equation analogous to Eq. (2.1) for mass flux. In the transfer of momentum in this manner there is of course no bulk flow of fluid from one layer to the other in the z direction. Instead, molecules in one layer, in the course of traveling in random directions, will move from a fast-moving layer to an adjacent, more slowly moving layer, thereby transmitting momentum corresponding to the difference in velocities of the layers. Diffusion in the z direction occurs by the same mechanism. At high molecular concentrations, such as in gases at high pressures or even more so in liquids, the molecular diameter becomes appreciable in comparison with the molecular movement between collisions, and momentum can be imagined as being transmitted directly through the molecules themselves [2]. Visualize, for example, a number of billiard balls arranged in a group in close contact with each other on a table. A moving cue ball colliding with one of the outermost balls of the packed group will transmit its momentum very rapidly to one of the balls on the opposite side of the group, which will then be propelled from its

original position. On the other hand, the cue ball is unlikely to move bodily through the group because of the large number of collisions it will experience. Thus, at high molecular concentrations the direct parallelism between molecular diffusivity and momentum diffusivity (or kinematic viscosity) breaks down: diffusion is much the slower process. It is interesting to note that a relatively simple kinetic theory predicts that both mass and momentum diffusivities are given by the same expression,

$$D_{AA} = \frac{\mu_A}{\rho_A} = \frac{w\lambda}{3} \quad (2.49)$$

where w is the average molecular velocity and λ is the mean free path of molecule. The Schmidt number, which is the dimensionless ratio of the two diffusivities, $Sc = \mu/\rho D$, should by this theory equal unity for a gas. A more sophisticated kinetic theory gives values from 0.67 to 0.83, which is just the range found experimentally at moderate pressures. For binary gas mixtures, Sc may range up to 5. For liquids, as might be expected, Sc is much higher: approximately 297 for self-diffusion in water at 25°C, for example, and ranging into the thousands for more viscous liquids and even for water with slowly diffusing solutes.

Heat Transfer

When a temperature gradient exists between the fluid and the plate, the rate of heat transfer in the laminar region of Fig. 2.6 is

$$q = -k \frac{dt}{dz} \quad (2.50)$$

where k is the thermal conductivity of the fluid. This can also be written

$$q = -\frac{k}{C_p \rho} \frac{d(t C_p \rho)}{dz} = -\alpha \frac{d(t C_p \rho)}{dz} \quad (2.51)$$

where C_p is the specific heat at constant pressure. The quantity $t C_p \rho$ may be looked upon as a volumetric thermal concentration, and $\alpha = k/C_p \rho$ is the thermal diffusivity, which, like momentum and mass diffusivities, has dimensions length²/time. Equation (2.51) is therefore a rate equation analogous to the corresponding equations for momentum and mass transfer.

In a gas at relatively low pressure the heat energy is transferred from one position to another by the molecules traveling from one layer to another at a lower temperature. A simplified kinetic theory leads to the expression

$$\alpha = \frac{k}{C_p \rho} = \frac{1}{3} w \lambda \frac{C_v}{C_p} \quad (2.52)$$

Equation (2.49) and (2.52) would give the dimensionless ratio $\nu/\alpha = C_p \mu/k$ equal to C_p/C_v . A more advanced kinetic theory modifies the size of the ratio, known as the Prandtl number Pr , and experimentally it has the range 0.65 to 0.9

for gases at low pressure, depending upon the molecular complexity of the gas. At high molecular concentrations, the process is modified. Thus for most liquids, Pr is larger ($Pr = 7.02$ for water at 20°C , for example).

The third dimensionless group, formed by dividing the thermal by the mass diffusivity, is the Lewis number, $Le = \alpha/D = Sc/Pr$, and it plays an important part in problems of simultaneous heat and mass transfer, as will be developed later.

We can summarize this brief discussion of the similarity between momentum, heat, and mass transfer as follows. An elementary consideration of the three processes leads to the conclusion that in certain simplified situations there is a direct analogy between them. In general, however, when three- rather than one-dimensional transfer is considered, the momentum-transfer process is of a sufficiently different nature for the analogy to break down. Modification of the simple analogy is also necessary when, for example, mass and momentum transfer occur simultaneously. Thus, if there were a net mass transfer toward the surface of Fig. 2.6, the momentum transfer of Eq. (2.48) would have to include the effect of the net diffusion. Similarly, mass transfer must inevitably have an influence on the velocity profile. Nevertheless, even the limited analogies which exist are put to important practical use.

NOTATION FOR CHAPTER 2

Consistent units in any system may be used, except in Eqs. (2.37) to (2.39) and (2.44).

c	concentration, mol/volume, mole L^3
C_p	heat capacity at constant pressure, FL/MT
C_v	heat capacity at constant volume, FL/MT
d	differential operator
D	diffusivity, L^2/Θ
D^0	diffusivity for a solute at infinite dilution, L^2/Θ
f	function
g_c	conversion factor, ML/FΘ^2
J	flux of diffusion relative to the molar average velocity, mole/L$^2\Theta$
k	thermal conductivity, FL/LTΘ
k	Boltzmann's constant, 1.38×10^{-16} erg/K
$\ln$	natural logarithm
Le	Lewis number = $k/\rho DC_p$, dimensionless
M	molecular weight, M/mole
n	number of moles, dimensionless
N	molar flux relative to a fixed surface, mole/L$^2\Theta$
p	vapor pressure, F/L^2
$\bar{p}$	partial pressure, F/L^2
p_t	total pressure, F/L^2
Pr	Prandtl number = $C_p\mu/k$, dimensionless
q	flux of heat, FL/L$^2\Theta$
r	molecular separation at collision, nm
R	universal gas constant, FL/mole T
R_i	rate of production of component i , mole/L$^3\Theta$
Sc	Schmidt number = $\mu/\rho D$, dimensionless

T	absolute temperature, T
T_b	normal boiling point, K
u	linear velocity, L/θ
v	liquid molar volume, $m^3/kmol$
V	volume, L^3
w	average molar velocity, L/θ
x	(with no subscript) distance in the x direction, L
x_i	mole-fraction concentration of component i in a liquid
y	(with no subscript) distance in the y direction, L
y_i	mole-fraction concentration of component i in a gas
y'_i	mole-fraction concentration of component i , diffusing-solute-free basis
z	distance in the z direction, L
α	thermal diffusivity, L^2/θ
γ	activity coefficient, dimensionless
∂	partial differential operator
Δ	difference
ϵ	energy of molecular attraction, ergs
θ	time
λ	mean free path of a molecule, L
μ	viscosity, $M/L\theta$
ν	kinematic viscosity or momentum diffusivity = μ/ρ , L^2/θ
ρ	density, M/L^3
τ	shearing stress, F/L^2
ϕ	dissociation factor for a solvent, dimensionless

Subscripts

A	component A
B	component B
i	component i
n	the last of n components
m	effective
M	mean
x	in the x direction
y	in the y direction
z	in the z direction
1	beginning of diffusion path
2	end of diffusion path

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PROBLEMS

2.1 In an oxygen-nitrogen gas mixture at 1 std atm, 25°C, the concentrations of oxygen at two planes 2 mm apart are 10 and 20 vol %, respectively. Calculate the flux of diffusion of the oxygen for the case where:

- (a) The nitrogen is nondiffusing. **Ans.**: 4.97×10^{-5} kmol/m² · s.
- (b) There is equimolar counterdiffusion of the two gases.

2.2 Repeat the calculations of Prob. 2.1 for a total pressure of 1000 kN/m².

2.3 Estimate the diffusivities of the following gas mixtures:

- (a) Acetone-air, STP.† **Ans.**: 9.25×10^{-6} m²/s.
- (b) Nitrogen-carbon dioxide, 1 std atm, 25°C.
- (c) Hydrogen chloride-air, 200 kN/m², 25°C. **Ans.**: 9.57×10^{-6} m²/s.
- (d) Toluene-air, 1 std atm, 30°C. Reported value [Gilliland, *Ind. Eng. Chem.*, **26**, 681 (1934)] = 0.088 cm²/s.
- (e) Aniline-air, STP. Observed value = 0.0610 cm²/s (Gilliland, *loc. cit.*).

2.4 The diffusivity of carbon dioxide in helium is reported to be 5.31×10^{-5} m²/s at 1 std atm, 3.2°C. Estimate the diffusivity at 1 std atm, 225°C. Reported value = 14.14×10^{-5} m²/s [Seager, Geertson, and Giddings, *J. Chem. Eng. Data*, **8**, 168 (1963)].

2.5 Ammonia is diffusing through a stagnant gas mixture consisting of one-third nitrogen and two-thirds hydrogen by volume. The total pressure is 30 lb_f/in² abs (206.8 kN/m²) and the temperature 130°F (54°C). Calculate the rate of diffusion of the ammonia through a film of gas 0.5 mm thick when the concentration change across the film is 10 to 5% ammonia by volume. **Ans.**: 2.05×10^{-4} kmol/m² · s.

2.6 Estimate the following liquid diffusivities:

- (a) Ethanol in dilute water solution, 10°C.

† STP (standard temperature and pressure is 0°C and 1 std atm.

(b) Carbon tetrachloride in dilute solution in methanol, 15°C (observed value = 1.69×10^{-5} cm²/s). **Ans.:** 1.49×10^{-9} m²/s.

2.7 The diffusivity of bromoform in dilute solution in acetone at 25°C is listed in Ref. 14, p. 63, as 2.90×10^{-5} cm²/s. Estimate the diffusivity of benzoic acid in dilute solution in acetone at 25°C. Reported value [Chang and Wilke, *J. Phys. Chem.*, **59**, 592 (1955)] = 2.62×10^{-5} cm²/s.

Ans.: 2.269×10^{-9} m²/s.

2.8 Calculate the rate of diffusion of NaCl at 18°C through a stagnant film of water 1 mm thick when the concentrations are 20 and 10%, respectively, on either side of the film. **Ans.:** 3.059×10^{-6} kmol/m² · s.

2.9 At 1 std atm, 100°C, the density of air = 0.9482 kg/m³, the viscosity = 2.18×10^{-5} kg/m · s, thermal conductivity = 0.0317 W/m · K, and specific heat at constant pressure = 1.047 kJ/kg · K. At 25°C, the viscosity = 1.79×10^{-5} kg/m · s.

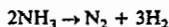
(a) Calculate the kinematic viscosity at 100°C, m²/s.

(b) Calculate the thermal diffusivity at 100°C, m²/s.

(c) Calculate the Prandtl number at 100°C.

(d) Assuming that for air at 1 std atm, $Pr = Sc$ and that $Sc = \text{const}$ with changing temperature, calculate D for air at 25°C. Compare with the value of D for the system O₂-N₂ at 1 std atm, 25°C, Table 2.1.

2.10 Ammonia is being cracked on a solid catalyst according to the reaction



At one place in the apparatus, where the pressure is 1 std atm abs and the temperature 200°C, the analysis of the bulk gas is 33.33% NH₃ (A), 16.67% N₂ (B), and 50.00% H₂ (C) by volume. The circumstances are such that NH₃ diffuses from the bulk-gas stream to the catalyst surface, and the products of the reaction diffuse back, as if by molecular diffusion through a gas film in laminar flow 1 mm thick. Estimate the local rate of cracking, kg NH₃/(m² catalyst surface) · s, which might be considered to occur if the reaction is diffusion-controlled (chemical reaction rate very rapid) with the concentration of NH₃ at the catalyst surface equal to zero.

Ans.: 0.0138 kg/m² · s.

2.11 A crystal of copper sulfate CuSO₄ · 5H₂O falls through a large tank of pure water at 20°C. Estimate the rate at which the crystal dissolves by calculating the flux of CuSO₄ from the crystal surface to the bulk solution. Repeat by calculating the flux of water. *Data and assumptions:* Molecular diffusion occurs through a film of water uniformly 0.0305 mm thick, surrounding the crystal. At the inner side of the film, adjacent to the crystal surface, the concentration of copper sulfate is its solubility value, 0.0229 mole fraction CuSO₄ (solution density = 1193 kg/m³). The outer surface of the film is pure water. The diffusivity of CuSO₄ is 7.29×10^{-10} m²/s.

MASS-TRANSFER COEFFICIENTS

We have seen that when a fluid flows past a solid surface under conditions such that turbulence generally prevails, there is a region immediately adjacent to the surface where the flow is predominantly laminar. With increasing distance from the surface, the character of the flow gradually changes, becoming increasingly turbulent, until in the outermost regions of the fluid fully turbulent conditions prevail. We have noted also that the rate of transfer of dissolved substance through the fluid will necessarily depend upon the nature of the fluid motion prevailing in the various regions.

In the turbulent region, particles of fluid no longer flow in the orderly manner found in the laminar sublayer. Instead, relatively large portions of the fluid, called *eddies*, move rapidly from one position to the other with an appreciable component of their velocity in the direction perpendicular to the surface past which the fluid is flowing. These eddies bring with them dissolved material, and the eddy motion thus contributes considerably to the mass-transfer process. Since the eddy motion is rapid, mass transfer in the turbulent region is also rapid, much more so than that resulting from molecular diffusion in the laminar sublayer. Because of the rapid eddy motion, the concentration gradients existing in the turbulent region will be smaller than those in the film, and Fig. 3.1 shows concentration gradients of this sort. In the experiment for which these are the data, air in turbulent motion flowed past a water surface, and water evaporated into the air. Samples of the air were taken at various distances from the surface, and the water-vapor concentration was determined by analysis. At the water surface, the water concentration in the gas was the same as the vapor pressure of pure water at the prevailing temperature. It was not possible to sample the gas very close to the water surface, but the rapid change in concentration in the region close to the surface and the slower change in the

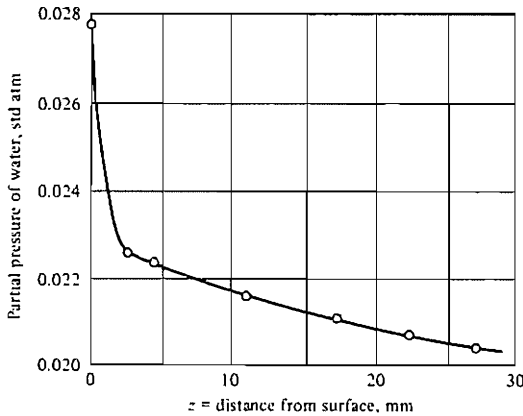


Figure 3.1 Evaporation of water into air [61].

outer turbulent region are nevertheless unmistakable. It is important also to note the general similarity of data of this sort to the velocity distribution shown in Fig. 2.6.

It is also useful to compare the data for mass transfer with similar data for heat transfer. Thus, in Fig. 3.2 are plotted the temperatures at various distances from the surface when air flowed past a heated plate. The large temperature gradient close to the surface and the lesser gradient in the turbulent region are again evident. It will generally be convenient to keep the corresponding heat-transfer process in mind when the mass-transfer process is discussed, since in many instances the methods of reasoning used to describe the latter are borrowed directly from those found successful with the former.

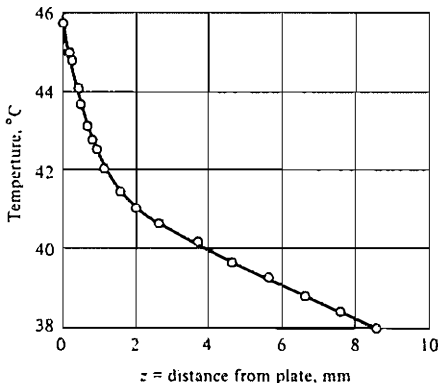


Figure 3.2 Heat transfer, flow of air past a heated plate [48].



c_{A1} is on surface on sugar

$c_{A2} = [conc]$ where constant

Mass-Transfer Coefficients

The mechanism of the flow process involving the movements of the eddies in the turbulent region is not thoroughly understood. On the other hand, the mechanism of molecular diffusion, at least for gases, is fairly well known, since it can be described in terms of a kinetic theory to give results which agree well with experience. It is therefore natural to attempt to describe the rate of mass transfer through the various regions from the surface to the turbulent zone in the same manner found useful for molecular diffusion. Thus, the $D_{AB}c/z$ of Eq. (2.22), which is characteristic of molecular diffusion, is replaced by F , a mass-transfer coefficient [5, 12]. For binary solutions,

$$N_A = \frac{N_A}{N_A + N_B} F \ln \frac{N_A / (N_A + N_B) - c_{A2}/c}{N_A / (N_A + N_B) - c_{A1}/c}$$

F vary along surface
use F_{av} then

(3.1)_N
multicomps: $\sum_{i=1}^N N_i$
replaced
 $N_A + N_B$

where c_A/c is the mole-fraction concentration, x_A for liquids, y_A for gases.† As in molecular diffusion, the ratio $N_A/(N_A + N_B)$ is ordinarily established by nondiffusional considerations.

Since the surface through which the transfer takes place may not be plane, so that the diffusion path in the fluid may be of variable cross section, N is defined as the flux at the phase interface, or boundary, where substance leaves or enters the phase for which F is the mass-transfer coefficient. N_A is positive when c_{A1} is at the beginning of the transfer path and c_{A2} at the end. In any case, one of these concentrations will be at the phase boundary. The manner of defining the concentration of A in the fluid will influence the value of F , and this is usually established arbitrarily. If mass transfer takes place between a phase boundary and a large quantity of unconfined fluid, as, for example, when a drop of water evaporates while falling through a great volume of air, the concentration of diffusing substance in the fluid is usually taken as the constant value found at large distances from the phase boundary. If the fluid is in a confining duct, so that the concentration is not constant along any position of the transfer path, the bulk-average concentration $\bar{c}_A$, as found by mixing all the fluid passing a given point, is usually used. In Fig. 3.3, where a liquid evaporates into the flowing gas, the concentration c_A of the vapor in the gas varies continuously from c_{A1} at the liquid surface to the value at $z = Z$. In this case c_{A2} in Eq. (3.1) would usually be taken as $\bar{c}_A$, defined by

$$\bar{c}_A = \frac{1}{\bar{u}_x S} \int_0^S u_x c_A dS \quad (3.2)$$

where $u_x(z)$ = velocity distribution in gas across the duct (= time average of u_x for turbulence)

$\bar{u}_x$ = bulk-average velocity (volumetric rate/duct cross section)

S = duct cross-sectional area

† Equation (3.1) is identical with the result obtained by combining Eqs. (21.4-11), (21.5-27), and (21.5-47) of Ref. 7 or by combining Eqs. (3.4) and (5.37) of Ref. 54, and is applicable to both low mass-transfer fluxes and to high fluxes as corrected through the film theory.

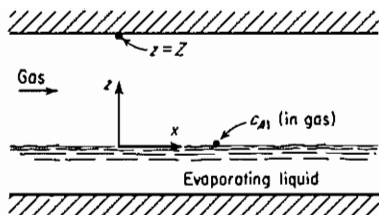


Figure 3.3 Mass transfer to a confined fluid.

In any case, one must know how the mass-transfer coefficient is defined in order to use it properly.

The E of Eq. (3.1) is a local mass-transfer coefficient, defined for a particular location on the phase-boundary surface. Since the value of E depends on the local nature of the fluid motion, which may vary along the surface, an average value E_{av} is sometimes used in Eq. (3.1) with constant c_{A1} and c_{A2} , which takes into account these variations in F . The effect of variation in c_{A1} and c_{A2} on the flux must be accounted for separately.

For multicomponent systems, there may be sufficiently important interaction between the components for the form of the binary system equation not to be exact [63]. Nevertheless, Eq. (3.1) can serve as a suitable approximation if $N_A + N_B$ is replaced by $\sum_{i=1}^n N_i$, where n is the number of components.

The two situations noted in Chap. 2, equimolar counterdiffusion and transfer of one substance through another which is not transferred, occur so frequently that special mass-transfer coefficients are usually used for them. These are defined by equations of the form

$$\text{Flux} = (\text{coefficient})(\text{concentration difference})$$

Since concentration may be defined in a number of ways and standards have not been established, we have a variety of coefficients for each situation:

Transfer of A through nontransferring B [$N_B = 0$, $N_A/(N_A + N_B) = 1$]:

$$N_A = \begin{cases} k_G(\bar{p}_{A1} - \bar{p}_{A2}) = k_y(y_{A1} - y_{A2}) = k_c(c_{A1} - c_{A2}) & \text{gases} \\ k_x(x_{A1} - x_{A2}) = k_L(c_{A1} - c_{A2}) & \text{liquids} \end{cases} \quad (3.3)$$

$$(3.4)$$

Equimolar countertransfer [$N_A = -N_B$, $N_A/(N_A + N_B) = \infty$]:

$$N_A = \begin{cases} k'_G(\bar{p}_{A1} - \bar{p}_{A2}) = k'_y(y_{A1} - y_{A2}) = k'_c(c_{A1} - c_{A2}) & \text{gases} \\ k'_x(x_{A1} - x_{A2}) = k'_L(c_{A1} - c_{A2}) & \text{liquids} \end{cases} \quad (3.5)$$

$$(3.6)$$

An expression of this sort was suggested as early as 1897 [45] for the dissolution of solids in liquids. These expressions are, of course, analogous to the definition of a heat-transfer coefficient h : $q = h(t_1 - t_2)$. Whereas the concept of the heat-transfer coefficient is generally applicable, at least in the absence of mass transfer, the coefficients of Eqs. (3.3) and (3.4) are more restricted. Thus, k_c

of Eq. (3.3) can be considered as a replacement of D_{AB}/z in an integration of Eq. (2.1), and the bulk-flow term of Eq. (2.4) has been ignored in equating this to N_A . The coefficients of Eqs. (3.3) and (3.4) are therefore generally useful only for low mass-transfer rates. Values measured under one level of transfer rate should be converted into F for use with Eq. (3.1) before being applied to another. To obtain the relation between F and the k 's, note that for gases, for example, F replaces $D_{AB}p_i/RTz$ in Eq. (2.25) and that k_G replaces $D_{AB}p_i/RTz\bar{p}_{B,M}$ in Eq. (2.30). From this it follows that $F = k_G\bar{p}_{B,M}$. In this manner the conversions of Table 3.1 have been obtained. Since the bulk-flow term $N_A + N_B$ of Eq. (2.4) is zero for equimolar countertransfer, $F = k'_y$ (gases) and $F = k'_x$ (liquids), and Eqs. (3.5) and (3.6) are identical with Eq. (3.1) for this case.

Table 3.1 Relations between mass-transfer coefficients

Rate equation		Units of coefficient
Equimolar counterdiffusion	Diffusion of A through nondiffusing B	
Gases		
$N_A = k'_G \Delta \bar{p}_A$	$N_A = k_G \Delta \bar{p}_A$	$\frac{\text{Moles transferred}}{(\text{Area})(\text{time})(\text{pressure})}$
$N_A = k'_y \Delta y_A$	$N_A = k_y \Delta y_A$	$\frac{\text{Moles transferred}}{(\text{Area})(\text{time})(\text{mole fraction})}$
$N_A = k'_c \Delta c_A$	$N_A = k_c \Delta c_A$	$\frac{\text{Moles transferred}}{(\text{Area})(\text{time})(\text{mol/vol})}$
	$W_A = k_Y \Delta Y_A$	$\frac{\text{Mass transferred}}{(\text{Area})(\text{time})(\text{mass A/mass B})}$
Conversions:		
$F = k_G \bar{p}_{B,M} = k_y \frac{\bar{p}_{B,M}}{p_i} = k_c \frac{\bar{p}_{B,M}}{RT} = \frac{k_y}{M_B} = k'_G p_i = k'_y = k'_c \frac{p_i}{RT} = k'_c c$		
Liquids		
$N_A = k'_L \Delta c_A$	$N_A = k_L \Delta c_A$	$\frac{\text{Moles transferred}}{(\text{Area})(\text{time})(\text{mol/vol})}$
$N_A = k'_x \Delta x_A$	$N_A = k_x \Delta x_A$	$\frac{\text{Moles transferred}}{(\text{Area})(\text{time})(\text{mole fraction})}$
Conversions:		
$F = k_x x_{B,M} = k_L x_{B,M} c = k'_L c = k'_L \frac{p}{M} = k'_x$		

Many data on mass transfer where $N_A/(N_A + N_B)$ is neither unity nor infinity have nevertheless been described in terms of the k -type coefficients. Before these can be used for other situations, they must be converted into F 's.

In a few limited situations mass-transfer coefficients can be deduced from theoretical principles. In the great majority of cases, however, we depend upon direct measurement under known conditions, for use later in design.

MASS-TRANSFER COEFFICIENTS IN LAMINAR FLOW

In principle, at least, we do not need mass-transfer coefficients for laminar flow, since molecular diffusion prevails, and the relationships of Chap. 2 can be used to compute mass-transfer rates. A uniform method of dealing with both laminar and turbulent flow is nevertheless desirable.

Mass-transfer coefficients for laminar flow should be capable of computation. To the extent that the flow conditions are capable of description and the mathematics remains tractable, this is so. These are, however, severe requirements, and frequently the simplification required to permit mathematical manipulation is such that the results fall somewhat short of reality. It is not our purpose to develop these methods in detail, since they are dealt with extensively elsewhere [6, 7]. We shall choose one relatively simple situation to illustrate the general technique and to provide some basis for considering turbulent flow.

Mass Transfer from a Gas into a Falling Liquid Film

Figure 3.4 shows a liquid falling in a thin film in laminar flow down a vertical flat surface while being exposed to a gas A, which dissolves in the liquid. The liquid contains a uniform concentration c_{A0} of A at the top. At the liquid

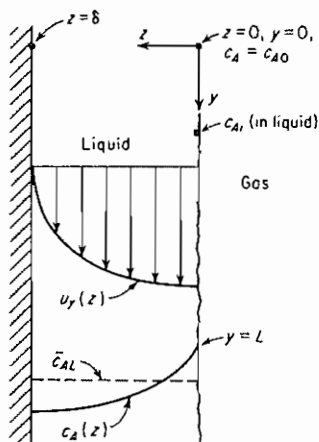


Figure 3.4 Falling liquid film.

surface, the concentration of the dissolved gas is $c_{A,i}$, in equilibrium with the pressure of A in the gas phase. Since $c_{A,i} > c_{A0}$, gas dissolves in the liquid. The problem is to obtain the mass-transfer coefficient k_L , with which the amount of gas dissolved after the liquid falls the distance L can be computed.

The problem is solved by simultaneous solution of the equation of continuity (2.17) for component A with the equations describing the liquid motion, the Navier-Stokes equations. The simultaneous solution of this formidable set of partial differential equations becomes possible only when several simplifying assumptions are made. For present purposes, assume the following:

1. There is no chemical reaction. R_A of Eq. (2.17) = 0.
2. Conditions do not change in the x direction (perpendicular to the plane of the paper, Fig. 3.4). All derivatives with respect to x of Eq. (2.17) = 0.
3. Steady-state conditions prevail. $\partial c_A / \partial \theta = 0$.
4. The rate of absorption of gas is very small. This means that u_z in Eq. (2.17) due to diffusion of A is essentially zero.
5. Diffusion of A in the y direction is negligible in comparison with the movement of A downward due to bulk flow. Therefore, $D_{AB} \partial^2 c_A / \partial y^2 = 0$.
6. Physical properties (D_{AB} , ρ , μ) are constant.

Equation (2.17) then reduces to

$$u_y \frac{\partial c_A}{\partial y} = D_{AB} \frac{\partial^2 c_A}{\partial z^2} \quad (3.7)$$

which states that any A added to the liquid running down at any location z , over an increment in y , got there by diffusion in the z direction. The equations of motion under these conditions reduce to

$$\mu \frac{d^2 u_y}{dz^2} + \rho g = 0 \quad (3.8)$$

The solution to Eq. (3.8), with the conditions that $u_y = 0$ at $z = \delta$ and that $du_y/dz = 0$ at $z = 0$, is well known

$$u_y = \frac{\rho g \delta^2}{2\mu} \left[1 - \left(\frac{z}{\delta} \right)^2 \right] = \frac{3}{2} \bar{u}_y \left[1 - \left(\frac{z}{\delta} \right)^2 \right] \quad (3.9)$$

where $\bar{u}_y$ is the bulk-average velocity. The film thickness is then

$$\delta = \left(\frac{3 \bar{u}_y \mu}{\rho g} \right)^{1/2} = \left(\frac{3 \mu \Gamma}{\rho^2 g} \right)^{1/3} \quad (3.10)$$

where Γ is the mass rate of liquid flow per unit of film width in the x direction. Substituting Eq. (3.9) into (3.7) gives

$$\frac{3}{2} \bar{u}_y \left[1 - \left(\frac{z}{\delta} \right)^2 \right] \frac{\partial c_A}{\partial y} = D_{AB} \frac{\partial^2 c_A}{\partial z^2} \quad (3.11)$$

which is to be solved under the following conditions:

1. At $z = 0$, $c_A = c_{A,i}$ at all values of y .
2. At $z = \delta$, $\partial c_A / \partial z = 0$ at all values of y , since no diffusion takes place into the solid wall.
3. At $y = 0$, $c_A = c_{A0}$ at all values of z .

The solution results in a general expression (an infinite series) giving c_A for any z and y , thus providing a concentration distribution $c_A(z)$ at $y = L$, as shown in Fig. 3.4. The bulk-average $\bar{c}_{A,L}$ at $y = L$ can then be found in the manner of Eq. (3.2). The result is [31]

$$\frac{c_{A,i} - \bar{c}_{A,L}}{c_{A,i} - c_{A0}} = 0.7857e^{-5.1213\eta} + 0.1001e^{-39.318\eta} + 0.03599e^{-105.64\eta} + \dots \quad (3.12)$$

where $\eta = 2D_{AB}L/3\delta^2\bar{u}_y$. The total rate of absorption is then $\bar{u}_y\delta(\bar{c}_{A,L} - c_{A0})$ per unit width of liquid film.

Alternatively, to obtain a local mass-transfer coefficient, we can combine Eq. (2.4) for the case of negligible bulk flow in the z direction ($N_A + N_B = 0$) with Eq. (3.4), keeping in mind that mass-transfer coefficients use fluxes at the phase interface ($z = 0$)

$$N_A = -D_{AB}\left(\frac{\partial c_A}{\partial z}\right)_{z=0} = k_L(c_{A,i} - \bar{c}_{A,L}) \quad (3.13)$$

In this case, however, because of the nature of the series which describes c_A , the derivative is undefined at $z = 0$. It is better therefore to proceed with an average coefficient for the entire liquid-gas surface. The rate at which A is carried by the liquid at any y , per unit width in the x direction, is $\bar{u}_y\delta\bar{c}_A$ mol/time. Over a distance δy , per unit width, therefore, the rate of solute absorption is, in mol/time,

$$\bar{u}_y\delta d\bar{c}_A = k_L(c_{A,i} - \bar{c}_A) \delta y \quad (3.14)$$

$$\bar{u}_y\delta \int_{\bar{c}_A=c_{A0}}^{\bar{c}_A=\bar{c}_{A,L}} \frac{d\bar{c}_A}{c_{A,i} - \bar{c}_A} = \int_0^L k_L \delta y = k_{L,av} \int_0^L \delta y \quad (3.15)$$

$$k_{L,av} = \frac{\bar{u}_y\delta}{L} \ln \frac{c_{A,i} - c_{A0}}{c_{A,i} - \bar{c}_{A,L}} \quad (3.16)$$

which defines the average coefficient. Now for small rates of flow or long times of contact of the liquid with the gas (usually for film Reynolds numbers $Re = 4\Gamma/\mu$ less than 100), only the first term of the series of Eq. (3.12) need be used. Substituting in Eq. (3.16) gives

$$k_{L,av} = \frac{\bar{u}_y\delta}{L} \ln \frac{e^{5.1213\eta}}{0.7857} = \frac{\bar{u}_y\delta}{L} (0.241 + 5.1213\eta) \approx 3.41 \frac{D_{AB}}{\delta} \quad (3.17)$$

$$\frac{k_{L,av}\delta}{D_{AB}} = Sh_{av} \approx 3.41 \quad (3.18)$$

where Sh represents the Sherwood number, the mass-transfer analog to the Nusselt number of heat transfer. A similar development for large Reynolds numbers or short contact time [53] leads to

$$k_{L,av} = \left(\frac{6D_{AB}\Gamma}{\pi\rho\delta L} \right)^{1/2} \quad (3.19)$$

$$Sh_{av} = \left(\frac{3}{2\pi} \frac{\delta}{L} Re Sc \right)^{1/2} \quad (3.20)$$

The product $Re Sc$ is the Péclet number Pe .

These average k_L 's can be used to compute the total absorption rate. Thus the average flux $N_{A,av}$ for the entire gas-liquid surface, per unit width, is the difference in rate of flow of A in the liquid at $y = L$ and at $y = 0$, divided by the liquid surface. This can be used with some mean concentration difference

$$N_{A,av} = \frac{\bar{u}_y \delta}{L} (\bar{c}_{A,L} - c_{A0}) = k_{L,av} (c_{A,i} - \bar{c}_A)_M \quad (3.21)$$

Substitution for $k_{L,av}$ from Eq. (3.16) shows that the logarithmic average of the difference at the top and bottom of the film is required

$$(c_{A,i} - \bar{c}_A)_M = \frac{(c_{A,i} - c_{A0}) - (c_{A,i} - \bar{c}_{A,L})}{\ln[(c_{A,i} - c_{A0}) / (c_{A,i} - \bar{c}_{A,L})]} \quad (3.22)$$

The experimental data show that the $k_{L,av}$ realized may be substantially larger than the theoretical values, even for low mass-transfer rates, owing to ripples and waves not considered in the analysis which form at values of Re beginning at about 25 [20]. The equations do apply for Re up to 1200 if ripples are suppressed by addition of wetting agents [37]. Rapid absorption, as for very soluble gases, produces important values of u_z , and this will cause further discrepancies due to alteration of the velocity profile in the film. The velocity profile may also be altered by flow of the gas, so that even in the simplest case, when both fluids move, k_L should depend on both flow rates.

Illustration 3.1 Estimate the rate of absorption of CO_2 into a water film flowing down a vertical wall 1 m long at the rate of 0.05 kg/s per meter of width at 25°C. The gas is pure CO_2 at 1 std atm. The water is essentially CO_2 -free initially.

SOLUTION The solubility of CO_2 in water at 25°C, 1 std atm, is $c_{A,i} = 0.0336$ kmol/m³ soln; $D_{AB} = 1.96 \times 10^{-9}$ m²/s; soln density $\rho = 998$ kg/m³; and viscosity $\mu = 8.94 \times 10^{-4}$ kg/m · s. $\Gamma = 0.05$ kg/m · s, $L = 1$ m.

$$\delta = \left(\frac{3\mu\Gamma}{\rho^2 g} \right)^{1/3} = \left[\frac{3(8.94 \times 10^{-4})(0.05)}{(998)^2(9.807)} \right]^{1/3} = 2.396 \times 10^{-4} \text{ m}$$

$$Re = \frac{4\Gamma}{\mu} = \frac{4(0.05)}{8.94 \times 10^{-4}} = 203$$

Consequently Eq. (3.19) should apply:

$$k_{L,av} = \left(\frac{6D_{AB}\Gamma}{\pi\rho\delta L} \right)^{1/2} = \left[\frac{6(1.96 \times 10^{-9})(0.05)}{\pi(998)(2.396 \times 10^{-4})(1)} \right]^{1/2}$$

$$= 2.798 \times 10^{-5} \text{ kmol/m}^2 \cdot \text{s} \cdot (\text{kmol/m}^3)$$

$$\bar{u}_y = \frac{\Gamma}{\rho\delta} = \frac{0.05}{998(2.396 \times 10^{-4})} = 0.209 \text{ m/s}$$

At the top, $c_{A,i} - \bar{c}_A = c_{A,i} - c_{A0} = c_{A,i} = 0.0336 \text{ kmol/m}^3$. At the bottom, $c_{A,i} - \bar{c}_{A,L} = 0.0336 - \bar{c}_{A,L} \text{ kmol/m}^3$. The flux of absorption is given by Eqs. (3.21) and (3.22):

$$\frac{0.209(2.396 \times 10^{-4})\bar{c}_{A,L}}{1} = \frac{(2.798 \times 10^{-5})[0.0336 - (0.0336 - \bar{c}_{A,L})]}{\ln[0.0336 / (0.0336 - \bar{c}_{A,L})]}$$

Therefore

$$\bar{c}_{A,L} = 0.01438 \text{ kmol/m}^3$$

and the rate of absorption is estimated to be

$$\bar{u}_y (\bar{c}_{A,L} - c_{A0}) = 0.209(2.396 \times 10^{-4})(0.01438 - 0)$$

$$= 7.2 \times 10^{-7} \text{ kmol/s} \cdot (\text{m of width})$$

The actual value may be substantially larger.

MASS-TRANSFER COEFFICIENTS IN TURBULENT FLOW

Most practically useful situations involve turbulent flow, and for these it is generally not possible to compute mass-transfer coefficients because we cannot describe the flow conditions mathematically. Instead, we rely principally on experimental data. The data are limited in scope, however, with respect to circumstances and situations as well as to range of fluid properties. Therefore it is important to be able to extend their applicability to situations not covered experimentally and to draw upon knowledge of other transfer processes (of heat, particularly) for help.

To this end, there are many theories which attempt to interpret or explain the behavior of mass-transfer coefficients, such as the film, penetration, surface-renewal, and other theories. They are all speculations, and are continually being revised. It is helpful to keep in mind that transfer coefficients, for both heat and mass transfer, are expedients used to deal with situations which are not fully understood. They include in one quantity effects which are the result of both molecular and turbulent diffusion. The relative contribution of these effects, and indeed the detailed character of the turbulent diffusion itself, differs from one situation to another. The ultimate interpretation or explanation of the transfer coefficients will come only when the problems of the fluid mechanics are solved, at which time it will be possible to abandon the concept of the transfer coefficient.

Eddy Diffusion

It will be useful first to describe briefly an elementary view of fluid turbulence, as a means of introducing the definitions of terms used in describing transfer

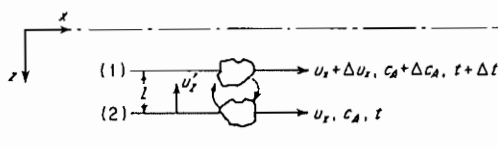


Figure 3.5 Eddy diffusion.

under turbulent conditions. Turbulence is characterized by motion of the fluid particles which is irregular with respect both to direction and time. Thus, for a fluid flowing turbulently in a duct (Fig. 3.5), the flow is, in the net, in the axial (or x) direction. At any location 2 within the central region of the cross section, the time average of the velocity may be u_x , but at any instant the velocity will actually be $u_x + u'_{ix}$, where u'_{ix} is the *deviating* or *fluctuating* velocity. Values of u'_{ix} will vary with time through a range of positive and negative values, the time average being zero, although u'_x , the square root of the time average of $(u'_{ix})^2$, will be finite. Although the time-average value of $u_z = 0$, since the net flow is axially directed, the deviating velocity in the z direction will be u'_{iz} at any instant.

The turbulent fluid is imagined to consist of lumps of fluid, or eddies, of a great size range [15, 28, 59]. The largest eddies, whose size in the case of flow in a pipe is of the order of the pipe radius, contain only perhaps 20 percent of the turbulent kinetic energy. These eddies, buffeted by smaller eddies, produce smaller and smaller eddies to which they transfer their energy. The medium, so-called *energy-containing eddies* make the largest contribution to the turbulent kinetic energy. The smallest eddies, which ultimately dissipate their energy through viscosity, are substantially reenergized by larger eddies, and a state of equilibrium is established. This range of the energy spectrum has lost all relationship to the means by which the turbulence was originally produced and depends only upon the rate at which energy is supplied and dissipated. For this reason it is called the *universal range*. Turbulence characteristic of this range is isotropic, *ie.*,

$$u'_x = u'_y = u'_z \quad (3.23)$$

Kolmogoroff [34] defined the velocity u'_d and length scale l_d of these small eddies in terms of the power input per unit mass of the fluid, P/m :

$$u'_d = \left(\frac{\nu P g_c}{m} \right)^{1/4} \quad (3.24)$$

$$l_d = \left(\frac{\nu^3 m}{P g_c} \right)^{1/4} \quad (3.25)$$

or, after elimination of ν ,

$$\frac{P g_c}{m} = \frac{u_d'^3}{l_d} \quad (3.26)$$

At equilibrium, when the energy lost from the medium-sized eddies must be at the same rate as that used to produce the small eddies, a similar relation should apply for the medium-sized eddies:

$$\frac{P_{G_c}}{m} = A \frac{u'^3}{l_c} \quad (3.27)$$

where A is a constant of the order of unity.

The size of the smallest eddies can be estimated in the case of pipe flow in the following manner.

Illustration 3.2 Consider the flow of water at 25°C in a 25-mm-ID tube at an average velocity $\bar{u} = 3 \text{ m/s}$. $\mu = 8.937 \times 10^{-4} \text{ kg/m} \cdot \text{s}$, $\rho = 997 \text{ kg/m}^3$.

$$d = 0.025 \text{ m} \quad \text{Re} = \frac{d\bar{u}\rho}{\mu} = 83 \text{ } 670$$

At this Reynolds number, the Fanning friction factor for smooth tubes ("The Chemical Engineers' Handbook," 5th ed., p. 5-22, McGraw-Hill, New York, 1973) is $f = 0.0047$.

$$\Delta p_f = \text{pressure drop} = \frac{2\rho f L \bar{u}^2}{dg_c}$$

where L = length of pipe. Take $L = 1 \text{ m}$, whence

$$\Delta p_f = \frac{2(997)(0.0047)(1)(3)^2}{0.025(1)} = 3374 \text{ N/m}^2$$

The power expended equals the product of volumetric flow rate and pressure drop:

$$P = \frac{\pi}{4} d^2 \bar{u} \Delta p_f = \frac{\pi}{4} (0.025)^2 (3)(3374) = 4.97 \text{ N} \cdot \text{m/s for 1-m pipe}$$

$$\text{Associated mass} = \frac{\pi}{4} d^2 L \rho = \frac{\pi}{4} (0.025)^2 (1)(997) = 0.489 \text{ kg}$$

$$\frac{P}{m} = \frac{4.97}{0.489} = 10.16 \text{ N} \cdot \text{m/kg} \cdot \text{s}$$

Eq. (3.25):

$$l_d = \left[\left(\frac{8.937 \times 10^{-4}}{997} \right)^3 \frac{1}{10.16(1)} \right]^{1/4} = 1.63 \times 10^{-5} \text{ m}$$

Eq. (3.24):

$$u'_d = \left[\frac{(8.937 \times 10^{-4})(10.16)(1)}{997} \right]^{1/4} = 0.0549 \text{ m/s}$$

A simpler view of turbulence, however, is helpful in understanding some of the concepts used in mass transfer. In Fig. 3.5, consider a second location 1, where the x velocity is larger than at 2 by an amount $\Delta u_x = -l du_x/dz$. The distance l , the *Prandtl mixing length*, is defined such that $u'_x = \Delta u_x = -l du_x/dz$. Owing to the z -directed fluctuating velocity, a particle of fluid, an eddy, may move from 2 to 1 at a velocity u'_{z2} , but it will be replaced by an eddy of equal volume moving from 1 to 2. It is imagined, in the Prandtl theory, that the eddies retain their identity during the interchange but blend into the fluid at their new location. This is clearly a great oversimplification in view of what was said before.

The mass velocity of eddy interchange is then $\rho u'_{iz}$, and thanks to the different velocities at 1 and 2 there will be a momentum transfer flux $\rho u'_{iz} u'_{ix}$. If u'_{iz} and u'_{ix} are essentially equal, there results an average shear stress due to the turbulent eddy interchange,

$$\tau_{\text{turb}} g_c = \rho |u'_{iz} u'_{ix}| = \rho u'_z u'_x = \rho (u'_x)^2 = \rho l^2 \left(\frac{du_x}{dz} \right)^2 \quad (3.28)$$

Molecular motion, of course, still also contributes to the shear stress, as given by Eq. (2.47), so that the *total* shear stress becomes

$$\tau g_c = -\mu \frac{du_x}{dz} + \rho l^2 \left(\frac{du_x}{dz} \right)^2 = - \left[\mu + \rho l^2 \left(-\frac{du_x}{dz} \right) \right] \frac{du_x}{dz} = - \left(\frac{\mu}{\rho} + E_v \right) \frac{d(u_x \rho)}{dz} \quad (3.29)$$

E_v is the eddy momentum diffusivity, $\text{length}^2/\text{time}$. While μ or $\nu = \mu/\rho$ is a constant for a given fluid at fixed temperature and pressure, E_v will depend on the local degree of turbulence. In the various regions of the duct, ν predominates near the wall, while in the turbulent core E_v predominates, to an extent depending upon the degree of turbulence.

The eddies bring about a transfer of dissolved solute, as we have mentioned before. The average concentration gradient between 1 and 2 in Fig. 3.5 is $\Delta c_A/l$, proportional to a local gradient, $-dc_A/dz$. The flux of A due to the interchange, $u'_z \Delta c_A$, and the concentration gradient can be used to define an eddy diffusivity of mass E_D , $\text{length}^2/\text{time}$

$$E_D = \frac{b_1 u'_z \Delta c_A}{\Delta c_A/l} = \frac{J_{A, \text{turb}}}{-dc_A/dz} \quad (3.30)$$

where b_1 is a proportionality constant. The *total* flux of A, due both to molecular and eddy diffusion, then will be

$$J_A = - (D_{AB} + E_D) \frac{dc_A}{dz} \quad (3.31)$$

As in the case of momentum transfer, D is a constant for a particular solution at fixed conditions of temperature and pressure, while E_D depends on the local turbulence intensity. D predominates in the region near the wall, E_D in the turbulent core.

In similar fashion, an eddy thermal diffusivity E_H , $\text{length}^2/\text{time}$, can be used to describe the flux of heat as a result of a temperature gradient,

$$E_H = \frac{b_2 u'_z \Delta(\rho C_p t)}{\Delta(\rho C_p t)/l} = \frac{q_{\text{turb}}}{-d(\rho C_p t)/dz} \quad (3.32)$$

where b_2 is a proportionality constant. The *total* heat flux due to conduction and eddy motion is

$$q = - (k + E_H \rho C_p) \frac{dt}{dz} = - (\alpha + E_H) \frac{d(t \rho C_p)}{dz} \quad (3.33)$$

As before, α for a given fluid is fixed for fixed conditions of temperature and pressure, but E_H varies with the degree of turbulence and hence with location.

The three eddy diffusivities of momentum, heat, and mass can be computed from measured velocity, temperature, and concentration gradients, respectively, in that order of increasing difficulty.

Momentum eddy diffusivity For turbulent flow in circular pipes, $Re = 50\,000$ to $350\,000$, the data of the turbulent core ($z_+ > 30$) yield [9]

$$\frac{2E_v}{d\bar{u}(f/2)^{0.5}} = 0.063 \left[1 + \left(\frac{2z}{d} \right)^2 - 2 \left(\frac{2z}{d} \right)^4 \right] \quad (3.34)$$

where z is the distance from the center and f is the Fanning friction factor. This shows that the maximum E_v occurs at a distance $z = d/4$. Thus, for water at 25°C in a smooth 5-cm-diameter tube at $Re = 150\,000$, the average velocity $\bar{u} = 2.69$ m/s, $f = 0.004$, and the maximum E_v calculates to be 2.1×10^{-4} m²/s, which is 238 times the kinematic viscosity ν . For the region closer to the wall ($z_+ < 5$), it is generally accepted that

$$\frac{E_v}{\nu} = \left(\frac{z_+}{K} \right)^3 \quad (3.35)$$

where

$$z_+ = \left(\frac{d}{2} - z \right) \frac{\bar{u}}{\nu} \left(\frac{f}{2} \right)^{0.5}$$

a dimensionless distance from the wall. The value of K has not been firmly established and may be in the range 8.9 [15] to 14.5 [40]. It is noteworthy that Eq. (3.35) indicates that eddy motion persists right up to the wall. It is also noteworthy that Murphree [43] deduced the proportionality between E_v and z_+^3 as early as 1932. For a *transition region* ($z_+ = 5$ to 30) [15],

$$\frac{E_v}{\nu} = \left(\frac{z_+}{11} \right)^2 \quad (3.36)$$

We can now estimate where the eddy and molecular momentum diffusivities are equal, a location which Levich suggested as the outer limit of the viscous sublayer [38]. For $E_v = \nu$, $z_+ = 11$, and for the same conditions as before (water at 25°C , $Re = 150\,000$, 5-cm tube), this occurs at $d/2 - z = 0.082$ mm from the wall. The small distance emphasizes the difficulty in obtaining reliable data.

Heat and mass eddy diffusivity The evidence is that with Prandtl and Schmidt numbers close to unity, as for most gases, the eddy diffusivities of heat and mass are equal to the momentum eddy diffusivity for all regions of turbulence [15]. For turbulent fluids where Prandtl and Schmidt numbers exceed unity, the ratios E_H/E_v and E_D/E_v will vary with location relative to the wall and in the turbulent core will lie generally in the range 1.2 to 1.3, with E_D and E_H essentially equal [44, 62]. For $z_+ = 0$ to 45, with Pr and $Sc > 1$, a critical analysis of the theoretical and experimental evidence [44] led to

$$\frac{E_D}{\nu} = \frac{E_H}{\nu} = \frac{0.00090z_+^3}{(1 + 0.0067z_+^2)^{0.5}} \quad (3.37)$$

which, as z_+ approaches zero, becomes

$$\frac{E_D}{\nu} = \frac{E_H}{\nu} = \left(\frac{z_+}{10.36} \right)^3 \quad (3.38)$$

The region close to the wall where molecular diffusivity exceeds eddy diffusivity of mass has been termed the *diffusional sublayer* [38], and we can estimate its extent in particular instances. For example, for flow of water at 25°C, in the pipe considered above, $\nu = 8.964 \times 10^{-7} \text{ m}^2/\text{s}$, and typically for solutes in water, D is of the order of $10^{-9} \text{ m}^2/\text{s}$. When E_D is set equal to D , Eq. (3.37) yields $z_+ = 1.08$; so the diffusional sublayer lies well within the viscous sublayer, the more so the smaller the D and the larger the Sc . At this z_+ for the pipe considered above and $Re = 150\,000$, the distance from the wall is 0.008 mm.

Mass-transfer coefficients covering the transfer of material from an interface to the turbulent zone will depend upon the total diffusivity $D + E_D$. Success in estimating this theoretically in the manner used for laminar flow will depend on knowledge of how the ratio E_D/D varies throughout the diffusion path. Knowledge of how the ratio E_v/ν varies will be of direct help only if $E_D = E_v$, which is evidently not true generally. But knowledge of the heat-transfer coefficients, which depend upon E_H/ν , should be useful in predicting the mass-transfer coefficients, since E_H and E_D are evidently the same except for the liquid metals.

Film Theory

This is the oldest and most obvious picture of the meaning of the mass-transfer coefficient, borrowed from a similar concept used for convective heat transfer. When a fluid flows turbulently past a solid surface, with mass transfer occurring from the surface to the fluid, the concentration-distance relation is as shown by the full curve of Fig. 3.6, the shape of which is controlled by the continuously varying ratio of E_D to D . The film theory postulates that the concentration will follow the broken curve of the figure, such that the entire concentration difference $c_{A1} - c_{A2}$ is attributed to molecular diffusion within an "effective" film of thickness z_f . It was early recognized [39] that if this theory were to have useful application, the film would have to be very thin, so that the

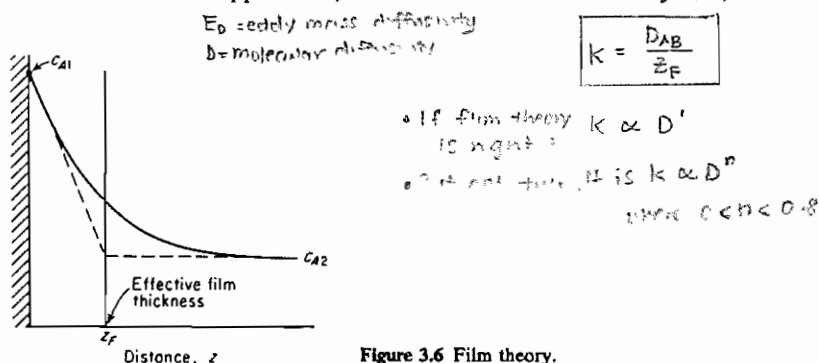


Figure 3.6 Film theory.

quantity of solute within the film would be small relative to the amount passing through it, or the concentration gradient would have to be set up quickly. The concentration gradient in the film is that characteristic of steady state.

In Eq. (3.1), it is clear that F has merely replaced the group $D_{AB}c/z$ of Eq. (2.22). The film theory states that z of Eq. (2.21) is z_F , the effective film thickness; this thickness depends upon the nature of the flow conditions. Similarly, the z 's of Eqs. (2.26), (2.30), (2.34), (2.40), (2.41), and (2.43) are interpreted to be z_F , incorporated into the k 's of Eqs. (3.3) to (3.6).

The film theory therefore predicts that F and the k -type mass-transfer coefficients for different solutes being transferred under the same fluid-flow conditions are directly proportional to the D 's for the solutes. On the other hand, we observe for turbulent flow a much smaller dependency, proportional to D^n , where n may be anything from nearly zero to 0.8 or 0.9, depending upon the circumstances. The simple film theory, in conflict with our knowledge of turbulent flow (see above), has therefore been largely discredited. Nevertheless it does well in handling the effect of high mass-transfer flux [Eq. (3.1)] and the effect of mass transfer on heat transfer and in predicting the effect of chemical reaction rate on mass transfer [14]. Mass- and heat-transfer coefficients are still frequently called *film coefficients*.

Mass Transfer at Fluid Surfaces

Because the velocity of a fluid surface, as in the contact of a gas and liquid, is not zero, a number of theories have been developed to replace the film theory.

Penetration theory Higbie [27] emphasized that in many situations the time of exposure of a fluid to mass transfer is short, so that the concentration gradient of the film theory, characteristic of steady state, would not have time to develop. His theory was actually conceived to describe the contact of two fluids, as in Fig. 3.7. Here, as Higbie depicted it in Fig. 3.7a, a bubble of gas rises through a liquid which absorbs the gas. A particle of the liquid b , initially at the top of the

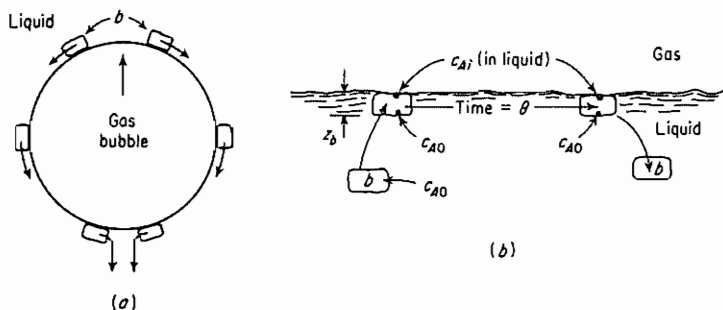


Figure 3.7 Penetration theory.

bubble, is in contact with the gas for the time θ required for the bubble to rise a distance equal to its diameter while the liquid particle slips along the surface of the bubble. An extension to cases where the liquid may be in turbulent motion, as in Fig. 3.7b shows an eddy b rising from the turbulent depths of the liquid and remaining exposed for a time θ to the action of the gas. In this theory the time of exposure is taken as constant for all such eddies or particles of liquid.

Initially, the concentration of dissolved gas in the eddy is uniformly c_{A0} , and internally the eddy is considered to be stagnant. When the eddy is exposed to the gas at the surface, the concentration in the liquid at the gas-liquid surface is $c_{A,i}$, which may be taken as the equilibrium solubility of the gas in the liquid. During the time θ , the liquid particle is subject to unsteady-state diffusion or penetration of solute in the z direction, and, as an approximation, Eq. (2.18) may be applied ν_{gas}

$$\frac{\partial c_A}{\partial \theta} = D_{AB} \frac{\partial^2 c_A}{\partial z^2} \quad (3.39)$$

For short exposure times, and with slow diffusion in the liquid, the molecules of dissolving solute are never able to reach the depth z_b corresponding to the thickness of the eddy, so that from the solute point of view, z_b is essentially infinite. The conditions on Eq. (3.39) then are

$$c_A = \begin{cases} c_{A0} & \text{at } \theta = 0 & \text{for all } z \\ c_{A,i} & \text{at } z = 0 & \text{for } \theta > 0 \\ c_{A0} & \text{at } z = \infty & \text{for all } \theta \end{cases}$$

Solving Eq. (3.39) and proceeding in the manner described earlier for a falling liquid film provides the average flux over the time of exposure

$$N_{A,av} = 2(c_{A,i} - c_{A0}) \sqrt{\frac{D_{AB}}{\pi\theta}} \quad (3.40)$$

and comparison with Eq. (3.4) shows

$$k_{L,av} = \sqrt{\frac{D_{AB}}{\pi\theta}} \propto \left(\frac{b}{\theta}\right)^{0.5} \quad (3.41)$$

with $k_{L,av}$ proportional to $D_{AB}^{0.5}$ for different solutes under the same circumstances. This indicated dependence on D is typical of short exposure times, where the depth of solute penetration is small relative to the depth of the absorbing pool [compare Eq. (3.19)]. As pointed out above, experience shows a range of exponents on D from nearly zero to 0.8 or 0.9.

Surface-renewal theories Danckwerts [13] pointed out that the Higbie theory with its constant time of exposure of the eddies of fluid at the surface is a special case of what may be a more realistic picture, where the eddies are exposed for varying lengths of time. The liquid-gas interface is then a mosaic of surface

elements of different exposure-time histories, and since the rate of solute penetration depends upon exposure time, the average rate for a unit surface area must be determined by summing up the individual values. On the assumption that the chance of a surface element's being replaced by another is quite independent of how long it has been in the surface, and if s is the fractional rate of replacement of elements, Danckwerts found

$$N_{A,av} = (c_{A,i} - c_{A0}) \sqrt{D_{AB}s} \quad (3.42)$$

and therefore $k_{L,av} = \sqrt{D_{AB}s} \quad k_L \propto \sqrt{s} \quad (3.43)$

Danckwerts pointed out that all theories of this sort, derived with the original boundary conditions on Eq. (3.39), will lead to $k_{L,av}$ proportional to $D_{AB}^{0.5}$, regardless of the nature of the surface-renewal rate s which may apply. Subsequently there have been many modifications of this approach [26, 35].

Combination film-surface-renewal theory Dobbins [17, 18], concerned with the rate of absorption of oxygen into flowing streams and rivers, pointed out that the film theory ($k_L \propto D_{AB}$) assumes a time of exposure of the surface elements sufficiently long for the concentration profile within the film to be characteristic of steady state, whereas the penetration and surface-renewal theories ($k_L \propto D_{AB}^{0.5}$) assume the surface elements to be essentially infinitely deep, the diffusing solute never reaching the region of constant concentration below. The observed dependence, $k_L \propto D_{AB}^n$, with n dependent upon circumstances, might be explained by allowing for a finite depth of the surface elements or eddies.

Accordingly, Dobbins replaced the third boundary condition on Eq. (3.39) by $c_A = c_{A0}$ for $z = z_b$, where z_b is finite. With Danckwerts' rate of renewal of the surface elements, he obtained

$$k_{L,av} = \sqrt{D_{AB}s} \coth \sqrt{\frac{sz_b^2}{D_{AB}}} \quad (3.44)$$

This is shown in Fig. 3.8. For rapid penetration (D_{AB} large), the rate of surface small (s small), or for thin surface elements, the mass-transfer coefficient takes on the character described by the film theory; whereas for slow penetration or

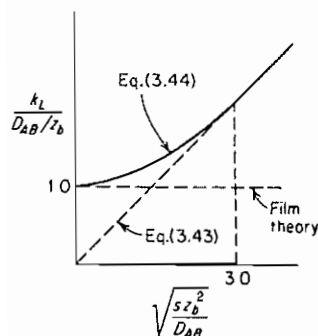


Figure 3.8 Combination film-surface-renewal theory. (After Dobbins [17].)

rapid renewal it follows Eq. (3.43). Consequently $k_L \propto D_{AB}^n$, where n may have any value between the limits 0.5 and 1.0, which could account for many observations. Toor and Marchello [59] have made similar suggestions.

King [33, 36] has proposed modifications which embody the notion of surface renewal with a dampening of the eddy diffusivities near the surface owing to surface tension, and these lead to values of n less than 0.5.

Surface-stretch theory Lightfoot and his coworkers [2, 3, 29, 50, 56] in a development which shows great promise, have extended the penetration-surface-renewal concepts to situations where the interfacial surface through which mass transfer occurs changes periodically with time. An example is a liquid drop such as ethyl acetate rising through a denser liquid such as water, with mass transfer of a solute such as acetic acid from the water to the drop. Such a drop, if relatively large, wobbles and oscillates, changing its shape as shown schematically in Fig. 3.9. If the central portion of the drop is thoroughly turbulent, the mass-transfer resistance of the drop resides in a surface layer of varying thickness. Similar situations occur while drops and bubbles form at nozzles and when liquid surfaces are wavy or rippled. For these cases the theory leads to

$$k_{L,av} = \frac{(A/A_r)\sqrt{D_{AB}/\pi\theta_r}}{\sqrt{\int_0^{\theta/\theta_r} \left(\frac{A}{A_r}\right)^2 d\theta}} \quad (3.45)$$

where A = time-dependent interface surface

A_r = reference value of A , defined for each situation

θ_r = const, with dimensions of time, defined for each situation; e.g., for drop formation θ_r might be drop-formation time

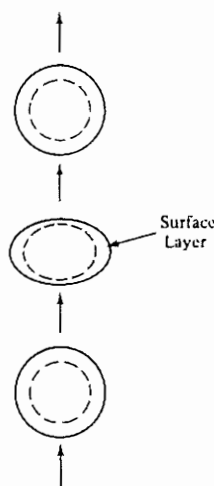


Figure 3.9 Liquid drop rising through another liquid. (After Rose and Kintner [50].)

The integral of Eq. (3.45) can be evaluated once the periodic nature of the surface variation is established, and this has been done for a number of situations.

Flow Past Solids; Boundary Layers

For the theories discussed above, where the interfacial surface is formed between two fluids, the velocity at that surface will not normally be zero. But when one of the phases is a solid, the fluid velocity parallel to the surface at the interface must of necessity be zero, and consequently the two circumstances are inherently different.

In Fig. 3.10 a fluid with uniform velocity u_0 and uniform solute concentration c_{A0} meets up with a flat solid surface AK . Since the velocity u_x is zero at the surface and rises to u_0 at some distance above the plate, the curve $ABCD$ separates the region of velocity u_0 from the region of lower velocity, called the *boundary layer*. The boundary layer may be characterized by laminar flow, as below curve AB , but if the velocity u_0 is sufficiently large, for values of $Re_x = xu_0\rho/\mu$ greater than approximately 5×10^5 , depending upon the turbulence intensity of the incident stream, the flow in the bulk of the boundary layer will be turbulent, as below the curve CD . Below the turbulent boundary layer there will be a thinner region, the viscous sublayer, extending from the plate to the curve FG . Boundary layers also occur for flow at the entrance to circular pipes, for flow along the outside of cylinders, and the like. When the surface over which the fluid passes is curved convexly in the direction of flow, as in flow at right angles to the outside of a cylinder and past a sphere, well-developed boundary layers form at very low rates of flow. At higher rates of flow, however, the boundary layer separates from the surface, and eddies form in the wake behind the object.

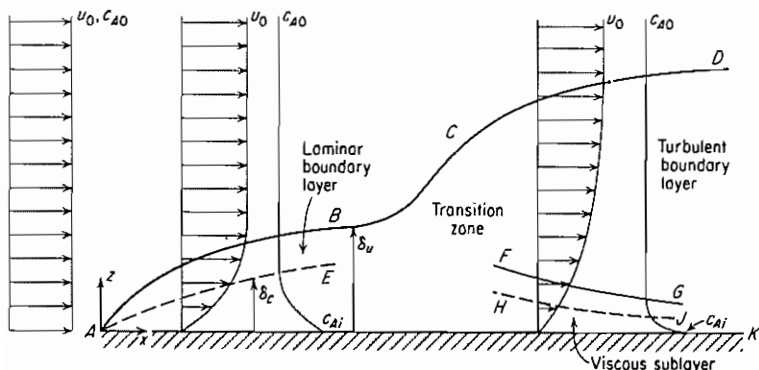


Figure 3.10 Boundary layer on a flat plate.

If mass transfer from the surface into the fluid occurs, as with a solid subliming into a gas or a solid dissolving into a liquid, the solute concentration in the fluid at the solid surface is everywhere $c_{A,i}$, greater than c_{A0} . There will be a curve AE and also one HJ , which separate regions of uniform concentration c_{A0} from regions of higher values of c_A , corresponding to a concentration boundary layer. In the region where only a laminar velocity boundary layer exists, the equations of motion and of mass transfer can be solved simultaneously to yield the concentration profile and, from its slope at the surface, the laminar mass-transfer coefficient [7]. This is a fairly complex problem particularly if account is taken of the influence of the flux of mass of A in the z direction on the velocity profile. If this influence is negligible, and if mass transfer begins at the leading edge A , it develops that the thickness of the velocity boundary layer δ_u and that of the concentration boundary layer δ_c are in the ratio $\delta_u/\delta_c = Sc^{1/2}$. Similar considerations apply if there is heat transfer between the plate and the fluid.

In laminar flow with low mass-transfer rates and constant physical properties past a solid surface, as for the two-dimensional laminar boundary layer of Fig. 3.10, the momentum balance or equation of motion (Navier-Stokes equation) for the x direction becomes [7]

$$u_x \frac{\partial u_x}{\partial x} + u_z \frac{\partial u_x}{\partial z} = \nu \left(\frac{\partial^2 u_x}{\partial x^2} + \frac{\partial^2 u_x}{\partial z^2} \right) \quad (3.46)$$

If there is mass transfer without chemical reaction, Eq. (2.17) gives the equation of continuity for substance A

$$u_x \frac{\partial c_A}{\partial x} + u_z \frac{\partial c_A}{\partial z} = D_{AB} \left(\frac{\partial^2 c_A}{\partial x^2} + \frac{\partial^2 c_A}{\partial z^2} \right) \quad (3.47)$$

and if there is heat transfer between the fluid and the plate, Eq. (2.19) provides

$$u_x \frac{\partial t}{\partial x} + u_z \frac{\partial t}{\partial z} = \alpha \left(\frac{\partial^2 t}{\partial x^2} + \frac{\partial^2 t}{\partial z^2} \right) \quad (3.48)$$

which are to be solved simultaneously, together with the equation of continuity (2.14). It is clear that these equations are all of the same form, with u_x , c_A , and t and the three diffusivities of momentum ν , mass D_{AB} , and heat α replacing each other in appropriate places.

In solving these, dimensionless forms of the variables are usually substituted

$$\frac{u_x - (u_{x,z=0} = 0)}{u_0 - (u_{x,z=0} = 0)} \quad \frac{c_A - c_{A,i}}{c_{A0} - c_{A,i}} \quad \text{and} \quad \frac{t - t_i}{t_0 - t_i}$$

Then the boundary conditions become identical. Thus, for Fig. 3.10, at $z = 0$ all three dimensionless variables are zero, and at $z = \infty$, all three equal unity. Consequently the form of the solutions, which provide dimensionless velocity, concentration, and temperature profiles, are the same. Indeed, if all three diffusivities are equal, so that $Sc = Pr = 1$, the profiles in dimensionless form are identical. The initial slopes of the concentration, temperature, and velocity

profiles provide the means of computing the corresponding local transfer coefficients

$$N_A = -D_{AB} \left(\frac{\partial c_A}{\partial z} \right)_{z=0} = k_L (c_{A,i} - c_{A0}) \quad (3.49)$$

$$q = -\alpha \left[\frac{\partial (t C_p \rho)}{\partial z} \right]_{z=0} = h (t_i - t_0) \quad (3.50)$$

$$\tau_i g_c = \nu \left[\frac{\partial (u_x \rho)}{\partial z} \right]_{z=0} = \frac{f}{2} u_0 (\rho u_0 - 0) \quad (3.51)$$

where f is the dimensionless friction factor and $f u_0 / 2$ might be considered a momentum-transfer coefficient. When the coefficients are computed and arranged as dimensionless groups, the results are all of the same form, as might be expected. In particular, for the flat plate of Fig. 3.10 at low mass-transfer rates,

$$\frac{Nu}{Re_x Pr^{1/3}} = \frac{Sh}{Re_x Sc^{1/3}} = \frac{f}{2} = 0.332 Re_x^{-1/2} \quad (3.52)$$

and the average coefficients provide Nu_{av} and Sh_{av} given by the same expressions with 0.332 replaced by 0.664. These show the mass-transfer coefficients to vary as $D_{AB}^{2/3}$, which is typical of the results of boundary-layer calculations. Equation (3.52) is applicable for cases of low mass-transfer rate, with heat or mass transfer beginning at the leading edge of the plate, and for Re_x up to approximately 80 000; for heat transfer, $Pr > 0.6$; and for constant t_0 and c_{A0} .

In the regions where a turbulent boundary layer and a viscous sublayer exist, the calculations for the transfer coefficients depend upon the expression chosen for the variation of the eddy diffusivities with distance from the wall. Alternatively, a great reliance is put upon experimental measurements. Results will be summarized later.

$$K = f(\tau_w, D_{AB}, \mu, u, D) \quad \text{Dimensional analysis}$$

MASS-, HEAT-, AND MOMENTUM-TRANSFER ANALOGIES

There are many more data available for pressure-drop friction and heat transfer than for mass transfer. The similarity of Eqs. (3.46) to (3.48) for the transfer processes of momentum, mass, and heat, and the identical solutions, Eq. (3.52), lead to the possibility of deducing the mass-transfer characteristics for other situations from knowledge of the other two.

For the case of the flat plate of Fig. 3.10 and other similar situations, the friction factor corresponds to a skin-friction drag along the surface. For flow where separation occurs, such as for flow past a sphere or at right angles to a cylinder or any bluff object, one might expect the friction factor based on total drag, which includes not only skin friction but form drag due to flow separation as well, to follow a function different from that of the mass- and heat-transfer groups.

In the case of turbulent flow, the differential equations will contain time-averaged velocities and in addition the eddy diffusivities of momentum, mass, and heat transfer. The resulting equations cannot be solved for lack of information about the eddy diffusivities, but one might expect results of the form

$$\frac{Sh}{Re Sc} = \psi_1\left(\frac{f}{2}, Sc, \frac{E_v}{E_D}\right) = \psi_2\left(\frac{f}{2}, Sc, \frac{E_D}{\nu}\right) \quad (3.53)$$

$$\frac{Nu}{Re Pr} = \psi_1\left(\frac{f}{2}, Pr, \frac{E_v}{E_H}\right) = \psi_2\left(\frac{f}{2}, Pr, \frac{E_H}{\nu}\right) \quad (3.54)$$

Successful completion of an analogy requires, therefore, knowledge of how the ratios E_v/ν , E_D/E_v , and E_H/E_v vary with distance from the fluid interface. It is customary arbitrarily to set $E_D/E_v = E_H/E_v = 1$, despite experimental evidence to the contrary and to make some arbitrary decision about E_v/ν . With these assumptions, however, experimental velocity profiles permit prediction of the profiles and coefficients for mass and heat transfer. Further, assuming only $E_D = E_H$ (which is much more reasonable) permits information on heat transfer to be converted directly for use in mass-transfer calculations and vice versa.

Let us sum up and further generalize what can be done with the analogies:

1. For analogous circumstances, temperature and concentration profiles in dimensionless form and heat- and mass-transfer coefficients in the form of dimensionless groups, respectively, are given by the same functions. To convert equations or correlations of data on heat transfer and temperatures to corresponding mass transfer and concentrations, the dimensionless groups of the former are replaced by the corresponding groups of the latter. Table 3.2 lists the commonly appearing dimensionless groups. The limitations are:
 - a. The flow conditions and geometry must be the same.
 - b. Most heat-transfer data are based on situations involving no mass transfer. Use of the analogy would then produce mass-transfer coefficients corresponding to no net mass transfer, in turn corresponding most closely to k'_G , k'_c , or k'_y ($= F$). Sherwood numbers are commonly written in terms of any of the coefficients, but when derived by replacement of Nusselt numbers for use where the net mass transfer is not zero, they should be taken as $Sh = Fl/cD_{AB}$, and the F used with Eq. (3.1). Further, the result will be useful only in the absence of chemical reaction.
 - c. The boundary conditions which would be used to solve the corresponding differential equations must be analogous. For example, in the case of the mass-transfer problem of the falling film of Fig. 3.4, the analogous circumstances for heat transfer would require heat transfer from the gas (not from the solid wall) to the liquid, the wall to be impervious to the transfer of heat, heat transfer beginning at the same value of y as mass transfer (in this case at $y = 0$), a constant temperature for the liquid-gas interface, and constant fluid properties.
 - d. For turbulent flow, $E_H = E_D$ at any location.

Table 3.2 Corresponding dimensionless groups of mass and heat transfer

No.	Mass transfer	Heat transfer
1	$\frac{c_A - c_{A1}}{c_{A2} - c_{A1}}$	$\frac{t - t_1}{t_2 - t_1}$
2	Reynolds number $Re = \frac{\rho u \mu}{\mu}$	Reynolds number $Re = \frac{\rho u \mu}{\mu}$
3	Schmidt number $Sc = \frac{\mu}{\rho D_{AB}} = \frac{\nu}{D_{AB}}$	Prandtl number $Pr = \frac{C_p \mu}{k} = \frac{\nu}{\alpha}$
4	Sherwood number $Sh = \frac{Fl}{c D_{AB}}, \frac{k_G \bar{p}_{B,M} R T l}{p_i D_{AB}},$ $\frac{k_c \bar{p}_{B,M} l}{p_i D_{AB}}, \frac{k'_c l}{D_{AB}}, \frac{k'_c R T l}{p_i D_{AB}}, \text{etc.}$	Nusselt number $Nu = \frac{h l}{k}$
5	ab† natural & convection Grashof number† $Gr_D = \frac{g l^3 \Delta \rho \left(\frac{\rho}{\mu}\right)^2}{\rho}$	Grashof number† $Gr_H = \frac{g l^3 \beta \Delta t \left(\frac{\rho}{\mu}\right)^2}{\rho}$
6	Peclet number $Pe_D = Re Sc = \frac{\rho u D_{AB}}{D_{AB}}$	Peclet number $Pe_H = Re Pr = \frac{C_p \rho u l}{k} = \frac{\rho u l}{\alpha}$
7	Stanton number $St_D = \frac{Sh}{Re Sc} = \frac{Sh}{Pe_H} = \frac{F}{c u},$ $\frac{F}{G}, \frac{k_G \bar{p}_{B,M} M_{A,v}}{\rho u}, \text{etc.}$	Stanton number $St_D = \frac{Nu}{Re Pr} = \frac{Nu}{Pe_H} = \frac{h}{C_p \rho u}$
8	$j_D = St_D Sc^{2/3}$	$j_H = St_H Pr^{2/3}$

† The Grashof number appears in cases involving natural convection; $\Delta \rho = |\rho_1 - \rho_2|$, $\Delta t = |t_1 - t_2|$, in the same phase.

- Friction factors and velocity profiles can be expected to correlate with the corresponding heat- and mass-transfer quantities only if $E_o = E_H = E_D$ in turbulent flow and in extension to viscous sublayers only if $E_o/\nu = E_H/\alpha = E_D/D_{AB}$. For either laminar or turbulent flow, the friction factors must indicate skin friction and not form drag as well. In general, it is safest to avoid the friction analogy to mass and heat transfer.

These analogies are particularly useful in extending the very considerable amount of information on heat transfer to yield corresponding information on

mass transfer, which is more likely to be lacking. Alternatively, local mass-transfer coefficients can be measured with relative ease through sublimation or dissolution of solids, and these can be converted into the analogous local heat-transfer coefficients, which are difficult to obtain [57].

Illustration 3.3 What is the heat-transfer analog to Eq. (3.12)?

SOLUTION The equation remains the same with the dimensionless concentration ratio replaced by $(t_i - t_L)/(t_i - t_0)$; the dimensionless group

$$\eta = \frac{2D_{AB}L}{3\delta^2 \bar{u}_y} = \frac{2}{3} \frac{D_{AB}}{\delta \bar{u}_y} \frac{L}{\delta} = \frac{2}{3} \frac{L}{Pe_D} \frac{L}{\delta}$$

for mass transfer is replaced by

$$\eta = \frac{2}{3} \frac{L}{Pe_H} \frac{L}{\delta} = \frac{2}{3} \frac{\alpha}{\delta \bar{u}_y} \frac{L}{\delta} = \frac{2\alpha L}{3\delta^2 \bar{u}_y} \quad \text{Ans.}$$

Illustration 3.4 For flow of a fluid at right angles to a circular cylinder, the average heat-transfer coefficient (averaged around the periphery) for fluid Reynolds numbers in the range 1 to 4000 is given by [19]

$$Nu_{av} = 0.43 + 0.532 Re^{0.5} Pr^{0.31}$$

where Nu and Re are computed with the cylinder diameter and fluid properties are taken at the mean of the cylinder and bulk-fluid temperatures.

Estimate the rate of sublimation of a cylinder of uranium hexafluoride, UF_6 , 6 mm diameter, exposed to an airstream flowing at a velocity of 3 m/s. The surface temperature of the solid is 43°C , at which temperature the vapor pressure of UF_6 is 400 mmHg (53.32 kN/m²). The bulk air is at 1 std atm pressure, 60°C .

SOLUTION The analogous expression for the mass-transfer coefficient is

$$Sh_{av} = 0.43 + 0.532 Re^{0.5} Sc^{0.31}$$

Along the mass-transfer path (cylinder surface to bulk air) the average temperature is 51.5°C , and the average partial pressure of UF_6 is 200 mmHg (26.66 kN/m²), corresponding to $26.66/101.33 = 0.263$ mole fraction UF_6 , 0.737 mole fraction air. The corresponding physical properties of the gas, at the mean temperature and composition, are estimated to be: density = 4.10 kg/m^3 , viscosity = $2.7 \times 10^{-5} \text{ kg/m} \cdot \text{s}$, diffusivity = $9.04 \times 10^{-6} \text{ m}^2/\text{s}$.

$$Re = \frac{d u \rho}{\mu} = \frac{(6 \times 10^{-3})(3)(4.10)}{2.7 \times 10^{-5}} = 2733 \quad Sc = \frac{\mu}{\rho D} = \frac{2.7 \times 10^{-5}}{4.10(9.04 \times 10^{-6})} = 0.728$$

$$Sh_{av} = 0.43 + 0.532(2733)^{0.5}(0.728)^{0.31} = 25.6 = \frac{F_{av} d}{c D_{AB}}$$

$$c = \frac{1}{22.41} \frac{273.2}{273.2 + 51.5} = 0.0375 \text{ kmol/m}^3$$

$$F_{av} = \frac{Sh_{av} c D_{AB}}{d} = \frac{25.6(0.0375)(9.04 \times 10^{-6})}{6 \times 10^{-3}} = 1.446 \times 10^{-3} \text{ kmol/m}^2 \cdot \text{s}$$

In this case, $N_B(\text{air}) = 0$, so that $N_A/(N_A + N_B) = 1.0$. Since $c_A/c = 53.32/101.33 = 0.526$ mole fraction, Eq. (3.1) provides

$$N_A = 1.44 \times 10^{-3} \ln \frac{1 - 0}{1 - 0.526} = 1.08 \times 10^{-3} \text{ kmol } UF_6/\text{m}^2 \cdot \text{s}$$

which is the mass-transfer flux based on the entire cylinder area.

Note 1 The calculated N_A is an instantaneous flux. Mass transfer will rapidly reduce the cylinder diameter, so that the Reynolds number and hence F will change with time. Furthermore, the surface will not remain in the form of a circular cylinder owing to the variation of the local F about the perimeter, so that the empirical correlation for Nu_{av} and Sh_{av} will then no longer apply.

Note 2 Use of k_c or k_G computed from a simple definition of Sh produces incorrect results because of the relatively high mass-transfer rate, which is not allowed for in the heat-transfer correlation. Thus, $k_c = k_G RT = Sh_{av} D_{AB}/d = 25.6(9.04 \times 10^{-6})/(6 \times 10^{-3}) = 0.0386$ kmol/m² · s · (kmol/m³) or $k_G = 0.0386/8.314(273.2 + 51.5) = 1.43 \times 10^{-5}$ kmol/m² · s · (kN/m²). This leads [Eq. (3.3)] to $N_A = (1.43 \times 10^{-5})(53.32 - 0) = 7.62 \times 10^{-4}$ kmol/m² · s, which is too low. However, if k_G is taken to be $1.43 \times 10^{-5} p_i/\bar{p}_{B,M}$, the correct answer will be given by Eq. (3.3).

Turbulent Flow in Circular Pipes

A prodigious amount of effort has been spent in developing the analogies between the three transport phenomena for this case, beginning with Reynolds' original concept of the analogy between momentum and heat transfer in 1874 [49]. The number of well-known relationships proposed is in the dozens, and new ones of increasing complexity are continually being suggested. Space limitations will not permit discussion of this development, and we shall confine ourselves to a few of the results.

Before any extensive measurements of the eddy diffusivities had been made, Prandtl had assumed that the fluid in turbulent flow consisted only of a viscous sublayer where E_v and E_H were both zero and a turbulent core where molecular diffusivities were negligibly important. The resulting analogy between heat and momentum transfer, as we might expect, is useful only for cases where Pr is essentially unity. A remarkably simple empirical modification of Prandtl's analogy by Colburn [11], on the other hand, represents the data even for large Pr very well (Colburn analogy)

$$St_{H,av} = \frac{h_{av}}{C_p \bar{u}_x \rho} = \frac{\frac{1}{2}f}{Pr^{2/3}} \quad (3.55)$$

The group $St_{H,av} Pr^{2/3}$ is called j_H

$$St_{H,av} Pr^{2/3} = \frac{h_{av}}{C_p \bar{u}_x \rho} Pr^{2/3} = j_H = \frac{1}{2}f = \psi(Re) \quad (3.56)$$

The mass-transfer (Chilton-Colburn) analogy [10] is

$$St_{D,av} Sc^{2/3} = \frac{F_{av}}{c \bar{u}_x} Sc^{2/3} = j_D = \frac{1}{2}f = \psi(Re) \quad (3.57)$$

which, as will be shown, agrees well with experimental data when Sc is not too large.

Analogies between the transport phenomena can be developed from the velocity, concentration, and temperature distributions from which the eddy diffusivities of Eqs. (3.34) to (3.38) were obtained. These data are not extremely

accurate near the wall, however, and the resulting analogies then depend upon the empirical expressions chosen to describe the data, as in the equations mentioned. One of the relatively simple, yet quite successful, of these is the pair [22]

$$St_{H,av} = \frac{f/2}{1.20 + 11.8(f/2)^{0.5}(Pr - 1)Pr^{-1/3}} \quad (3.58)$$

$$St_{D,av} = \frac{f/2}{1.20 + 11.8(f/2)^5(Sc - 1)Sc^{-1/3}} \quad (3.59)$$

which are useful for $Pr = 0.5$ to 600 and Sc up to 3000.

We have seen, however, that the ratios E_D/D and E_H/α are not equal to E_c/ν , and particularly for large Sc one would not anticipate an analogy between heat and mass transfer on the one hand and momentum transfer on the other. The experimental data bear this out.

Experimental data have come from so-called wetted-wall towers (Fig. 3.11) and flow of liquids through soluble pipes. In Fig. 3.11, a volatile pure liquid is permitted to flow down the inside surface of a circular pipe while a gas is blown upward or downward through the central core. Measurement of the rate of evaporation of the liquid into the gas stream over the known surface permits calculation of the mass-transfer coefficients for the gas phase. Use of different gases and liquids provides variation of Sc . In this way, Sherwood and Gilliland

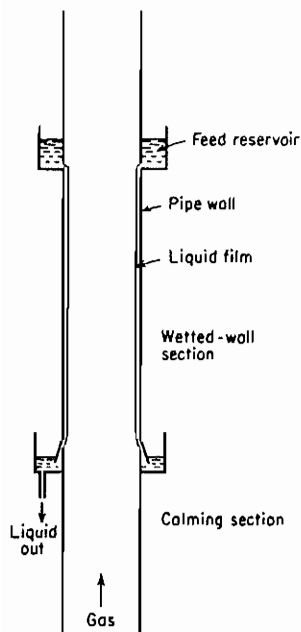


Figure 3.11 Wetted-wall tower.

[52] covered values of Re from 2000 to 35 000, Sc from 0.6 to 2.5, and gas pressures from 0.1 to 3 atm. Linton and Sherwood [41] caused water to flow through a series of pipes made by casting molten benzoic acid and other sparingly soluble solids. In this way the range of Sc was extended to 3000. Those data are empirically correlated by

$$Sh_{av} = 0.023 Re^{0.83} Sc^{1/3} \quad (3.60)$$

although the data for gases only are better correlated with 0.44 replacing $\frac{1}{3}$ as the exponent on Sc , perhaps because of the influence of ripples in the liquid film affecting the gas mass transfer [24]. Over the range $Re = 5000$ to 200 000, the friction factor in smooth pipes can be expressed as

$$\frac{1}{2}f = 0.023 Re^{-0.2} \quad (3.61)$$

Substitution in Eq. (3.60) then gives

$$\frac{Sh_{av}}{Re^{1.03} Sc} Sc^{2/3} = \frac{1}{2}f \quad (3.62)$$

which can be compared with Eq. (3.57). The heat-transfer analog of Eq. (3.60) is satisfactory for $Pr = 0.7$ to 120 [15].

As turbulence intensity increases, theory predicts that the exponent on Re should approach unity [38]. Thus [44], for liquids with $Sc > 100$,

$$Sh_{av} = 0.0149 Re^{0.88} Sc^{1/3} \quad (3.63)$$

and the heat- mass-transfer analogy applies [Eq. (3.63) yields values of Sh_{av} slightly low for $Sc = 20\ 000$ to 100 000].

MASS-TRANSFER DATA FOR SIMPLE SITUATIONS

Table 3.3 provides a few of the correlations taken from the literature for relatively simple situations. Others more appropriate to particular types of mass-transfer equipment will be introduced as needed.

Experimental data are usually obtained by blowing gases over various shapes wet with evaporating liquids or causing liquids to flow past solids which dissolve. Average, rather than local, mass-transfer coefficients are usually obtained. In most cases, the data are reported in terms of k_G , k_c , k_L , and the like, coefficients applicable to the binary systems used with $N_B = 0$, without details concerning the actual concentrations of solute during the experiments. Fortunately, the solute concentrations during the experiments are usually fairly low, so that if necessary, conversion of the data to the corresponding F is usually at least approximately possible by simply taking $\bar{p}_B, M/p_i, x_{B, M}$, etc., equal to unity (see Table 3.1).

Particularly when fluids flow past immersed objects, the local mass-transfer coefficient varies with position on the object, due especially to the separation of the boundary layer from the downstream surfaces to form a wake. This phenomenon has been studied in great detail for some shapes, e.g., cylinders [8]. The average mass-transfer coefficient in these cases can sometimes best be correlated by adding the contributions of the laminar boundary layer and the wake. This is true for the second entry of item 5, Table 3.3, for example, where these contributions correspond respectively to the two Reynolds-number terms.

For immersed objects the turbulence level of the incident fluid also has an important effect [8]. Thus, for example, a sphere falling through a quiet fluid will result in a mass-transfer coefficient different from that where the fluid flows past a stationary sphere. In most cases the distinction has not yet been fully established. Surface roughness, which usually increases the coefficient, has also been incompletely studied, and the data of Table 3.3 are for smooth surfaces, for the most part.

When mass or heat transfer occurs, it necessarily follows that the physical properties of the fluid vary along the transfer path. This problem has been dealt with theoretically (see, for example, Refs. 25 and 47), but in most cases empirical correlations use properties which are the average of those at the ends of the transfer path. The problem is most serious for natural convection, less so for forced.

In many cases, particularly when only gases have been studied, the range of Schmidt numbers covered is relatively small; sometimes only one Sc (0.6 for air-water vapor) is studied. The j_D of Eq. (3.57) has been so successful in dealing with turbulent flow in pipes that many data of small Sc range have been put in this form without actually establishing the validity of the $\frac{2}{3}$ exponent on Sc for the situation at hand. Extension to values of Sc far outside the original range of the experiments, particularly for liquids, must therefore be made with the understanding that a considerable approximation may be involved. Some data, as for flow past spheres, for example, cannot be put in the j_D form, and it is likely that many of the correlations using j_D may not be very general.

For situations not covered by the available data, it seems reasonable to assume that the mass-transfer coefficients can be estimated as functions of Re from the corresponding heat-transfer data, if available. If necessary, in the absence of a known effect of Pr , the heat-transfer data can be put in the j_H form and the analog completed by equating j_D to j_H at the same Re . This should serve reasonably well if the range of extrapolation of Sc (or Pr) is not too large.

Illustration 3.5 We wish to estimate the rate at which water will evaporate from a wetted surface of unusual shape when hydrogen at 1 std atm, 38°C, is blown over the surface at a superficial velocity of 15 m/s. No mass-transfer measurements have been made, but heat-transfer measurements indicate that for air at 38°C, 1 std atm, the heat-transfer coefficient h between air and the surface is given empirically by

$$h = 2.3G'^{0.6}$$

where G' is the superficial air mass velocity, 21.3 kg/m² · s. Estimate the required mass-transfer

Table 3.3 Mass transfer† for simple situations

Fluid motion	Range of conditions	Equation	Ref.
1. Inside circular pipes	$Re = 4000-60\ 000$	$j_D = 0.023 Re^{-0.17}$	41,
	$Sc = 0.6-3000$	$Sh = 0.023 Re^{0.83} Sc^{1/3}$	
	$Re = 10\ 000-400\ 000$	$j_D = 0.0149 Re^{-0.12}$	52
	$Sc > 100$	$Sh = 0.0149 Re^{0.88} Sc^{1/3}$	
2. Unconfined flow parallel to flat plates‡	Transfer begins at leading edge		32
	$Re_x < 50\ 000$	$j_D = 0.664 Re_x^{-0.5}$	
	$Re_x = 5 \times 10^5-3 \times 10^7$	$Nu = 0.037 Re_x^{0.8} Pr_0^{0.43} \left(\frac{Pr_0}{Pr_i} \right)^{0.25}$	65
	$Pr = 0.7-380$	Between above and	
3. Confined gas flow parallel to a flat plate in a duct	$Re_c = 2 \times 10^4-5 \times 10^5$	$Nu = 0.0027 Re_x Pr_0^{0.43} \left(\frac{Pr_0}{Pr_i} \right)^{0.25}$	46
	$Pr = 0.7-380$		
4. Liquid film in wetted-wall tower, transfer between liquid and gas	$\frac{4\Gamma}{\mu} = 0-1200$, ripples suppressed	Eqs. (3.18)-(3.22)	20, 37
	$\frac{4\Gamma}{\mu} = 1300-8300$	$Sh = (1.76 \times 10^{-5}) \left(\frac{4\Gamma}{\mu} \right)^{1.506} Sc^{0.5}$	

$$Re_x = \frac{\rho v_x x}{\mu}$$

$$Pr = \frac{c_p \mu}{k}$$

$$Sc = \frac{\mu}{\rho D}$$

$$Nu = \frac{h x}{k}$$

$$Sh = \frac{h x}{D}$$

$$j_D = \frac{h}{G} = \frac{h x}{G x}$$

$$j_D = \frac{h}{G} = \frac{h x}{G x}$$

5. Perpendicular to single cylinders	Re = 400-25 000 Sc = 0.6-2.6 Re' = 0.1-10 ⁵ Pr = 0.7-1500	$\frac{k_G P_i}{G_M} Sc^{0.56} = 0.281 Re^{0.4}$ Nu = (0.35 + 0.34 Re ^{0.5} + 0.15 Re ^{0.38}) Pr ^{0.3}	5 16, 21, 42
6. Past single spheres	Sc = 0.6-3200 Re'' Sc ^{0.5} = 1.8-600 000 $Re = \frac{d \rho u'}{\mu} (local \rho, \mu)$	Sh = Sh ₀ + 0.347(Re'' Sc ^{0.5}) ^{0.62} Sh ₀ = $\begin{cases} 2.0 + 0.569(Gr_D Sc)^{0.250} & Gr_D Sc < 10^8 \\ 2.0 + 0.0254(Gr_D Sc)^{0.333} Sc^{0.244} & Gr_D Sc > 10^8 \end{cases}$	55
7. Through fixed beds of pellets§	Re'' = 90-4000 Sc = 0.6	$j_D = j_H = \frac{2.06}{\epsilon} Re''^{-0.575}$ $j_D = 0.95 j_H = \frac{20.4}{\epsilon} Re''^{-0.815}$ $j_D = \frac{1.09}{\epsilon} Re''^{-2/3}$ $j_D = \frac{0.250}{\epsilon} Re''^{-0.31}$	4, 23, 64

† Average mass-transfer coefficients throughout, for constant solute concentrations at the phase surface. Generally, fluid properties are evaluated at the average conditions between the phase surface and the bulk fluid. The heat-mass-transfer analogy is valid throughout.

‡ Mass-transfer data for this case scatter badly but are reasonably well represented by setting $j_D = j_H$.

§ For fixed beds, the relation between ϵ and d_p is $a = 6(1 - \epsilon)/d_p$, where a is the specific solid surface, surface per volume of bed. For mixed sizes [58]

$$d_p = \frac{\sum_{i=1}^n n_i d_{pi}^3}{\sum_{i=1}^n n_i d_{pi}^2}$$

coefficient. Physical-property data are:

	38°C, 1 std atm	
	Air	H ₂
Density ρ , kg/m ³	0.114	0.0794
Viscosity μ , kg/m · s	1.85×10^{-5}	9×10^{-6}
Thermal conductivity k W/m · K	0.0273	0.1850
Heat capacity C_p , kJ/kg · K	1.002	14.4
Diffusivity with water vapor, m ² /s		7.75×10^{-5}

SOLUTION The experimental heat-transfer data do not include effects of changing Prandtl number. It will therefore be necessary to assume that the j_H group will satisfactorily describe the effect.

$$j_H = \frac{h}{C_p \rho u} \text{Pr}^{2/3} = \frac{h}{C_p G'} \text{Pr}^{2/3} = \psi(\text{Re})$$

$$h = \frac{C_p G'}{\text{Pr}^{2/3}} \psi(\text{Re}) = 21.3 G'^{0.6} \quad \text{for air}$$

The $\psi(\text{Re})$ must be compatible with $21.3 G'^{0.6}$. Therefore, let $\psi(\text{Re}) = b \text{Re}^n$ and define Re as lG'/μ , where l is a characteristic linear dimension of the solid body.

$$h = \frac{C_p G'}{\text{Pr}^{2/3}} b \text{Re}^n = \frac{C_p G'}{\text{Pr}^{2/3}} b \left(\frac{lG'}{\mu} \right)^n = \frac{b C_p}{\text{Pr}^{2/3}} \left(\frac{l}{\mu} \right)^{G'^{1+n}} = 21.3 G'^{0.6}$$

$$1 + n = 0.6 \quad n = -0.4$$

$$\frac{b C_p}{\text{Pr}^{2/3}} \left(\frac{l}{\mu} \right)^{-0.4} = 21.3 \quad \text{and}$$

$$b = 21.3 \frac{\text{Pr}^{2/3}}{C_p} \left(\frac{l}{\mu} \right)^{0.4}$$

Using the data for air at 38°C, 1 std atm, gives

$$\text{Pr} = \frac{C_p \mu}{k} = \frac{1002(1.85 \times 10^{-5})}{0.0273} = 0.68$$

$$b = \frac{21.3(0.68)^{2/3}}{1002} \left(\frac{1}{1.85 \times 10^{-5}} \right)^{0.4} = 1.285 l^{0.4}$$

$$j_H = \frac{h}{C_p G'} \text{Pr}^{2/3} = \frac{1.285 l^{0.4}}{\text{Re}^{0.4}} = \psi(\text{Re})$$

The heat-mass-transfer analogy will be used to estimate the mass-transfer coefficient ($j_D = j_H$):

$$j_D = \frac{k_G \bar{p}_{B,M} M_{av}}{\rho \mu} \text{Sc}^{2/3} = \psi(\text{Re}) = \frac{1.285 l^{0.4}}{\text{Re}^{0.4}}$$

$$k_G \bar{p}_{B,M} = F = \frac{1.285 l^{0.4} \rho u}{\text{Re}^{0.4} M_{av} \text{Sc}^{2/3}} = \frac{1.285 (\rho u)^{0.6} \mu^{0.4}}{M_{av} \text{Sc}^{2/3}}$$

For H₂-H₂O, 38°C, 1 std atm, $\text{Sc} = \mu/\rho D = (9 \times 10^{-6})/(0.0794(7.75 \times 10^{-5})) = 1.465$. At $u = 15$ m/s, and assuming the density, molecular weight, and viscosity of the gas are

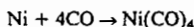
essentially those of H_2 ,

$$k_G \bar{p}_{B,M} = F = \frac{1.285[0.0794(15)]^{0.6}(9 \times 10^{-6})^{0.4}}{2.02(1.465)^{2/3}} = 0.00527 \text{ kmol/m}^2 \cdot \text{s} \quad \text{Ans.}$$

Flux Variation with Concentration

In many situations the concentrations of solute in the bulk fluid, and even at the fluid interface, may vary in the direction of flow. Further, the mass-transfer coefficients depend upon fluid properties and rate of flow, and if these vary in the direction of flow, the coefficients will also. The flux N_A of Eqs. (3.1) and (3.3) to (3.6) is therefore a local flux and will generally vary with distance in the direction of flow. This problem was dealt with, in part, in the development leading to Illustration 3.1. In solving problems where something other than the local flux is required, allowance must be made for these variations. This normally requires some considerations of material balances, but there is no standard procedure. An example is offered below, but it must be emphasized that generally some sort of improvisation for the circumstances at hand will be required.

Illustration 3.6 Nickel carbonyl is to be produced by passing carbon monoxide gas downward through a bed of nickel spheres, 12.5 mm diam. The bed is 0.1 m^2 in cross section and is packed so that there are 30% voids. Pure CO enters at 50°C , 1 std atm, at the rate of $2 \times 10^{-3} \text{ kmol/s}$. The reaction is



For present purposes, the following simplifying assumptions will be made:

1. The reaction is very rapid, so that the partial pressure of CO at the metal surface is essentially zero. The carbonyl forms as a gas, which diffuses as fast as it forms from the metal surface to the bulk-gas stream. The rate of reaction is controlled entirely by the rate of mass transfer of CO from the bulk gas to the metal surface and that of the $\text{Ni}(\text{CO})_4$ to the bulk gas.
2. The temperature remains at 50°C and the pressure at 1 std atm throughout.
3. The viscosity of the gas = $2.4 \times 10^{-5} \text{ kg/m} \cdot \text{s}$, and the Schmidt number = 2.0 throughout.
4. The size of the nickel spheres remains constant.

Estimate the bed depth required to reduce the CO content of the gas to 0.5%.

SOLUTION Let A = CO, B = $\text{Ni}(\text{CO})_4$. $N_B = -N_A/4$ and $N_A/(N_A + N_B) = \frac{4}{5}$. In Eq. (3.1), $c_{A2}/c = y_{A,1}$ at the metal interface = 0, and $c_{A1}/c = y_A$ = mole fraction CO in the bulk gas. Equation (3.1) therefore becomes

$$N_A = \frac{4}{3} F \ln \frac{4/3}{4/3 - y_A} = \frac{4}{3} F \ln \frac{4}{4 - 3y_A} \quad (3.64)$$

where N_A is the mass-transfer flux based on the metal surface. Since y_A varies with position in the bed, Eq. (3.64) gives a local flux, and to determine the total mass transfer, allowance must be made for the variation of y_A and F with bed depth.

Let $G = \text{kmol gas}/(\text{m}^2 \text{ bed cross section}) \cdot \text{s}$ flowing at any depth z from the top. The CO content of this gas is $y_A G \text{ kmol CO}/(\text{m}^2 \text{ bed}) \cdot \text{s}$. The change in CO content in passing through

a depth dz of bed is $-d(y_A G)$ kmol/m² · s. If a is the specific metal surface, m²/m³ bed, a depth dz and a cross section of 1 m² (volume = dz m³) has a metal surface $a dz$ m². Therefore, $N_A = -d(y_A G)/a dz$ kmol/(m² metal surface) · s. For each kmol of CO consumed, $\frac{1}{3}$ kmol Ni(CO)₄ forms, representing a net loss of $\frac{2}{3}$ kmol per kilomole of CO consumed. The CO consumed through bed depth dz is therefore $(G_0 - G)(\frac{1}{3})$ kmol, where G_0 is the molar superficial mass velocity at the top, and the CO content at depth z is $G_0 - (G_0 - G)(\frac{1}{3})$. Therefore

$$y_A = \frac{G_0 - (G_0 - G)(4/3)}{G} \quad G = \frac{G_0}{4 - 3y_A} \quad d(y_A G) = \frac{4G_0 dy_A}{(4 - 3y_A)^2}$$

Substituting in Eq. (3.64) gives

$$-\frac{4G_0 dy_A}{(4 - 3y_A)^2 a dz} = \frac{4}{3} F \ln \frac{4}{4 - 3y_A} \quad (3.65)$$

F is given by item 7 of Table 3.3 and is dependent upon y_A . At depth z , the CO mass velocity = $[G_0 - (G_0 - G)(\frac{1}{3})]28.0$, and the Ni(CO)₄ mass velocity = $[(G_0 - G)(\frac{1}{3})]170.7$, making a total mass velocity $G' = 47.6G_0 - 19.6G$ kg/m² · s. Substituting for G leads to

$$G' = 47.6G_0 - \frac{19.6G_0}{4 - 3y_A} = G_0 \left(47.6 - \frac{19.6}{4 - 3y_A} \right) \text{ kg/m}^2 \cdot \text{s}$$

With 12.5-mm spheres, $d_p = 0.0125$ m; $\mu = 2.4 \times 10^{-5}$ kg/m · s.

$$\text{Re}'' = \frac{d_p G'}{\mu} = \frac{0.0125 G_0 [47.6 - 19.6/(4 - 3y_A)]}{2.4 \times 10^{-5}} = G_0 \left(2.48 - \frac{1.02}{4 - 3y_A} \right) (10^4)$$

With $G_0 = (2 \times 10^{-3})/0.1 = 0.02$ kmol/m² · s and y_A in the range 1 to 0.005, the range of Re is 292 to 444. Therefore, Table 3.3,

$$j_D = \frac{F}{G} \text{Sc}^{2/3} = \frac{2.06}{\epsilon} \text{Re}''^{-0.575}$$

For $\text{Sc} = 2$ and $\epsilon = 0.3$ void fraction, this becomes

$$F = \frac{2.06}{0.3(2)^{2/3}} \frac{G_0}{4 - 3y_A} \left[G_0 \left(2.48 - \frac{1.02}{4 - 3y_A} \right) (10^4) \right]^{-0.575} \quad (3.66)$$

Equation (3.66) can be substituted in Eq. (3.65). Since $a = 6(1 - \epsilon)/d_p = 6(1 - 0.3)/0.0125 = 336$ m²/m³ and $G_0 = 0.02$ kmol/m² · s, the result, after consolidation and rearrangement for integration, is

$$Z = \int_0^Z dz = -0.0434 \int_{1.0}^{0.005} \frac{[2.48 - 1.02/(4 - 3y_A)]^{0.575}}{(4 - 3y_A) \ln[4/(4 - 3y_A)]} dy_A$$

The integration is most readily done numerically, or graphically by plotting the integrand as ordinate vs. y_A as abscissa and determining the area under the resulting curve between the indicated limits. As a result, $Z = 0.154$ m. Ans.

SIMULTANEOUS MASS AND HEAT TRANSFER

Mass transfer may occur simultaneously with the transfer of heat, either as a result of an externally imposed temperature difference or because of the absorption or evolution of heat which generally occurs when a substance is transferred from one phase to another. In such cases, within one phase, the heat transferred is a result not only of the conduction (convection) by virtue of the temperature

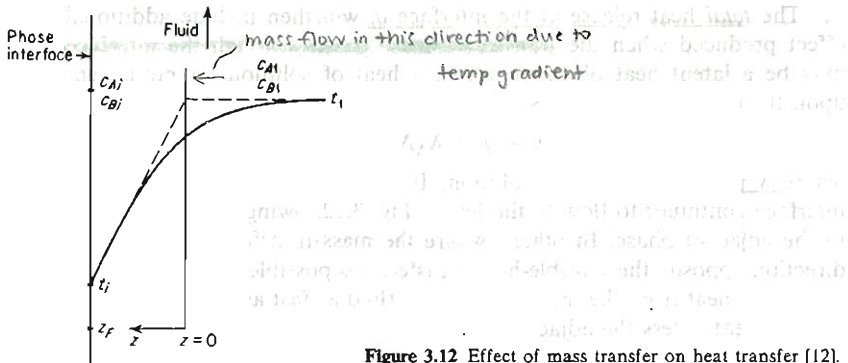


Figure 3.12 Effect of mass transfer on heat transfer [12].

difference which would happen in the absence of mass transfer but also includes the sensible heat carried by the diffusing matter.

Consider the situation shown in Fig. 3.12. Here a fluid consisting of substances A and B flows past a second phase under conditions causing mass transfer. The total mass transferred is given by the following variation of Eq. (3.1):

$$N_A + N_B = F \ln \frac{N_A / (N_A + N_B) - c_{A,i} / c}{N_A / (N_A + N_B) - c_{A1} / c} \quad (3.67)$$

As usual, the relationship between N_A and N_B is fixed by other considerations. As a result of the temperature difference, there is a heat flux described by the ordinary heat-transfer coefficient h in the absence of mass transfer. If we think in terms of the film theory, this heat flux is $h(-dt/dz)z_F$. The total sensible heat flux q_s to the interface must include in addition the sensible heat brought there by the movement of matter through the temperature difference. Thus,

$$q_s = h \left(-\frac{dt}{dz} \right) z_F + (N_A M_A C_{p,A} + N_B M_B C_{p,B})(t - t_i) \quad (3.68)$$

Rearranging and integrating gives

$$\int_{t_i}^{t_1} \frac{dt}{q_s - (N_A M_A C_{p,A} + N_B M_B C_{p,B})(t - t_i)} = \frac{1}{h z_F} \int_{z_f}^0 dz \quad (3.69)$$

$$q_s = \frac{N_A M_A C_{p,A} + N_B M_B C_{p,B}}{1 - e^{-(N_A M_A C_{p,A} + N_B M_B C_{p,B})/h}(t_1 - t_i)} \quad (3.70)$$

The term multiplying the temperature difference of Eq. (3.70) may be considered as a heat-transfer coefficient corrected for mass transfer [1, 12]; it will be larger if the mass transfer is in the same direction as the heat transfer, smaller if the two are in opposite directions. Equation (3.70) is corrected for high mass-transfer flux. It can be applied for condensation of component A in the presence of noncondensing B, ($N_B = 0$) or for multicomponent mixtures [51], for which $\sum N_i M_i C_{p,i}$ is then used in the corrected coefficient.

The total heat release at the interface q_i will then include additionally the effect produced when the transferred mass passes through the interface. This may be a latent heat of vaporization, a heat of solution, or both, depending upon the circumstances. Thus,

$$\overset{\text{at interface}}{q_i} = \overset{\text{flux}}{q_s} + \lambda_A N_A + \lambda_B N_B \quad (3.71)$$

where λ is the molar heat evolution. In some cases the heat released at the interface continues to flow to the left in Fig. 3.12, owing to a temperature drop in the adjacent phase. In others, where the mass transfer in the fluid is in the direction opposite the sensible-heat transfer, it is possible that the diffusing mass may carry heat from the interface into the fluid as fast as it is released, in which case no heat enters the adjacent phase.

Illustration 3.7 An air–water–vapor mixture flows upward through a vertical copper tube, 25.4 mm OD, 1.65 mm wall thickness, which is surrounded by flowing cold water. As a result, the water vapor condenses and flows as a liquid down the inside of the tube. At one level in the apparatus, the average velocity of the gas is 4.6 m/s, its bulk-average temperature 66°C, the pressure 1 std atm, and the bulk-average partial pressure of water vapor = 0.24 std atm. The film of condensed liquid is such that its heat-transfer coefficient = 11 400 W/m² · K. The cooling water has a bulk-average temperature of 24°C and a heat-transfer coefficient = 570 W/m² · K (see Fig. 3.13). Compute the local rate of condensation of water from the airstream.

SOLUTION For the metal tube, ID = 25.4 – 1.65(2) = 22.1 mm = 0.0221 m. $\Delta v_{\text{diam}} = (0.0254 + 0.0221)/2 = 0.0238$ m.

For the gas mixture, A = water, B = air. $N_B = 0$, $N_A/(N_A + N_B) = 1$; $y_{A1} = 0.24$, $y_{A,i} = p_{A,i} =$ vapor pressure of water at the interface temperature t_i . $M_{\text{av}} = 0.24(18.02) + 0.76(29) = 26.4$; $\rho = (26.4/22.41)(273/(273 + 66)) = 0.950$ kg/m³; $\mu = 1.75 \times 10^{-5}$ kg/m · s. $C_{p,A} = 1880$, C_p of the mixture = 1145 J/kg · K. $Sc = 0.6$, $Pr = 0.75$.

$G' =$ mass velocity $= u\rho = 4.6(0.950) = 4.37$ kg/m² · s

$$G = \text{molar mass velocity} = \frac{G'}{M_{\text{av}}} = \frac{4.37}{26.4} = 0.1652$$

$$Re = \frac{dG'}{\mu} = \frac{0.0221(4.37)}{1.75 \times 10^{-5}} = 5510$$

The mass-transfer coefficient is given by item 1, Table 3.3:

$$j_D = St_D Sc^{2/3} = \frac{F}{G} Sc^{2/3} = 0.023 Re^{-0.17} = 0.023(5510)^{-0.17} = 0.00532$$

$$F = \frac{0.00532 G}{Sc^{2/3}} = \frac{0.00532(0.1652)}{(0.60)^{2/3}} = 1.24 \times 10^{-3} \text{ kmol/m}^2 \cdot \text{s}$$

The heat-transfer coefficient in the absence of mass transfer will be estimated through $j_D = j_H$.

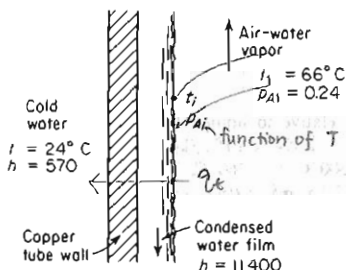
$$j_H = St_H Pr^{2/3} = \frac{h}{C_p G'} Pr^{2/3} = 0.00532$$

$$h = \frac{0.00532(1145)(4.37)}{(0.75)^{2/3}} = 32.3 \text{ W/m}^2 \cdot \text{K}$$

The sensible-heat transfer flux to the interface is given by Eq. (3.70), with $N_B = 0$. This is combined with Eq. (3.71) to produce

$$q_i = \frac{N_A(18.02)(1880)}{1 - e^{-N_A(18.02)(1880)/32.3}} (66 - t_i) + \lambda_{A,i} N_A \quad (3.72)$$

where $\lambda_{A,i}$ is the molal latent heat of vaporization of water at t_i . All the heat arriving at the interface is carried to the cold water. U is the overall heat-transfer coefficient, interface-to-cold.



only water moving

∴ A transferred

no + B.

 $N_A = k_a (P_{A1} - P_{A2})$ Pg 48

Use Pg 74 Relation (1) =

⇒ If I know t_i , use average T evaluate p

Figure 3.13 Illustration 3.7.

water, based on the inside tube surface

$$\frac{1}{U} = \frac{1}{11400} + \frac{0.00165}{381} \frac{22.1}{23.8} + \frac{1}{570} \frac{22.1}{25.4}$$

where 381 is the thermal conductivity of the copper. $\frac{D_i + D_o}{2} \wedge h_{\text{cold water}}$

$$U = 618 \text{ W/m}^2 \cdot \text{K} \quad q_i = 618(t_i - 24) \quad (3.73)$$

The rate of mass transfer is given by Eq. (3.67) with $N_B = 0$, $c_{A,i}/c = p_{A,i}$, $c_{A1}/c = y_{A1} = 0.24$.

$$N_A = 1.24 \times 10^{-3} \ln \frac{1 - p_{A,i}}{1 - 0.24} \quad (3.74)$$

Equations (3.72) to (3.74) are solved simultaneously by trial with the vapor-pressure curve for water, which relates $p_{A,i}$ and t_i , and the latent-heat data. It is easiest to assume t_i , which is checked when q_i from Eqs. (3.72) and (3.73) agree.

As a final trial, assume $t_i = 42.2^\circ\text{C}$, whence $p_{A,i} = 0.0806$ std atm, $\lambda_{A,i} = 43.4 \times 10^6 \text{ J/kmol}$, and [Eq. (3.74)] $N_A = 2.45 \times 10^{-4} \text{ kmol/m}^2 \cdot \text{s}$. By Eq. (3.73) $q_i = 11240$; by Eq. (3.72) $q_i = 11470 \text{ W/m}^2$, a sufficiently close check. The local rate of condensation of water is therefore $2.45 \times 10^{-4} \text{ kmol/m}^2 \cdot \text{s}$. Ans.

Note In this case, the true q_i [Eq. (3.68)] = 875 W/m^2 , whereas the uncorrected h gives $32.3(66 - 42.2) = 770 \text{ W/m}^2$.

Equation (3.70) can also be used for calculations of *transpiration cooling*, a method of cooling porous surfaces which are exposed to very hot gases by forcing a cold gas or evaporating liquid through the surface into the gas stream.

Illustration 3.8 Air at 600°C , 1 std atm, flows past a flat, porous surface. Saturated steam at 1.2 std atm flows through the surface into the airstream in order to keep the temperature at the surface at 260°C . The air velocity is such that h would be $1100 \text{ W/m}^2 \cdot \text{K}$ if no steam were used. (a) What rate of steam flow is required? (b) Repeat if instead of steam liquid water at 25°C is forced through the surface.

SOLUTION (a) A = water, B = air, $N_B = 0$. The positive direction is taken to be from the bulk gas to the surface. Let $H_{A,i}$ = enthalpy of steam at the interface and $H_{A,s}$ = initial enthalpy of steam. Equation (3.70) becomes†

$$q_s = N_A M_A (H_{A,s} - H_{A,i}) = \frac{N_A M_A C_{p,A} (t_1 - t_i)}{1 - e^{-N_A M_A C_{p,A} / h}}$$

† Radiation contributions to the heat transfer from the gas to the surface are negligible; it is assumed that the emissivity of the air and its small water-vapor content is negligible.

which reduces to

$$N_A = \frac{-h}{M_A C_{p,A}} \ln \left[1 - \frac{C_{p,A}(t_1 - t_i)}{H_{A,S} - H_{A,i}} \right] \quad (3.75)$$

$H_{A,S}$ = enthalpy of saturated steam at 1.2 std atm, relative to liquid at $0^\circ\text{C} = 2.684 \times 10^6$ J/kg. $H_{A,i}$ = enthalpy of steam at 1 std atm, $260^\circ\text{C} = 2.994 \times 10^6$ J/kg. $M_A = 18.02$, $h = 1100$, $C_{p,A}$ from 260 to $600^\circ\text{C} = 2090$ J/kg $\cdot$ K, $t_1 = 600^\circ\text{C}$, $t_i = 260^\circ\text{C}$. Substitution into Eq. (3.75) provides $N_A = -0.0348$ kmol/m² $\cdot$ s or -0.627 kg/m² $\cdot$ s (the negative sign indicates that the mass flux is in the negative direction, into the gas). **Ans.**

The convective heat-transfer coefficient has been reduced from 1100 to 572 W/m² $\cdot$ K.

(b) The enthalpy of liquid water at 25°C relative to liquid at $0^\circ\text{C} = 1.047 \times 10^5$ J/kg. Substitution of this instead of 2.684×10^6 into Eq. (3.75) yields $N_A = -6.42 \times 10^{-3}$ kmol/m² $\cdot$ s or -0.1157 kg/m² $\cdot$ s. **Ans.**

NOTATION FOR CHAPTER 3

Any consistent set of units may be used.

a	specific surface of a fixed bed of pellets, pellet surface/volume of bed, L^2/L^3
A	mass-transfer surface, L^2
b_1, b_2	const
c	solute concentration (if subscripted), molar density of a solution (if not subscripted), mole/ L^3
$\bar{c}$	bulk-average concentration, mole/ L^3
C_p	heat capacity at constant pressure, FL/MT
d	differential operator
	diameter, L
d_c	diameter of a cylinder, L
d_e	equivalent diameter of a noncircular duct = $4(\text{cross-sectional area})/\text{perimeter}$, L
d_p	diameter of a sphere; for a nonspherical particle, diameter of a sphere of the same surface as the particle, L
D	molecular diffusivity, L^2/θ
E_D	eddy mass diffusivity, L^2/θ
E_H	eddy thermal diffusivity, L^2/θ
E_v	eddy momentum diffusivity, L^2/θ
f	friction factor; Fanning friction factor for flow through pipes; dimensionless
F	mass-transfer coefficient, mole/ $\text{L}^2\theta$
g	acceleration due to gravity, L^2/θ
g_c	conversion factor, $\text{ML}/\text{F}\theta^2$
G	molar mass velocity, mole/ $\text{L}^2\theta$
G'	mass velocity, $\text{M}/\text{L}^2\theta$
Gr	Grashof number, dimensionless
h	heat-transfer coefficient, $\text{FL}/\text{L}^2\theta$
H	enthalpy, FL/M
j_D	mass-transfer dimensionless group, $\text{St}_D \text{Sc}^{2/3}$
j_H	heat-transfer dimensionless group, $\text{St}_H \text{Pr}^{2/3}$
J	mass-transfer flux relative to molar average velocity, mole/ $\text{L}^2\theta$
k	thermal conductivity, $\text{FL}^2/\text{L}^2\theta$
k_c, k_G, k_x, k_y , etc.	mass-transfer coefficients mole/ $\text{L}^2\theta(\text{concentration difference})$ (see Table 3.1)
K	const
l	length; Prandtl mixing length; L
l_d	length scale of eddies in the universal range, L

l_e	length scale of medium-sized eddies, L
L	length of a wetted-wall tower, L
m	mass, M
M	molecular weight, M/mole
n	a number, dimensionless
N	mass-transfer flux at, and relative to, a phase boundary, $\text{mole}/L^2\Theta$
Nu	Nusselt number, dimensionless
p	vapor pressure, F/L^2
$\bar{p}$	partial pressure, F/L^2
p_t	total pressure, F/L^2
P	power, FL/Θ
Pe	Péclet number, dimensionless
Pr	Prandtl number, dimensionless
q	heat-transfer flux, $FL/L^2\Theta$
R	universal gas constant, $FL/\text{mole } T$
Re	Reynolds number, dimensionless
Re'	Reynolds number for flow outside a cylinder, $d_e G'/\mu$, dimensionless
Re''	Reynolds number for flow past a sphere, $d_p G'/\mu$, dimensionless
Re_e	Reynolds number for flow in a noncircular duct, $d_e G'/\mu$, dimensionless
Re_x	Reynolds number computed with x as the length dimension, dimensionless
s	fractional rate of surface-element replacement, Θ^{-1}
S	cross-sectional area of a duct, L^2
Sc	Schmidt number, dimensionless
Sh	Sherwood number, dimensionless
St	Stanton number, dimensionless
t	temperature, T
u	local velocity, L/Θ
$\bar{u}$	bulk-average velocity, L/Θ
u'	root-mean-square deviating velocity, L/Θ
u'_i	instantaneous deviating velocity, L/Θ
u'_e	velocity of eddies in the universal range, L/Θ
x, y, z (no subscript)	distance in the x, y , and z direction, respectively, L
x_A	concentration of component A in a liquid, mole fraction
y_A	concentration of component A in a gas, mole fraction
z	distance from the center of a pipe, L
z_b	depth of penetration (film and surface-renewal theories), L
z_F	effective film thickness (film theory), L
z_+	dimensionless distance from the wall $\left(\frac{d}{2} - z\right) \frac{u}{\nu} \left(\frac{f}{2}\right)^{0.5}$ for a pipe
α	thermal diffusivity = $k/C_p\rho$, L^2/Θ
β	volumetric coefficient of expansion, T^{-1}
Γ	mass rate of flow/unit width, $M/L\Theta$
δ	thickness of a layer, L
θ	time, Θ
λ	molar heat evolution on passing through a surface, FL/mole
μ	viscosity, $M/L\Theta$
ρ	density, M/L^3
τ	shear stress, F/L^2
φ	area of surface elements (surface-renewal theory), $L^2/L^2\Theta$
ψ_1, ψ_2	functions

Subscripts

A	component A
B	component B

c	concentration
D	for mass transfer
H	for heat transfer
i	interface; instantaneous when used with velocity
M	logarithmic mean
r	reference condition
s	sensible heat
t	total
turb	turbulent
x, y, z	in the x, y, z directions, respectively
0	approach, or initial, value
1	at beginning of mass-transfer path
2	at end of mass-transfer path

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PROBLEMS

3.1 Estimate the mass-transfer coefficient and effective film thickness to be expected in the absorption of ammonia from air by a 2 N sulfuric acid solution in a wetted-wall tower under the following circumstances:

- Airflow = 41.4 g/min (air only)
- A_v partial pressure ammonia in air = 30.8 mmHg
- Total pressure = 760 mmHg
- A_v gas temp = 25°C
- A_v liquid temp = 25°C
- Tower diam = 1.46 cm

For absorption of ammonia in sulfuric acid of this concentration; the entire mass-transfer resistance lies within the gas, and the partial pressure of ammonia at the interface is negligible. *Note:* The circumstances correspond to run 47 of Chambers and Sherwood [*Trans. AIChE*, 33, 579 (1937)], who observed $d/z_F = 16.6$.

3.2 Powell [*Trans. Inst. Chem. Eng. Lond.*, 13, 175 (1935); 18, 36 (1940)] evaporated water from the outside of cylinders into an airstream flowing parallel to the axes of the cylinders. The air temperature was 25°C, and the total pressure standard atmospheric. The results are given by

$$\frac{wl}{p_w - \bar{p}_A} = 3.17 \times 10^{-8} (ul)^{0.8}$$

where w = water evaporated, g/cm² · s

$\bar{p}_A$ = partial pressure of water in airstream, mmHg

p_w = water-vapor pressure at surface temperature, mmHg

u = velocity of airstream, cm/s

l = length of cylinder, cm

(a) Transform the equation into the form $j_D = \psi(Re_l)$, where Re_l is a Reynolds number based on the cylinder length.

(b) Calculate the rate of sublimation from a cylinder of naphthalene 0.075 m diam by 0.60 m long (3 in diam by 24 in long) into a stream of pure carbon dioxide at a velocity of 6 m/s (20 ft/s) at 1 std atm, 100°C. The vapor pressure of naphthalene at the surface temperature may be taken as 1330 N/m² (10 mmHg) and the diffusivity in carbon dioxide as 5.15×10^{-6} m²/s (0.0515 cm²/s) at STP. Express the results as kilograms naphthalene evaporated per hour. *Ans.*: 0.386 kg/h.

3.3 Water flows down the inside wall of a wetted-wall tower of the design of Fig. 3.11, while air flows upward through the core. In a particular case, the ID is 25 mm (or 1 in), and dry air enters at the rate 7.0 kg/m² · s of inside cross section (or 5000 lb/ft² · h). Assume the air is everywhere at its average temperature, 36°C, the water at 21°C, and the mass-transfer coefficient constant. Pressure = 1 std atm. Compute the average partial pressure of water in the air leaving if the tower is 1 m (3 ft) long. *Ans.*: 1620 N/m².

3.4 Winding and Cheney [*Ind. Eng. Chem.*, 40, 1087 (1948)] passed air through a bank of rods of naphthalene. The rods were of "streamline" cross section, arranged in staggered array, with the air flowing at right angles to the axes of the rods. The mass-transfer coefficient was determined by measuring the rate of sublimation of the naphthalene. For a particular shape, size, and spacing of the rods, with air at 37.8°C (100°F), 1 std atm, the data could be correlated by†

$$k_G = 3.58 \times 10^{-9} G'^{0.56}$$

where G' is the superficial mass velocity, kg/m² · s, and k_G the mass-transfer coefficient, kmol/m² · s · (N/m²). Estimate the mass-transfer coefficient to be expected for evaporation of water into hydrogen gas for the same geometrical arrangement when the hydrogen flows at a superficial velocity of 15.25 m/s (50 ft/s), 37.8°C (100°F), 2 std atm pressure. D for naphthalene-air, 1 std atm, 37.8°C = 7.03×10^{-6} m²/s; for water-hydrogen, 0°C, 1 std atm, 7.5×10^{-5} m²/s.

3.5 A mixing tank 3 ft (0.915 m) in diameter contains water at 25°C to a depth of 3 ft (0.915 m). The liquid is agitated with a rotating impeller at an intensity of 15 hp/1000 gal (2940 W/m³). Estimate the scale of the smallest eddies in the universal range.

3.6 The free-fall terminal velocity of water drops in air at 1 std atm pressure is given by [53]:

Diam, mm	0.05	0.2	0.5	1.0	2.0	3.0
Velocity, m/s	0.055	0.702	2.14	3.87	5.86	7.26
ft/s	0.18	2.3	7.0	12.7	19.2	23.8

† For G in lb/ft² · h and k_G in lb mol/ft² · h · atm the coefficient is 0.00663.

A water drop of initial diameter 1.0 mm falls in quiet dry air at 1 std atm, 38°C (100°F). The liquid temperature may be taken as 14.4°C (58°F). Assume that the drop remains spherical and that the atmospheric pressure remains constant at 1 std atm.

(a) Calculate the initial rate of evaporation.

(b) Calculate the time and distance of free fall for the drop to evaporate to a diameter of 0.2 mm. Ans.: 103.9 s, 264 m.

(c) Calculate the time for the above evaporation, assuming that the drop is suspended without motion (as from a fine thread) in still air. Ans.: 372 s.

3.7 The temperature variation of rate of a chemical reaction which involves mass transfer is sometimes used to determine whether the rate of mass transfer or that of the chemical reaction controls, or is the dominating mechanism. Consider a fluid flowing through a 25-mm-ID (1.0-in) circular tube, where the transferred solute is ammonia in dilute solution. Compute the mass-transfer coefficient for each of the following cases:

(a) The fluid is a dilute solution of ammonia in air, 25°C, 1 std atm, flowing at a Reynolds number = 10 000. $D_{AB} = 2.26 \times 10^{-5} \text{ m}^2/\text{s}$ ($0.226 \text{ cm}^2/\text{s}$).

(b) Same as part (a) (same mass velocity), but temperature = 35°C.

(c) The fluid is a dilute solution of ammonia in liquid water, 25°C, flowing at a Reynolds number = 10 000. $D_{AB} = 2.65 \times 10^{-9} \text{ m}^2/\text{s}$ ($2.65 \times 10^{-5} \text{ cm}^2/\text{s}$).

(d) Same as part (c) (same mass velocity) but temperature = 35°C.

In the case of both gas and liquid, assuming that the mass-transfer coefficient follows an Arrhenius-type equation, compute the "energy of activation" of mass transfer. Are these high or low in comparison with the energy of activation of typical chemical reactions? Note that, for dilute solutions, the identity of the diffusing solute need not have been specified in order to obtain the "energy of activation" of mass transfer. What other method might be used to determine whether reaction rate or mass-transfer rate controls?

3.8 A gas consisting of 50% air and 50% steam, by volume, flows at 93.0°C (200°F), 1 std atm pressure, at 7.5 m/s (25 ft/s) average velocity through a horizontal square duct 0.3 m (1 ft) wide. A horizontal copper tube, 25.4 mm (1 in) OD, 1.65 mm (0.065 in) wall thickness, passes through the center of the duct from one side to the other at right angles to the duct axis, piercing the duct walls. Cold water flows inside the tube at an average velocity 3.0 m/s (10 ft/s), av temp = 15.6°C (60°F). Estimate the rate of condensation on the outside of the copper tube. Ans.: $6.92 \times 10^{-4} \text{ kg/s}$.

Data: McAdams [42, p. 338] gives the heat-transfer coefficient for the condensate film on the tube as

$$h_{av} = 1.51 \left(\frac{k^3 \rho^2 g}{\mu^2} \right)^{1/3} \left(\frac{\mu L}{4W} \right)^{1/3}$$

where, in any consistent unit system, W = mass rate of condensate and L = tube length. The fluid properties are to be evaluated at the mean temperature of the condensate film. Data for the group $(k^3 \rho^2 g / \mu^2)^{1/3}$ for water, in English engineering units, are listed by McAdams [42, table A-27, p. 484] for convenience. They must be multiplied by 5.678 for use with SI units.

3.9 A hollow, porous cylinder, OD 25 mm (1 in), ID 15 mm, is fed internally with liquid diethyl ether at 20°C. The ether flows radially outward and evaporates on the outer surface. Nitrogen, initially ether-free, at 100°C, 1 std atm, flows at right angles to the cylinder at 3 m/s (10 ft/s), carrying away the evaporated ether. The ether flow is to be that which will just keep the outer surface of the cylinder wet with liquid (since mass-transfer rates vary about the periphery, the cylinder will be rotated slowly to keep the surface uniformly wet). Compute the surface temperature and the rate of ether flow, kg/s per meter of cylinder length.

Ans.: Flow rate = $9.64 \times 10^{-4} \text{ kg/s} \cdot \text{m}$.

Data: The heat capacity C_p for liquid ether = 2282, vapor = 1863 J/kg · K. Thermal conductivity of the vapor at 0°C = 0.0165, at 85°C = 0.0232 W/m · K. Latent heat of vaporization at -17.8°C = 397.8, at 4.4°C = 391.7, at 32.2°C = 379.2 kJ/kg. Other data can be obtained from "The Chemical Engineers' Handbook."

CHAPTER
FOUR

DIFFUSION IN SOLIDS

It was indicated in Chap. 1 that certain of the diffusional operations such as leaching, drying, and adsorption and membrane operations such as dialysis, reverse osmosis, and the like involve contact of fluids with solids. In these cases, some of the diffusion occurs in the solid phase and may proceed according to any of several mechanisms. While in none of these is the mechanism as simple as in the diffusion through solutions of gases and liquids, nevertheless with some exceptions it is usually possible to describe the transfer of diffusing substance by means of the same basic law used for fluids, Fick's law.

FICK'S-LAW DIFFUSION

When the concentration gradient remains unchanged with passage of time, so that the rate of diffusion is constant, Fick's law can be applied in the form used in Chap. 2 when the diffusivity is independent of concentration and there is no bulk flow. Thus N_A , the rate of diffusion of substance A per unit cross section of solid, is proportional to the concentration gradient in the direction of diffusion, $-dc_A/dz$,

$$N_A = -D_A \frac{dc_A}{dz} \quad (4.1)$$

where D_A is the diffusivity of A through the solid. If D_A is constant, integration of Eq. (4.1) for *diffusion through a flat slab* of thickness z results in

$$N_A = \frac{D_A(c_{A1} - c_{A2})}{z} \quad (4.2)$$

which parallels the expressions obtained for fluids in a similar situation. Here

c_{A1} and c_{A2} are the concentrations at opposite sides of the slab. For other solid shapes, the rate is given by

$$w = N_A S_{av} = \frac{D_A S_{av} (c_{A1} - c_{A2})}{z} \quad (4.3)$$

with appropriate values of the average cross section for diffusion S_{av} to be applied. Thus, for *radial diffusion through a solid cylinder* of inner and outer radii a_1 and a_2 , respectively, and of length l ,

$$S_{av} = \frac{2\pi l(a_2 - a_1)}{\ln(a_2/a_1)} \quad (4.4)$$

and
$$z = a_2 - a_1 \quad (4.5)$$

For *radial diffusion through a spherical shell* of inner and outer radii a_1 and a_2 ,

$$S_{av} = 4\pi a_1 a_2 \quad (4.6)$$

$$z = a_2 - a_1 \quad (4.7)$$

Illustration 4.1 Hydrogen gas at 2 std atm, 25°C, flows through a pipe made of unvulcanized neoprene rubber, with ID and OD 25 and 50 mm, respectively. The solubility of the hydrogen is reported to be $0.053 \text{ cm}^3 \text{ (STP)/cm}^3 \cdot \text{atm}$, and the diffusivity of hydrogen through the rubber to be $1.8 \times 10^{-10} \text{ cm}^2/\text{s}$. Estimate the rate of loss of hydrogen by diffusion per meter of pipe length.

SOLUTION At 2 std atm hydrogen pressure the solubility is $0.053(2) = 0.106 \text{ m}^3 \text{ H}_2 \text{ (STP)/m}^3 \text{ (rubber)}$. Therefore concentration c_{A1} at the inner surface of the pipe = $0.106/22.41 = 4.73 \times 10^{-3} \text{ kmol H}_2/\text{m}^3$. At the outer surface, $c_{A2} = 0$, assuming the resistance to diffusion of H_2 away from the surface is negligible.

$$D_A = 1.8 \times 10^{-10} \text{ m}^2/\text{s} \quad l = 1 \text{ m}$$

$$z = a_2 - a_1 = \frac{50 - 25}{2(1000)} = 0.0125 \text{ m}$$

Eq. (4.4):

$$S_{av} = \frac{2\pi(1)(50 - 25)}{2(1000) \ln(50/25)} = 0.1133 \text{ m}^2$$

Eq. (4.3):

$$w = \frac{(1.8 \times 10^{-10})(0.1133)[(4.73 \times 10^{-3}) - 0]}{0.0125} \\ = 7.72 \times 10^{-12} \text{ kmol H}_2/\text{s for 1 m length}$$

which corresponds to $5.6 \times 10^{-5} \text{ g H}_2/\text{m} \cdot \text{h}$. Ans.

Unsteady-State Diffusion

Since solids are not so readily transported through equipment as fluids, the application of batch and semibatch processes and consequently unsteady-state diffusional conditions arise much more frequently than with fluids. Even in continuous operation, e.g., a continuous drier, the history of each solid piece as it passes through equipment is representative of the unsteady state. These cases are therefore of considerable importance.

Where there is no bulk flow, and in the absence of chemical reaction, Fick's second law, Eq. (2.18), can be used to solve problems of unsteady-state diffusion by integration with appropriate boundary conditions. For some simple cases, Newman [10] has summarized the results most conveniently.

1. *Diffusion from a slab with sealed edges.* Consider a slab of thickness $2a$, with sealed edges on four sides, so that diffusion can take place only toward and from the flat parallel faces, a cross section of which is shown in Fig. 4.1. Suppose initially the concentration of solute throughout the slab is uniform, c_{A0} , and that the slab is immersed in a medium so that the solute will diffuse out of the slab. Let the concentration at the surfaces be $c_{A,\infty}$, invariant with passage of time. If the diffusion were allowed to continue indefinitely, the concentration would fall to the uniform value $c_{A,\infty}$, and $c_{A0} - c_{A,\infty}$ is a measure of the amount of solute removed. On the other hand, if diffusion from the slab were stopped at time θ , the distribution of solute would be given by the curve marked c , which by internal diffusion would level off to the uniform concentration $c_{A,\theta}$, where $c_{A,\theta}$ is the average concentration at time θ . The quantity $c_{A,\theta} - c_{A,\infty}$ is a measure of the amount of solute still unremoved. The fraction unremoved E is given by integration of Eq. (2.18),

$$E = \frac{c_{A,\theta} - c_{A,\infty}}{c_{A0} - c_{A,\infty}} = f\left(\frac{D\theta}{a^2}\right)$$

$$= \frac{8}{\pi^2} \left(e^{-D\theta\pi^2/4a^2} + \frac{1}{9} e^{-9D\theta\pi^2/4a^2} + \frac{1}{25} e^{-25D\theta\pi^2/4a^2} + \dots \right) = E_a \quad (4.8)$$

The function is shown graphically in Fig. 4.2.

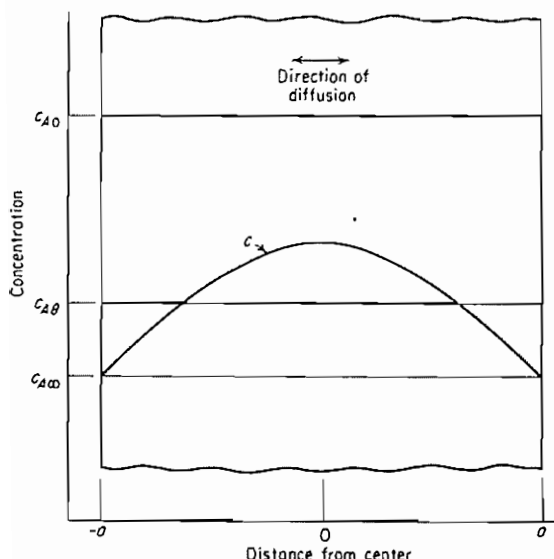


Figure 4.1 Unsteady-state diffusion in a slab.

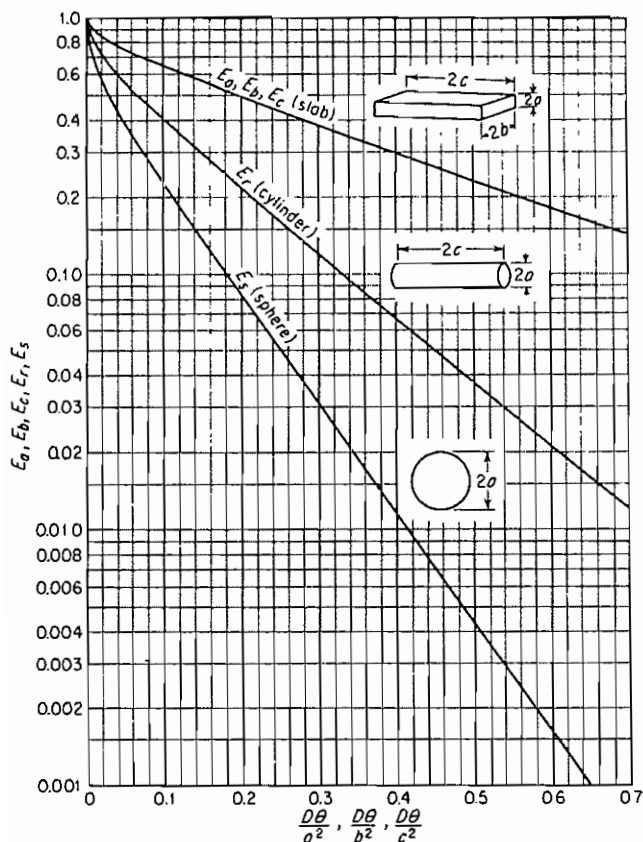


Figure 4.2 Unsteady-state diffusion.

2. *Diffusion from a rectangular bar with sealed ends.* For a rectangular bar of thickness $2a$ and width $2b$, with sealed ends,

$$E = f\left(\frac{D\theta}{a^2}\right)f\left(\frac{D\theta}{b^2}\right) = E_a E_b \quad (4.9)$$

3. *Diffusion for a rectangular parallelepiped.* For a brick-shaped bar, of dimensions $2a$, $2b$, and $2c$, with diffusion from all six faces,

$$E = f\left(\frac{D\theta}{a^2}\right)f\left(\frac{D\theta}{b^2}\right)f\left(\frac{D\theta}{c^2}\right) = E_a E_b E_c \quad (4.10)$$

4. *Diffusion from a sphere.* For a sphere of radius a ,

$$E = f'\left(\frac{D\theta}{a^2}\right) = E_s \quad (4.11)$$

5. *Diffusion from a cylinder with sealed ends.* For a cylinder of radius a , with plane ends sealed,

$$E = f''\left(\frac{D\theta}{a^2}\right) = E_r \quad (4.12)$$

6. *Diffusion from a cylinder.* For a cylinder of radius a and length $2c$, with diffusion from both ends as well as from the cylindrical surface,

$$E = f\left(\frac{D\theta}{c^2}\right)f''\left(\frac{D\theta}{a^2}\right) = E_c E_r \quad (4.13)$$

The functions $f'(D\theta/a^2)$ and $f''(D\theta/a^2)$ are also shown in Fig. 4.2.

For solid shapes where the diffusion takes place from one rather than two opposite faces, the functions are calculated as if the thickness were twice the true value. For example, if diffusion occurs through only one face of the flat slab of thickness $2a$, edges sealed, the calculation is made with $D\theta/4a^2$. The equations can also be used for diffusion into, as well as out of, the various shapes. Concentrations c may be expressed as mass solute/volume or mass solute/mass solid.

It is important to note that Eqs. (4.8) to (4.13) and Fig. 4.2 all assume constant diffusivity, initially uniform concentration within the solid, and constancy of the edge concentration $c_{A,\infty}$. The last is the same as assuming (1) that there is no resistance to diffusion in the fluid surrounding the solid and (2) that the quantity of such fluid is so large that its concentration does not change with time or (3) that the fluid is continuously replenished. In many instances, the diffusional resistance within the solid is so large that the assumption of absence of any in the fluid is quite reasonable. In any case, however, integrations of Eq. (2.18) involving varying D [18] and the effect of added diffusional resistance [1, 3, 8, 10] have been developed. It is useful to note that Eq. (2.18) is of the same form as Fourier's equation for heat conduction, with molecular rather than thermal diffusivity and concentration rather than temperature. Consequently the lengthy catalog of solutions to the problems of heat transfer of Carslaw and Jaeger [2] can be made applicable to diffusion by appropriate substitutions. Crank's book [3] deals particularly with problems of this sort for diffusion.

Illustration 4.2 A 5% agar gel containing a uniform urea concentration of 5 g/100 cm³ is molded in the form of a 3-cm cube. One face of the cube is exposed to a running supply of fresh water, into which the urea diffuses. The other faces are protected by the mold. The temperature is 5°C. At the end of 68 h, the average urea concentration in the gel has fallen to 3 g/100 cm³. The resistance to diffusion may be considered as residing wholly within the gel. (a) Calculate the diffusivity of the urea in the gel. (b) How long would it have taken for the average concentration to fall to 1 g/100 cm³? (c) Repeat part (b) for the case where two opposite faces of the cube are exposed.

SOLUTION (a) c_A can be calculated in terms of g/100 cm³. $c_{A0} = 5$ g/100 cm³; $c_{A,\infty} = 3$, $c_{A,\infty} = 0$ since pure water was the leaching agent. $a = \frac{3}{2} = 1.5$ cm; $\theta = 68(3600) = 245\,000$ s.

$$\frac{c_{A,\theta} - c_{A,\infty}}{c_{A0} - c_{A,\infty}} = \frac{3}{5} = 0.6 = E$$

The abscissa read from Fig. 4.2, which is $D\theta/4a^2$ for diffusion from only one exposed face, is 0.128

$$D = \frac{0.128(4)(a^2)}{\theta} = \frac{0.128(4)(1.5)^2}{245\,000} = 4.70 \times 10^{-6} \text{ cm}^2/\text{s} = 4.70 \times 10^{-10} \text{ m}^2/\text{s}$$

(b) For $c_{A,\theta} = 1 \text{ g}/100 \text{ cm}^3$

$$\frac{c_{A,\theta} - c_{A,\infty}}{c_{A0} - c_{A,\infty}} = \frac{1}{5} = 0.20 \quad \frac{D\theta}{4a^2} = 0.568 \quad \text{from Fig. 4.2}$$

$$\theta = \frac{0.568(4)(a^2)}{D} = \frac{0.568(4)(1.5)^2}{4.70 \times 10^{-6}} = 1\,087\,000 \text{ s} = 302 \text{ h}$$

(c) For two opposite faces exposed, $a = 1.5 \text{ cm}$, $c_{A,\theta}/c_{A0} = 0.2$, and $D\theta/a^2 = 0.568$.

$$\theta = \frac{0.568a^2}{D} = \frac{0.568(1.5)^2}{4.70 \times 10^{-6}} = 222\,000 \text{ s} = 61.5 \text{ h}$$

TYPES OF SOLID DIFFUSION

The structure of the solid and its interaction with the diffusing substance have a profound influence on how diffusion occurs and on the rate of transport.

Diffusion through Polymers

In many respects, diffusion of solutes through certain types of polymeric solids is more like diffusion through liquid solutions than any of the other solid-diffusion phenomena, at least for the permanent gases as solutes. Imagine two bodies of a gas (e.g., H_2) at different pressures separated by a polymeric membrane (e.g., polyethylene). The gas dissolves in the solid at the faces exposed to the gas to an extent usually describable by Henry's law, concentration directly proportional to pressure. The gas then diffuses from the high- to the low-pressure side in a manner usually described as *activated*: the polymeric chains are in a state of constant thermal motion, and the diffusing molecules jump from one position to another over a potential barrier [4, 11]. A successful jump requires that a hole or passage of sufficient size be available, and this in turn depends on the thermal motion of the polymer chains. The term "activated" refers to the temperature dependence of the diffusivity, which follows an Arrhenius-type expression,

$$D_A = D_0 e^{-H_D/RT} \quad (4.14)$$

where H_D is the energy of activation and D_0 is a constant. For simple gases, D_A is usually reasonably independent of concentration. It may, however, be a strong function of pressure of molding the polymer [5]. For the permanent gases diffusivities may be of the order of $10^{-10} \text{ m}^2/\text{s}$.

Particularly for large molecules, the size and shape of the diffusing molecules (as measured by molecular volume, branched as opposed to straight-chain structure, etc.) determine the hole size required. Solvents sometimes diffuse by *plasticizing* the polymers†, and consequently the rate may be very much higher

† Plasticizers are organic compounds added to high polymers which solvate the polymer molecule. This results in easier processing and a product of increased flexibility and toughness.

for good solvents, e.g., benzene or methyl ethyl ketone in polymeric rubbers, than for the permanent gases. Such diffusivities are frequently strongly dependent upon solute concentration in the solid. With certain oxygenated polymers, e.g., cellulose acetate, such solutes as water, ammonia, and alcohols form hydrogen bonds with the polymer and move from one set of bonding sites to another. Solutes which cannot form hydrogen bonds are then excluded.

Unfortunately (because it necessarily adds complicated dimensions and awkward units) it has become the practice to describe the diffusional characteristics in terms of a quantity P , the permeability. Since at the two faces of a membrane the equilibrium solubility of the gas in the polymer is directly proportional to the pressure, Eq. (4.2) can be converted into†

$$V_A = \frac{D_A s_A (\bar{p}_{A1} - \bar{p}_{A2})}{z} \quad (4.15)$$

where V_A = diffusional flux, $\text{cm}^3 \text{ gas (STP)}/\text{cm}^2 \cdot \text{s}$

D_A = diffusivity of A, cm^2/s

$\bar{p}_A$ = partial pressure of diffusing gas, cmHg

s_A = solubility coefficient or Henry's law constant, $\text{cm}^3 \text{ gas (STP)}/(\text{cm}^3 \text{ solid}) \cdot \text{cmHg}$

z = thickness of polymeric membrane, cm

Permeability is then defined as†

$$P = D_A s_A \quad (4.16)$$

where P = permeability, $\text{cm}^3 \text{ gas (STP)}/\text{cm}^2 \cdot \text{s} \cdot (\text{cmHg}/\text{cm})$.

Commercial application of these principles has been made for separating hydrogen from waste refinery gases in shell-and-tube devices which resemble in part the common heat exchanger. However, in this use the polymeric-fiber tubes are only $30 \mu\text{m}$ OD, and there are 50 million of them in a shell roughly 0.4 m in diameter [6].

Illustration 4.3 Calculate the rate of diffusion of carbon dioxide, CO_2 , through a membrane of vulcanized rubber 1 mm thick at 25°C if the partial pressure of the CO_2 is 1 cmHg on one side and zero on the other. Calculate also the permeability of the membrane for CO_2 . At 25°C , the solubility coefficient is $0.90 \text{ cm}^3 \text{ gas (STP)}/\text{cm}^3 \text{ atm}$. The diffusivity is $1.1 \times 10^{-10} \text{ m}^2/\text{s}$.

SOLUTION Since 1 std atm = 76.0 cmHg pressure, the solubility coefficient in terms of cmHg is $0.90/76.0 = 0.01184 \text{ cm}^3 \text{ gas (STP)}/\text{cm}^3 \cdot \text{cmHg} = s_A$. $z = 0.1 \text{ cm}$, $\bar{p}_{A1} = 1 \text{ cmHg}$, $\bar{p}_{A2} = 0$, $D_A = (1.1 \times 10^{-10})(10^4) = 1.1 \times 10^{-6} \text{ cm}^2/\text{s}$. Eq. (4.15):

$$\begin{aligned} V_A &= \frac{(1.1 \times 10^{-6})(0.01184)(1.0 - 0)}{0.1} \\ &= 0.13 \times 10^{-6} \text{ cm}^3 \text{ (STP)}/\text{cm}^2 \cdot \text{s} \quad \text{Ans.} \end{aligned}$$

† Equations (4.15) and (4.16) are actually dimensionally consistent and may be used with any set of consistent units. Thus if SI units are used [$D = \text{m}^2/\text{s}$, $s = \text{m}^3/\text{m}^3 \cdot (\text{N}/\text{m}^2)$, $\bar{p} = \text{N}/\text{m}^2$, $z = \text{m}$, $V = \text{m}^3/\text{m}^2 \cdot \text{s}$], the units of P will be $\text{m}^3/\text{m}^2 \cdot \text{s} \cdot (\text{N}/\text{m}^2)/\text{m}$. Similarly in English engineering units [$D = \text{ft}^2/\text{h}$, $s = \text{ft}^3/\text{ft}^3 \cdot (\text{lb}_f/\text{ft}^2)$, $\bar{p} = \text{lb}_f/\text{ft}^2$, $z = \text{ft}$, $V = \text{ft}^3/\text{ft}^2 \cdot \text{s}$] the units of P will be $\text{ft}^3/\text{ft}^2 \cdot \text{s} \cdot [(\text{lb}_f/\text{ft}^2)/\text{ft}]$. But nowhere do consistent units of any well-known system ever seem to be used, unfortunately.

Eq. (4.16):

$$P = (1.1 \times 10^{-6})(0.01184) = 0.13 \times 10^{-7} \text{ cm}^3 \text{ (STP)/cm}^2 \cdot \text{s} \cdot (\text{cmHg/cm})$$

Diffusion through Crystalline Solids

The mechanisms of diffusion vary greatly depending upon the crystalline structure and the nature of the solute [16, 17]. For crystals with lattices of cubic symmetry, the diffusivity is isotropic, but not so for noncubic crystals. Particularly, but not necessarily, in metals the principal mechanisms are the following:

1. *Interstitial mechanism.* Interstitial sites are places between the atoms of a crystal lattice. Small diffusing solute atoms may pass from an interstitial site to the next when the matrix atoms of the crystal lattice move apart temporarily to provide the necessary space. Carbon diffuses through α - and γ -iron in this manner.
2. *Vacancy mechanism.* If lattice sites are unoccupied (vacancies), an atom in an adjacent site may jump into such a vacancy.
3. *Interstitialcy mechanism.* In this case a large atom occupying an interstitial site pushes one of its lattice neighbors into an interstitial position and moves into the vacancy thus produced.
4. *Crowd-ion mechanism.* An extra atom in a chain of close-packed atoms can displace several atoms in the line from their equilibrium position, thus producing a diffusion flux.
5. *Diffusion along grain boundaries.* The diffusivity in a single-crystal metal is always substantially smaller than that for a multicrystalline sample because the latter has diffusion along the grain boundaries (crystal interfaces) and dislocations.

Diffusion in Porous Solids

The solid may be in the form of a porous barrier or membrane separating two bodies of fluid, as in the case of gaseous diffusion. Here the solute movement may be by *diffusion* from one fluid body to the other by virtue of a concentration gradient, or it may be *hydrodynamic* as a result of a pressure difference. Alternatively, in the case of adsorbents, catalyst pellets [13], solids to be dried, ore particles to be leached, and the like, the solid is normally completely surrounded by a single body of fluid, and inward and outward movement of solute through the pores of the solid is solely by diffusion. Diffusive movement may be within the fluid filling the pores or may also involve *surface* diffusion of adsorbed solute.

The pores of the solid may be *interconnected* (accessible to fluid from both ends of the pores), *dead-end* (connected to the outside of the solid only from one end), or *isolated* (inaccessible to external fluid). The pores in most solids are neither straight nor of constant diameter. Catalyst particles manufactured by pressing powders containing *micropores* into pellets, with *macropores* surrounding the powder particles of a different order of magnitude in size, are said to be *bidisperse* [19].

Diffusion, constant total pressure Imagine the pores of a solid to be all of the same length and diameter, filled with a binary solution at constant pressure, with a concentration gradient of the components over the length of the pores. Then within the solution filling the pores, Eq. (2.26) for gases or Eq. (2.40) for liquids would apply. The fluxes N_A and N_B would be based upon the cross-sectional area of the pores, and the distance would be the length of the pores l . The ratio of fluxes $N_A/(N_A + N_B)$ would, as usual, depend upon nondiffusional matters, fixed perhaps because in leaching of a solute A the solvent B would be nondiffusing, or fixed by the stoichiometry of a catalyzed reaction on the surface of the pores. Since the length of the various pores and their cross-sectional area are not constant, it is more practical to base the flux upon the gross external surface of the membrane, pellet, or whatever the nature of the solid, and the length upon some arbitrary but readily measured distance z such as membrane thickness, pellet radius, and the like. Since the fluxes will be smaller than the true values based on true fluid cross section and length, an *effective* diffusivity $D_{AB, \text{eff}}$, smaller than the true D_{AB} , must be used, ordinarily determined by experiment [13]. For the same type of processes in a given solid, presumably the ratio $D_{AB}/D_{AB, \text{eff}}$ will be constant, and once measured it can be applied to all solutes. When $N_B = 0$, Eqs. (4.1) to (4.13) would be expected to apply with $D_{AB, \text{eff}}$ replacing D_A .

Illustration 4.4 Porous alumina spheres, 10 mm diameter, 25% voids, were thoroughly impregnated with an aqueous potassium chloride, KCl, solution, concentration 0.25 g/cm³. When immersed in pure running water, the spheres lost 90% of their salt content in 4.75 h. The temperature was 25°C. At this temperature the average diffusivity of KCl in water over the indicated concentration range is 1.84×10^{-9} m²/s.

Estimate the time for removal of 90% of the dissolved solute if the spheres had been impregnated with potassium chromate, K₂CrO₄, solution at a concentration 0.28 g/cm³, when immersed in a running stream of water containing 0.02 g K₂CrO₄/cm³. The average diffusivity of K₂CrO₄ in water at 25°C is 1.14×10^{-9} m²/s.

SOLUTION For these spheres, $a = 0.005$ m, and for the KCl diffusion, $\theta = 4.75(3600) = 17\,000$ s. When the spheres are surrounded by pure water, the ultimate concentration in the spheres $c_{A, \infty} = 0$.

$$\frac{c_{A, \theta} - c_{A, \infty}}{c_{A0} - c_{A, \infty}} = 0.1 \quad \text{for 90\% removal of KCl}$$

From Fig. 4.2, $D_{\text{eff}} \theta / a^2 = 0.18$, where D_{eff} is the effective diffusivity.

$$D_{\text{eff}} = \frac{0.18a^2}{\theta} = \frac{0.18(0.005)^2}{17\,000} = 2.65 \times 10^{-10} \text{ m}^2/\text{s}$$

$$D_{AB}/D_{\text{eff}} = \frac{1.84 \times 10^{-9}}{2.65 \times 10^{-10}} = 6.943$$

For the K₂CrO₄ diffusion, $c_{A0} = 0.28$, $c_{A, \infty} = 0.02$, and $c_{A, \theta} = 0.1(0.28) = 0.028$ g/cm³.

$$E = \frac{c_{A, \theta} - c_{A, \infty}}{c_{A0} - c_{A, \infty}} = \frac{0.028 - 0.02}{0.28 - 0.02} = 0.0308 = E_i$$

Fig. 4.2:

$$\begin{aligned}
 D_{\text{eff}} \frac{\theta}{a^2} &= 0.30 \\
 D_{\text{eff}} &= \frac{D_{AB}}{6.943} = \frac{1.14 \times 10^{-9}}{6.943} = 1.642 \times 10^{-10} \text{ m}^2/\text{s} \\
 \theta &= \frac{0.30 a^2}{D_{\text{eff}}} = \frac{0.30(0.005)^2}{1.642 \times 10^{-10}} = 45\,700 \text{ s} = 12.7 \text{ h} \quad \text{Ans.}
 \end{aligned}$$

In steady-state diffusion of gases, there are two types of diffusive movement, depending on the ratio of pore diameter d to the mean free path of the gas molecules λ . If the ratio d/λ is greater than approximately 20, ordinary molecular diffusion predominates and

$$N_A = \frac{N_A}{N_A + N_B} \frac{D_{AB, \text{eff}} p_t}{RTz} \ln \frac{N_A/(N_A + N_B) - y_{A2}}{N_A/(N_A + N_B) - y_{A1}} \quad (2.26)$$

$D_{AB, \text{eff}}$, like D_{AB} , varies inversely as p_t and approximately directly as $T^{3/2}$ (see Chap. 2). If, however, the pore diameter and the gas pressure are such that the molecular mean free path is relatively large, d/λ less than about 0.2, the rate of diffusion is governed by the collisions of the gas molecules with the pore walls and follows Knudsen's law. Since molecular collisions are unimportant under these conditions, each gas diffuses independently. In a straight circular pore of diameter d and length l

$$N_A = \frac{d \bar{u}_A}{3RTl} (\bar{p}_{A1} - \bar{p}_{A2}) \quad (4.17)$$

where u_A is the mean molecular velocity of A. Since the kinetic theory of gases provides

$$u_A = \left(\frac{8g_c RT}{\pi M_A} \right)^{1/2} \quad (4.18)$$

$$\text{we have } N_A = \left(\frac{8g_c RT}{\pi M_A} \right)^{1/2} \frac{d}{3RTl} (\bar{p}_{A1} - \bar{p}_{A2}) = \frac{D_{K,A} (\bar{p}_{A1} - \bar{p}_{A2})}{RTl} \quad (4.19)$$

where $D_{K,A}$ is the Knudsen diffusion coefficient

$$D_{K,A} = \frac{d}{3} \left(\frac{8g_c RT}{\pi M_A} \right)^{1/2} \quad (4.20)$$

Since normally d is not constant and the true l is unknown, l in Eq. (4.19) is ordinarily replaced by z , the membrane thickness, and $D_{K,A}$ by $D_{K,A, \text{eff}}$, the effective Knudsen diffusivity, which is determined by experiment. $D_{K,A, \text{eff}}$ is independent of pressure and varies as $(T/M)^{1/2}$. For binary gas mixtures,

$$\frac{N_B}{N_A} = - \left(\frac{M_A}{M_B} \right)^{1/2} \quad (4.21)$$

In addition, for a given solid [15], $D_{AB, \text{eff}}/D_{K, A, \text{eff}} = D_{AB}/D_{K, A}$. The mean free path λ can be estimated from the relation

$$\lambda = \frac{3.2\mu}{p_i} \left(\frac{RT}{2\pi g_c M} \right)^{0.5} \quad (4.22)$$

In the range d/λ from roughly 0.2 to 20, a transition range, both molecular and Knudsen diffusion have influence, and the flux is given by [7, 12, 15]†

$$N_A = \frac{N_A}{N_A + N_B} \frac{D_{AB, \text{eff}} p_i}{RTz} \ln \frac{\frac{N_A}{N_A + N_B} \left(1 + \frac{D_{AB, \text{eff}}}{D_{K, A, \text{eff}}} \right) - y_{A2}}{\frac{N_A}{N_A + N_B} \left(1 + \frac{D_{AB, \text{eff}}}{D_{K, A, \text{eff}}} \right) - y_{A1}} \quad (4.23)$$

It has been shown [15] that Eq. (4.23) reverts to Eq. (2.26) for conditions under which molecular diffusion prevails ($D_{K, A, \text{eff}} \gg D_{AB, \text{eff}}$) and to Eq. (4.19) when Knudsen diffusion prevails ($D_{AB, \text{eff}} \gg D_{K, A, \text{eff}}$). Further, for open-ended pores, Eq. (4.21) applies throughout the transition range for solids whose pore diameters are of the order of 10 μm or less.

Illustration 4.5 The effective diffusivities for passage of hydrogen and nitrogen at 20°C through a 2-mm-thick piece of unglazed porcelain were measured by determining the countercurrent diffusion fluxes at 1.0 and 0.01 std atm pressure [15]. Equation (4.23), solved simultaneously for the two measurements, provided the diffusivities, $D_{H_2-N_2, \text{eff}} = 5.3 \times 10^{-6} \text{ m}^2/\text{s}$ at 1.0 std atm and $D_{K, H_2, \text{eff}} = 1.17 \times 10^{-5} \text{ m}^2/\text{s}$. Estimate (a) the equivalent pore diameter of the solid and (b) the diffusion fluxes for O_2-N_2 mixtures at a total pressure of 0.1 std atm, 20°C, with mole fractions of $O_2 = 0.8$ and 0.2 on either side of the porcelain.

SOLUTION (a) $D_{H_2-N_2}$ at 20°C, 1 std atm = $7.63 \times 10^{-5} \text{ m}^2/\text{s}$. Therefore $D_{\text{true}}/D_{\text{eff}} = (7.63 \times 10^{-5})/(5.3 \times 10^{-6}) = 14.4$. Since this ratio is strictly a matter of the geometry of the solid, it should apply to all gas mixtures at any condition. Therefore $D_{K, H_2} = (1.17 \times 10^{-5})(14.4) = 1.684 \times 10^{-4} \text{ m}^2/\text{s}$. From Eq. (4.20)

$$d = 3D_{K, H_2} \left(\frac{\pi M_{H_2}}{8g_c RT} \right)^{0.5} = 3(1.684 \times 10^{-4}) \left[\frac{\pi(2.02)}{8(1)(8314)(293)} \right]^{0.5} \\ = 2.88 \times 10^{-7} \text{ m} = 0.288 \mu\text{m} \quad \text{Ans.}$$

(b) Let A = O_2 , B = N_2 . Table 2.1: $D_{AB, \text{STP}} = 1.81 \times 10^{-5} \text{ m}^2/\text{s}$. Therefore at 0.1 std atm, 20°C,

$$D_{AB} = 1.81 \times 10^{-5} \frac{1}{0.1} \left(\frac{293}{273} \right)^{1.5} = 2.01 \times 10^{-4} \text{ m}^2/\text{s}$$

$$D_{AB, \text{eff}} = \frac{2.01 \times 10^{-4}}{14.4} = 1.396 \times 10^{-5} \text{ m}^2/\text{s}$$

Eq. (4.20):

$$D_{K, A} = D_{K, H_2} \left(\frac{M_{H_2}}{M_A} \right)^{0.5} = 1.684 \times 10^{-4} \left(\frac{2.02}{32} \right)^{0.5} = 4.23 \times 10^{-5} \text{ m}^2/\text{s}$$

$$D_{K, A, \text{eff}} = \frac{4.23 \times 10^{-5}}{14.4} = 2.94 \times 10^{-6} \text{ m}^2/\text{s}$$

† Equation (4.23) is known as the “dusty gas” equation because a porous medium consisting of a random grouping of spheres in space was used as a model for its derivation.

Since the equivalent pore diameter is less than $10\text{ }\mu\text{m}$, Eq. (4.21) will apply.

$$\frac{N_B}{N_A} = - \left(\frac{M_A}{M_B} \right)^{0.5} = - \left(\frac{32}{28.02} \right)^{0.5} = -1.069$$

$$\frac{N_A}{N_A + N_B} = \frac{1}{1 + N_B/N_A} = \frac{1}{1 - 1.069} = -14.49$$

$y_{A1} = 0.8$, $y_{A2} = 0.2$, $p_i = 10\text{ }133\text{ N/m}^2$, $z = 0.002\text{ m}$, $D_{AB, \text{eff}}/D_{K, A, \text{eff}} = (1.396 \times 10^{-5})/(2.94 \times 10^{-6}) = 4.75$. Eq. (4.23):

$$N_{O_2} = \frac{-14.49(1.396 \times 10^{-5})(10\text{ }133)}{8314(293)(0.002)} \ln \frac{-14.49(1 + 4.75) - 0.2}{-14.49(1 + 4.75) - 0.8}$$

$$= 3.01 \times 10^{-6} \text{ kmol/m}^2 \cdot \text{s}$$

$$N_{N_2} = (3.01 \times 10^{-6})(-1.069) = -3.22 \times 10^{-6} \text{ kmol/m}^2 \cdot \text{s} \quad \text{Ans.}$$

Knudsen diffusion is not known for liquids, but important reductions in diffusion rates occur when the molecular size of the diffusing solute becomes significant relative to the pore size of the solid [14].

Surface diffusion is a phenomenon accompanying adsorption of solutes onto the surface of the pores of the solid. It is an activated diffusion [see Eq. (4.14)], involving the jumping of adsorbed molecules from one adsorption site to another. It can be described by a two-dimensional analog of Fick's law, with surface concentration expressed, for example, as mol/area instead of mol/volume. Surface diffusivities are typically of the order of 10^{-7} to $10^{-9} \text{ m}^2/\text{s}$ at ordinary temperatures for physically adsorbed gases [13] (see Chap. 11). For liquid solutions in adsorbent resin particles, surface diffusivities may be of the order of $10^{-12} \text{ m}^2/\text{s}$ [9].

Hydrodynamic flow of gases If there is a difference in absolute pressure across a porous solid, a hydrodynamic flow of gas through the solid will occur. Consider a solid consisting of uniform straight capillary tubes of diameter d and length l reaching from the high- to low-pressure side. At ordinary pressures, the flow of gas in the capillaries may be either laminar or turbulent, depending upon whether the Reynolds number $du\rho/\mu$ is below or above 2100. For present purposes, where velocities are small, flow will be laminar. For a single gas, this can be described by Poiseuille's law for a compressible fluid obeying the ideal gas law

$$N_A = \frac{d^2 g_c}{32 \mu l R T} p_{i, \text{av}} (p_{i1} - p_{i2}) \quad (4.24)$$

where

$$p_{i, \text{av}} = \frac{p_{i1} - p_{i2}}{2} \quad (4.25)$$

This assumes that the entire pressure difference is the result of friction in the pores and ignores entrance and exit losses and kinetic-energy effects, which is satisfactory for present purposes. Since the pores are neither straight nor of constant diameter, just as with diffusive flow it is best to base N_A on the gross

external cross section of the solid and write Eq. (4.24) as†

$$N_A = \frac{k}{RTz} p_{t,av} (p_{i1} - p_{i2}) \quad (4.26)$$

If conditions of pore diameter and pressure occur for which Knudsen flow prevails ($d/\lambda < 0.2$), the flow will be described by Knudsen's law, Eqs. (4.17) to (4.20). There will be of course a range of conditions for a transition from hydrodynamic to Knudsen flow. If the gas is a mixture with different compositions and different total pressure on either side of the porous solid, the flow may be a combination of hydrodynamic, Knudsen, and diffusive. Younquist [20] reviews these problems.

Illustration 4.6 A porous carbon diaphragm 1 in (25.4 mm) thick of average pore diameter 0.01 cm permitted the flow of nitrogen at the rate of 9.0 ft³ (measured at 1 std atm, 80°F)/ft² · min (0.0457 m³/m² · s, 26.7°C) with a pressure difference across the diaphragm of 2 inH₂O (50.8 mmH₂O). The temperature was 80°F and the downstream pressure 0.1 std atm. Calculate the flow to be expected at 250°F (121°C) with the same pressure difference.

SOLUTION The viscosity of nitrogen is 1.8×10^{-5} kg/m · s at 26.7°C (300 K). At 0.1 std atm, $p_t = 10\,133$ N/m². $M_{N_2} = 28.02$. Eq. (4.22):

$$\lambda = \frac{3.2\mu}{p_t} \left(\frac{RT}{2\pi g_c M} \right)^{0.5} = \frac{3.2(1.8 \times 10^{-5})}{10\,133} \left[\frac{8314(300)}{2\pi(1)(28.02)} \right]^{0.5}$$

$$= 6.77 \times 10^{-7} \text{ m}$$

With $d = 10^{-4}$ m, $d/\lambda = 148$, Knudsen flow will not occur, and Poiseuille's law, Eq. (4.26), applies and will be used to calculate k . N₂ flow corresponding to 9.0 ft³/ft² · min at 300 K, 1 std atm = 0.0457 m³/m² · s or

$$0.0457 \frac{273}{300} \frac{1}{22.41} = 1.856 \times 10^{-3} \text{ kmol/m}^2 \cdot \text{s} = N_A$$

$$p_{i1} - p_{i2} = (2 \text{ inH}_2\text{O}) \frac{3386}{13.6} = 498 \text{ N/m}^2$$

$$p_{t,av} = 10\,133 + \frac{498}{2} = 10\,382 \text{ N/m}^2$$

Eq. (4.26):

$$k = \frac{N_A RTz}{p_{t,av}(p_{i1} - p_{i2})} = \frac{(1.856 \times 10^{-3})(8314)(300)(0.0254)}{10\,382(498)}$$

$$= 2.27 \times 10^{-5} \text{ m}^4/\text{N} \cdot \text{s}$$

At 250°F (121°C = 393 K), the viscosity of nitrogen = 2.2×10^{-5} kg/m · s. This results in a new k [Eq. (4.24)]:

$$k = \frac{(2.27 \times 10^{-5})(1.8 \times 10^{-5})}{2.2 \times 10^{-5}} = 1.857 \times 10^{-5} \text{ m}^4/\text{N} \cdot \text{s}$$

Eq. (4.26):

$$N_A = \frac{(1.857 \times 10^{-5})(10\,382)(498)}{8314(393)(0.0254)} = 1.157 \times 10^{-3} \text{ kmol/m}^2 \cdot \text{s}$$

$$= 7.35 \text{ ft}^3/\text{ft}^2 \cdot \text{min, measured at } 121^\circ\text{C, } 1 \text{ std atm} \quad \text{Ans.}$$

† Equation (4.26) is sometimes written in terms of volume rate V of gas flow at the average pressure per unit cross section of the solid and a permeability P

$$p_{t,av} V = \frac{P p_{t,av} (p_{i1} - p_{i2})}{z}$$

NOTATION FOR CHAPTER 4

Any consistent set of units may be used, except as noted.

a	one-half thickness; radius, L
b	one-half width, L
c	concentration, mole/L ³
	one-half length, L
d	pore diameter, L
	differential operator
D	molecular diffusivity, L ² /Θ [in Eqs. (4.15) to (4.16), cm ² /s]
D_{eff}	effective diffusivity, L ² /Θ
D_K	Knudsen diffusivity, L ² /Θ
E	fraction of solute removal, dimensionless
f, f', f''	functions
g_c	conversion factor, ML/FΘ ²
H_D	energy of activation, FL/mole
k	const, L ⁴ /FΘ
l	length, pore length, L
M	molecular weight, M/mole
N	molar flux, mole/L ² Θ
p	partial pressure, F/L ² [in Eq. (4.15), cmHg]
p_i	total pressure, F/L ²
P	permeability, L ³ /L ² Θ(F/L ²)/L [in Eq. (4.16), cm ³ gas (STP)/cm ² · s · (cmHg/cm)]
R	universal gas constant, FL/mole T
s	solubility coefficient or Henry's-law constant, L ³ /L ³ (F/L ²) [in Eqs. (4.15) to (4.16), cm ³ gas (STP) cm ³ · (cmHg)]
S	cross-sectional area, L ²
T	absolute temperature, T
u	average velocity, L/Θ
$\bar{u}$	mean molecular velocity, L/Θ
V	flux of diffusion, L ³ /L ² Θ [in Eq. (4.15), cm ³ gas (STP)/cm ² · s]
w	rate of diffusion, mole/Θ
y	concentration, mole fraction
z	thickness of membrane or pellet, L
	distance in direction of diffusion, L
θ	time, Θ
λ	molecular mean free path, L
μ	viscosity, M/LΘ
π	3.1416
ρ	density, M/L ³

Subscripts

av	average
A, B	components A, B
θ	at time Θ
0	initial (at time zero)
$1, 2$	positions 1, 2
∞	at time ∞; at equilibrium

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PROBLEMS

4.1 Removal of soybean oil impregnating a porous clay by contact with a solvent for the oil has been shown to be a matter of internal diffusion of the oil through the solid [Boucher, Brier, and Osburn, *Trans. AIChE*, **38**, 967 (1942)]. Such a clay plate, $\frac{1}{8}$ in thick, 1.80 in long, and 1.08 in wide (1.588 by 45.7 by 27.4 mm), thin edges sealed, was impregnated with soybean oil to a uniform concentration 0.229 kg oil/kg dry clay. It was immersed in a flowing stream of pure tetrachloroethylene at 120°F (49°C), whereupon the oil content of the plate was reduced to 0.048 kg oil/kg dry clay in 1 h. The resistance to diffusion may be taken as residing wholly within the plate, and the final oil content of the clay may be taken as zero when contacted with pure solvent for an infinite time.

(a) Calculate the effective diffusivity.

(b) A cylinder of the same clay 0.5 in (12.7 mm) diameter, 1 in (25.4 mm) long, contains an initial uniform concentration of 0.17 kg oil/kg clay. When immersed in a flowing stream of pure tetrachloroethylene at 49°C, to what concentration will the oil content fall in 10 h? **Ans.:** 0.0748.

(c) Recalculate part (b) for the cases where only one end of the cylinder is sealed and where neither end is sealed.

(d) How long will it take for the concentration to fall to 0.01 kg oil/kg clay for the cylinder of part (b) with neither end sealed? **Ans.:** 41 h.

4.2 A slab of clay, like that used to make brick, 50 mm thick, was dried from both flat surfaces with the four thin edges sealed, by exposure to dry air. The initial uniform moisture content was 15%. The drying took place by internal diffusion of the liquid water to the surface, followed by evaporation at the surface. The diffusivity may be assumed to be constant with varying water concentration and uniform in all directions. The surface moisture content was 3%. In 5 h the average moisture content had fallen to 10.2%.

(a) Calculate the effective diffusivity.

(b) Under the same drying conditions, how much longer would it have taken to reduce the average water content to 6%?

(c) How long would be required to dry a sphere of 150 mm radius from 15 to 6% under the same drying conditions?

(d) How long would be required to dry a cylinder 0.333 m long by 150 mm diameter, drying from all surfaces, to a moisture content of 6%? **Ans.:** 47.5 h.

4.3 A spherical vessel of steel walls 2 mm thick is to contain 1 l of pure hydrogen at an absolute pressure of $1.3 \times 10^6 \text{ N/m}^2$ (189 lb_f/in²) and temperature 300°C. The internal surface will be at the saturation concentration of hydrogen; the outer surface will be maintained at zero hydrogen content. The solubility is proportional to $p_i^{1/2}$, where p_i is the hydrogen pressure; and at 1 std atm, 300°C, the solubility is 1 ppm (parts per million) by weight. At 300°C, the diffusivity of hydrogen in the steel = $5 \times 10^{-10} \text{ m}^2/\text{s}$. The steel density is 7500 kg/m^3 (468 lb/ft³).

(a) Calculate the rate of loss of hydrogen when the internal pressure is maintained at $1.3 \times 10^6 \text{ N/m}^2$, expressed as kg/h.

(b) If no hydrogen is admitted to the vessel, how long will be required for the internal pressure to fall to one-half its original value? Assume the linear concentration gradient in the steel is always maintained and the hydrogen follows the ideal-gas law at prevailing pressures. Ans.: 7.8 h.

4.4 An unglazed porcelain plate 5 mm thick has an average pore diameter of 0.2 μm . Pure oxygen gas at an absolute pressure of 20 mmHg, 100°C, on one side of the plate passed through at a rate 0.093 cm^3 (at 20 mmHg, 100°C)/ $\text{cm}^2 \cdot \text{s}$ when the pressure on the downstream side was so low as to be considered negligible. Estimate the rate of passage of hydrogen gas at 25°C and a pressure of 10 mmHg abs, with a negligible downstream pressure. Ans.: $1.78 \times 10^{-6} \text{ kmol/m}^2 \cdot \text{s}$.

INTERPHASE MASS TRANSFER

Thus far we have considered only the diffusion of substances within a single phase. In most of the mass-transfer operations, however, two insoluble phases are brought into contact in order to permit transfer of constituent substances between them. Therefore we are now concerned with the simultaneous application of the diffusional mechanism for each phase to the combined system. We have seen that the rate of diffusion within each phase is dependent upon the concentration gradient existing within it. At the same time the concentration gradients of the two-phase system are indicative of the departure from equilibrium which exists between the phases. Should equilibrium be established, the concentration gradients and hence the rate of diffusion will fall to zero. It is necessary, therefore, to consider both the diffusional phenomena and the equilibria in order to describe the various situations fully.

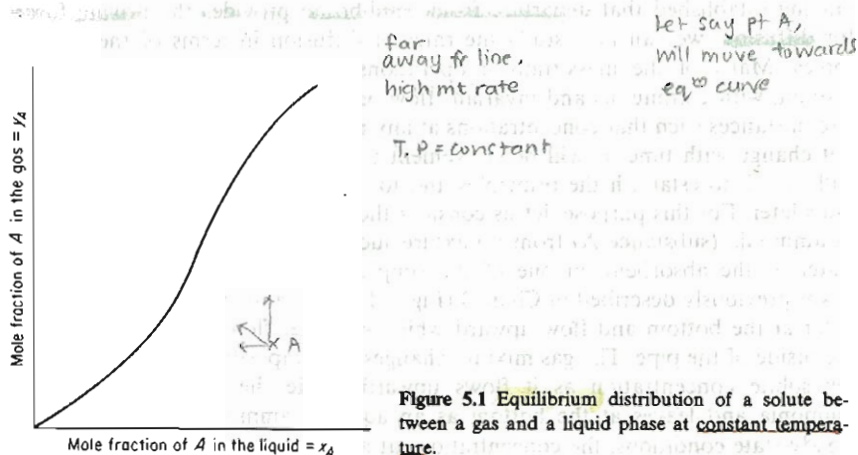
EQUILIBRIUM

It is convenient first to consider the equilibrium characteristics of a particular operation and then to generalize the result for others. As an example, consider the gas-absorption operation which occurs when ammonia is dissolved from an ammonia-air mixture by liquid water. Suppose a fixed amount of liquid water is placed in a closed container together with a gaseous mixture of ammonia and air, the whole arranged so that the system can be maintained at constant temperature and pressure. Since ammonia is very soluble in water, some ammonia molecules will instantly transfer from the gas into the liquid, crossing the interfacial surface separating the two phases. A portion of the ammonia

molecules escapes back into the gas, at a rate proportional to their concentration in the liquid. As more ammonia enters the liquid, with consequent increase in concentration within the liquid, the rate at which ammonia returns to the gas increases, until eventually the rate at which it enters the liquid exactly equals that at which it leaves. At the same time, through the mechanism of diffusion, the concentrations throughout each phase become uniform. A dynamic equilibrium now exists, and while ammonia molecules continue to transfer back and forth from one phase to the other, the net transfer falls to zero. The concentrations within each phase no longer change. To the observer who cannot see the individual molecules the diffusion has apparently stopped.

If we now inject additional ammonia into the container, a new set of equilibrium concentrations will eventually be established, with higher concentrations in each phase than were at first obtained. In this manner we can eventually obtain the complete relationship between the equilibrium concentrations in both phases. If the ammonia is designated as substance A, the equilibrium concentrations in the gas and liquid, y_A and x_A mole fractions, respectively, give rise to an equilibrium-distribution curve of the type shown in Fig. 5.1. This curve results irrespective of the amounts of water and air that we start with and is influenced only by the conditions, such as temperature and pressure, imposed upon the three-component system. It is important to note that at equilibrium the concentrations in the two phases are not equal; instead the chemical potential of the ammonia is the same in both phases, and it will be recalled (Chap. 2) that it is equality of chemical potentials, not concentrations, which causes the net transfer of solute to stop.

The curve of Fig. 5.1 does not of course show all the equilibrium concentrations existing within the system. For example, the water will partially vaporize into the gas phase, the components of the air will also dissolve to a small extent in the liquid, and equilibrium concentrations for these substances will also be established. For the moment we need not consider these equilibria, since they



are of minor importance to the discussion at hand. Obviously also, concentration units other than mole fractions may be used to describe the equilibria.

Generally speaking, whenever a substance is distributed between two insoluble phases, a dynamic equilibrium of this type can be established. The various equilibria are peculiar to the particular system considered. For example, replacement of the water in the example considered above with another liquid such as benzene or with a solid adsorbent such as activated carbon or replacement of the ammonia with another solute such as sulfur dioxide will each result in new curves not at all related to the first. The equilibrium resulting for a two-liquid-phase system bears no relation to that for a liquid-solid system. A discussion of the characteristic shapes of the equilibrium curves for the various situations and the influence of conditions such as temperature and pressure must be left for the studies of the individual unit operations. Nevertheless the following principles are common to all systems involving the distribution of substances between two insoluble phases:

1. At a fixed set of conditions, referring to temperature and pressure, there exists a set of equilibrium relationships which can be shown graphically in the form of an equilibrium-distribution curve for each distributed substance by plotting the equilibrium concentrations of the substance in the two phases against each other.
2. For a system in equilibrium, there is no net diffusion of the components between the phases.
3. For a system not in equilibrium, diffusion of the components between the phases will occur so as to bring the system to a condition of equilibrium. If sufficient time is available, equilibrium concentrations will eventually prevail.

DIFFUSION BETWEEN PHASES

Having established that departure from equilibrium provides the driving force for diffusion, we can now study the rates of diffusion in terms of the driving forces. Many of the mass-transfer operations are carried out in steady-flow fashion, with continuous and invariant flow of the contacted phases and under circumstances such that concentrations at any position in the equipment used do not change with time. It will be convenient to use one of these as an example with which to establish the principles and to generalize respecting other operations later. For this purpose, let us consider the absorption of a soluble gas such as ammonia (substance A) from a mixture such as air and ammonia, by liquid water as the absorbent, in one of the simplest of apparatus, the wetted-wall tower previously described in Chap. 3 (Fig. 3.11). The ammonia-air mixture may enter at the bottom and flow upward while the water flows downward around the inside of the pipe. The gas mixture changes its composition from a high- to a low-solute concentration as it flows upward, while the water dissolves the ammonia and leaves at the bottom as an aqueous ammonia solution. Under steady-state conditions, the concentrations at any point in the apparatus do not change with passage of time.

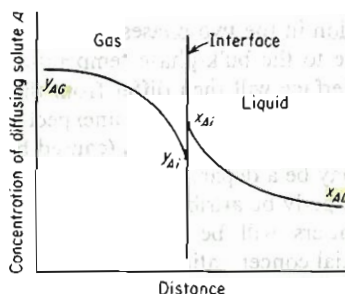


Figure 5.2 The two-resistance concept.

Local Two-Phase Mass Transfer

Let us investigate the situation at a particular level along the tower, e.g., at a point midway between top and bottom. Since the solute is diffusing from the gas phase into the liquid, there must be a concentration gradient in the direction of mass transfer within each phase. This can be shown graphically in terms of the distance through the phases, as in Fig. 5.2, where a section through the two phases in contact is shown. It will be assumed that no chemical reaction occurs. The concentration of A in the main body of the gas is $y_{A,G}$ mole fraction, and it falls to $y_{A,i}$ at the interface. In the liquid, the concentration falls from $x_{A,i}$ at the interface to $x_{A,L}$ in the bulk liquid. The bulk concentrations $y_{A,G}$ and $x_{A,L}$ are clearly not equilibrium values, since otherwise diffusion of the solute would not occur. At the same time, these bulk concentrations cannot be used directly with a mass-transfer coefficient to describe the rate of interphase mass transfer, since the two concentrations are differently related to the chemical potential, which is the real "driving force" of mass transfer.

To get around this problem, Lewis and Whitman [8, 17] assumed that the only diffusional resistances are those residing in the fluids themselves. There is then no resistance to solute transfer across the interface separating the phases, and as a result the concentrations $y_{A,i}$ and $x_{A,i}$ are equilibrium values, given by the system's equilibrium-distribution curve.†

The reliability of this theory has been the subject of a great amount of study. A careful review of the results indicates that departure from concentration equilibrium at the interface must be a rarity. It has been shown theoretically [11] that departure from equilibrium can be expected if mass-transfer rates are very high, much higher than are likely to be encountered in any practical situation. Careful measurements of real situations when interfaces are clean and conditions carefully controlled [4, 12] verify the validity of the assumption.

Unexpectedly large and small transfer rates between two phases nevertheless occur, and these have frequently been incorrectly attributed to departure from the equilibrium assumption. For example, the heat of transfer of a solute,

† This has been called the "two-film theory," improperly so, because it is not related to, or dependent upon, the film theory of mass-transfer coefficients described in Chap. 3. A more appropriate name would be the "two-resistance theory."

resulting from a difference in the heats of solution in the two phases, will either raise or lower the interface temperature relative to the bulk-phase temperature [13], and the equilibrium distribution at the interface will then differ from that which can be assumed for the bulk-phase temperature. Because of unexpected phenomena near the interface in one or both of the contacted phases (caused by the presence of surfactants and the like) there may be a departure of the transfer rates from the expected values which may improperly be attributed to departure from equilibrium at the interface. These matters will be considered later. Consequently, in ordinary situations the interfacial concentrations of Fig. 5.2 are those corresponding to a point on the equilibrium-distribution curve.

Referring again to Fig. 5.2, it is clear that the concentration rise at the interface, from $y_{A,i}$ to $x_{A,i}$, is not a barrier to diffusion in the direction gas to liquid. They are equilibrium concentrations and hence correspond to equal chemical potentials of substance A in both phases at the interface.

The various concentrations can also be shown graphically, as in Fig. 5.3, whose coordinates are those of the equilibrium-distribution curve. Point P represents the two bulk-phase concentrations and point M those at the interface. For steady-state mass transfer, the rate at which A reaches the interface from the gas must equal that at which it diffuses to the bulk liquid, so that no accumulation or depletion of A at the interface occurs. We can therefore write the flux of A in terms of the mass-transfer coefficients for each phase and the concentration changes appropriate to each (the development will be done in terms of the k -type coefficients, since this is simpler, and the results for F -type coefficients will be indicated later). Thus, when k_y and k_x are the locally applicable coefficients,

$$N_A = k_y(y_{A,G} - y_{A,i}) = k_x(x_{A,i} - x_{A,L}) \quad (5.1)$$

and the differences in y 's or x 's are considered the driving forces for the mass transfer. Rearrangement as

$$\frac{y_{A,G} - y_{A,i}}{x_{A,L} - x_{A,i}} = -\frac{k_x}{k_y} \quad (5.2)$$

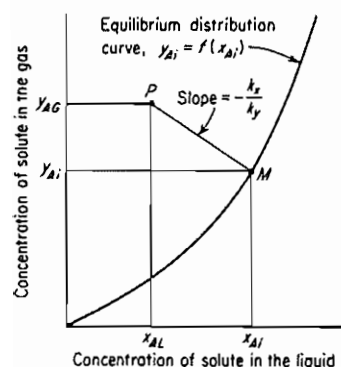


Figure 5.3 Departure of bulk-phase concentrations from equilibrium.

provides the slope of the line PM . If the mass-transfer coefficients are known, the interfacial concentrations and hence the flux N_A can be determined, either graphically by plotting the line PM or analytically by solving Eq. (5.2) with an algebraic expression for the equilibrium-distribution curve,

$$y_{A,i} = f(x_{A,i}) \quad (5.3)$$

Local Overall Mass-Transfer Coefficients

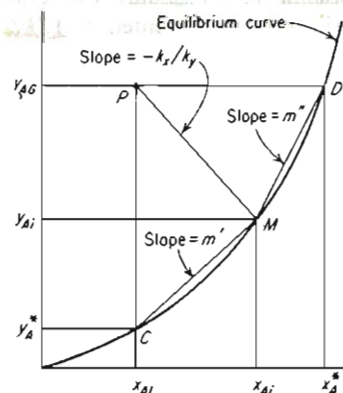
In experimental determinations of the rate of mass transfer, it is usually possible to determine the solute concentrations in the bulk of the fluids by sampling and analyzing. Successful sampling of the fluids at the interface, however, is ordinarily impossible, since the greatest part of the concentration differences, such as $y_{A,G} - y_{A,i}$, takes place over extremely small distances. Any ordinary sampling device will be so large in comparison with this distance that it is impossible to get close enough to the interface. Sampling and analyzing, then, will provide $y_{A,G}$ and $x_{A,L}$ but not $y_{A,i}$ and $x_{A,i}$. Under these circumstances, only an overall effect, in terms of the bulk concentrations, can be determined. The bulk concentrations are, however, not by themselves on the same basis in terms of chemical potential.

Consider the situation shown in Fig. 5.4. Since the equilibrium-distribution curve for the system is unique at fixed temperature and pressure, then y_A^* in equilibrium with $x_{A,L}$ is as good a measure of $x_{A,L}$ as $x_{A,L}$ itself, and moreover it is on the same basis as $y_{A,G}$. The entire two-phase mass-transfer effect can then be measured in terms of an overall mass-transfer coefficient K_y ,

$$N_A = K_y(y_{A,G} - y_A^*) \quad (5.4)$$

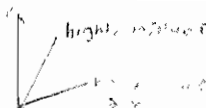
From the geometry of the figure,

$$y_{A,G} - y_A^* = (y_{A,G} - y_{A,i}) + (y_{A,i} - y_A^*) = (y_{A,G} - y_{A,i}) + m'(x_{A,i} - x_{A,L}) \quad (5.5)$$



$$\begin{aligned} y_A^* - x_{A,L} &= (x_A^* - x_{A,i}) + (x_{A,i} - x_{A,L}) \\ &= \frac{y_{A,G} - y_{A,i}}{m} + (x_{A,i} - x_{A,L}) \\ \therefore \frac{N_A}{K_y x} &= \frac{N_A}{K_y m} + \frac{N_A}{K_y} \end{aligned}$$

Figure 5.4 Overall concentration differences.



where m' is the slope of the chord CM . Substituting for the concentration differences the equivalents (flux/coefficient) as given by Eqs. (5.1) and (5.4) gives

$$\frac{N_A}{K_y} = \frac{N_A}{k_y} + \frac{m' N_A}{k_x} \quad (5.6)$$

or

$$\text{overall resistance} \quad \frac{1}{K_y} = \frac{1}{k_y} + \frac{m'}{k_x} \quad (5.7)$$

This shows that the relationship between the individual-phase transfer coefficients and the overall coefficient takes the form of addition of resistances (hence "two-resistance" theory). In similar fashion, x_A^* is a measure of $y_{A,G}$ and can be used to define another overall coefficient K_x

$$N_A = K_x(x_A^* - x_{A,L}) \quad (5.8)$$

and it is readily shown that

$$\frac{1}{K_x} = \frac{1}{m'' k_y} + \frac{1}{k_x} \quad (5.9)$$

where m'' is the slope of the chord MD in Fig. 5.4. Equations (5.7) and (5.9) lead to the following relationships between the mass-transfer resistances:

$$\frac{\text{Resistance in gas phase}}{\text{Total resistance, both phases}} = \frac{1/k_y}{1/K_y} \quad (5.10)$$

$$\frac{\text{Resistance in liquid phase}}{\text{Total resistance, both phases}} = \frac{1/k_x}{1/K_x} \quad (5.11)$$

Assuming that the numerical values of k_x and k_y are roughly the same, the importance of the slope of the equilibrium-curve chords can readily be demonstrated. If m' is small (equilibrium-distribution curve very flat), so that at equilibrium only a small concentration of A in the gas will provide a very large concentration in the liquid (solute A is very soluble in the liquid), the term m'/k_x of Eq. (5.7) becomes minor, the major resistance is represented by $1/k_y$, and it is said that the rate of mass transfer is gas-phase-controlled. In the extreme, this becomes

$$\frac{1}{K_y} \approx \frac{1}{k_y} \quad (5.12)$$

or

$$y_{A,G} - y_A^* \approx y_{A,G} - y_{A,i} \quad (5.13)$$

Under such circumstances, even fairly large percentage changes in k_x will not significantly affect K_y , and efforts to increase the rate of mass transfer would best be directed toward decreasing the gas-phase resistance. Conversely, when m'' is very large (solute A relatively insoluble in the liquid), with k_x and k_y nearly equal, the first term on the right of Eq. (5.9) becomes minor and the major resistance to mass transfer resides within the liquid, which is then said to

control the rate. Ultimately, this becomes

$$\frac{1}{K_x} \approx \frac{1}{k_x} \quad (5.14)$$

$$x_A^* - x_{A,L} \approx x_{A,i} - x_{A,L} \quad (5.15)$$

In such cases efforts to effect large changes in the rate of mass transfer are best directed to conditions influencing the liquid coefficient k_x . For cases where k_x and k_y are not nearly equal, Fig. 5.4 shows that it will be the relative size of the ratio k_x/k_y and of m' (or m'') which will determine the location of the controlling mass-transfer resistance.

It is sometimes useful to note that the effect of temperature is much larger for liquid mass-transfer coefficients than for those for gases (see Prob. 3.7, for example). Consequently a large effect of temperature on the overall coefficient, when it is determined experimentally, is usually a fairly clear indication that the controlling mass-transfer resistance is in the liquid phase.

For purposes of establishing the nature of the two-resistance theory and the overall coefficient concept, gas absorption was chosen as an example. The principles are applicable to any of the mass-transfer operations, however, and can be applied in terms of k -type coefficients using any concentration units, as listed in Table 3.1 (the F -type coefficients are considered separately below). For each case, the values of m' and m'' must be appropriately defined and used consistently with the coefficients. Thus, if the generalized phases are termed E (with concentrations expressed as i) and R (with concentrations expressed as j),

$$\frac{1}{K_E} = \frac{1}{k_E} + \frac{m'}{k_R} \quad (5.16)$$

$$\frac{1}{K_R} = \frac{1}{m''k_E} + \frac{1}{k_R} \quad (5.17)$$

$$m' = \frac{i_{A,i} - i_A^*}{j_{A,i} - j_{A,R}} \quad m'' = \frac{i_{A,E} - i_{A,i}}{j_A^* - j_{A,i}} \quad (5.18)$$

$$i_A^* = f(j_{A,R}) \quad i_{A,E} = f(j_A^*) \quad (5.19)$$

where f is the equilibrium-distribution function.

Local Coefficients—General Case

When we deal with situations which do not involve either diffusion of only one substance or equimolar counterdiffusion, or if mass-transfer rates are large, the F -type coefficients should be used. The general approach is the same, although the resulting expressions are more cumbersome than those developed above. Thus, in a situation like that shown in Figs. 5.2 to 5.4, the mass-transfer flux is

$$N_A = \frac{N_A}{\Sigma N} F_G \ln \frac{N_A/\Sigma N - y_{A,i}}{N_A/\Sigma N - y_{A,G}} = \frac{N_A}{\Sigma N} F_L \ln \frac{N_A/\Sigma N - x_{A,L}}{N_A/\Sigma N - x_{A,i}} \quad (5.20)$$

where F_G and F_L are the gas- and liquid-phase coefficients for substance A and

$\Sigma N = N_A + N_B + N_C + \dots$. Equation (5.20) then becomes [5]

$$\frac{N_A/\Sigma N - y_{A,i}}{N_A/\Sigma N - y_{A,G}} = \left(\frac{N_A/\Sigma N - x_{A,L}}{N_A/\Sigma N - x_{A,i}} \right)^{F_L/F_G} \quad (5.21)$$

The interfacial compositions $y_{A,i}$ and $x_{A,i}$ can be found by plotting Eq. (5.21) (with $y_{A,i}$ replaced by y_A and $x_{A,i}$ by x_A) on the distribution diagram (Fig. 5.3) and determining the intersection of the resulting curve with the distribution curve. This is, in general, a trial-and-error procedure, since $N_A/\Sigma N$ may not be known, and must be done in conjunction with Eq. (5.20). In the special cases for diffusion only of A and of equimolar counterdiffusion in two-components phases, no trial and error is required.

We can also define the overall coefficients F_{OG} and F_{OL} as

$$N_A = \frac{N_A}{\Sigma N} F_{OG} \ln \frac{N_A/\Sigma N - y_A^*}{N_A/\Sigma N - y_{A,G}} = \frac{N_A}{\Sigma N} F_{OL} \ln \frac{N_A/\Sigma N - x_{A,L}}{N_A/\Sigma N - x_A^*} \quad (5.22)$$

By a procedure similar to that used for the K 's, it can be shown that the overall and individual phase F 's are related:

$$\exp \left[\frac{N_A}{(N_A/\Sigma N)F_{OG}} \right] = \exp \left[\frac{N_A}{(N_A/\Sigma N)F_G} \right] + m' \frac{N_A/\Sigma N - x_{A,L}}{N_A/\Sigma N - y_{A,G}} \left\{ 1 - \exp \left[- \frac{N_A}{(N_A/\Sigma N)F_L} \right] \right\} \quad (5.23)$$

$$\exp \left[- \frac{N_A}{(N_A/\Sigma N)F_{OL}} \right] = \frac{1}{m''} \left(\frac{N_A/\Sigma N - y_{A,G}}{N_A/\Sigma N - x_{A,L}} \right) \left\{ 1 - \exp \left[\frac{N_A}{(N_A/\Sigma N)F_{DG}} \right] \right\} + \exp \left[- \frac{N_A}{(N_A/\Sigma N)F_L} \right] \quad (5.24)$$

where $\exp Z$ means e^Z . These fortunately simplify for the two important special cases:

1. Diffusion of one component ($\Sigma N = N_A$, $N_A/\Sigma N = 1.0$)†

$$e^{N_A/F_{OG}} = e^{N_A/F_G} + m' \frac{1 - x_{A,L}}{1 - y_{A,G}} (1 - e^{-N_A/F_L}) \quad (5.25)$$

$$e^{-N_A/F_{OL}} = \frac{1}{m''} \frac{1 - y_{A,G}}{1 - x_{A,L}} (1 - e^{N_A/F_G}) + e^{-N_A/F_L} \quad (5.26)$$

† Equations (5.25) and (5.26) can also be written in the following forms:

$$\frac{1}{F_{OG}} = \frac{1}{F_G} \frac{(1 - y_A)_{IM}}{(1 - y_A)_{*M}} + \frac{m'(1 - x_A)_{IM}}{F_L(1 - y_A)_{*M}} \quad (5.25a)$$

$$\frac{1}{F_{OL}} = \frac{1}{m''F_G} \frac{(1 - y_A)_{IM}}{(1 - x_A)_{*M}} + \frac{1}{F_L} \frac{(1 - x_A)_{IM}}{(1 - x_A)_{*M}} \quad (5.26a)$$

where

$(1 - y_A)_{IM}$ = logarithmic mean of $1 - y_{A,G}$ and $1 - y_{A,i}$

$(1 - y_A)_{*M}$ = logarithmic mean of $1 - y_{A,G}$ and $1 - y_A^*$

$(1 - x_A)_{IM}$ = logarithmic mean of $1 - x_{A,L}$ and $1 - x_{A,i}$

$(1 - x_A)_{*M}$ = logarithmic mean of $1 - x_{A,L}$ and $1 - x_A^*$

2. Equimolar counterdiffusion ($\Sigma N = 0$ ($F_G = k'_y$, $F_L = k'_x$))

$$\frac{1}{F_{OG}} = \frac{1}{F_G} + \frac{m'}{F_L} \quad (5.27)$$

$$\frac{1}{F_{OL}} = \frac{1}{m'' F_G} + \frac{1}{F_L} \quad (5.28)$$

Use of Local Overall Coefficients

The concept of overall mass-transfer coefficients is in many ways similar to that of overall heat-transfer coefficients in heat-exchanger design. And, as is the practice in heat transfer, the overall coefficients of mass transfer are frequently synthesized through the relationships developed above from the individual coefficients for the separate phases. These can be taken, for example, from the correlations of Chap. 3 or from those developed in later chapters for specific types of mass-transfer equipment. It is important to recognize the limitations inherent in this procedure [6].

The hydrodynamic circumstances must be the same as those for which the correlations were developed. Especially in the case of two fluids, where motion in one may influence motion in the other, the individual correlations may make inadequate allowances for the effect on the transfer coefficients. There are, however, other important effects. Sometimes the transfer of a solute, e.g., the transfer of ethyl ether from water into air, results in large interfacial-tension gradients at the interface, and these in turn may result in an astonishingly rapid motion of the liquid at the interface (interfacial turbulence or Marangoni effect) [10]. In such a case the bulk-phase Reynolds numbers used in Chap. 3 no longer represent the turbulence level at the interface, and the mass-transfer rates are likely to be substantially larger than expected. In other cases, surface-active agents in a liquid, even when at extraordinarily low average concentrations, are concentrated at the interface, where they may (1) partly block the interface to solute transfer [15], (2) make the interfacial liquid layers more rigid [1], or (3) interact with a transferring solute [3]. In any case, the mass-transfer rates are reduced.

The mere presence of a mass-transfer resistance in one phase must have no influence on that in the other: the resistances must not interact, in other words. This may be important, since the individual-phase coefficient correlations are frequently developed under conditions where mass-transfer resistance in the second phase is nil, as when the second phase is a pure substance. The analysis of such situations is dependent upon the mechanism assumed for mass transfer (film, surface-renewal, etc., theories), and has been incompletely developed.

Average Overall Coefficients

As will be shown, in practical mass-transfer apparatus the bulk concentrations in the contacted phases normally vary considerably from place to place, so that a point such as P in Figs. 5.3 and 5.4 is just one of an infinite number which forms a curve on these figures. Thus, in the countercurrent wetted-wall gas

absorber considered earlier, the solute concentrations in both fluids are small at the top and large at the bottom. In such situations, we can speak of an average overall coefficient applicable to the entire device. Such an average coefficient can be synthesized from constituent individual-phase coefficients by the same equations as developed above for local overall coefficients only, provided the quantity $m'k_E/k_R$ (or $m''k_R/k_E$) remains everywhere constant [6]. Of course, variations in values of the coefficients and slope m' may occur in such a way that they are self-compensating. But in practice, since average phase coefficients, assumed constant for the entire apparatus, are the only ones usually available, average overall coefficients generally will have meaning only when $m' = m'' = \text{const}$, in other words, for cases of straight-line equilibrium-distribution curves. Obviously, also, the same hydrodynamic regime must exist throughout the apparatus for an average overall coefficient to have meaning.

Illustration 5.1 A wetted-wall absorption tower, 1 in (2.54 cm) ID, is fed with water as the wall liquid and an ammonia-air mixture as the central-core gas. At a particular level in the tower, the ammonia concentration in the bulk gas is 0.80 mole fraction, that in the bulk liquid 0.05 mole fraction. The temperature = 80°F (26.7°C), the pressure 1 std atm. The rates of flow are such that the local mass-transfer coefficient in the liquid, from a correlation obtained with dilute solutions, is $k_L = 0.34 \text{ mol/ft}^2 \cdot \text{h} \cdot (\text{lb mol/ft}^3) = 2.87 \times 10^{-5} \text{ kmol/m}^2 \cdot \text{s} \cdot (\text{kmol/m}^3)$, and the local Sherwood number for the gas is 40. The diffusivity of ammonia in air = $0.890 \text{ ft}^2/\text{h} = 2.297 \times 10^{-5} \text{ m}^2/\text{s}$. Compute the local mass-transfer flux for the absorption of ammonia, ignoring the vaporization of water.

$N_B = 0$

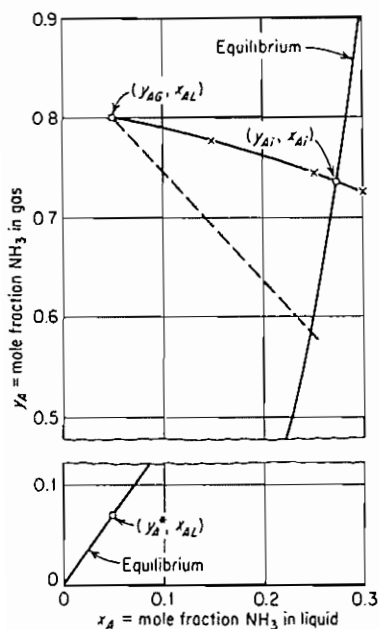


Figure 5.5 Construction for Illustration 5.1.

SOLUTION $y_{A,G} = 0.80$, $x_{A,L} = 0.05$ mole fraction ammonia. Because of the large concentration of ammonia in the gas, F 's rather than k 's will be used. Notation is that of Chap. 3.

Liquid From Table 3.1, $F_L = k_L x_{B,M}$. Since the molecular weight of ammonia and water are so nearly the same and the density of a dilute solution is practically that of water, the molar density $c = 1000/18 = 55.5 \text{ kmol/m}^3$. Since the k_L was determined for dilute solutions, where $x_{B,M}$ is practically 1.0, $\rightarrow \frac{K_A}{m^2} \times 10^3$ (for NH_3 in water very small) blue for ammonia

$$F_L = (2.87 \times 10^{-5})(1.0)(55.5) = 1.590 \times 10^{-3} \text{ kmol/m}^2 \cdot \text{s}$$

Gas

$$\text{Sh} = \frac{F_G d}{c D_A} = 40 \quad \leftarrow \text{10 obsorp. tower} \quad \text{molar } \rho \quad \text{molal } \rho$$

$$d = 0.254 \text{ m} \quad \left(G = \frac{273}{22.41} \frac{1}{273 + 26.7} = 0.04065 \text{ kmol/m}^3 \right)$$

$$D_A = 2.297 \times 10^{-5} \text{ m}^2/\text{s} \quad \left(D_{\text{gas}} \propto T^{1.5} \right) \quad 273 \text{ K} \sim 22.41 \text{ m}^3/\text{mol}$$

$$F_G = \frac{40 c D_A}{d} = \frac{40(0.04065)(2.297 \times 10^{-5})}{0.0254} = 1.471 \times 10^{-3} \text{ kmol/m}^2 \cdot \text{s}$$

Mass-transfer flux The equilibrium-distribution data for ammonia are taken from those at 26.7°C in "The Chemical Engineers' Handbook," 5th ed., p. 3-68:

NH ₃ mole fraction x_A	NH ₃ partial pressure		$y_A = \frac{\bar{p}_A}{1.0133 \times 10^5}$
	lb _f /in ²	N/m ² = $\bar{p}_A$	
0	0	0	0
0.05	1.04	7 171	0.0707
0.10	1.98	13 652	0.1347
0.25	8.69	59 917	0.591
0.30	13.52	93 220	0.920

These are plotted as the equilibrium curve in Fig. 5.5.

For transfer of only one component, $N_A/\Sigma N = 1.0$. Equation (5.21), with $y_{A,i}$ and $x_{A,i}$ replaced by y_A and x_A , becomes

$$y_A = 1 - (1 - y_{A,G}) \left(\frac{1 - x_{A,L}}{1 - x_A} \right)^{F_L/F_G} = 1 - (1 - 0.8) \left(\frac{1 - 0.05}{1 - x_A} \right)^{1.078}$$

bulk liq. gas

from which the following are computed:

x_A	0.05	0.15	0.25	0.30
y_A	0.80	0.780	0.742	0.722

These are plotted as x 's on Fig. 5.5, and the resulting curve intersects the equilibrium curve to give the interface compositions, $x_{A,i} = 0.274$, $y_{A,i} = 0.732$. Therefore Eq. (5.20) provides

$$\sqrt{N_A} = 1(1.590 \times 10^{-3}) \ln \frac{1 - 0.05}{1 - 0.274} = 1(1.471 \times 10^{-3}) \ln \frac{1 - 0.732}{1 - 0.80}$$

same.

$$= 4.30 \times 10^{-4} \text{ kmol NH}_3 \text{ absorbed/m}^2 \cdot \text{s, local flux} \quad \text{Ans.}$$

Overall coefficient Although it is not particularly useful in this case, the overall coefficient will be computed to demonstrate the method. The gas concentration in equilibrium with the bulk

liquid ($x_{A,L} = 0.05$) is, from Fig. 5.5, $y_A^* = 0.0707$. The chord slope m' is therefore

$$m' = \frac{y_{A,i} - y_A^*}{x_{A,i} - x_{A,L}} = \frac{0.732 - 0.0707}{0.274 - 0.05} = 2.95$$

Note that unless the equilibrium curve is straight, this cannot be obtained without first obtaining ($y_{A,i}$, $x_{A,i}$), in which case there is no need for the overall coefficient. In Eq. (5.25), normally we would obtain F_{OG} by trial, by assuming N_A . Thus (with an eye on the answer obtained above), assume $N_A = 4.30 \times 10^{-4}$. Then Eq. (5.25) becomes

$$\begin{aligned} \frac{e^{4.30 \times 10^{-4}}}{F_{OG}} &= e^{(4.30 \times 10^{-4})/(1.471 \times 10^{-3})} \\ &\quad + 2.95 \frac{1 - 0.05}{1 - 0.80} [1 - e^{-(4.30 \times 10^{-4})/(1.590 \times 10^{-3})}] \\ &= 4.66 \\ F_{OG} &= 2.78 \times 10^{-4} \text{ kmol/m}^2 \cdot \text{s} \end{aligned}$$

As a check of the trial value of N_A , Eq. (5.22) is

$$N_A = 1(2.78 \times 10^{-4}) \ln \frac{1 - 0.0707}{1 - 0.80} = 4.30 \times 10^{-4} \quad (\text{check})$$

Use of k -type coefficients There are, of course, k 's which are consistent with the F 's and which will produce the correct result. Thus, $k_y = F_{OG}/\bar{p}_{B,M} = 6.29 \times 10^{-3}$ and $k_x = F_L/x_{B,M} = 1.885 \times 10^{-3}$ will produce the same result as above. But these k 's are specific for the concentration levels at hand, and the $\bar{p}_{B,M}$ and $x_{B,M}$ terms, which correct for the bulk-flow flux [the $N_A + N_B$ term of Eq. (2.4)], cannot be obtained until $x_{A,i}$ and $y_{A,i}$ are first obtained as above.

However, if it had been assumed that the concentrations were dilute and that the bulk-flow terms were negligible, the Sherwood number *might* have been (incorrectly) interpreted as

$$\begin{aligned} \text{Sh} = \frac{k_y RTd}{P_r D_A} &= 40 = \frac{k_y (8314)(273.2 + 26.7)(0.0254)}{(1.0133 \times 10^5)(2.297 \times 10^{-5})} \\ k_y &= 1.47 \times 10^{-3} \text{ kmol/m}^2 \cdot \text{s} \cdot (\text{mole fraction}) \end{aligned}$$

and, in the case of the liquid,

$$k_x = k_L c = (2.87 \times 10^{-5})(55.5) = 1.59 \times 10^{-3} \text{ kmol/m}^2 \cdot \text{s} \cdot (\text{mole fraction})$$

These k 's are suitable for small driving forces but are unsuitable here. Thus

$$-\frac{k_x}{k_y} = -\frac{1.59 \times 10^{-3}}{1.47 \times 10^{-3}} = -1.08$$

and a line of this slope (the dashed line of Fig. 5.5) drawn from ($x_{A,L}$, $y_{A,G}$) intersects the equilibrium curve at ($x_A = 0.250$, $y_A = 0.585$). If this is interpreted to be ($x_{A,i}$, $y_{A,i}$), the calculated flux would be

$$N_A = k_x(x_{A,i} - x_{A,L}) = (1.59 \times 10^{-3})(0.250 - 0.05) = 3.17 \times 10^{-4}$$

or

$$N_A = k_y(y_{A,G} - y_{A,i}) = (1.47 \times 10^{-3})(0.8 - 0.585) = 3.17 \times 10^{-4}$$

The corresponding overall coefficient, with $m' = (0.585 - 0.0707)/(0.250 - 0.05) = 2.57$, would be

$$\begin{aligned} \frac{1}{K_y} &= \frac{1}{k_y} + \frac{m'}{k_x} = \frac{1}{1.47 \times 10^{-3}} + \frac{2.57}{1.59 \times 10^{-3}} \\ K_y &= 4.35 \times 10^{-4} \text{ kmol/m}^2 \cdot \text{s} \cdot (\text{mole fraction}) \\ N_A &= K_y(y_{A,G} - y_A^*) = (4.35 \times 10^{-4})(0.8 - 0.0707) = 3.17 \times 10^{-4} \end{aligned}$$

This value of N_A is of course incorrect. However, had the gas concentration been low, say 0.1 mole fraction NH_3 , these k 's would have been satisfactory and would have given the same flux as the F 's.

MATERIAL BALANCES

The concentration-difference driving forces discussed above, as we have said, are those existing at one position in the equipment used to contact the immiscible phases. In the case of a steady-state process, because of the transfer of solute from one phase to the other, the concentration within each phase changes as it moves through the equipment. Similarly, in the case of a batch process, the concentration in each phase changes with time. These changes produce corresponding variations in the driving forces, and these can be followed with the help of material balances. In what follows, all concentrations are the bulk-average values for the indicated streams.

Steady-State Cocurrent Processes

Consider any mass-transfer operation whatsoever conducted in a steady-state cocurrent fashion, as in Fig. 5.6, in which the apparatus used is represented simply as a rectangular box. Let the two insoluble phases be identified as phase E and phase R , and for the present consider only the case where a single substance A diffuses from phase R to phase E during their contact. The other constituents of the phases, solvents for the diffusing solutes, are then considered not to diffuse.

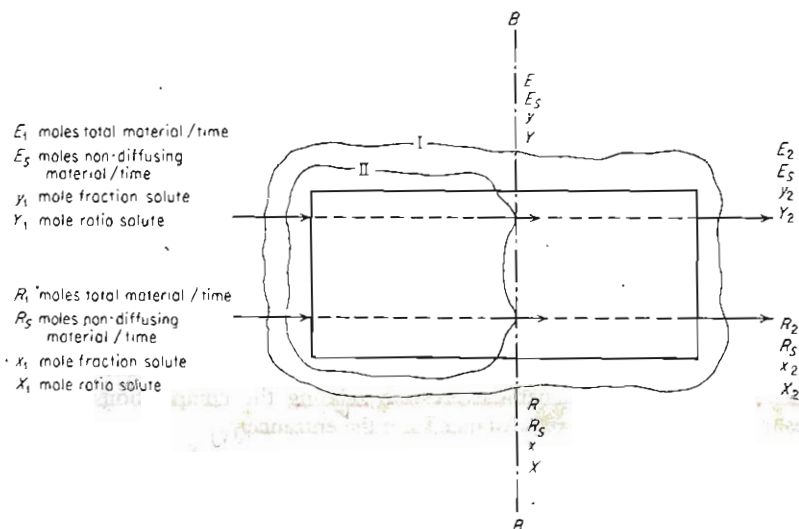


Figure 5.6 Steady-state cocurrent processes.

At the entrance to the device in which the phases are contacted, phase R contains R_1 mol per unit time of total substances, consisting of nondiffusing solvent R_S mol per unit time and diffusing solute A , whose concentration is x_1 mole fraction. As phase R moves through the equipment, A diffuses to phase E and consequently the total quantity of R falls to R_2 mol per unit time at the exit, although the rate of flow of nondiffusing solvent R_S is the same as at the entrance. The concentration of A has fallen to x_2 mole fraction. Similarly, phase E at the entrance contains E_1 mol per unit time total substances, of which E_S mol is nondiffusing solvent, and an A concentration of y_1 mole fraction. Owing to the accumulation of A to a concentration y_2 mole fraction, phase E increases in amount to E_2 mol per unit time at the exit, although the solvent content E_S has remained constant.

Envelope I, the closed, irregular line drawn about the equipment, will help to establish a material balance for substance A , since an accounting for substance A must be made wherever the envelope is crossed by an arrow representing a flowing stream. The A content of the entering R phase is $R_1 x_1$; that of the E phase is $E_1 y_1$. Similarly the A content of the leaving streams is $R_2 x_2$ and $E_2 y_2$, respectively. Thus

$$R_1 x_1 + E_1 y_1 = R_2 x_2 + E_2 y_2 \quad (5.29)$$

$$\text{or} \quad R_1 x_1 - R_2 x_2 = E_2 y_2 - E_1 y_1 \quad (5.30)$$

$$\text{but} \quad R_1 x_1 = R_S \frac{x_1}{1 - x_1} = R_S X_1 \quad (5.31)$$

where X_1 is the mole-ratio concentration of A at the entrance, mol A /mol non- A . The other terms can be similarly described, and Eq. (5.30) becomes

$$R_S(X_1 - X_2) = E_S(Y_2 - Y_1) \quad (5.32)$$

The last is the equation of a straight line on X, Y coordinates, of slope $-R_S/E_S$, passing through two points whose coordinates are (X_1, Y_1) and (X_2, Y_2) , respectively.

At any section $B-B$ through the apparatus, the mole fractions of A are x and y and the mole ratios X and Y , in phases R and E , respectively, and if an envelope II is drawn to include all the device from the entrance to section $B-B$, the A balance becomes

$$R_S(X_1 - X) = E_S(Y - Y_1) \quad (5.33)$$

This is also the equation of a straight line on X, Y coordinates, of slope $-R_S/E_S$, through the points (X_1, Y_1) and (X, Y) . Since the two straight lines have the same slope and a point in common, they are the same straight line and Eq. (5.33) is therefore a general expression relating the compositions of the phases in the equipment at any distance from the entrance.†

† Masses, mass fractions, and mass ratios can be substituted consistently for moles, mole fractions, and mole ratios in Eqs. (5.29) to (5.33).

Since X and Y represent concentrations in the two phases, the equilibrium relationship may also be expressed in terms of these coordinates. Figure 5.7 shows a representation of the equilibrium relationship as well as the straight line QP of Eqs. (5.32) and (5.33). The line QP , called an *operating line*, should not be confused with the driving-force lines of the earlier discussion. At the *entrance* to the apparatus, for example, the *mass-transfer coefficients* in the *two phases* may give rise to the *driving-force line* KP , where K represents the *interface compositions at the entrance* and the distances KM and MP represent, respectively, the *driving forces in phase E* and *phase R*. Similarly, at the exit, point L may represent the *interface composition* and LQ the line *representative of the driving forces*. If the apparatus were longer than that indicated in Fig. 5.6, so that eventually an equilibrium between the two phases is established, the corresponding equilibrium compositions X_e and Y_e would be given by an *extension* of the operating line to the *intersection with the equilibrium curve at T*. The *driving forces*, and therefore the diffusion rate, at this point will have fallen to *zero*. Should the *diffusion* be in the opposite direction, i.e., from phase *E* to phase *R*, (y to x) the operating line will fall on the opposite side of the equilibrium curve, as in Fig. 5.8.

It must be emphasized that the graphical representation of the operating line as a *straight line* is greatly dependent upon the units in which the concentrations of the material balance are expressed. The representations of Figs. 5.7 and 5.8 are straight lines because the *mole-ratio concentrations* are based on the *unchanging quantities* E_s and R_s . If Eq. (5.30) were plotted on mole-fraction coordinates, or if any concentration unit proportional to mole fractions such as partial pressure were used, the nature of the operating curve obtained would be

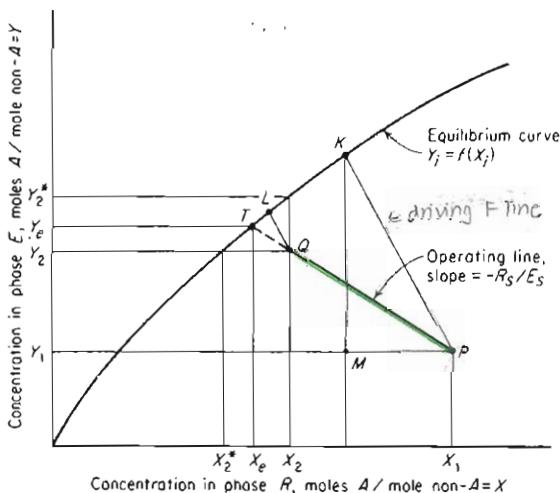


Figure 5.7 Steady-state cocurrent process, transfer of solute from phase R to phase E.

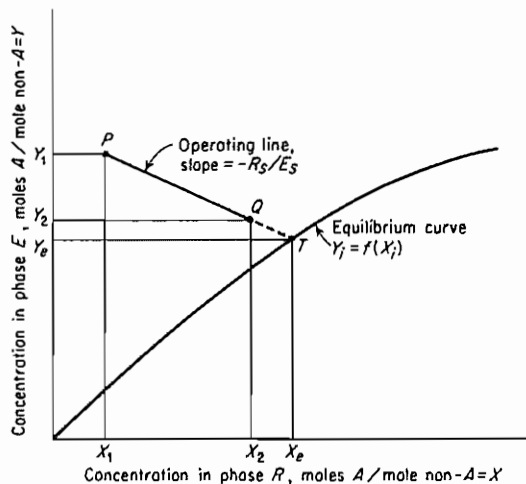


Figure 5.8 Steady-state cocurrent process, transfer of solute from phase E to phase R.

indicated as in Fig. 5.9. Extrapolation to locate the ultimate equilibrium conditions at T is of course much more difficult than for mole-ratio coordinates. On the other hand, in any operation where the total quantities of each of the phases E and R remain constant while the compositions change owing to diffusion of several components, a diagram in terms of mole fraction will provide a straight-line operating line, as Eq. (5.30) indicates (let $E_2 = E_1 = E$; $R_2 = R_1 = R$). If all the components diffuse so that the total quantities of each phase do not remain constant, the operating line will generally be curved.

To sum up, the cocurrent operating line is a graphical representation of the material balance. A point on the line represents the bulk-average concentrations

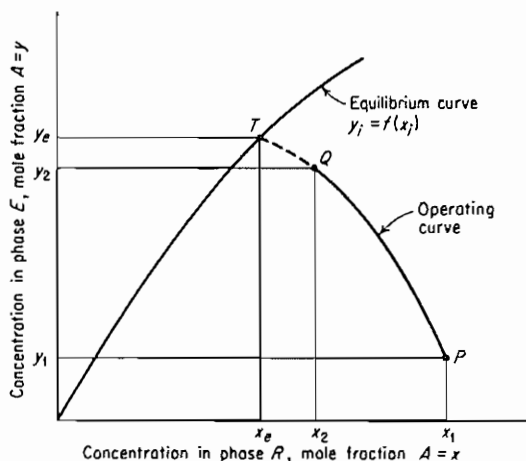


Figure 5.9 Steady-state cocurrent process, transfer of solute from phase R to phase E.

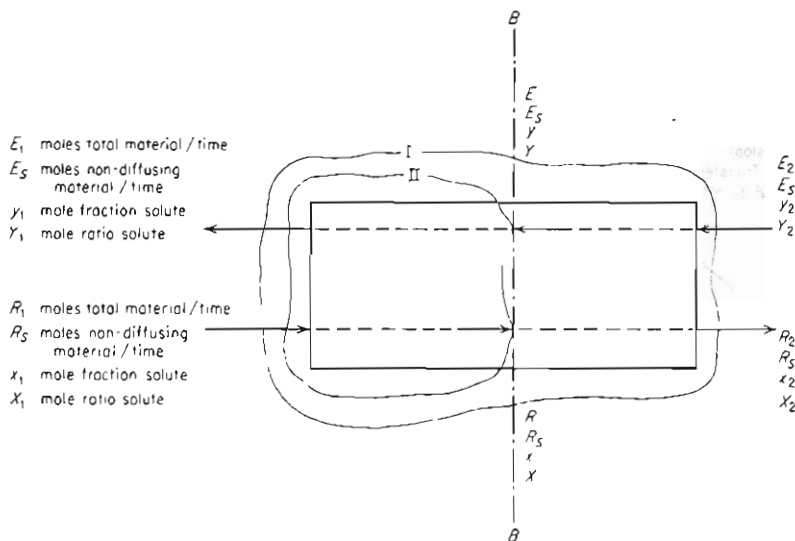


Figure 5.10 Steady-state countercurrent process.

of the streams in contact with each other at any section in the apparatus. Consequently the line passes from the point representing the streams entering the apparatus to that representing the effluent streams.

Steady-State Countercurrent Processes

If the same process as previously considered is carried out in countercurrent fashion, as in Fig. 5.10, where the subscripts 1 indicate that end of the apparatus where phase R enters and 2 that end where phase R leaves, the material balances become, by envelope I,[†]

$$E_2 y_2 + R_1 x_1 = E_1 y_1 + R_2 x_2 \quad (5.34)$$

$$\text{and} \quad R_S (X_1 - X_2) = E_S (Y_1 - Y_2) \quad (5.35)$$

and, for envelope II,

$$E y + R_1 x_1 = E_1 y_1 + R x \quad (5.36)$$

$$\text{and} \quad R_S (X_1 - X) = E_S (Y_1 - Y) \quad (5.37)$$

Equations (5.36) and (5.37) give the general relationship between concentrations in the phases at any section, while Eqs. (5.34) and (5.35) establish the entire material balance. Equation (5.35) is that of a straight line on X, Y coordinates,

[†] Masses, mass ratios, and mass fractions can be substituted consistently for moles, mole ratios, and mole fractions in Eqs. (5.34) to (5.37).

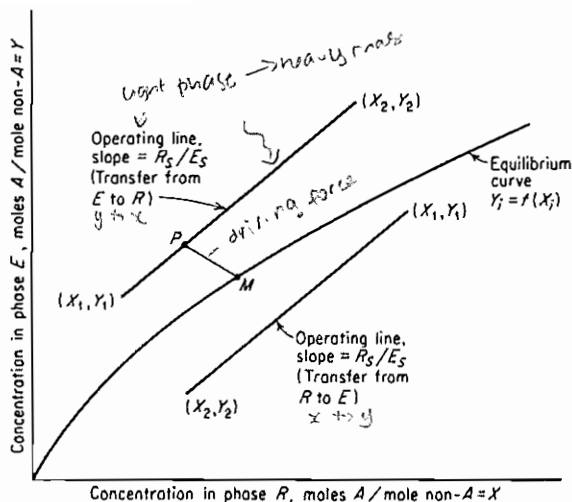


Figure 5.11 Steady-state countercurrent process.

of slope R_S/E_S , through points of coordinates (X_1, Y_1) , (X_2, Y_2) , as shown in Fig. 5.11. The line will lie above the equilibrium-distribution curve if diffusion proceeds from phase E to phase R , below for diffusion in the opposite direction. For the former case, at a point where the concentrations in the phases are given by the point P , the driving-force line may be indicated by line PM , whose slope depends upon the relative diffusional resistances of the phases. The driving forces obviously change in magnitude from one end of the equipment to the other. If the operating line should touch the equilibrium curve anywhere, so that the contacted phases are in equilibrium, the driving force and hence the rate of mass transfer would become zero and the time required for a finite material transfer would be infinite. This can be interpreted in terms of a limiting ratio of flow rates of the phases for the concentration changes specified.

As in the cocurrent case, linearity of the operating line depends upon the method of expressing the concentrations. The operating lines of Fig. 5.11 are straight because the mole-ratio concentrations X and Y are based on the quantities R_S and E_S , which are stipulated to be constant. If for this situation mole fractions (or quantities such as partial pressures, which are proportional to mole fractions) are used, the operating lines are curved, as indicated in Fig. 5.12. However, for some operations should the total quantity of each of the phases E and R be constant while the compositions change, the mole-fraction diagram would provide the straight-line operating lines, as Eq. (5.34) would indicate (let $E = E_1 = E_2$; $R = R_1 = R_2$). As in the cocurrent case, however, the operating line will generally be curved if all the components diffuse so that the total quantities of each phase do not remain constant.

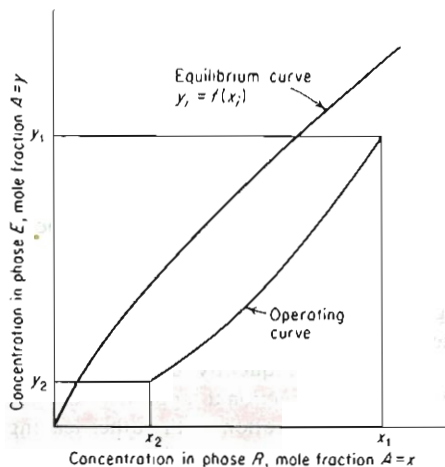


Figure 5.12 Steady-state countercurrent process, transfer of solute from phase R to phase E.

The countercurrent operating line is a graphical representation of the material balance, passing from the point representing the streams at one end of the apparatus to the point representing the streams at the other. A point on the line represents the bulk-average concentrations of the streams passing each other at any section in the apparatus. It is useful to rearrange Eq. (5.36) to read

$$Ey - Rx = E_1 y_1 - R_1 x_1 \quad (5.38)$$

The left-hand side represents the *net* flow rate of solute A to the left at section B-B in Fig. 5.10. The right-hand side is the net flow rate of solute out at the left end of the apparatus, or the difference in solute flow, out - in, and for steady state this is constant. Since the section B-B was taken to represent any section of the apparatus, at every point on the operating line the *net* flow rate of solute is constant, equal to the net flow out at the end.

The advantage of the countercurrent method over the cocurrent is that, for the former, the average driving force for a given situation will be larger, resulting in either smaller equipment for a given set of flow conditions or smaller flows for a given equipment size.

$$n \text{ Hstages} = f(k) = \text{function of mass transfer}$$

STAGES

A stage is defined as any device or combination of devices in which two insoluble phases are brought into intimate contact, where mass transfer occurs between the phases tending to bring them to equilibrium, and where the phases are then mechanically separated. A process carried out in this manner is a single-stage process. An equilibrium, ideal, or theoretical, stage is one where the time of contact between phases is sufficient for the effluents indeed to be in

equilibrium, and although in principle this cannot be attained, in practice we can frequently approach so close to equilibrium if the cost warrants it, that the difference is unimportant.

Continuous Cocurrent Processes

Quite obviously the cocurrent process and apparatus of Fig. 5.6 is that of a single stage, and if the stage were ideal, the effluent compositions would be at point T on Fig. 5.7 or 5.8.

A *stage efficiency* is defined as the fractional approach to equilibrium which a real stage produces. Referring to Fig. 5.7, this might be taken as the fraction which the line QP represents of the line TP , or the ratio of actual solute transfer to that if equilibrium were attained. The most frequently used expression, however, is the *Murphree stage efficiency*, the fractional approach of one leaving stream to equilibrium with the actual concentration in the other leaving stream [9]. Referring to Fig. 5.7, this can be expressed in terms of the concentrations in phase E or in phase R ,

$$E_{ME} = \frac{Y_2 - Y_1}{Y_2^* - Y_1} \quad E_{MR} = \frac{X_1 - X_2}{X_1 - X_2^*} \quad (5.39)$$

continuous process *discontinuous process*

These definitions are somewhat arbitrary since, in the case of a truly cocurrent operation such as that of Fig. 5.7, it would be impossible to obtain a leaving concentration in phase E higher than Y_e or one in phase R lower than X_e . They are nevertheless useful, as will be developed in later chapters. The two Murphree efficiencies are not normally equal for a given stage, and they can be simply related only when the equilibrium relation is a straight line. Thus, for a straight equilibrium line of slope $m = (Y_2^* - Y_2)/(X_2 - X_2^*)$, it can be shown that

$$E_{ME} = \frac{E_{MR}}{E_{MR}(1 - S) + S} = \frac{E_{MR}}{E_{MR}(1 - 1/A) + 1/A} \quad (5.40)$$

The derivation is left to the student (Prob. 5.6). Here $A = R_S/mE_S$ is called the *absorption factor*, and its reciprocal, $S = mE_S/R_S$, is the *stripping factor*. As will be shown, both have great economic importance.

Batch Processes

It is characteristic of batch processes that while there is no flow of the phases into and out of the equipment used, the concentrations within each phase change with time. When initially brought into contact, the phases will not be of equilibrium compositions, but they will approach equilibrium with passage of time. The material-balance equation (5.33) for the cocurrent steady-state operation then shows the relation between the concentrations X and Y in the phases which coexist at any time after the start of the operation, and Figs. 5.7 and 5.8 give the graphical representation of these compositions. Point T on these figures represents the ultimate compositions which are obtained at equilibrium. The batch operation is a single stage.

Cascades

A group of stages interconnected so that the various streams flow from one to the other is called a cascade. Its purpose is to increase the extent of mass transfer over and above that which is possible with a single stage. The fractional overall stage efficiency of a cascade is then defined as the number of equilibrium stages to which the cascade is equivalent divided by the number of real stages.

Two or more stages connected so that the flow is cocurrent between stages will, of course, never be equivalent to more than one equilibrium stage, although the overall stage efficiency can thereby be increased. For effects greater than one equilibrium stage, they can be connected for cross flow or countercurrent flow.

Cross-Flow Cascades

In Fig. 5.13 each stage is represented simply by a circle, and within each the flow is cocurrent. The R phase flows from one stage to the next, being contacted in each stage by a fresh E phase. There may be different flow rates of the E phase to each stage, and each stage may have a different Murphree stage efficiency. The material balances are obviously merely a repetition of that for a single stage and the construction on the distribution diagram is obvious. Cross flow is used sometimes in adsorption, leaching, drying, and extraction operations but rarely in the others.

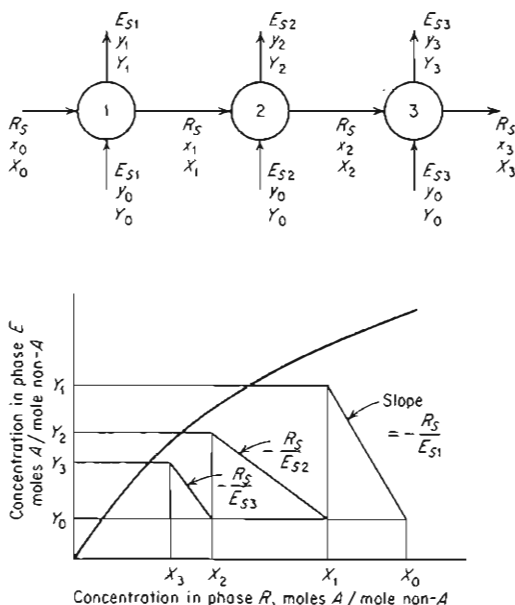


Figure 5.13 A cross-flow cascade of three real stages.

Countercurrent Cascades

These are the most efficient arrangements, requiring the fewest stages for a given change of composition and ratio of flow rates, and they are therefore most frequently used. Refer to Fig. 5.14, where a cascade of N_p equilibrium stages is shown. The flow rates and compositions are numbered corresponding to the effluent from a stage, so that Y_2 is the concentration in the E phase leaving stage 2, etc. Each stage is identical in its action to the cocurrent process of Fig. 5.6; yet the cascade as a whole has the characteristics of the countercurrent process of Fig. 5.10. The cocurrent operating lines for the first two stages are written beneath them on the figure, and since the stages are ideal, the effluents are in equilibrium (Y_2 in equilibrium with X_2 , etc.). The graphical relations are shown in Fig. 5.15. Line PQ is the operating line for stage 1, MN for stage 2, etc., and the coordinates (X_1, Y_1) fall on the equilibrium curve because the stage is ideal. Line ST is the operating line for the entire cascade, and points such as B, C , etc., represent compositions of streams passing each other between stages. We can therefore determine the number of equilibrium stages required for a countercurrent process by drawing the stairlike construction $TQBNC \cdots S$ (Fig. 5.15). If anywhere the equilibrium curve and cascade operating line should touch, the stages become *pinched* and an infinite number are required to bring about the desired composition change. If the solute transfer is from E to R , the entire construction falls above the equilibrium curve of Fig. 5.15.

For most cases, because of either a curved operating line or equilibrium curve, the relation between number of stages, compositions, and flow ratio must be determined graphically, as shown. For the special case where both are straight, however, with the equilibrium curve continuing straight to the origin of the distribution graph, an analytical solution can be developed which will be most useful.

In Fig. 5.14, a solute balance for stages $n+1$ through N_p is

$$E_S(Y_{n+1} - Y_{N_p+1}) = R_S(X_n - X_{N_p}) \quad (5.41)$$

If the equilibrium-curve slope is $m = Y_{n+1}/X_{n+1}$, and if the absorption factor $A = R_S/mE_S$, then

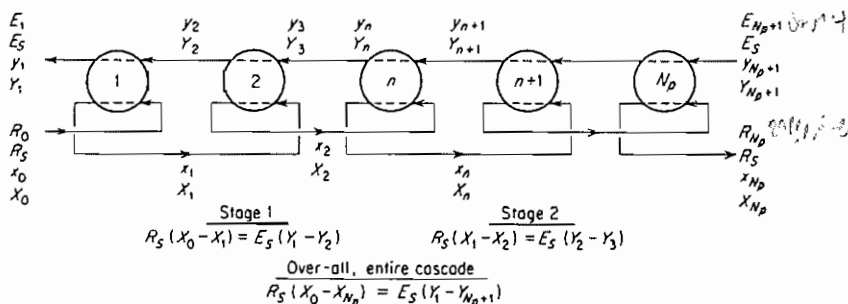
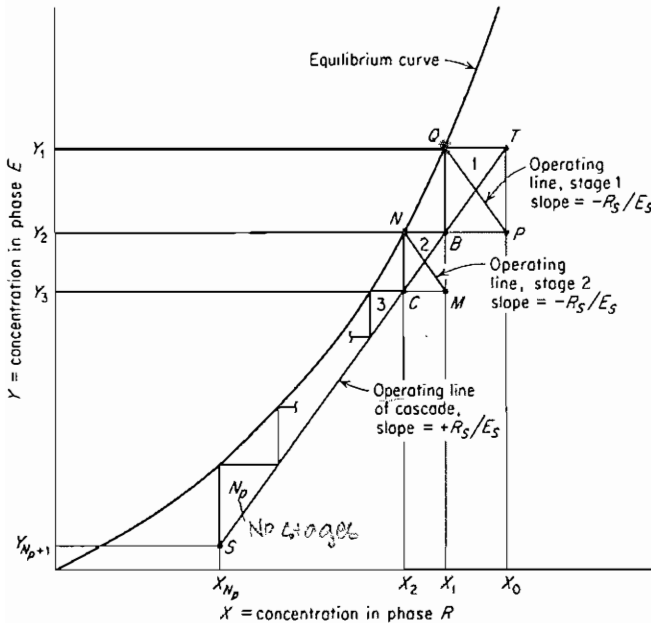


Figure 5.14 Countercurrent multistage cascade.



When up line tangent
to curve, eqⁿ
stages too many
(not possible)

Figure 5.15 Countercurrent multistage cascade, solute transfer from phase R to phase E.

by substitution and rearrangement, Eq. (5.41) becomes

$$X_{n+1} - AX_n = \frac{Y_{N_p+1}}{m} - AX_{N_p} \quad (5.42)$$

This is a linear first-order finite-difference equation, whose solution is handled much like that of ordinary differential equations [18]. Thus, putting it in operator form gives

$$(D - A)X_n = \frac{Y_{N_p+1}}{m} - AX_{N_p} \quad (5.43)$$

where the operator D indicates the finite difference. The characteristic equation is then

$$M - A = 0 \quad (5.44)$$

from which $M = A$. Hence the general solution (here a little different from ordinary differential equations) is

$$X_n = C_1 A^n \quad (5.45)$$

with C_1 a constant. Since the right-hand side of Eq. (5.43) is a constant, the particular solution is $X = C_2$, where C_2 is a constant. Substituting this into the original finite-difference equation (5.42) provides

$$C_2 - AC_2 = \frac{Y_{N_p+1}}{m} - AX_{N_p} \quad (5.46)$$

from which

$$C_2 = \frac{Y_{N_p+1}/m - AX_{N_p}}{1 - A} \quad (5.47)$$

The complete solution is therefore

$$X_n = C_1 A^n + \frac{Y_{N_p+1}/m - AX_{N_p}}{1-A} \quad (5.48)$$

To determine C_1 , we set $n = 0$:

$$C_1 = X_0 - \frac{Y_{N_p+1}/m - AX_{N_p}}{1-A}$$

and therefore
$$X_n = \left(X_0 - \frac{Y_{N_p+1}/m - AX_{N_p}}{1-A} \right) A^n + \frac{Y_{N_p+1}/m - AX_{N_p}}{1-A} \quad (5.49)$$

This result is useful to get the concentration X_n at any stage in the cascade, knowing the terminal concentrations. Putting $n = N_p$ and rearranging provide the very useful forms which follow.

For transfer from R to E (stripping of R)

$A \neq 1$:

Strip liquid not gas
$$\frac{X_0 - X_{N_p}}{X_0 - Y_{N_p+1}/m} = \frac{(1/A)^{N_p+1} - 1/A}{(1/A)^{N_p+1} - 1} \quad (5.50)$$

$$N_p = \frac{\log \left[\frac{X_0 - Y_{N_p+1}/m}{X_{N_p} - Y_{N_p+1}/m} (1-A) + A \right]}{\log 1/A} \quad (5.51)$$

$A = 1$:

$$\frac{X_0 - X_{N_p}}{X_0 - Y_{N_p+1}/m} = \frac{N_p}{N_p + 1} \quad (5.52)$$

$$N_p = \frac{X_0 - X_{N_p}}{X_{N_p} - Y_{N_p+1}/m} \quad (5.53)$$

For transfer from E to R (absorption into R). A similar treatment yields:

$A \neq 1$:

Gas not liquid
$$\frac{Y_{N_p+1} - Y_1}{Y_{N_p+1} - mX_0} = \frac{A^{N_p+1} - A}{A^{N_p+1} - 1} \quad (5.54)$$

$$N_p = \frac{\log \left[\frac{Y_{N_p+1} - mX_0}{Y_1 - mX_0} \left(1 - \frac{1}{A} \right) + \frac{1}{A} \right]}{\log A} \quad (5.55)$$

$A = 1$:

$$\frac{Y_{N_p+1} - Y_1}{Y_{N_p+1} - mX_0} = \frac{N_p}{N_p + 1} \quad (5.56)$$

$$N_p = \frac{Y_{N_p+1} - Y_1}{Y_1 - mX_0} \quad (5.57)$$

These are called the Kremser-Brown-Souders (or simply Kremser) equations, after those who derived them for gas absorption [7, 14] although apparently Turner [16] had used them earlier for leaching and solids washing. They are plotted in Fig. 5.16, which then becomes very convenient for quick solutions. We

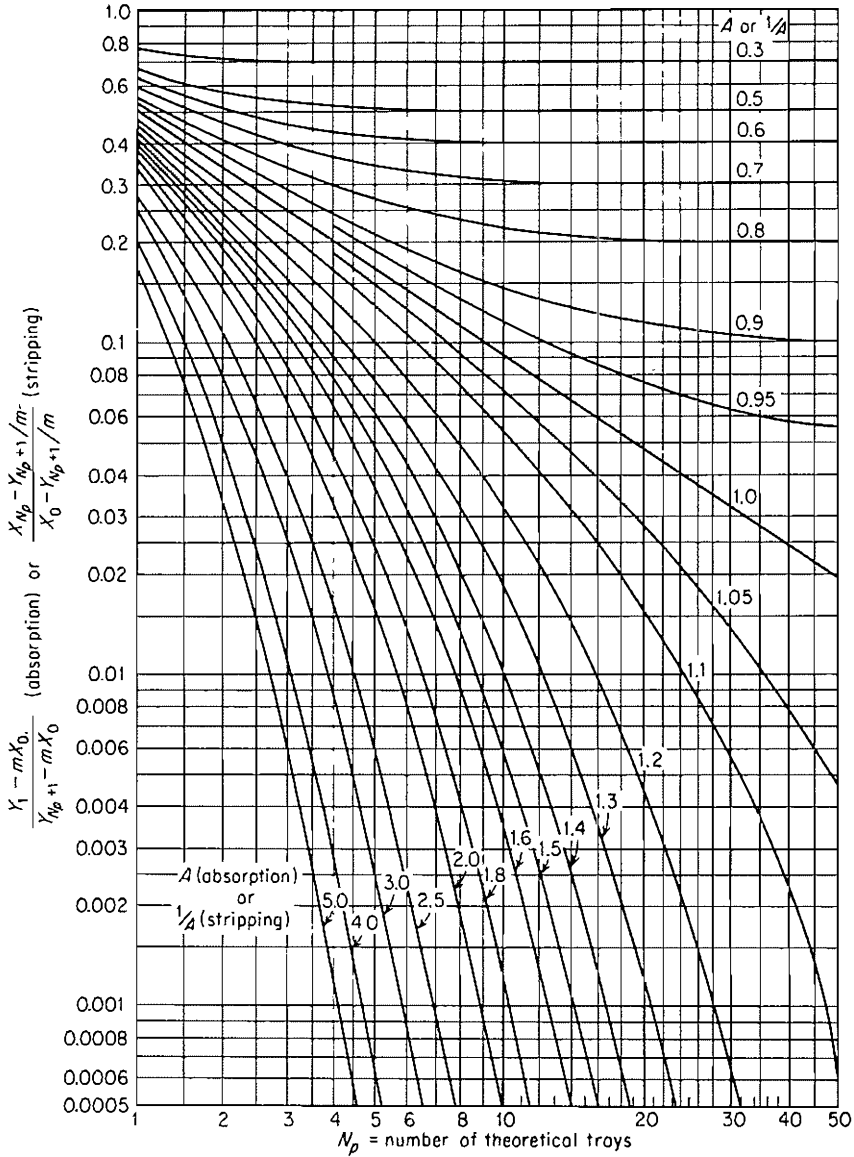


Figure 5.16 Number of theoretical stages for countercurrent cascades, with Henry's law equilibrium and constant absorption or stripping factors. [After Hachmuth and Vance, *Chem. Eng. Prog.*, 48, 523, 570, 617 (1952).]

shall have many opportunities to use them for different operations. In order to make them so generally useful, it is important to note that in the derivation, $A = R_S/mE_S = \text{const}$, where R_S and E_S are the nonsolute molar flow rates with concentrations (X in R and Y in E) defined as mole ratios and $m = Y/X$ at equilibrium. However, as necessary to keep A constant, A can be defined as R/mE , with R and E either as total molar flow rates, concentrations (x in R and y in E) as mole fractions, $m = y/x$, or as total mass flow rates and weight fractions, respectively.

We shall delay using the equations until specific opportunity arises.

Stages and Mass-Transfer Rates

It is clear from the previous discussion that each process can be considered either in terms of the number of stages it represents or in terms of the appropriate mass-transfer rates. A batch or continuous cocurrent operation, for example, is a single-stage operation, but the stage efficiency realized in the available contact time will depend upon the average mass-transfer rates prevailing. A change of composition greater than that possible with one stage can be brought about by repetition of the cocurrent process, where one of the effluents from the first stage is brought again into contact with a fresh treating phase. Alternatively, a countercurrent multistage cascade can be arranged. If, however, the countercurrent operation is carried out in continuous-contact fashion without repeated separation and recontacting of the phases in a stepwise manner, it is still possible to describe the operation in terms of the number of equilibrium stages to which it is equivalent. But in view of the differential changes in composition which occur in such cases, it is more correct to characterize them in terms of average mass-transfer coefficients or equivalent. Integration of the point mass-transfer rate equations must be delayed until the characteristics of each operation can be considered, but the computation of the number of equilibrium stages requires only the equilibrium and material-balance relationships.

Illustration 5.2 When a certain sample of moist soap is exposed to air at 75°C, 1 std atm pressure, the equilibrium distribution of moisture between the air and soap is as follows:

Wt % moisture in soap	0	2.40	3.76	4.76	6.10	7.83	9.90	12.63	15.40	19.02
Partial pressure water in air, mmHg	0	9.66	19.20	28.4	37.2	46.4	55.0	63.2	71.9	79.5

(a) Ten kilograms of wet soap containing 16.7% moisture by weight is placed in a vessel containing 10 m³ moist air whose initial moisture content corresponds to a water-vapor partial pressure of 12 mmHg. After the soap has reached a moisture content of 13.0%, the air in the vessel is entirely replaced by fresh air of the original moisture content and the system is then allowed to reach equilibrium. The total pressure and temperature are maintained at 1 std atm and 75°C, respectively. What will be the ultimate moisture content of the soap?

(b) It is desired to dry the soap from 16.7 to 4% moisture continuously in a countercurrent stream of air whose initial water-vapor partial pressure is 12 mmHg. The pressure and

temperature will be maintained throughout at 1 std atm and 75°C. For 1 kg initial wet soap per hour, what is the minimum amount of air required per hour?

(c) If 30% more air than that determined in part (b) is used, what will be the moisture content of the air leaving the drier? To how many equilibrium stages will the process be equivalent?

SOLUTION (a) The process is a batch operation. Since air and soap are mutually insoluble, with water distributing between them, Eq. (5.32) applies. Let the *E* phase be air-water and the *R* phase be soap-water. Define *Y* as kg water/kg dry air and *X* as kg water/kg dry soap. It will be necessary to convert the equilibrium data to these units.

For a partial pressure of moisture = $\bar{p}$ mmHg and a total pressure of 760 mmHg, $\bar{p}/(760 - \bar{p})$ is the mole ratio of water to dry air. Multiplying this by the ratio of molecular weights of water to air then gives $Y = [\bar{p}/(760 - \bar{p})](18.02/29)$. Thus, at $\bar{p} = 71.9$ mmHg,

$$Y = \frac{71.9}{760 - 71.9} \frac{18.02}{29} = 0.0650 \text{ kg water/kg dry air}$$

In similar fashion the percent moisture content of the soap is converted into units of *X*. Thus, at 15.40% moisture,

$$X = \frac{15.40}{100 - 15.40} = 0.182 \text{ kg water/kg dry soap}$$

In this manner all the equilibrium data are converted into these units and the data plotted as the equilibrium curve of Fig. 5.17.

First operation Initial air, $\bar{p} = 12$ mmHg:

$$Y_1 = \frac{12}{760 - 12} \frac{18.02}{29} = 0.00996 \text{ kg water/kg dry air}$$

Initial soap, 16.7% water:

$$S_1 = \frac{16.7}{100 - 16.7} = 0.2 \text{ kg water/kg dry soap}$$

Final soap, 13% water:

$$X_2 = \frac{13.0}{100 - 13.0} = 0.1493 \text{ kg water/kg dry soap}$$

$$R_S = 10(1 - 0.167) = 8.33 \text{ kg dry soap}$$

E_S , the mass of dry air, is found by use of the ideal-gas law,

$$E_S = 10 \frac{760 - 12}{760} \frac{273}{273 + 75} \frac{29}{18.02} = 10.1 \text{ kg dry air}$$

$$\text{Slope of operating line} = \frac{-R_S}{E_S} = \frac{-8.33}{10.1} = -0.825$$

From point *P*, coordinates (X_1 , Y_1), in Fig. 5.17, the operating line of slope -0.825 is drawn, to reach the abscissa $X_2 = 0.1493$ at point *Q*. The conditions at *Q* correspond to the end of the first operation.

Second operation $X_1 = 0.1493$, $Y_1 = 0.00996$, which locate point *S* in the figure. The operating line is drawn parallel to the line *PQ* since the ratio $-R_S/E_S$ is the same as in the first operation. Extension of the line to the equilibrium curve at *T* provides the final value of $X_2 = 0.103$ kg water/kg dry soap.

$$\text{Final moisture content of soap} = \frac{0.103}{1.103} \times 100 = 9.33\%$$

(b) Equation (5.35) and Fig. 5.10 apply

$$R_S = 1(1 - 0.167) = 0.833 \text{ kg dry soap/h}$$

Entering soap: $X_1 = 0.20 \text{ kg water/kg dry soap}$

Leaving soap: $X_2 = \frac{0.04}{1 - 0.04} = 0.0417 \text{ kg water/kg dry soap}$

Entering air: $Y_2 = 0.00996 \text{ kg water/kg dry air}$

$$\text{Slope of operating line} = \frac{R_S}{E_S} = \frac{\text{kg dry soap/h}}{\text{kg dry air/h}}$$

Point D , coordinates (X_2, Y_2) , is plotted in Fig. 5.17. The operating line of least slope giving rise to equilibrium conditions will indicate the least amount of air usable. Such a line is DG , which provides equilibrium conditions at $X_1 = 0.20$. The corresponding value of $Y_1 = 0.068 \text{ kg water/kg dry air}$ is read from the figure at G . Eq. (5.35):

$$R_S(Y_1 - Y_2) = E_S(Y_2 - Y_1) \quad 0.833(0.20 - 0.0417) = E_S(0.068 - 0.00996)$$

$$E_S = 2.27 \text{ kg dry air/h}$$

This corresponds to

$$\frac{2.27}{29} \cdot 22.41 \cdot \frac{760}{760 - 12} \cdot \frac{273 + 75}{273} = 2.27 \text{ m}^3/\text{kg dry soap}$$

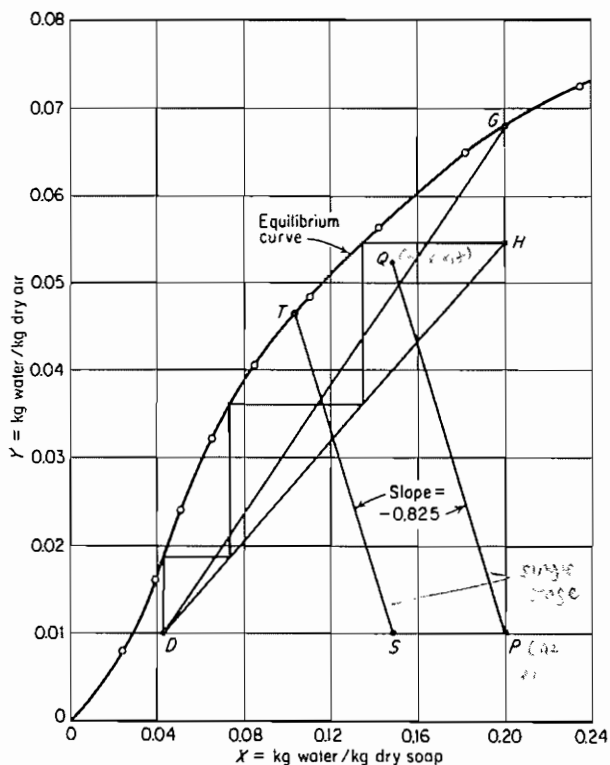


Figure 5.17 Solution to Illustration 5.2.

$$(c) E_S = 1.30(2.27) = 2.95 \text{ kg dry air/h. Eq. (5.35):}$$

$$0.833(0.20 - 0.0417) = 2.95(Y_1 - 0.00996)$$

$$Y_1 = 0.0547 \text{ kg water/kg dry air}$$

This corresponds to operating line DH , where H has the coordinates $X_1 = 0.20$, $Y_1 = 0.0547$. The equilibrium stages are marked on the figure and total three.

NOTATION FOR CHAPTER 5

Any consistent set of units may be used.

A	substance A, the diffusing solute
A	absorption factor, dimensionless
e	2.7183
E	phase E; total rate of flow of phase E, mole/ Θ
E_S	rate of flow of nondiffusing solvent in phase E, mole/ Θ
E_{ME}	fractional Murphree stage efficiency for phase E
E_{MR}	fractional Murphree stage efficiency for phase R
f	equilibrium-distribution function
F_G	gas-phase mass-transfer coefficient, mole/ $L^2\Theta$
F_L	liquid-phase mass-transfer coefficient, mole/ $L^2\Theta$
F_{OG}	overall gas mass-transfer coefficient, mole/ $L^2\Theta$
F_{OL}	overall liquid mass-transfer coefficient, mole/ $L^2\Theta$
i	generalized concentration in phase R
j	generalized concentration in phase E
k_x	liquid mass-transfer coefficient, mole/ $L^2\Theta$ (mole fraction)
k_y	gas mass-transfer coefficient, mole/ $L^2\Theta$ (mole fraction)
K_x	overall liquid mass-transfer coefficient, mole/ $L^2\Theta$ (mole fraction)
K_y	overall gas mass-transfer coefficient, mole/ $L^2\Theta$ (mole fraction)
m	equilibrium-curve slope, dimensionless
m', m''	slopes of chords of the equilibrium curve, dimensionless
n	stage n
N	mass-transfer flux, mole/ $L^2\Theta$
N_p	total number of ideal stages in a cascade
R	phase R; total rate of flow of phase R, mole/ Θ
R_S	rate of flow of nondiffusing solvent in phase R, mole/ Θ
S	stripping factor, dimensionless
x	concentration of A in a liquid or in phase R, mole fraction
X	concentration of A in phase R, mole A/mole non-A
y	concentration of A in a gas or in phase E, mole fraction
Y	concentration of A in phase E, mole A/mole non-A

Subscripts

A	substance A
e	equilibrium
E	phase E
G	gas
i	interface
L	liquid
n	stage n
O	overall
R	phase R

- 0 entering stage 1
1, 2 positions 1 and 2; stages 1 and 2

Superscript

- * in equilibrium with bulk concentration in other phase

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PROBLEMS

5.1 Repeat the calculations for the local mass-transfer flux of ammonia in Illustration 5.1 on the assumption that the gas pressure is 2 std atm. The gas mass velocity, bulk-average gas and liquid concentration, liquid flow rate, and temperature are unchanged.

5.2 In a certain apparatus used for the absorption of sulfur dioxide, SO_2 , from air by means of water, at one point in the equipment the gas contained 10% SO_2 by volume and was in contact with liquid containing 0.4% SO_2 (density = $990 \text{ kg/m}^3 = 61.8 \text{ lb/ft}^3$). The temperature was 50°C and the total pressure 1 std atm. The overall mass-transfer coefficient based on gas concentrations was $K_G = 7.36 \times 10^{-10} \text{ kmol/m}^2 \cdot \text{s} \cdot (\text{N/m}^2) = 0.055 \text{ lb mol SO}_2/\text{ft}^2 \cdot \text{h} \cdot \text{atm}$. Of the total diffusional resistance, 47% lay in the gas phase, 53% in the liquid. Equilibrium data at 50°C are:

kg SO_2 /100 kg water	0.2	0.3	0.5	0.7
Partial pressure SO_2 , mmHg	29	46	83	119

- (a) Calculate the overall coefficient based on liquid concentrations in terms of mol/vol.
(b) Calculate the individual mass-transfer coefficient for the gas, expressed as k_G mol/(area)(time)(pressure), k_y mol/(area)(time)(mole fraction), and k_c mol/(area)(time)(mole/vol),

and for the liquid expressed as k_L mole/(area)(time)(mole/vol) and k_x mol/(area)(time)(mole fraction).

(c) Determine the interfacial compositions in both phases.

5.3 The equilibrium partial pressure of water vapor in contact with a certain silica gel on which water is adsorbed is, at 25°C, as follows:

Partial pressure of water, mmHg	0	2.14	4.74	7.13	9.05	10.9	12.6	14.3	16.7
lb water/100 lb dry gel = kg/100 kg	0	5	10	15	20	25	30	35	40

(a) Plot the equilibrium data as $\bar{p}$ = partial pressure of water vapor against x = wt fraction water in the gel.

(b) Plot the equilibrium data as X = mol water/mass dry gel, Y = mol water vapor/mol dry air, for a total pressure of 1 std atm.

(c) When 10 lb (4.54 kg) of silica gel containing 5 wt% adsorbed water is placed in a flowing airstream containing a partial pressure of water vapor = 12 mmHg, the total pressure is 1 std atm and the temperature 25°C. When equilibrium is reached, what mass of additional water will the gel have adsorbed? Air is not adsorbed.

(d) One pound mass (0.454 kg) of silica gel containing 5 wt% adsorbed water is placed in a vessel in which there are 400 ft³ (11.33 m³) moist air whose partial pressure of water is 15 mmHg. The total pressure and temperature are kept at 1 std atm and 25°C, respectively. At equilibrium, what will be the moisture content of the air and gel and the mass of water adsorbed by the gel? Ans.: 0.0605 kg water.

(e) Write the equation of the operating line for part (d) in terms of X and Y . Convert this into an equation in terms of $\bar{p}$ and x , and plot the operating curve on $\bar{p}$, x coordinates.

(f) If 1 lb (0.454 kg) of silica gel containing 18% adsorbed water is placed in a vessel containing 500 ft³ (14.16 m³) dry air and the temperature and pressure are maintained at 25°C and 1 std atm, respectively, what will be the final equilibrium moisture content of the air and the gel.

(g) Repeat part (f) for a total pressure of 2 std atm. Note that the equilibrium curve in terms of X and Y previously used is not applicable.

5.4 The equilibrium adsorption of benzene vapor on a certain activated charcoal at 33.3°C is reported as follows:

Benzene vapor adsorbed, cm ³ (STP)/g charcoal	15	25	40	50	65	80	90	100
Partial pressure benzene, mmHg	0.0010	0.0045	0.0251	0.115	0.251	1.00	2.81	7.82

(a) A nitrogen-benzene vapor mixture containing 1.0% benzene by volume is to be passed countercurrently at the rate 100 ft³/min (4.72×10^{-2} m³/s) to a moving stream of the activated charcoal so as to remove 95% of the benzene from the gas in a continuous process. The entering charcoal contains 15 cm³ benzene vapor (at STP) adsorbed per gram charcoal. The temperature and total pressure are to be maintained at 33.3°C and 1 std atm, respectively, throughout. Nitrogen is not adsorbed. What is the least amount of charcoal which can be used per unit time? If twice as much is used, what will be the concentration of benzene adsorbed upon the charcoal leaving?

(b) Repeat part (a) for a cocurrent flow of gas and charcoal.

(c) Charcoal which has adsorbed upon it 100 cm³ (at STP) benzene vapor per gram is to be stripped at the rate 100 lb/h (45.4 kg/h) of its benzene content to a concentration of 55 cm³ adsorbed benzene per gram charcoal by continuous countercurrent contact with a stream of pure nitrogen gas at 1 std atm. The temperature will be maintained at 33.3°C. What is the minimum rate

of nitrogen flow? What will be the benzene content of the exit gas if twice as much nitrogen is fed? What will be the number of stages? Ans.: 3 theoretical stages.

5.5 A mixture of hydrogen and air, 4 std atm pressure, bulk-average concentration 50% H_2 by volume, flows through a circular reinforced tube of vulcanized rubber with respective ID and OD of 25.4 and 28.6 mm (1.0 and 1.125 in) at an average velocity 3.05 m/s (10 ft/s). Outside the pipe, hydrogen-free air at 1 std atm flows at right angles to the pipe at a velocity 3.05 m/s (10 ft/s). The temperature is everywhere 25°C. The solubility of hydrogen in the rubber is reported as 0.053 cm³ H_2 (STP)/(cm³ rubber) · (atm partial pressure), and its diffusivity in the rubber as 0.18×10^{-5} cm²/s. The diffusivity of H_2 -air = 0.611 cm²/s at 0°C, 1 std atm. Estimate the rate of loss of hydrogen from the pipe per unit length of pipe. Ans.: 4.5×10^{-11} kmol/(m of pipe length) · s.

5.6 Assuming that the equilibrium curve of Fig. 5.7 is straight, and of slope m , derive the relation between the Murphree stage efficiencies E_{ME} and E_{MR} .

5.7 If the equilibrium-distribution curve of the cross-flow cascade of Fig. 5.13 is everywhere straight, and of slope m , make a solute material balance about stage $n + 1$ and by following the procedure used to derive Eq. (5.51) show that

$$N_p = \frac{\log[(X_0 - Y_0/m)/(X_{N_p} - Y_0/m)]}{\log(S + 1)}$$

where S is the stripping factor, mE_S/R_S , constant for all stages, and N_p is the total number of stages.

5.8 A single-stage liquid-extraction operation with a linear equilibrium-distribution curve ($Y = mX$ at equilibrium) operates with a stripping factor $S = mE_S/R_S = 1.0$ and has a Murphree stage efficiency $E_{ME} = 0.6$. In order to improve the extraction with a minimum of plant alteration, the effluents from the stage will both be led to another stage identical in construction, also of efficiency $E_{ME} = 0.6$. The combination, two stages in series, is in effect a single stage. It is desired to compute the overall Murphree stage efficiency E_{MEO} of the combination. A direct derivation of the relation between E_{ME} and E_{MEO} is difficult. The result is most readily obtained as follows:

(a) Refer to Fig. 5.7. Define a stage efficiency for one stage as $E = (Y_2 - Y_1)/(Y_e - Y_1)$. For a linear equilibrium-distribution curve, derive the relation between E_{ME} and E .

(b) Derive the relation between E_O , the value for the combined stages in series, and E .

(c) Compute E_{MEO} . Ans. = 0.8824.

PART TWO

GAS-LIQUID OPERATIONS

The operations which include humidification and dehumidification, gas absorption and desorption, and distillation in its various forms all have in common the requirement that a gas and a liquid phase be brought into contact for the purpose of a diffusional interchange between them.

The order listed above is in many respects that of increasing complexity of the operations, and this is therefore the order in which they will be considered. Since in humidification the liquid is a pure substance, concentration gradients exist and diffusion of matter occurs only in the gas phase. In absorption, concentration gradients exist in both the liquid and the gas, and diffusion of at least one component occurs in both. In distillation, all the substances comprising the phases diffuse. These operations are also characterized by an especially intimate relationship between heat and mass transfer. The evaporation or condensation of a substance introduces consideration of latent heats of vaporization, sometimes heats of solution as well. In distillation, the new phase necessary for mass-transfer separation is created from the original by addition or withdrawal of heat. Our discussion must necessarily include consideration of these important heat quantities and their effects.

In all these operations, the equipment used has as its principal function the contact of the gas and liquid in as efficient a fashion as possible, commensurate with the cost. In principle, at least, any type of equipment satisfactory for one of these operations is also suitable for the others, and the major types are indeed used for all. For this reason, our discussion begins with equipment.

EQUIPMENT FOR GAS-LIQUID OPERATIONS

The purpose of the equipment used for the gas-liquid operations is to provide intimate contact of the two fluids in order to permit interphase diffusion of the constituents. The rate of mass transfer is directly dependent upon the interfacial surface exposed between the phases, and the nature and degree of dispersion of one fluid in the other are therefore of prime importance. The equipment can be broadly classified according to whether its principal action is to disperse the gas or the liquid, although in many devices both phases become dispersed.

GAS DISPERSED

In this group are included those devices, such as sparged and agitated vessels and the various types of tray towers, in which the gas phase is dispersed into bubbles or foams. Tray towers are the most important of the group, since they produce countercurrent, multistage contact, but the simpler vessel contactors have many applications.

Gas and liquid can conveniently be contacted, with gas dispersed as bubbles, in agitated vessels whenever multistage, countercurrent effects are not required. This is particularly the case when a chemical reaction between the dissolved gas and a constituent of the liquid is required. The carbonation of a lime slurry, the chlorination of paper stock, the hydrogenation of vegetable oils, the aeration of fermentation broths, as in the production of penicillin, the production of citric acid from beet sugar by action of microorganisms, and the

aeration of activated sludge for biological oxidation are all examples. It is perhaps significant that in most of them solids are suspended in the liquids. Because the more complicated countercurrent towers have a tendency to clog with such solids and because solids can be suspended in the liquids easily in agitated vessels, the latter are usually more successful in such service.†

The gas-liquid mixture can be mechanically agitated, as with an impeller, or, as in the simplest design, agitation can be accomplished by the gas itself in sparged vessels. The operation may be batch, semibatch with continuous flow of gas and a fixed quantity of liquid, or continuous with flow of both phases.

SPARGED VESSELS (BUBBLE COLUMNS)

A *sparger* is a device for introducing a stream of gas in the form of small bubbles into a liquid. If the vessel diameter is small, the sparger, located at the bottom of the vessel, may simply be an open tube through which the gas issues into the liquid. For vessels of diameter greater than roughly 0.3 m, it is better to use several orifices for introducing the gas to ensure better gas distribution. In that case, the orifices may be holes, from 1.5 to 3 mm ($\frac{1}{16}$ to $\frac{1}{4}$ in) in diameter, drilled in a pipe distributor placed horizontally at the bottom of the vessel. Porous plates made of ceramics, plastics, or sintered metals are also used, but their fine pores are more readily plugged with solids which may be present in the gas or the liquid.

The purpose of the sparging may be contacting the sparged gas with the liquid. On the other hand, it may be simply a device for agitation. It can provide the gentlest of agitation, used for example in washing nitroglycerin with water; it can provide vigorous agitation as in the *pachuca tank* (which see). Air agitation in the extraction of radioactive liquids offers the advantage of freedom from moving parts, but it may require decontamination of the effluent air. There is no standardization of the depth of liquid; indeed, very deep tanks, 15 m (50 ft) or more, may be advantageous [117] despite the large work of compression required for the gas.

Gas-Bubble Diameter

The size of gas bubbles depends upon the rate of flow through the orifices, the orifice diameter, the fluid properties, and the extent of turbulence prevailing in the liquid. What follows is for cases where turbulence in the liquid is solely that generated by the rising bubbles and when orifices are horizontal and sufficiently separated to prevent bubbles from adjacent orifices from interfering with each other (at least approximately $3d_o$ apart).

† For cases where very low gas-pressure drop is required, as in absorption of sulfur dioxide from flue gas by limestone slurries, venturi scrubbers (which see) are frequently used.

Very slow gas flow rate [118], $Q_{Go} < [20(\sigma d_o g_c)^5 / (g \Delta \rho)^2 \rho_L^3]^{1/6}$ For waterlike liquids, the diameter can be computed by equating the bouyant force on the immersed bubble, $(\pi/6)d_p^3 \Delta \rho g / g_c$, which tends to lift the bubble away from the orifice, to the force $\pi d_o \sigma$ due to surface tension, which tends to retain it at the orifice. This provides

$$d_p = \left(\frac{6 d_o \sigma g_c}{g \Delta \rho} \right)^{1/3} \quad (6.1)$$

which has been tested for orifice diameters up to 10 mm. For large liquid viscosities, up to 1 kg/m · s (1000 cP) [32],

$$d_p = 2.312 \left(\frac{\mu_L Q_{Go}}{\rho_L g} \right)^{1/4} \quad (6.2)$$

Intermediate flow rates, $Q_{Go} > [20(\sigma d_o g_c)^5 / (g \Delta \rho)^2 \rho_L^3]^{1/6}$ but $Re_o < 2100$ These bubbles are larger than those described above, although still fairly uniform, and they form in chains rather than separately. For air-water [71]

$$d_p = 0.0287 d_o^{1/2} Re_o^{1/3} \quad (6.3)$$

where d_p and d_o are in meters† and $Re_o = d_o V_o \rho_G / \mu_G = 4 w_o / \pi d_o \mu_G$. For other gases and liquids [118]

$$d_p = \left(\frac{72 \rho_L}{\pi^2 g \Delta \rho} \right)^{1/5} Q_{Go}^{0.4} \quad (6.4)$$

Large gas rates [71], $Re_o = 10\,000$ to $50\,000$ Jets of gas which rise from the orifice break into bubbles at some distance from the orifice. The bubbles are smaller than those described above and nonuniform in size. For air-water and orifice diameters 0.4 to 1.6 mm

$$d_p = 0.0071 Re_o^{-0.05} \quad (6.5)$$

with d_p in meters.‡ For the *transition range* ($Re_o = 2100$ to $10\,000$) there is no correlation of data. It is suggested that d_p for air-water can be approximated by the straight line on log-log coordinates between the points given by d_p at $Re_o = 2100$ and at $Re_o = 10\,000$.

Rising Velocity (Terminal Velocity) of Single Bubbles [83]

Typically, the steady-state rising velocity of single gas bubbles, which occurs when the bouyant force equals the drag force on the bubbles, varies with the bubble diameter as shown in Fig. 6.1.

† For d_p and d_o in feet, the coefficient of Eq. (6.3) is 0.052.

‡ For d_p in feet, the coefficient of Eq. (6.5) is 0.0233.

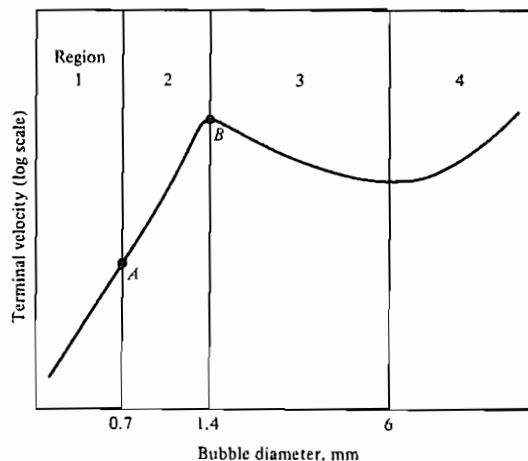


Figure 6.1 Rising (terminal) velocity of single gas bubbles.

Region 1, $d_p < 0.7$ mm The bubbles are spherical, and they behave like rigid spheres, for which the terminal velocity is given by Stokes' law

$$V_t = \frac{g d_p^2 \Delta \rho}{18 \mu_L} \quad (6.6)$$

Region 2, $0.7 \text{ mm} < d_p < 1.4 \text{ mm}$ The gas within the bubble circulates, so that the surface velocity is not zero. Consequently the bubble rises faster than rigid spheres of the same diameter. There is no correlation of data; it is suggested that V_t may be estimated as following the straight line on Fig. 6.1 drawn between points A and B. Coordinates of these points are given respectively by Eqs. (6.6) and (6.7).

Regions 3 ($1.4 \text{ mm} < d_p < 6 \text{ mm}$) and 4 ($d_p > 6 \text{ mm}$) The bubbles are no longer spherical and in rising follow a zigzag or helical path. For region 4 the bubbles have a spherically shaped cap. In both these regions for liquids of low viscosity [80]

$$V_t = \sqrt{\frac{2 \sigma g_c}{d_p \rho_L} + \frac{g d_p}{2}} \quad (6.7)$$

Swarms of Gas Bubbles

The behavior of large numbers of bubbles crowded together is different from that of isolated bubbles. Rising velocities are smaller because of crowding, and bubble diameter may be altered by liquid turbulence, which causes bubble breakup, and by coalescence of colliding bubbles.

Gas Holdup

By *gas holdup* ϕ_G is meant the volume fraction of the gas-liquid mixture in the vessel which is occupied by the gas. If the superficial gas velocity, defined as the volume rate of gas flow divided by the cross-sectional area of the vessel, is V_G , then V_G/ϕ_G can be taken as the true gas velocity relative to the vessel walls. If the liquid flows upward, cocurrently with the gas, at a velocity relative to the vessel walls $V_L/(1 - \phi_G)$, the relative velocity of gas and liquid, or *slip velocity*, is

$$V_s = \frac{V_G}{\phi_G} - \frac{V_L}{1 - \phi_G} \quad (6.8)$$

Equation (6.8) will also give the slip velocity for countercurrent flow of liquid if V_L for the downward liquid flow is assigned a negative sign.

The holdup for sparged vessels, correlated through the slip velocity, is shown in Fig. 6.2. This is satisfactory for no liquid flow ($V_L = 0$), for cocurrent

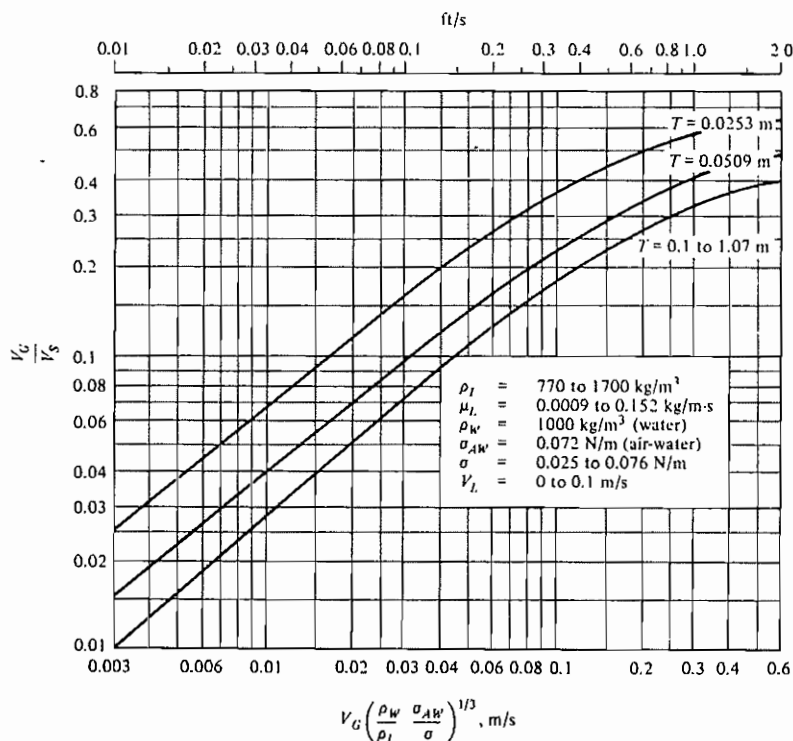


Figure 6.2 Slip velocity, sparged vessels. [G. A. Hughmark, *Ind. Eng. Chem. Process Des. Dev.*, 6, 218 (1967), with permission of the American Chemical Society.]

liquid flow up to $V_L = 0.1$ m/s (0.3 ft/s), and very probably also for small countercurrent liquid flow [1]. For large countercurrent velocities, holdup is best treated differently [17]. The holdup increases with increasing liquid viscosity, passes through a maximum at $\mu_L = 0.003$ kg/m · s (3 cP), and then decreases [40].

Specific Interfacial Area

If unit volume of a gas-liquid mixture contains a gas volume φ_G made up of n bubbles of diameter d_p , then $n = \varphi_G / (\pi d_p^3 / 6)$. If the interfacial area in the unit volume is a , then $n = a / \pi d_p^2$. Equating the two expressions for n provides the specific area

$$a = \frac{6\varphi_G}{d_p} \quad (6.9)$$

For low velocities, the bubble size may be taken as that produced at the orifices of the sparger, corrected as necessary for pressure. If the liquid velocity is large, the bubble size will be altered by turbulent breakup and coalescence of bubbles. Thus, for air-water, and in the ranges $\varphi_G = 0.1$ to 0.4 and $V_L / (1 - \varphi_G) = 0.15$ to 15 m/s (0.5 to 5 ft/s), the bubble size is approximated by [95]

$$d_p = \frac{2.344 \times 10^{-3}}{[V_L / (1 - \varphi_G)]^{0.67}} \quad (6.10)$$

where d_p is in meters and V_L in m/s.†

Mass Transfer

In practically all the gas-bubble liquid systems, the liquid-phase mass-transfer resistance is strongly controlling, and gas-phase coefficients are not needed. The liquid-phase coefficients, to within about 15 percent, are correlated by [61]

$$\text{Sh}_L = \frac{F_L d_p}{c D_L} = 2 + b' \text{Re}_G^{0.779} \text{Sc}_L^{0.546} \left(\frac{d_p g^{1/3}}{D_L^{2/3}} \right)^{0.116} \quad (6.11)$$

$$\text{where } b' = \begin{cases} 0.061 & \text{single gas bubbles} \\ 0.0187 & \text{swarms of bubbles} \end{cases}$$

The gas Reynolds number must be calculated with the slip velocity: $\text{Re}_G = d_p V_s \rho_L / \mu_L$. For single bubbles, $\varphi_G = 0$ and $V_s = V_L$. In most cases the liquid will be stirred well enough to permit solute concentrations in the liquid to be considered uniform throughout the vessel.

† For d_p in feet and V_L in ft/s, the coefficient of Eq. (6.10) is 0.01705.

Power

The power supplied to the vessel contents, which is responsible for the agitation and creation of large interfacial area, is derived from the gas flow. Bernoulli's equation, a mechanical-energy balance written for the gas between location o (just above the sparger orifices) and location s (at the liquid surface), is

$$\frac{V_s^2 - V_o^2}{2g_c} + (Z_s - Z_o)\frac{g}{g_c} + \int_o^s \frac{d_p}{\rho_G} + W + H_f = 0 \quad (6.12)$$

The friction loss H_f and V_s can be neglected, and the gas density can be described by the ideal-gas law, whereupon

$$W = \frac{V_o^2}{2g_c} + \frac{p_o}{\rho_{Go}} \ln \frac{p_o}{p_s} + (Z_o - Z_s)\frac{g}{g_c} \quad (6.13)$$

W is the work done by the gas on the vessel contents, per unit mass of gas. Work for the gas compressor will be larger, to account for friction in the piping and orifices, any dust filter for the gas that may be used, and compressor inefficiency.

Illustration 6.1 A vessel 1.0 m in diameter and 3.0 m deep (measured from gas sparger at the bottom to liquid overflow at the top) is to be used for stripping chlorine from water by sparging with air. The water will flow continuously upward at the rate $8 \times 10^{-4} \text{ m}^3/\text{s}$ ($1.7 \text{ ft}^3/\text{min}$). Airflow will be 0.055 kg/s (7.28 lb/min) at 25°C . The sparger is in the form of a ring, 25 cm in diameter, containing 50 orifices, each 3.0 mm in diameter. Estimate the specific interfacial area and the mass-transfer coefficient.

SOLUTION The gas flow rate per orifice = $0.055/50 = 0.0011 \text{ kg/s} = w_o \cdot d_o = 0.003 \text{ m}$, and $\mu_G = 1.8 \times 10^{-5} \text{ kg/m} \cdot \text{s}$. $\text{Re}_o = 4w_o/\pi d_o \mu_G = 26\,000$; Eq. (6.5): $d_p = 4.27 \times 10^{-3} \text{ m}$ at the orifice. The pressure at the orifice, ignoring the gas holdup, is that corresponding to 3.0 m of water (density 1000 kg/m^3) or $101.3 \text{ kN/m}^2 \text{ std atm} + 3.0(1000)(9.81)/1000 = 130.8 \text{ kN/m}^2$.

The average pressure in the column (for 1.5 m of water) is 116.0 kN/m^2 . Therefore

$$\rho_G = \frac{29}{22.41} \frac{273}{298} \frac{116.0}{101.3} = 1.358 \text{ kg/m}^3$$

$T = 1.0 \text{ m}$, and the vessel cross-sectional area = $\pi T^2/4 = 0.785 \text{ m}^2$.

$$V_G = \frac{0.055}{0.785(1.358)} = 0.0516 \text{ m/s} \quad \text{and} \quad V_L = \frac{8 \times 10^{-4}}{0.785} = 0.00102 \text{ m/s}$$

Since $\rho_L = 1000 \text{ kg/m}^3$ and $\sigma = 0.072 \text{ N/m}$ for the liquid, the abscissa of Fig. 6.2 is 0.0516 m/s . Hence (Fig. 6.2), $V_G/V_S = 0.11$ and $V_S = V_G/0.11 = 0.469 \text{ m/s}$. Eq. (6.8):

$$0.469 = \frac{0.0516}{\Phi_G} - \frac{0.00102}{1 - \Phi_G}$$

$$\Phi_G = 0.1097$$

Equation (6.10) is not to be used in view of the small V_L . Therefore the bubble diameter at the average column pressure, calculated from d_p at the orifice, is

$$d_p = \left[(0.00427)^3 \frac{130.8}{116.0} \right]^{1/3} = 0.00447 \text{ m}$$

Eq. (6.9): $a = 147.2 \text{ m}^2/\text{m}^3$. Ans.

At 25°C, D_L for Cl_2 in H_2O = 1.44×10^{-9} m²/s; g = 9.81 m/s². In Eq. (6.11), $d_p g^{1/3} / D_L^{2/3} = 7500$. The viscosity of water $\mu_1 = 8.937 \times 10^{-4}$ kg/m · s, and $\text{Re}_G = d_p V_S \rho_L / \mu_1 = 2346$, $\text{Sc}_L = \mu_L / \rho_L D_L = 621$, and hence $\text{Sh}_L = 746$. For dilute solutions of Cl_2 in H_2O the molar density $c = 1000/18.02 = 55.5$ kmol/m³. Therefore $F_L = (c D_L \text{Sh}_L) / d_p = 0.0127$ kmol/m² · s. Ans.

MECHANICALLY AGITATED VESSELS

Mechanical agitation of a liquid, usually by a rotating device, is especially suitable for dispersing solids, liquids, or gases into liquids, and it is used for many of the mass-transfer operations. Agitators can produce very high turbulence intensities (u'/u averages as high as 0.35 [104] ranging up to 0.75 in the vicinity of an agitating impeller [31, 74], contrasted with about 0.04 for turbulent flow in pipes), which not only produce good mass-transfer coefficients but are necessary for effective dispersion of liquids and gases. High liquid velocities, particularly desirable for suspending solids, are readily obtained.

It will be useful first to establish the characteristics of these vessels when used simply for stirring single-phase liquids, as in blending, for example. Applications to particular mass-transfer operations will be considered separately later.

MECHANICAL AGITATION OF SINGLE-PHASE LIQUIDS

Typical agitated vessels are vertical circular cylinders; rectangular tanks are unusual, although not uncommon in certain liquid-extraction applications. The liquids are usually maintained at a depth of one to two tank diameters.

Impellers

There are literally scores of designs. The discussion here is limited to the most popular, as shown in Fig. 6.3. These are usually mounted on an axially arranged, motor-driven shaft, as in Fig. 6.4. In the smaller sizes, particularly, the impeller and shaft may enter the vessel at an angle to the vessel axis, with the motor drive clamped to the rim of the vessel.

The *marine-type propeller*, Fig. 6.3a, is characteristically operated at relatively high speed, particularly in low-viscosity liquids, and is especially useful for its high liquid-circulating capacity. The ratio of impeller diameter to vessel diameter, d_i/T , is usually set at 1 : 5 or less. The liquid flow is axial, and the propeller is turned so that it produces downward flow toward the bottom of the vessel. In describing the propeller, *pitch* refers to the ratio of the distance advanced per revolution by a free propeller operating without slip to the propeller diameter; *square pitch*, which is most common in agitator designs, means a pitch equal to unity. Propellers are more frequently used for liquid-blending operations than for mass-transfer purposes.

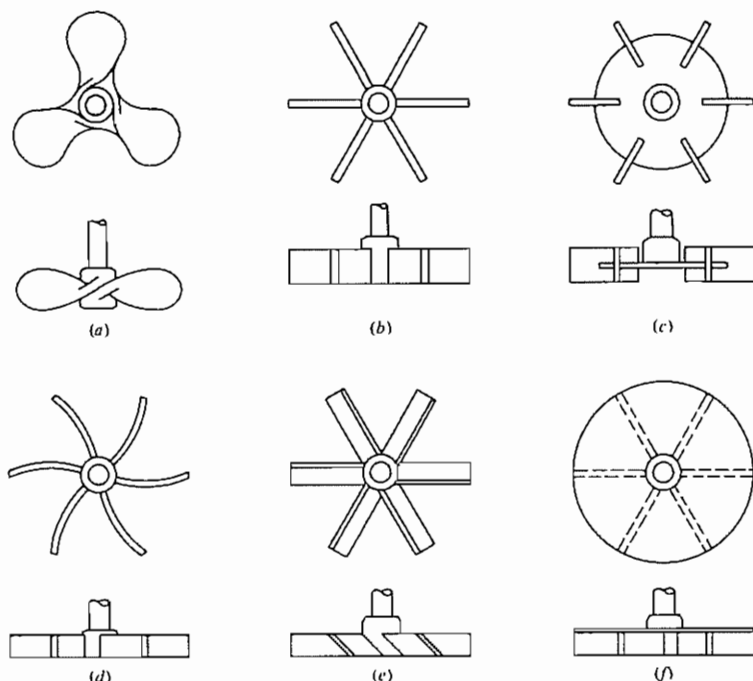


Figure 6.3 Impellers, with typical proportions: (a) marine-type propellers; (b) flat-blade turbine, $w = d_i/5$; (c) disk flat-blade turbine, $w = d_i/5$, $d_d = 2d_i/3$, $B = d_i/4$; (d) curved-blade turbine, $w = d_i/8$; (e) pitched-blade turbine, $w = d_i/8$; and (f) shrouded turbine, $w = d_i/8$.

Turbines, particularly the flat-blade designs of Fig. 6.3b and c, are frequently used for mass-transfer operations.† The curved blade design (Fig. 6.3d) is useful for suspension of fragile pulps, crystals, and the like and the pitched-blade turbine (Fig. 6.3e) more frequently for blending liquids. The shrouded impeller of Fig. 6.3f has limited use in gas-liquid contracting. Flow of liquid from the impeller is radial except for the pitched-blade design, where it is axial. Although the best ratio d_i/T depends upon the application, a value of 1 : 3 is common. Turbines customarily operate with peripheral speeds of the order of 2.5 to 4.6 m/s (450 to 850 ft/min), depending upon the service.

Vortex Formation and Prevention

Typical flow patterns for single-phase Newtonian liquids of moderate viscosity are shown in Fig. 6.4. For an axially located impeller operating at low speeds in

† These designs have been in use for many years. G. Agricola's "De re metallica" (H. C. and L. H. Hoover, translators, from the Latin of 1556, book 8, p. 299, Dover, New York, 1950) shows a woodcut of six-bladed flat-blade turbine impellers being used to suspend a gold ore in water.

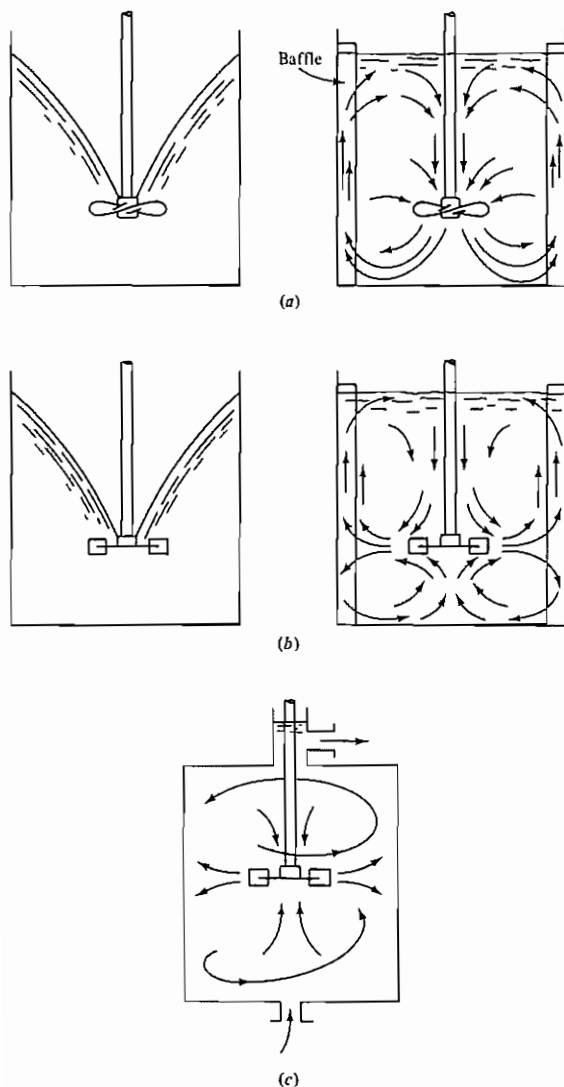


Figure 6.4 Liquid agitation in presence of a gas-liquid interface, with and without wall baffles: (a) marine impeller and (b) disk flat-blade turbines; (c) in full vessels without a gas-liquid interface (continuous flow) and without baffles.

an open vessel with a gas-liquid surface, the liquid surface is level and the liquid circulates about the axis. As the impeller speed is increased to produce turbulent conditions, the power required to turn the impeller increases and a vortex begins to form around the shaft. At higher speeds the vortex eventually reaches the impeller, as in the left-hand sketches of Fig. 6.4*a* and *b*. Air is drawn into the liquid, the impeller operates partly in air, and the power required drops. The drawing of gas into the liquid is frequently undesirable, and, in addition, vortex formation leads to difficulties in scaling up of model experiments and pilot-plant studies, so that steps are usually taken to prevent vortices. A possible exception is when a solid that is difficult to wet is to be dissolved in a liquid, e.g., powdered boric acid in water, when it is useful to pour the solid into the vortex.

Vortex formation can be prevented by the following means:

1. *Open tanks with gas-liquid surfaces.*[†]
 - a. Operation only in the laminar range for the impeller (Re 10 to 20). This is usually impractically slow for mass-transfer purposes.
 - b. Off-center location of the impeller on a shaft entering the vessel at an angle to the vessel axis. This is used relatively infrequently in permanent installations, and is used mostly in small-scale work with impellers where the agitator drive is clamped to the rim of the vessel.
 - c. Installation of baffles. This is by far the most common method. Standard baffling [78] consists of four flat, vertical strips arranged radially at 90° intervals around the tank wall, extending for the full liquid depth, as in the right-hand sketches of Fig. 6.4*a* and *b*. The standard baffle width is usually $T/12$, less frequently $T/10$ ("10 percent baffles"). Baffles are sometimes set at a clearance from the vessel wall of about one-sixth the baffle width to eliminate stagnant pockets in which the solids can accumulate. They may be used as supports for helical heating or cooling coils immersed in the liquid. The condition of *fully baffled turbulence* is considered to exist for these vessels at impeller Reynolds numbers above 10 000.[‡] The presence of baffles reduces swirl and increases the vertical liquid currents in the vessel, as shown on the right of Fig. 6.4. The power required to turn the impeller is also increased.
2. *Closed tanks, operated full, with no gas-liquid surface* [70]. This is especially convenient for cases where the liquid flows continuously through the vessel, as in Fig. 6.4*c*. A circular flow pattern is superimposed upon the axial flow directed toward the center of the impeller. The arrangement is not practical, of course, for gas-liquid contact, where baffles should be used.

[†] Addition of minute amounts of certain polymers to the liquid can inhibit vortex formation during bottom drainage of unagitated tanks [R. J. Gordon, *et al.*, *Nature Phys. Sci.*, **231**, 25 (1971); *J. Appl. Polymer Sci.*, **16**, 1629 (1972); *AIChE J.*, **22**, 947 (1976)].

[‡] At very high speeds, gas can still be drawn into the impeller despite the presence of baffles. For flat-blade turbines with four blades in tanks with standard baffles, this will occur when [27]

$$\frac{d_i N^2}{g} \frac{d_w^2}{T^2 Z} \left(\frac{C}{Z} \right)^{2/3} > 0.005$$

Fluid Mechanics

Discussions of the characteristics of agitated vessels frequently involve considerations of similarity, and this is particularly important when it is desired to obtain simple laws describing vessels of different sizes. For example, with proper regard to similarity and by describing results in terms of dimensionless groups, it is possible to experiment with a vessel filled with air and expect the results for power, degree of turbulence, and the like, to be applicable to vessels filled with liquid [52]. Three types of similarity are significant when dealing with liquid motion:

1. *Geometric similarity* refers to linear dimensions. Two vessels of different sizes are geometrically similar if the ratios of corresponding dimensions on the two scales are the same, and this refers to tank, baffles, impellers, and liquid depths. If photographs of two vessels are completely superimposable, they are geometrically similar.
2. *Kinematic similarity* refers to motion and requires geometric similarity and the same ratio of velocities for corresponding positions in the vessels. This is especially important for mass-transfer studies.
3. *Dynamic similarity* concerns forces and requires all force ratios for corresponding positions to be equal in kinematically similar vessels.

The complex motion in an agitated vessel cannot be put into complete analytical form, but the equations of motion can be described in terms of dimensionless groups as follows [5]:

$$f\left(\frac{\bar{L}u\rho}{\mu}, \frac{u^2}{\bar{L}g}, \frac{\Delta p}{\rho u^2} g_c\right) = 0 \quad (6.14)$$

where $\bar{L}$ is some characteristic length and Δp is a pressure difference. The first of these groups is the ratio of inertial to viscous forces, the familiar Reynolds number. If the characteristic length is taken as d_i and the velocity proportional to the impeller-tip speed πNd_i , when the π is omitted the group becomes the impeller Reynolds number $Re = d_i^2 N \rho_L / \mu_L$.

The second group, the ratio of inertial to gravity forces, important when the liquid surface is wavy or curved, as with a vortex, is the *Froude number*. With u and $\bar{L}$ defined as before, it becomes $Fr = d_i N^2 / g$.

The third is the ratio of pressure differences resulting in flow to inertial forces. If Δp is taken as force/area, the group can be recognized as a *drag coefficient*, which is usually written $F_D / A(\rho u^2 / 2g_c)$. The group can be developed into a dimensionless *power number* as follows. Power required by the impeller is the product of Δp and the volumetric liquid-flow rate Q_L , so that

$$\Delta p = \frac{P}{Q_L} \quad (6.15)$$

The fluid velocity is proportional [104] to Nd_i^2/T or, for geometrically similar vessels, to Nd_i , so that Q_L is proportional to Nd_i^3 [52]. Substitution of these transforms the group into $Po = P g_c / \rho N^3 d_i^5$.

For geometrically similar vessels, therefore, Eq. (6.14) can be written

$$f(\text{Re}, \text{Fr}, \text{Po}) = 0 \quad (6.16)$$

For nongeometrical similarity, a variety of dimension ratios is also required [4, 102]

$$\frac{T}{d_i}, \frac{Z}{d_i}, \frac{C}{d_i}, \frac{1}{d_i}, \frac{w}{d_i}, \frac{nb}{T}$$

For two-phase dispersions, other groups such as the *Weber number* $We = \rho u^2 d_p / \sigma_{gc}$, the ratio of inertial to surface forces, may be significant. *Sherwood* and *Schmidt numbers* are of course important in mass transfer. For dynamic similarity in two sizes of vessels operated with a vortex, the Reynolds and Froude numbers must be the same for both vessels. Since the impellers would be geometrically similar but unequally sized, it becomes impossible to specify the impeller speeds in the two vessels containing the same liquid to accomplish this. Thus, equal Reynolds numbers require

$$\frac{N_1}{N_2} = \left(\frac{d_{i2}}{d_{i1}} \right)^2$$

while equal Froude numbers require

$$\frac{N_1}{N_2} = \left(\frac{d_{i2}}{d_{i1}} \right)^{0.5}$$

where 1 and 2 designate the two sizes, and the two conditions cannot be met simultaneously. For this reason most work with open vessels has been done with baffles to prevent vortex formation. Closed vessels operated full cannot form a vortex, and baffles for these are unnecessary. For these nonvortexing cases, the Froude number is no longer important; Eq. (6.16) becomes

$$f(\text{Re}, \text{Po}) = 0 \quad (6.17)$$

Power in nonvortexing systems For geometrically similar vessels operated with single-phase newtonian liquids, the power characteristics of the impellers of Fig. 6.3 are shown in Fig. 6.5. These curves represent Eq. (6.17) and were obtained experimentally. At Reynolds numbers less than about 10, the curves have a slope of -1 , and at very high Reynolds numbers the power number becomes constant. The curves are thus quite analogous to those for the drag coefficient of an immersed body or the friction factor for pipe flow.

All the energy in the liquid stream from an impeller is dissipated through turbulence and viscosity to heat [93]. The velocity gradient in the liquid, which sets the shear rate and hence the turbulence intensity, is greatest at the impeller tip, less at a distance relatively removed. At high Reynolds numbers for all the curves of Fig. 6.5, where the power number becomes constant, the impeller power becomes independent of liquid viscosity and varies only as $\rho_L N^3 d_i^5$. Since it has already been shown that for geometrically similar vessels Q varies as $N d_i^3$, at constant power number P is proportional to $Q N^2 d_i^2$ for a given liquid. Consequently a small impeller operating at high speed produces less flow and

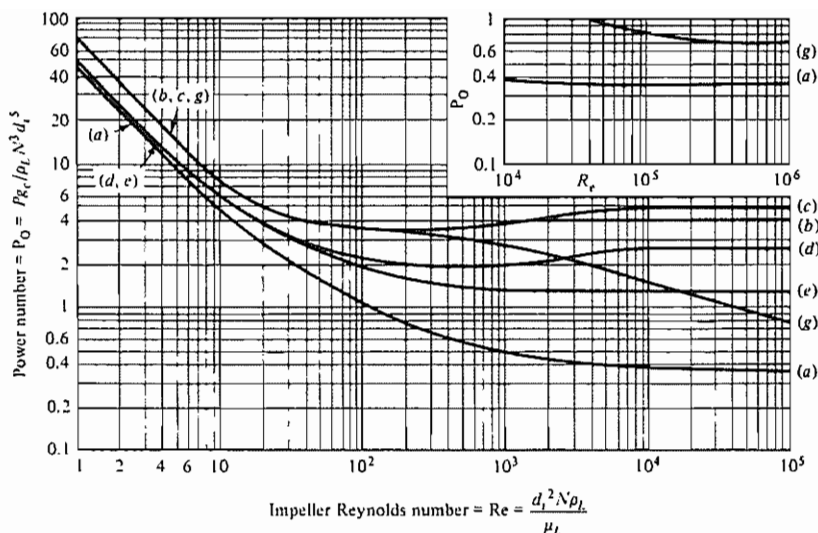


Figure 6.5 Power for agitation impellers immersed in single-phase liquids, baffled vessels with a gas-liquid surface [except curves (c) and (g)]. Curves correspond to impellers of Fig. 6.3: (a) marine impellers, (b) flat-blade turbines, $w = d_i/5$, (c) disk flat-blade turbines with and without a gas-liquid surface, (d) curved-blade turbines, (e) pitched-blade turbines, (g) flat-blade turbines, no baffles, no gas-liquid interface, no vortex.

Notes on Fig. 6.5

1. The power P is only that imparted to the liquid by the impeller. It is not that delivered to the motor drive, which additionally includes losses in the motor and speed-reducing gear. These may total 30 to 40 percent of P . A stuffing box where the shaft enters a covered vessel causes additional losses.
2. All the curves are for axial impeller shafts, with liquid depth Z equal to the tank diameter T .
3. Curves a to e are for open vessels, with a gas-liquid surface, fitted with four baffles, $b = T/10$ to $T/12$.
4. Curve a is for marine propellers, $d_i/T \approx \frac{1}{3}$, set a distance $C = d_i$ or greater from the bottom of the vessel. The effect of changing d_i/T is apparently felt only at very high Reynolds numbers and is not well established.
5. Curves b to e are for turbines located at a distance $C = d_i$ or greater from the bottom of the vessel. For disk flat-blade turbines, curve c , there is essentially no effect of d_i/T in the range 0.15 to 0.50. For open types, curve b , the effect of d_i/T may be strong, depending upon the group nb/T [8].
6. Curve g is for disk flat-blade turbines operated in unbaffled vessels filled with liquid, covered, so that no vortex forms [70]. If baffles are present, the power characteristics at high Reynolds numbers are essentially the same as curve b for baffled open vessels, with only a slight increase in power [88].
7. For very deep tanks, two impellers are sometimes mounted on the same shaft, one above the other. For the flat-blade disk turbines, at a spacing equal to $1.5d_i$ or greater, the combined power for both will approximate twice that for a single turbine [4].

more turbulence than a large impeller operating at slow speed at the same power. Smaller higher-speed impellers are thus especially suited for dispersing liquids and gases, while the larger impellers are particularly effective for suspending solids.

MECHANICAL AGITATION, GAS-LIQUID CONTACT

A typical arrangement is shown in Fig. 6.6*a*. The baffled vessel, with liquid depth approximating the tank diameter, must be provided with adequate free-board to allow for the gas holdup during gas flow. Gas is best introduced below the impeller through a ring-shaped sparger of diameter equal to, or somewhat smaller than, that of the impeller [92], arranged with holes on top. Holes may be 3 to 6.5 mm ($\frac{1}{8}$ to $\frac{1}{4}$ in) in diameter, in number to provide a hole Reynolds number Re_o of the order of 10 000 or more, although this is not especially important. The distance between holes should not be less than d_p , the gas-bubble diameter as given for example by Eq. (6.5). When time of contact must be relatively large, deep vessels can be used, as in Fig. 6.6*b*, whereupon multiple impellers (preferably not more than two [93]) are used to redisperse gas bubbles which coalesce, thereby maintaining a large interfacial area. In very large aeration tanks for activated sludge and other fermentations, where volumes may

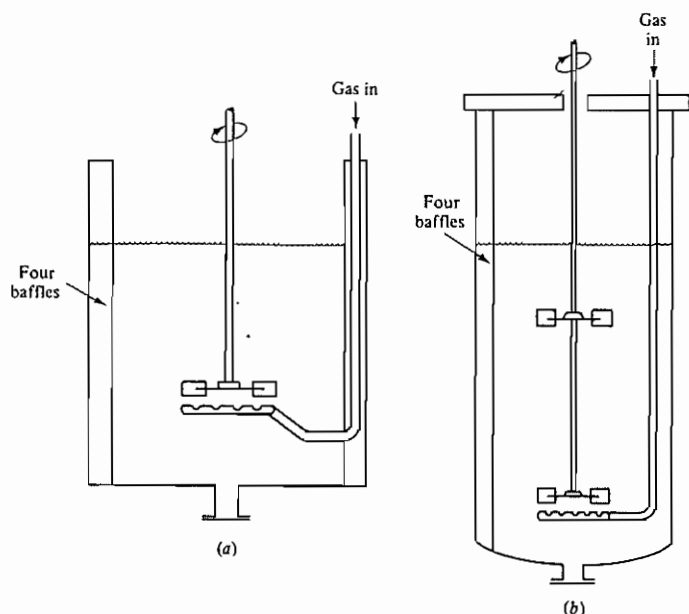


Figure 6.6 Mechanically agitated vessels for gas-liquid contact: (a) standard, (b) multiple impellers for deep tanks.

be as large as 90 m³ (24 000 gal), liquid depths must usually be kept not larger than about 3 to 5 m to save on air-compression costs. The large tank-diameter-liquid-depth ratios which then develop require multiple mixers and points for introducing the gas.

Impellers

Both open and disk flat-blade turbines are used extensively, particularly because of the high discharge velocities normal to the flow of gas which they maintain. Especially in the larger sizes, the disk type is preferred.† They are best specified with $d_i/T = 0.25$ to 0.4 and set off the bottom of the vessel a distance equal to the impeller diameter. In some cases, especially designed impellers can be used to induce the gas flow from the space above the liquid down into the agitated mass [81, 115].

For production of effective gas dispersions with disk flat-blade turbines, the impeller speed should exceed that given by [122]

$$\frac{Nd_i}{(\sigma g g_c / \rho_L)^{0.25}} = 1.22 + 1.25 \frac{T}{d_i} \quad (6.18)$$

Gas Flow

If the gas flow rate is too large, especially for liquids of high viscosity [20], bubbles of gas become trapped below the eye of the impeller, thus blocking the flow of liquid from below to the impeller. Such an effect should be minimized for impeller speeds exceeding those of Eq. (6.18). In any event, for most installations the superficial gas velocity based on the vessel cross-sectional area does not exceed $V_G = 0.08$ m/s (0.25 ft/s).

Impeller Power

The presence of gas in the vessel contents results in lowering the power required to turn an impeller at a given speed, probably because of the lowered mean density of the mixture. Of the many correlations which have been attempted [86], that shown in Fig. 6.7, despite some shortcomings [20], is recommended for disk flat-blade turbines [24] in water and aqueous solutions of nonelectrolytes.‡ It has been found applicable to vessels ranging in capacity up to 4 m³ (140 ft³) and larger [5]. The broken lines of the figure adequately represent the data, and

† See Miller [84] for characteristics of four-bladed open turbines.

‡ It is anticipated that the superficial gas velocity $Q_G/(\pi T^2/4)$, rather than Q_G , would better correlate data for a range of tank diameters, which leads to a correlating group Q_G/Nd_iT^2 instead of that of the figure. If the range of the ratio d_i/T is relatively narrow, however, the group of Fig. 6.7 will serve.

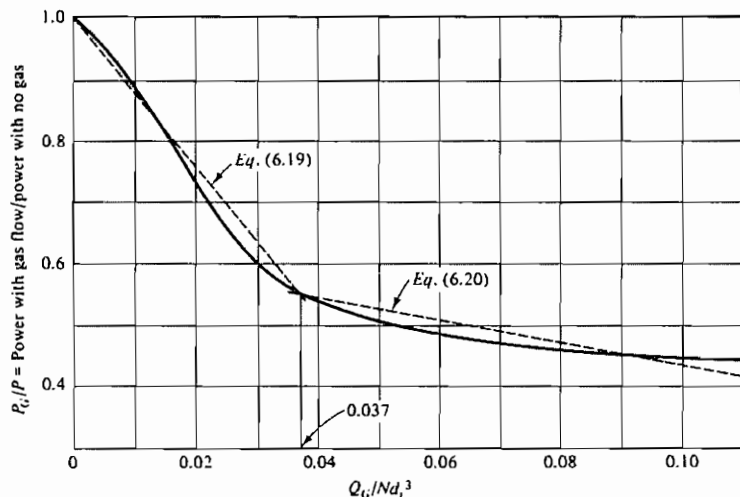


Figure 6.7 Impeller power for gassed vessels, disk flat-blade turbines, $d_i/T = 0.25$ to 0.33 . [Adapted from Calderbank, *Trans. Inst. Chem. Eng., Lond.*, 36, 443 (1958).]

their equations are

$$\frac{P_G}{P} = \begin{cases} 1 - 12.2 \frac{Q_G}{Nd_i^3} & \frac{Q_G}{Nd_i^3} < 0.037 \\ 0.62 - 1.85 \frac{Q_G}{Nd_i^3} & \frac{Q_G}{Nd_i^3} > 0.037 \end{cases} \quad (6.19)$$

$$(6.20)$$

where P_G is the power with gas and P that with no gas flowing. Gassed power levels of 600 to 1000 W/m³ (12.5 to 20 ft · lb_f/ft³ · s or 3 to 5 hp per 1000 gal) are common, although values as large as 4 times as much have been used. Furthermore, the ratio of power required to volume of vessel decreases with vessel size. Power characteristics for two impellers, as in Fig. 6.6*b*, have not been studied; a conservative estimate would be that for one gassed plus that for one ungassed impeller [5]. In many instances an appreciable amount of power will be supplied to the liquid by way of the gas [see Eq. 6.13]. Aqueous electrolytes, because of the smaller bubble-coalescence rates which they produce, exhibit different power requirements [55].

To safeguard against overloading the agitator motor if the flow of gas should suddenly be stopped, the motor and drive should be sized as if there were no gas. An excellent review provides many details [57].

Bubble Diameter, Gas Holdup, and Interfacial Area

If it is turning at suitable speed, the impeller will break the gas up into very fine bubbles, which are swept out radially from the tip. In the immediate vicinity of the impeller tip, the specific interfacial area is consequently very large. Coalescence of the bubbles in the slower-moving liquid in other parts of the vessel increases the bubble diameter and lowers the local area. For moderate impeller speeds, only the gas fed to the impeller produces the specific interfacial area designated as a_o , but at higher impeller speeds, especially in small vessels, gas drawn from the space above the liquid contributes importantly to produce the larger mean specific area a . For disk flat-blade turbines, the specific area can be estimated from [23, 24, 84]:

$$\text{Re}^{0.7} \left(\frac{Nd_i}{V_G} \right)^{0.3} < 30\,000:$$

$$a_o = 1.44 \left[\left(\frac{P_G}{v_L} \right)^{0.4} \left(\frac{\rho_L}{\sigma^3 g_c} \right)^{0.2} \right] \left(\frac{V_G}{V_t} \right)^{1/2} \quad (6.21)$$

$$\text{Re}^{0.7} \left(\frac{Nd_i}{V_G} \right)^{0.3} > 30\,000 \text{ (surface aeration significant):}$$

$$\frac{a}{a_o} = 8.33 \times 10^{-5} \text{Re}^{0.7} \left(\frac{Nd_i}{V_G} \right)^{0.3} - 1.5 \quad (6.22)$$

The quantity in the square brackets of Eq. (6.21) has the net dimension L^{-1} . The impeller Reynolds number is computed with liquid density and viscosity.

The mean bubble diameter for disk flat-blade turbines is given by

$$d_p = K \left[\left(\frac{v_L}{P_G} \right)^{0.4} \left(\frac{\sigma^3 g_c}{\rho_L} \right)^{0.2} \right] \varphi_G^m \left(\frac{\mu_G}{\mu_L} \right)^{0.25} \quad (6.23)$$

where for electrolyte solutions $K = 2.25$, $m = 0.4$ and for aqueous solutions of alcohols and tentatively for other organic solutes in water $K = 1.90$, $m = 0.65$.

The gas holdup can then be estimated by substituting Eqs. (6.21) to (6.23) into Eq. (6.9). Thus, for $\text{Re}^{0.7} (Nd_i/V_G)^{0.3} < 30\,000$, we obtain

$$\varphi_G = \left[0.24 K \left(\frac{\mu_G}{\mu_L} \right)^{0.25} \left(\frac{V_G}{V_t} \right)^{1/2} \right]^{1/(1-m)} \quad (6.24)$$

The use of Eqs. (6.21) to (6.24) requires a trial-and-error procedure. To initiate this, it is frequently satisfactory to try $V_t = 0.21$ m/s, or the equivalent in other units, to be corrected through Eqs. (6.6) to (6.7).

Mass-Transfer Coefficients

The evidence is that the mass-transfer resistance lies entirely within the liquid phase [23, 24, 101], and for gas bubbles of diameter likely to be encountered in agitated vessels

$$\text{Sh}_L = 2.0 + 0.31 \text{Ra}^{1/3} \quad (6.25)$$

where Ra is the Rayleigh number $d_p^3 \Delta \rho g / D_L \mu_L$. Equation (6.25) indicates that k_L is independent of agitator power. This is because the $\Delta \rho$ is so large for gas-liquid mixtures that bubble motion due to g rather than to P_G determines the rate of mass transfer. There is some evidence to the contrary [69].

Illustration 6.2 Water containing dissolved air is to be deoxygenated by contact with nitrogen in an agitated vessel 1 m (3.28 ft) in diameter. The water will enter continuously at the bottom of the vessel and leave through an overflow pipe set in the side of the vessel at an appropriate height. Assume that the ungasged water would have a depth of 1 m (3.28 ft).

$$\text{Water rate} = 9.5 \times 10^{-4} \text{ m}^3/\text{s} (1.99 \text{ ft}^3/\text{min})$$

$$\text{Gas rate} = 0.061 \text{ kg/s} (8.07 \text{ lb/min}) \quad \text{temp} = 25^\circ\text{C}$$

Specify the arrangement to be used and estimate the average gas-bubble diameter, gas holdup, specific interfacial area, and mass-transfer coefficient to be expected.

SOLUTION Use four vertical wall baffles, 100 mm wide, set at 90° intervals. Use a 305-mm-diameter (1-ft), six-bladed disk flat-blade turbine impeller, arranged axially, 300 mm from the bottom of the vessel.

The sparger underneath the impeller will be in the form of a 240-mm-diameter ring made of 12.7-mm ($\frac{1}{2}$ -in) tubing drilled in the top with 3.18-mm-diameter ($\frac{1}{8}$ -in) holes ($d_o = 0.00316$ m). The viscosity of N_2 is 1.8×10^{-5} kg/m $\cdot$ s at 25°C . For an orifice Reynolds number of 35 000 = $4w_o/\pi d_o \mu_G = 4w_o/\pi(0.00316)(1.8 \times 10^{-5})$, $w_o = 1.564 \times 10^{-3}$ kg/s gas flow through each orifice. Therefore, use $0.061/(1.564 \times 10^{-3}) = 39$ holes at approx $\pi(240)/39 = 16$ -mm intervals around the sparger ring.

At 25°C , the viscosity of water is 8.9×10^{-4} kg/m $\cdot$ s, the surface tension is 0.072 N/m, and the density is 1000 kg/m³. In Eq. (6.18) $d_i = 0.305$ m, $T = 1$ m, $g = 9.81$ m/s², and $g_c = 1$, whence $N = 2.84$ r/s (170 r/min) minimum impeller speed for effective dispersion. Use $N = 5$ r/s (300 r/min).

$Re = d_i^2 N \rho_L / \mu_L = 523$ 000. Fig. 6.5: $Po = 5.0$, and the no-gas power = $P = 5.0 \rho_L N^3 d_i^5 / g_c = 1650$ W:

The gas at the sparger, and hence at the impeller, is at a head of 0.7 m of water, or $101\,330 + 0.7(1000)(9.81) = 108.2 \times 10^3$ N/m² abs. Mol wt $N_2 = 28.02$.

$$Q_G = \frac{0.061}{28.02} (22.41) \frac{298}{273} \frac{101.3}{108.2} = 0.050 \text{ m}^3/\text{s}$$

With $N = 5$ r/s, $Q_G / Nd_i^3 = 0.35$. In Fig. 6.7 the abscissa is off scale. Take $P_G/P = 0.43$, whence $P_G = 0.43(1650) = 710$ W (0.95 hp) delivered by the impeller to the gassed mixture. Note: Power for the ungasged mixture is 1650 W (2.2 hp), and allowance for inefficiencies of motor and speed reducer may bring the power input to the motor to 2250 W (3 hp), which should be specified.

$V_G = 0.050/[\pi(1)^2/4] = 0.0673$ m/s superficial gas velocity. $Re^{0.7}(Nd_i/V_G)^{0.3} = 26$ 100. The interfacial area is given by Eq. (6.21).

$P_G = 710$ W, $v_L = \pi(1)^3/4 = 0.785$ m³, $g_c = 1$. Equation (6.21) provides

$$a = \frac{106.8}{V_i^{0.5}} \quad (6.26)$$

with a in m²/m³ and V_i in m/s. For d_p [Eq. (6.23)], use $K = 2.25$, $m = 0.4$, and $\mu_G = 1.8 \times 10^{-5}$ kg/m $\cdot$ s. Equation (6.23) becomes

$$d_p = 2.88 \times 10^{-3} \varphi_G^{0.4} \quad (6.27)$$

Substitution of Eqs. (6.26) and (6.27) into (6.9) provides

$$\varphi_G = \frac{7.067 \times 10^{-3}}{V_i^{0.833}} \quad (6.28)$$

Equations (6.27) and (6.28) are to be solved simultaneously with the relationship for V_i . After several trials, choose $V_i = 0.240$ m/s, whence $\phi_G = 0.0232$, $d_p = 6.39 \times 10^{-4}$ m = 0.639 mm, whence Eq. (6.6) provides $V_i = 0.250$ m/s, a satisfactory check. From Eq. (6.26), $a = 214$ m²/m³.

For N_2 in water at 25°C, $D_L = 1.9 \times 10^{-9}$ m²/s, and $Ra = 1.514 \times 10^{-6}$. Equation (6.25) provides $Sh_L = 37.6 = F_L d_p / c D_L$. Since $c = 1000/18.02 = 55.5$ kmol/m³, $F_L = 6.20 \times 10^{-3}$ kmol/m² · s.

Since a well-designed turbine impeller provides very thorough mixing of the liquid, it can reasonably be assumed that the concentration throughout the liquid phase in such devices is quite uniform. Therefore the benefits of counter-current flow for continuous operation cannot be had with a single tank and agitator.

Multistage Absorption

Multistage arrangements, with counterflow of gas and liquid, can be made with multiple vessels and agitators piped to lead the gas from the top of one tank to the bottom of the next and the liquid from vessel to vessel in the opposite direction. A more acceptable alternative is to place the agitated vessels one atop the other, with the agitators on a single shaft. Towers of this sort are regularly used for liquid extraction, as shown in Fig. 10.52. However a laboratory tower of that design containing 12 agitated compartments was studied for the absorption of ammonia from air into water [113]. The absorption rates were comparable to those obtained with towers packed with 25-mm (1-in) Raschig rings. The design should be particularly advantageous where liquid rates are low, where packing is inadequately wet, or where the liquid contains suspended solids tending to clog conventional packed and tray towers.

TRAY TOWERS

Tray towers are vertical cylinders in which the liquid and gas are contacted in stepwise fashion on trays or plates, as shown schematically for one type (bubble-cap trays) in Fig. 6.8. The liquid enters at the top and flows downward by gravity. On the way, it flows across each tray and through a downspout to the tray below. The gas passes upward through openings of one sort or another in the tray, then bubbles through the liquid to form a froth, disengages from the froth, and passes on to the next tray above. The overall effect is a multiple countercurrent contact of gas and liquid, although each tray is characterized by a cross flow of the two. Each tray of the tower is a stage, since on the tray the fluids are brought into intimate contact, interphase diffusion occurs, and the fluids are separated.

The number of equilibrium stages (theoretical trays) in a column or tower is dependent only upon the difficulty of the separation to be carried out and is determined solely from material balances and equilibrium considerations. The stage or tray efficiency, and therefore the number of real trays, is determined by

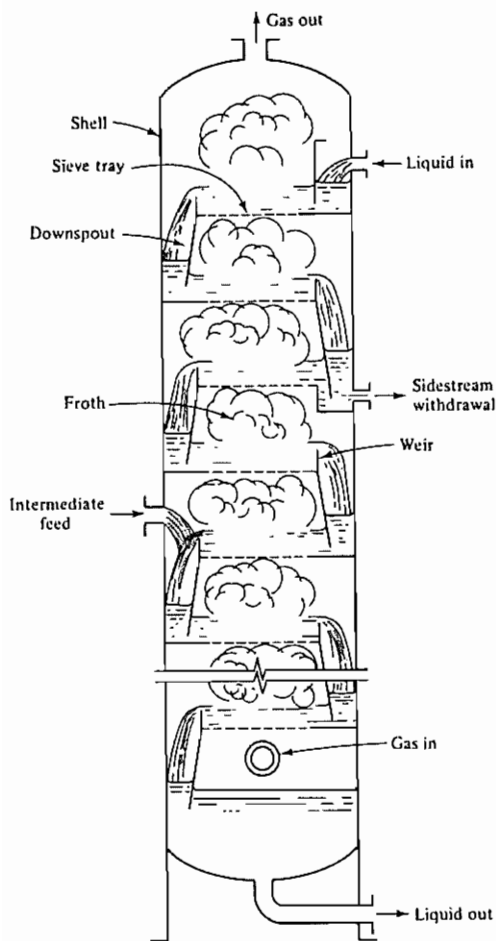


Figure 6.8 Schematic section through sieve-tray tower.

the mechanical design used and the conditions of operation. The diameter of the tower, on the other hand, depends upon the quantities of liquid and gas flowing through the tower per unit time. Once the number of equilibrium stages, or theoretical trays, required has been determined, the principal problem in the design of the tower is to choose dimensions and arrangements which will represent the best compromise between several opposing tendencies, since it is generally found that conditions leading to high tray efficiencies will ultimately lead to operational difficulties.

For stage or tray efficiencies to be high the time of contact should be long to permit the diffusion to occur, the interfacial surface between phases must be made large, and a relatively high intensity of turbulence is required to obtain

high mass-transfer coefficients. In order to provide long contact time, the liquid pool on each tray should be deep, so that bubbles of gas will require a relatively long time to rise through the liquid. When the gas bubbles only slowly through the openings on the tray, the bubbles are large, the interfacial surface per unit of gas volume is small, the liquid is relatively quiescent, and much of it may even pass over the tray without having contacted the gas. On the other hand, when the gas velocity is relatively high, it is dispersed very thoroughly into the liquid, which in turn is agitated into a froth. This provides large interfacial surface areas. For high tray efficiencies, therefore, we require deep pools of liquid and relatively high gas velocities.

These conditions, however, lead to a number of difficulties. One is the mechanical entrainment of droplets of liquid in the rising gas stream. At high gas velocities, when the gas is disengaged from the froth, small droplets of liquid will be carried by the gas to the tray above. Liquid carried up the tower in this manner reduces the concentration change brought about by the mass transfer and consequently adversely affects the tray efficiency. And so the gas velocity may be limited by the reduction in tray efficiency due to liquid entrainment.

Furthermore, great liquid depths on the tray and high gas velocities both result in high pressure drop for the gas in flowing through the tray, and this in turn leads to a number of difficulties. In the case of absorbers and humidifiers, high pressure drop results in high fan power to blow or draw the gas through the tower, and consequently high operating cost. In the case of distillation, high pressure at the bottom of the tower results in high boiling temperatures, which in turn may lead to heating difficulties and possibly damage to heat-sensitive compounds.

Ultimately, purely mechanical difficulties arise. High pressure drop may lead directly to a condition of *flooding*. With a large pressure difference in the space between trays, the level of liquid leaving a tray at relatively low pressure and entering one of high pressure must necessarily assume an elevated position in the downspouts, as shown in Fig. 6.8. As the pressure difference is increased due to the increased rate of flow of either gas or liquid, the level in the downspout will rise further to permit the liquid to enter the lower tray. Ultimately the liquid level may reach that on the tray above. Further increase in either flow rate then aggravates the condition rapidly, and the liquid will fill the entire space between the trays. The tower is then flooded, the tray efficiency falls to a low value, the flow of gas is erratic, and liquid may be forced out of the exit pipe at the top of the tower.

For liquid-gas combinations which tend to foam excessively, high gas velocities may lead to a condition of *priming*, which is also an inoperative situation. Here the foam persists throughout the space between trays, and a great deal of liquid is carried by the gas from one tray to the tray above. This is an exaggerated condition of entrainment. The liquid so carried recirculates between trays, and the added liquid-handling load increases the gas pressure drop sufficiently to lead to flooding.

We can summarize these opposing tendencies as follows. Great depths of liquid on the trays lead to high tray efficiencies through long contact time but

also to high pressure drop per tray. High gas velocities, within limits, provide good vapor-liquid contact through excellence of dispersion but lead to excessive entrainment and high pressure drop. Several other undesirable conditions may occur. If liquid rates are too low, the gas rising through the openings of the tray may push the liquid away (*coning*), and contact of the gas and liquid is poor. If the gas rate is too low, much of the liquid may rain down through the openings of the tray (*weeping*), thus failing to obtain the benefit of complete flow over the trays; and at very low gas rates, none of the liquid reaches the downspouts (*dumping*). The relations between these conditions are shown schematically in Fig. 6.9, and all types of trays are subject to these difficulties in some form. The various arrangements, dimensions, and operating conditions chosen for design are those which experience has proved to be reasonably good compromises. The general design procedure involves a somewhat empirical application of them, followed by computational check to ensure that pressure drop and flexibility, i.e., ability of the tower to handle more or less than the immediately expected flow quantities, are satisfactory.

A great variety of tray designs have been and are being used. During the first half of this century, practically all towers were fitted with bubble-cap trays, but new installations now use either sieve trays or one of the proprietary designs which have proliferated since 1950.

General Characteristics

Certain design features common to the most frequently used tray designs will be dealt with first.

Shell and trays The tower may be made of any number of materials, depending upon the corrosion conditions expected. Glass, glass-lined metal, impervious carbon, plastics, even wood but most frequently metals are used. For metal towers, the shells are usually cylindrical for reasons of cost. In order to facilitate cleaning, small-diameter towers are fitted with hand holes, large towers with manways about every tenth tray [66].

The trays are usually made of sheet metals, of special alloys if necessary, the thickness governed by the anticipated corrosion rate. The trays must be stiffened and supported (see, for example, Fig. 6.14) and must be fastened to the shell to prevent movement owing to surges of gas, with allowance for thermal expansion. This can be arranged by use of tray-support rings with slotted bolt holes to which the trays are bolted. Large trays must be fitted with manways (see Fig. 6.14) so that a person can climb from one tray to another during repair and cleaning [126]. Trays should be installed level to within 6 mm ($\frac{1}{4}$ in) to promote good liquid distribution.

Tray spacing Tray spacing is usually chosen on the basis of expediency in construction, maintenance, and cost and later checked to be certain that adequate insurance against flooding and excessive entrainment is present. For special cases where tower height is an important consideration, spacings of

Table 6.1 Recommended general conditions and dimensions for tray towers**1. Tray spacing**

Tower diameter T		Tray spacing t	
m	ft	m	in
1 or less	4 or less	0.15	6 minimum
1-3	4-10	0.50	20
3-4	10-12	0.60	24
4-8	12-24	0.75	30
		0.90	36

2. Liquid flow

- a. Not over $0.015 \text{ m}^3/(\text{m diam}) \cdot \text{s}$ ($0.165 \text{ ft}^3/\text{ft} \cdot \text{s}$) for single-pass cross-flow trays
 b. Not over $0.032 \text{ m}^3/(\text{m weir length}) \cdot \text{s}$ ($0.35 \text{ ft}^3/\text{ft} \cdot \text{s}$) for others

3. Downspout seal

- a. Vacuum, 5 mm minimum, 10 mm preferred ($\frac{1}{4}$ – $\frac{1}{2}$ in)
 b. Atmospheric pressure and higher, 25 mm minimum, 40 mm preferred (1–1.5 in)

4. Weir length for straight, rectangular weirs, cross-flow trays, $0.6T$ to $0.8T$, $0.7T$ typical

Weir length W	Distance from center of tower	Tower area used by one downspout, %
$0.55T$	$0.4181T$	3.877
$0.60T$	$0.3993T$	5.257
$0.65T$	$0.2516T$	6.899
$0.70T$	$0.3562T$	8.808
$0.75T$	$0.3296T$	11.255
$0.80T$	$0.1991T$	14.145

5. Typical pressure drop per tray

Total pressure	Pressure drop
35 mmHg abs	3 mmHg or less
1 std atm	500–800 N/m ² (0.07–0.12 lb _f /in ²)
$2 \times 10^6 \text{ N/m}^2$	1000 N/m ²
300 lb _f /in ²	0.15 lb _f /in ²

15 cm (6 in) have been used. For all except the smallest tower diameters, 50 cm (20 in) would seem to be a more workable minimum from the point of view of cleaning the trays. Table 6.1 summarizes recommended values.

Tower diameter The tower diameter and consequently its cross-sectional area must be sufficiently large to handle the gas and liquid rates within the region of satisfactory operation of Fig. 6.9. For a given type of tray at flooding, the

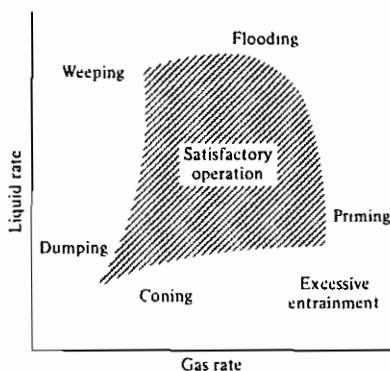


Figure 6.9 Operating characteristics of sieve trays.

superficial velocity of the gas V_F (volumetric rate of gas flow Q per net cross section for flow A_n) is related to fluid densities by†

$$V_F = C_F \left(\frac{\rho_L - \rho_G}{\rho_G} \right)^{1/2} \quad (6.29)$$

The net cross section A_n is the tower cross section A , minus the area taken up by the downspouts (A_d in the case of a cross-flow tray as in Fig. 6.8). C_F is an empirical constant, the value of which depends on the tray design. Some appropriately smaller value of V is used for actual design; for nonfoaming liquids this is typically 80 to 85 percent of V_F (75 percent or less for foaming liquids), subject to check for entrainment and pressure-drop characteristics. Ordinarily the diameter so chosen will be adequate, although occasionally the liquid flow may be limiting. A well-designed single-pass cross-flow tray can ordinarily be expected to handle up to $0.015 \text{ m}^3/\text{s}$ of liquid per meter of tower diameter ($q/T = 0.015 \text{ m}^3/\text{m} \cdot \text{s} = 0.165 \text{ ft}^3/\text{ft} \cdot \text{s}$). For most installations, considerations of cost make it impractical to vary the tower diameter from one end of the tower to the other to accommodate variations in gas or liquid flow, and maximum flow quantities are used to set a uniform diameter. When variation in flows are such that a 20 percent difference in diameter is indicated for the upper and lower sections, two diameters will probably be economical [108].

The tower diameter required may be decreased by use of increased tray spacing, so that tower cost, which depends on height as well as diameter, passes through a minimum at some optimum tray spacing.

Downspouts The liquid is led from one tray to the next by means of downspouts, or downcomers. These may be circular pipes or preferably portions of the tower cross section set aside for liquid flow by vertical plates, as in Fig. 6.8. Since the liquid is agitated into a froth on the tray, adequate residence time must

† Equation (6.29) is empirical; the values of C_F depend on the units used as well as the tray design.

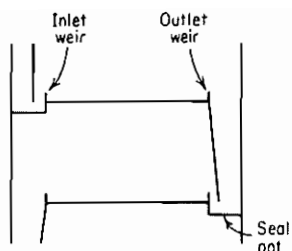


Figure 6.10 Seal-pot arrangement.

be allowed in the downspout to permit disengaging the gas from the liquid, so that only clear liquid enters the tray below. The downspout must be brought close enough to the tray below to seal into the liquid on that tray (item 3, Table 6.1), thus preventing gas from rising up the downspout to short-circuit the tray above. Seal pots and seal-pot dams (inlet weirs) may be used [126], as in Fig. 6.10, but they are best avoided (see below), especially if there is a tendency to accumulate sediment. If they are used, weep holes (small holes through the tray) in the seal pot should be used to facilitate draining the tower on shutdown.

Weirs The depth of liquid on the tray required for gas contacting is maintained by an overflow (outlet) weir, which may or may not be a continuation of the downspout plate. Straight weirs are most common; multiple V-notch weirs maintain a liquid depth which is less sensitive to variations in liquid flow rate and consequently also from departure of the tray from levelness; circular weirs, which are extensions of circular pipes used as downspouts, are not recommended. Inlet weirs (Fig. 6.10) may result in a hydraulic jump of the liquid [67] and are not generally recommended. In order to ensure reasonably uniform distribution of liquid flow on a single-pass tray, a weir length of from 60 to 80 percent of the tower diameter is used. Table 6.1 lists the percentage of the tower cross section taken up by downspouts formed from such weir plates.

Liquid flow Several of the schemes commonly used for directing the liquid flow on trays are shown in Fig. 6.11. Reverse flow can be used for relatively small towers, but by far the most common arrangement is the single-pass cross-flow tray of Fig. 6.11*b*. For large-diameter towers, radial or split flow (Fig. 6.11*c* and *d*) can be used, although every attempt is usually made to use the cross-flow tray because of its lower cost. A recent variant of Fig. 6.11*c* uses ring-shaped downspouts concentric with the tower shell instead of the pipes of that figure [110]. For trays fitted with bubble caps (see below) requiring the liquid to flow great distances leads to undesirable liquid-depth gradients. For towers of very large diameter containing such trays, cascade designs of several levels each with its own weir (Fig. 6.11*e*) have been used, but their cost is considerable. Commercial columns up to 15 m (50 ft) in diameter have been built. Two-pass

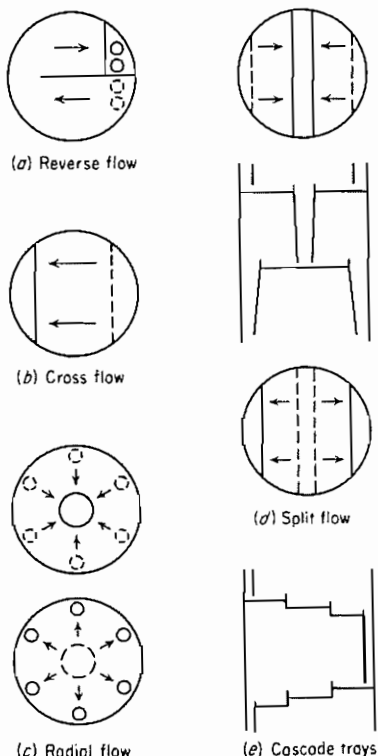


Figure 6.11 Tray arrangements. Arrows show direction of liquid flow.

trays are common for diameters of 3 to 6 m, and more passes for larger diameters.

Bubble-Cap Trays

On these trays (Figs. 6.12 and 6.13) chimneys or risers lead the gas through the tray and underneath caps surmounting the risers. A series of slots is cut into the rim or skirt of each cap, and the gas passes through them to contact the liquid which flows past the caps. The liquid depth is such that the caps are covered or nearly so. Detailed design characteristics are available [13, 46, 119].

Bubble caps have been used for over 150 years, and during 1920–1950 practically all new tray towers used them. They offer the distinct advantage of being able to handle very wide ranges of liquid and gas flow rates satisfactorily (the ratio of design rate to minimum rate is the *turndown ratio*). They have now been abandoned for new installations because of their cost, which is roughly double that for sieve, counterflow, and valve trays.

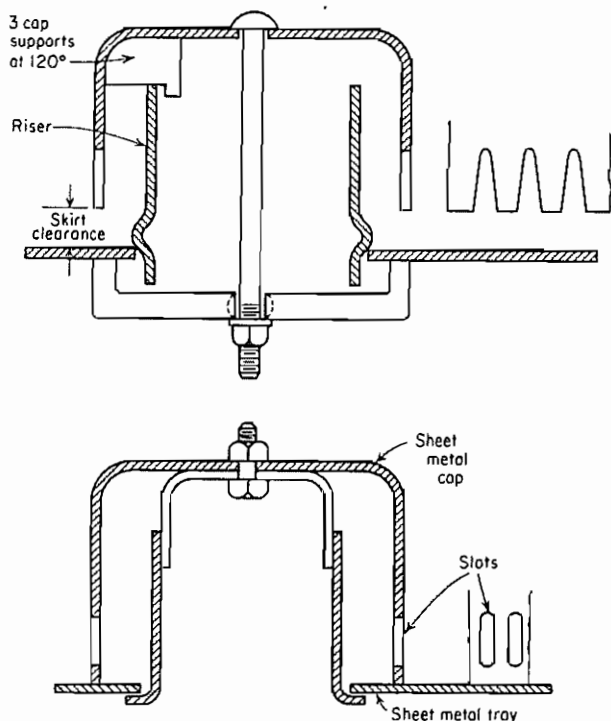


Figure 6.12 Typical bubble-cap designs. (With permission of the Pressed Steel Company.)

Sieve (perforated) trays

These trays have been known almost as long as bubble-cap trays, but they fell out of favor during the first half of this century. Their low cost, however, has now made them the most important of tray devices.

A vertical section and a plan view are shown schematically in Fig. 6.14. The principal part of the tray is a horizontal sheet of perforated metal, across which the liquid flows, with the gas passing upward through the perforations. The gas, dispersed by the perforations, expands the liquid into a turbulent froth, characterized by a very large interfacial surface for mass transfer. The trays are subject to flooding because of backup of liquid in the downspouts or excessive entrainment (priming), as described earlier.

Design of Sieve Trays

The diameter of the tower must be chosen to accommodate the flow rates, the details of the tray layout must be selected, estimates must be made of the gas-pressure drop and approach to flooding, and assurance against excessive weeping and entrainment must be established.

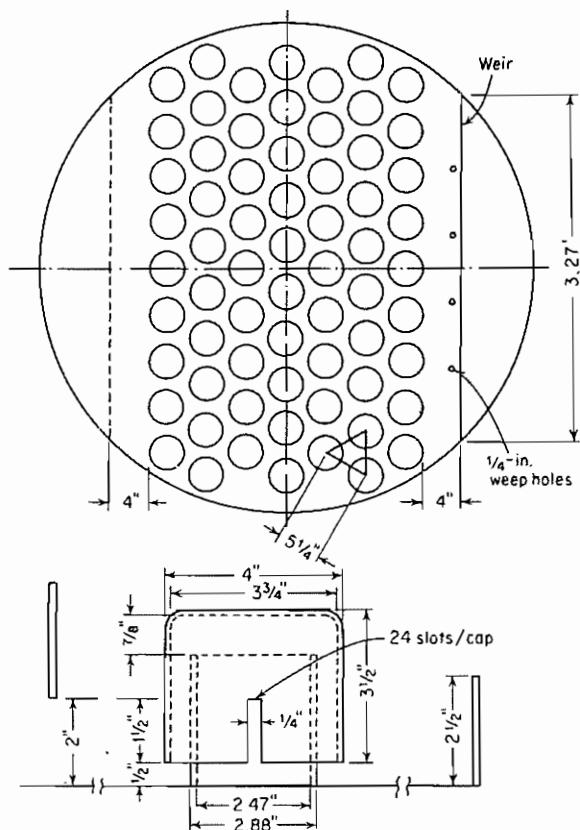


Figure 6.13 Typical bubble-cap tray arrangement.

Tower diameter The flooding constant C_F of Eq. (6.29) has been correlated for the data available on flooding [44, 47]. The original curves can be represented by†

$$C_F = \left[\alpha \log \frac{1}{(L'/G')(\rho_G/\rho_L)^{0.5}} + \beta \right] \left(\frac{\sigma}{0.020} \right)^{0.2} \quad (6.30)$$

Typically, for perforations of 6 mm diameter, the best tray efficiencies result from values of $V\sqrt{\rho_G}$ in the range 0.7 to 2.2 (with V and ρ_G in SI units; for V in ft/s and ρ_G in lb/ft³, the range is 0.6 to 1.8) [2].

Perforations and active area Hole diameters from 3 to 12 mm ($\frac{1}{8}$ to $\frac{1}{2}$ in) are commonly used, 4.5 mm ($\frac{1}{8}$ in) most frequently [45], although holes as large as 25 mm have been successful [72]. For most installations, stainless steel or other alloy perforated sheet is used, rather than carbon steel, even though not necessarily required for corrosion resistance. Sheet thickness [25] is usually less than

† Equation (6.30) is empirical. See Table 6.2 for values of α and β .

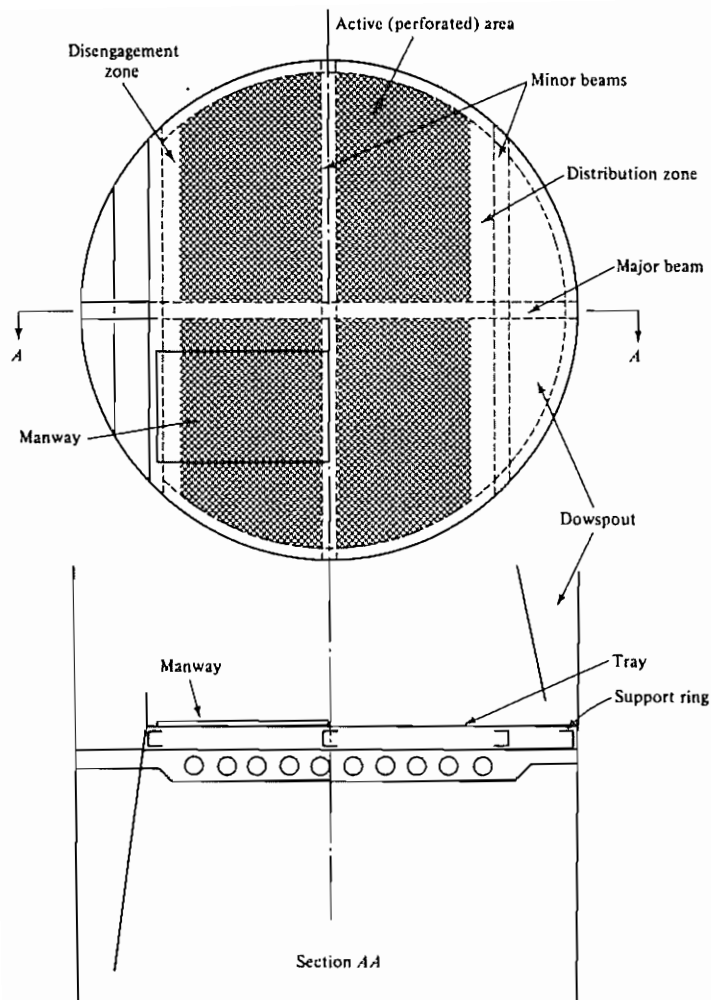


Figure 6.14 Cross-flow sieve tray.

one-half the hole diameter for stainless steel, less than one diameter for carbon steel or copper alloys. Table 6.2 lists typical values.

The holes are placed in the corners of equilateral triangles at distances between centers (pitch) of from 2.5 to 5 hole diameters. For such an arrangement

$$\frac{A_o}{A_a} = \frac{\text{hole area}}{\text{active area}} = 0.907 \left(\frac{d_o}{p'} \right)^2 \quad (6.31)$$

Typically, as in Fig. 6.14, the peripheral tray support, 25 to 50 mm (1 to 2 in) wide, and the beam supports will occupy up to 15 percent of the cross-sectional area of the tower; the distribution

Table 6.2 Recommended dimensions for sieve-tray towers**1. Flooding constant C_F [Eqs. (6.29) and (6.30)], $d_o < 6 \text{ mm}$ ($\frac{1}{4} \text{ in}$)**

Range of $\frac{A_o}{A_a}$	Range of $\frac{L'}{G'} \left(\frac{\rho_G}{\rho_L} \right)^{0.5}$	Units of t	Units of σ	Units of V_F	α, β
> 0.1	0.01–0.1, use values at 0.1				
	0.1–1.0	m	N/m	m/s	$\alpha = 0.0744t + 0.01173$ $\beta = 0.0304t + 0.015$
		in	dyn/cm $\times 10^{-3}$	ft/s	$\alpha = 0.0062t + 0.0385$ $\beta = 0.00253t + 0.050$
< 0.1 Multiply α and β by $5A_o/A_a + 0.5$					

2. Hole diameter and plate thickness

Hole diameter		Plate thickness/hole diameter	
mm	in	Stainless steel	Carbon steel
3.0	$\frac{1}{8}$	0.65	
4.5	$\frac{3}{16}$	0.43	
6.0	$\frac{1}{4}$	0.32	
9.0	$\frac{3}{8}$	0.22	0.5
12.0	$\frac{1}{2}$	0.16	0.38
15.0	$\frac{5}{8}$	0.17	0.3
18.0	$\frac{3}{4}$	0.11	0.25

3. Liquid depth

50 mm (2 in) minimum, 100 mm (4 in) maximum

4. Typical active area

Tower diameter		$\frac{A_a}{A_t}$
m	ft	
1	3	0.65
1.25	4	0.70
2	6	0.74
2.5	8	0.76
3	10	0.78

zone for liquid entering the tray and the disengagement zone for disengaging foam (which are sometimes omitted) use 5 percent or more [47, 66]; and downspouts require additional area (Table 6.1). The remainder is available for active perforations (active area A_a). Typical values of A_a are listed in Table 6.2.

Liquid depth Liquid depths should not ordinarily be less than 50 mm (2 in), to ensure good froth formation; depths of 150 mm (6 in) have been used [126], but 100 mm is a more common maximum. These limits refer to the sum of the weir height h_w plus the crest over the weir h_1 , calculated as clear liquid although in the perforated area the equivalent clear-liquid depth will be smaller than this. Liquid depths as great as 1.5 m have been studied [56] but are not used in ordinary operations.

Weirs Refer to Fig. 6.15, where a schematic representation of a cross-flow tray is shown. The crest of liquid over a straight rectangular weir can be estimated by the well-known Francis formula†

$$\frac{q}{W_{eff}} = 1.839 h_1^{3/2} \quad (6.32)$$

where q = rate of liquid flow, m^3/s

W_{eff} = effective length of the weir, m

h_1 = liquid crest over the weir, m

Because the weir action is hampered by the curved sides of the circular tower, it is recommended [39] that W_{eff} be represented as a chord of the circle of diameter T , a distance h_1 farther from the center than the actual weir, as in Fig. 6.16. Equation (6.32) can then be rearranged to‡

$$h_1 = 0.666 \left(\frac{q}{W} \right)^{2/3} \left(\frac{W}{W_{eff}} \right)^{2/3} \quad (6.33)$$

The geometry of Fig. 6.16 leads to

$$\left(\frac{W_{eff}}{W} \right)^2 = \left(\frac{T}{W} \right)^2 - \left[\left(\frac{T}{W} \right)^2 - 1 \right]^{0.5} + \frac{2h_1}{T} \left(\frac{T}{W} \right)^2 \quad (6.34)$$

For $W/T = 0.7$, which is typical, Eq. (6.33) can be used with $W_{eff} = W$ for $h_1/W = 0.055$ or less with a maximum error of only 2 percent in h_1 , which is negligible.

Pressure drop for the gas For convenience, all gas-pressure drops will be expressed as heads of clear liquid of density ρ_L on the tray. The pressure drop for the gas h_G is the sum of the effects for flow of gas through the dry plate and those caused by the presence of liquid:

$$h_G = h_D + h_L + h_R \quad (6.35)$$

where h_D = dry-plate pressure drop

h_L = pressure drop resulting from depth of liquid on tray

h_R = residual pressure drop

Although Fig. 6.15 shows a change in liquid depth Δ , half of which should be used in Eq. (6.35), in practice this is so small, except for the very largest trays, that it can reasonably be neglected. It can be roughly estimated, if necessary [62]. Estimates of the gas-pressure drop summarized here are based on a critical study [79] of all available data and the generally accepted methods. Most of the data are for the air-water system, and extension to others is somewhat uncertain.

Dry pressure drop h_D This is calculated on the basis that it is the result to a loss in pressure on entrance to the perforations, friction within the short tube formed by the perforation owing to plate

† The coefficient 1.839 applies only for meter-second units. For q in ft^3/s , W_{eff} in feet, and h_1 in inches the coefficient is 0.0801.

‡ The coefficient 0.666 applies only for meter-second units. For q in ft^3/s , W and W_{eff} in feet, and h_1 in inches the coefficient is 5.38.

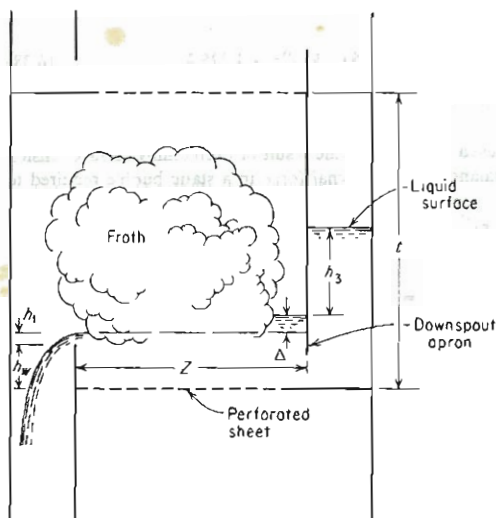


Figure 6.15 Schematic diagram of a cross-flow sieve tray.

thickness, and an exit loss [64]

$$\frac{2h_D g \rho_L}{V_o^2 \rho_G} = C_o \left[0.40 \left(1.25 - \frac{A_o}{A_n} \right) + \frac{4lf}{d_o} + \left(1 - \frac{A_o}{A_n} \right)^2 \right] \quad (6.36)$$

The Fanning friction factor f is taken from a standard chart [15]. C_o is an orifice coefficient which depends upon the ratio of plate thickness to hole diameter [82]. Over the range $l/d_o = 0.2$ to 2.0

$$C_o = 1.09 \left(\frac{d_o}{l} \right)^{0.25} \quad (6.37)$$

Hydraulic head h_L In the perforated region of the tray, the liquid is in the form of a froth. The equivalent depth of clear liquid h_L is an estimate of that which would obtain if the froth collapsed. That is usually less than the height of the outlet weir, decreasing with increased gas rate. Some methods of estimating h_L use a specific *aeration factor* to describe this [87]. In Eq. (6.38), which is the recommended relationship [49], the effect of the factor is included as a function of the variables

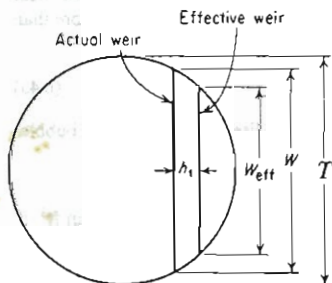


Figure 6.16 Effective weir length.

which influence it†

$$h_L = 6.10 \times 10^{-3} + 0.725h_W - 0.238h_W V_a \rho_G^{0.5} + 1.225 \frac{q}{z} \quad (6.38)$$

where z is the average flow width, which can be taken as $(T + W)/2$.

Residual gas-pressure drop h_R This is believed to be largely the result of overcoming surface tension as the gas issues from a perforation. A balance of the internal force in a static bubble required to overcome surface tension is

$$\frac{\pi d_p^2}{4} \Delta p_B = \pi d_p \sigma \quad (6.39)$$

or

$$\Delta p_B = \frac{4\sigma}{d_p} \quad (6.40)$$

where Δp_B is the excess pressure in the bubble due to surface tension. But the bubble of gas grows over a finite time when the gas flows, and by averaging over time [43], it develops that the appropriate value is Δp_R

$$\Delta p_R = \frac{6\sigma}{d_p} \quad (6.41)$$

Since the bubbles do not really issue singly from the perforations into relatively quiet liquid, we substitute as an approximation the diameter of the perforations d_o , which leads to

$$h_R = \frac{\Delta p_R g_c}{\rho_L g} = \frac{6\sigma g_c}{\rho_L d_o g} \quad (6.42)$$

Other methods of dealing with h_L + h_R have been proposed (see, for example, Ref. 33).

Comparison of observed data with values of h_G calculated by these methods shows a standard deviation of 14.7 percent [79]

Pressure loss at liquid entrance h_2 The flow of liquid under the downspout apron as it enters the tray results in a pressure loss which can be estimated as equivalent to three velocity heads [26, 47]

$$h_2 = \frac{3}{2g} \left(\frac{q}{A_{da}} \right)^2 \quad (6.43)$$

where A_{da} is the smaller of two areas, the downspout cross section or the free area between the downspout apron and the tray. Friction in the downspout is negligible.

Backup in the downspout Refer to Fig. 6.15. The distance h_3 , the difference in liquid level inside and immediately outside the downspout, will be the sum of the pressure losses resulting from liquid and gas flow for the tray above

$$h_3 = h_G + h_2 \quad (6.44)$$

Since the mass in the downspout will be partly froth carried over the weir from the tray above, not yet disengaged, whose average density can usually be estimated roughly as half that of the clear liquid, safe design requires that the level of equivalent clear liquid in the downspout be no more than half the tray spacing. Neglecting Δ , the requirement is

$$h_W + h_1 + h_3 < \frac{t}{2} \quad (6.45)$$

For readily foaming systems or where high liquid viscosity hampers disengagement of gas bubbles the backup should be less.

† SI units must be used in Eq. (6.38). For h_L and h_W in inches, V_a in ft/s, ρ_G in lb/ft³, q in ft³/s, and z in feet

$$h_L = 0.24 + 0.725h_W - 0.29h_W V_a \rho_G^{0.5} + 4.48 \frac{q}{z}$$

Weeping If the gas velocity through the holes is too small, liquid will drain through them and contact on the tray for that liquid will be lost. In addition, for cross-flow trays, such liquid does not flow the full length of the tray below. The data on incipient weeping are meager, particularly for large liquid depths, and in all likelihood there will always be some weeping. A study of the available data [79] led to the following as the best representation of V_{ow} , the minimum gas velocity through the holes below which excessive weeping is likely:

$$\frac{V_{ow}\mu_G}{\sigma_{LG}} = 0.0229 \left(\frac{\mu_G^2}{\sigma_{LG}\rho_G d_o} \frac{\rho_L}{\rho_G} \right)^{0.379} \left(\frac{l}{d_o} \right)^{0.293} \left(\frac{2A_o d_o}{\sqrt{3} p^{1/3}} \right)^{2.8/(2/d_o)^{0.724}} \quad (6.46)$$

Available data for h_L in the range 23 to 48 mm (0.9 to 1.9 in) do not indicate h_L to be of important influence. It may be for larger depths.

Liquid entrainment When liquid is carried by the gas up to the tray above, the entrained liquid is caught in the liquid on the upper tray. The effect is cumulative, and liquid loads on the upper trays of a tower can become excessive. A convenient definition of the degree of entrainment is the fraction of the liquid entering a tray which is carried to the tray above [44]

$$\text{Fractional entrainment} = E = \frac{\text{moles liquid entrained}/(\text{area})(\text{time})}{L + \text{moles liquid entrained}/(\text{area})(\text{time})}$$

The important influence of entrainment on tray efficiency will be considered later. Figure 6.17 represents a summary of sieve-tray entrainment data [44, 47] with an accuracy of ± 20 percent.

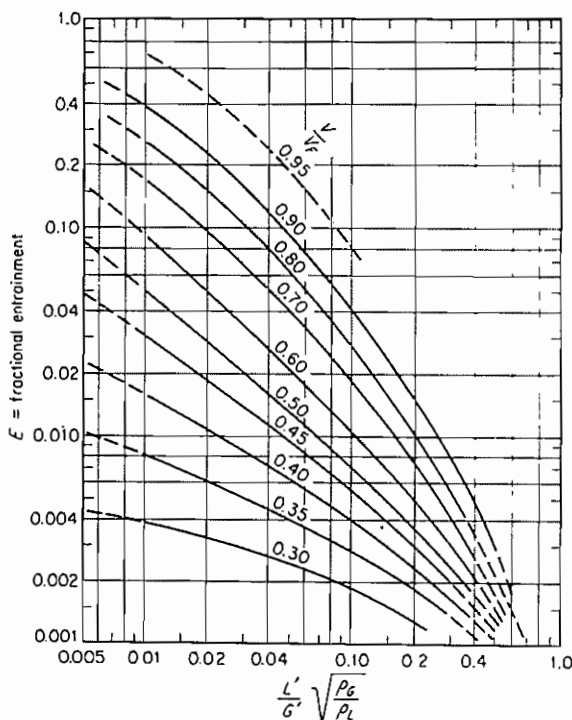


Figure 6.17 Entrainment, sieve trays [44]. (From Petro/Chemical Engineering, with permission.)

Under certain conditions, sieve trays are subject to lateral oscillations of the liquid, which may slosh from side to side or from the center to the sides and back. The latter type especially may seriously increase entrainment. The phenomenon is related to froth height relative to tower diameter [8, 9] and a change from frothing to bubbling [96] but is incompletely characterized.

Illustration 6.3 A dilute aqueous solution of methanol is to be stripped with steam in a sieve-tray tower. The conditions chosen for design are:

Vapor. 0.100 kmol/s (794 lb mol/h), 18 mol % methanol

Liquid. 0.25 kmol/s (1984 lb mol/h), 15 mass % methanol

Temperature. 95°C

Pressure. 1 std atm

Design a suitable cross-flow tray.

SOLUTION Mol wt methanol = 32, mol wt water = 18. Av mol wt gas = $0.18(32) + 0.82(18) = 20.5$ kg/kmol.

$$\rho_G = \frac{20.5}{22.41} \frac{273}{273 + 95} = 0.679 \text{ kg/m}^3 = \text{gas density}$$

$$Q = 0.10(22.41) \frac{273 + 95}{273} = 3.02 \text{ m}^3/\text{s} = \text{vapor rate}$$

$$\rho_L = 961 \text{ kg/m}^3 = \text{liquid density}$$

$$\text{Av mol wt liquid} = \frac{100}{15/32 + 85/18} = 19.26 \text{ kg/kmol}$$

$$q = \frac{0.25(19.26)}{961} = 5.00 \times 10^{-3} \text{ m}^3/\text{s} = \text{liquid rate}$$

Perforations Take $d_o = 4.5$ mm on an equilateral-triangular pitch 12 mm between hole centers, punched in sheet metal 2 mm thick (0.078 in, 14 U.S. Standard gauge). Eq. (6.31):

$$\frac{A_o}{A_a} = \frac{0.907(0.0045)^2}{(0.012)^2} = 0.1275$$

Tower diameter Tentatively take $t = 0.50$ m tray spacing.

$$\begin{aligned} \frac{L'}{G'} \left(\frac{\rho_G}{\rho_L} \right)^{0.5} &= \frac{q \rho_L}{Q \rho_G} \left(\frac{\rho_G}{\rho_L} \right)^{0.5} = \frac{q}{Q} \left(\frac{\rho_L}{\rho_G} \right)^{0.5} \\ &= \frac{5.00 \times 10^{-3}}{3.02} \left(\frac{961}{0.679} \right)^{0.5} = 0.0622 \end{aligned}$$

Table 6.2:

$$\alpha = 0.0744(0.50) + 0.01173 = 0.0489$$

$$\beta = 0.0304(0.50) + 0.015 = 0.0302$$

In Eq. (6.30), since $(L'/G')(\rho_G/\rho_L)^{0.5}$ is less than 0.1, use 0.1. The surface tension is estimated as 40 dyn/cm, or $\sigma = 0.040$ N/m.

$$C_F = \left(0.04893 \log \frac{1}{0.1} + 0.0302 \right) \left(\frac{0.040}{0.020} \right)^{0.2} = 0.0909$$

Eq. (6.29):

$$V_F = 0.0909 \left(\frac{961 - 0.679}{0.679} \right)^{0.5} = 3.42 \text{ m/s at flooding}$$

Use 80% of flooding velocity. $V = 0.8(3.42) = 2.73$ m/s, based on A_n .

$$A_n = \frac{Q}{V} = \frac{3.02}{2.73} = 1.106 \text{ m}^2$$

Tentatively choose a weir length $W = 0.7T$. Table 6.1: the tray area used by one downspout = 8.8%,

$$A_t = \frac{1.106}{1 - 0.088} = 1.213 \text{ m}^2$$

$T = [4(1.213)/\pi]^{0.5} = 1.243$ m, say 1.25 m. Corrected $A_t = \pi(1.25)^2/4 = 1.227 \text{ m}^2$

$$W = 0.7(1.25) = 0.875 \text{ m (final)}$$

$$A_d = 0.088(1.227) = 0.1080 \text{ m}^2 \text{ downspout cross section}$$

$$A_a = A_t - 2A_d - \text{area taken by (tray support + disengaging and distributing zones)}$$

For a design similar to that shown in Fig. 6.14, with a 40-mm-wide support ring, beams between downspouts, and 50-mm-wide disengaging and distributing zones, these areas total 0.222 m².

$$A_a = 1.227 - 2(0.1080) - 0.222 = 0.789 \text{ m}^2 \text{ for perforated sheet}$$

$$\frac{q}{W} = \frac{5.00 \times 10^{-3}}{0.875} = 5.71 \times 10^{-3} \text{ m}^3/\text{m} \cdot \text{s} \quad \text{OK}$$

Weir crest h_1 and weir height h_w Try $h_1 = 25$ mm = 0.025 m. $h_1/T = 0.025/1.25 = 0.02$. $T/W = 1/0.7 = 1.429$. Eq. (6.34): $W_{eff}/W = 0.938$.

Eq. (6.33): $h_1 = 0.666[(5.00 \times 10^{-3})/0.875]^{2/3}(0.938)^{2/3} = 0.0203$ m. Repeat with $h_1 = 0.0203$: $W_{eff}/W = 0.9503$; $h_1[\text{Eq. (6.33)}] = 0.0205$ m. OK. Set weir height $h_w = 50$ mm = 0.05 m.

Dry pressure drop h_D Eq. (6.37): $C_o = 1.09(0.0045/0.002)^{0.25} = 1.335$.

$$A_o = 0.1275A_a = 0.1275(0.789) = 0.1006 \text{ m}^2$$

$$V_o = \frac{Q}{A_o} = \frac{3.02}{0.1006} = 30.0 \text{ m/s} \quad \mu_G = 0.0125 \text{ cP} = 1.25 \times 10^{-5} \text{ kg/m} \cdot \text{s}$$

$$\text{Hole Reynolds number} = \frac{d_o V_o \rho_G}{\mu_G} = \frac{0.0045(30.0)(0.679)}{1.25 \times 10^{-5}} = 7330$$

$f = 0.008$ ("The Chemical Engineers' Handbook," 5th ed., fig. 5.26). $g = 9.807 \text{ m}^2/\text{s}$; $l = 0.002$ m; Eq. (6.36): $h_D = 0.0564$ m.

Hydraulic head h_L

$$V_a = \frac{Q}{A_a} = \frac{3.01}{0.789} = 3.827 \text{ m/s}$$

$$z = \frac{T + W}{2} = \frac{1.25 + 0.875}{2} = 1.063 \text{ m}$$

Eq. (6.38):

$$h_L = 0.0106 \text{ m}$$

Residual pressure drop h_R Eq. (6.42):

$$h_R = \frac{6(0.04)(1)}{961(0.0045)(9.807)} = 5.66 \times 10^{-3} \text{ m}$$

Total gas-pressure drop h_G

Eq. (6.35):

$$h_G = 0.0564 + 0.0106 + 5.66 \times 10^{-3} = 0.0727 \text{ m}$$

Pressure loss at liquid entrance h_2 The downspout apron will be set at $h_W - 0.025 = 0.025$ m above the tray. The area for liquid flow under the apron $= 0.025W = 0.0219$ m². Since this is smaller than A_d , $A_{da} = 0.0219$ m². Eq. (6.43):

$$h_2 = \frac{3}{2(9.807)} \left(\frac{5.0 \times 10^{-3}}{0.0219} \right)^2 = 7.97 \times 10^{-3} \text{ m}$$

Backup in downspout Eq. (6.44): $h_3 = 0.0727 + 7.97 \times 10^{-3} = 0.0807$ m.

Check on flooding $h_W + h_1 + h_3 = 0.1512$, which is well below $t/2 = 0.25$ m. Therefore the chosen t is satisfactory.

Weeping velocity For $W/T = 0.7$, the weir is set $0.3296T = 0.412$ m from the center of the tower. Therefore $Z = 2(0.412) = 0.824$ m. All other quantities in Eq. (6.46) have been evaluated, and the equation then yields $V_{ow} = 8.71$ m/s. The tray will not weep excessively until the gas velocity through the holes V_o is reduced to close to this value.

Entrainment

$$\frac{V}{V_F} = 0.8 \quad \frac{L'}{G'} \left(\frac{\rho_G}{\rho_L} \right)^{0.5} = 0.0622 \text{ (see tower diameter)}$$

Fig. 6.17: $E = 0.05$. The recycling of liquid resulting from such entrainment is too small to influence the tray hydraulics appreciably.

The mass-transfer efficiency is estimated in Illustration 6.4.

Proprietary Trays

Although for many decades the basic design of tray-tower internals remained relatively static, recent years have seen a great many innovations, only a few of which can be mentioned here.

Linde trays These designs have involved improvements both in the perforation design and the tray arrangements [108, 120]. Figure 6.18a shows the *slotted tray*, an alteration in the perforation pattern to influence the flow of liquid. The slots, distributed throughout the tray, not only reduce the hydraulic gradient in large trays (over 10 m diameter [34]) but are also so deployed that they influence the direction of liquid flow to eliminate stagnant areas and achieve, as nearly as possible, desirable plug flow of liquid across the tray. Figure 6.18b shows a *bubbling promoter*, or inclined perforated area at the liquid entrance to the tray. This reduces excessive weeping and produces more uniform froth throughout the tray. The multiple downspouts of Fig. 6.19a are not sealed in the liquid on the tray below; instead the liquid is delivered through slots in the bottom closure to spaces between the downspouts on the tray below. The parallel-flow tray of Fig. 6.19b is so designed that the liquid on all trays in one-half the tower flows from right to left, and on the trays in the other half from left to right. Such an arrangement approximates the so-called Lewis case II [75] and results in an improved Murphree tray efficiency.

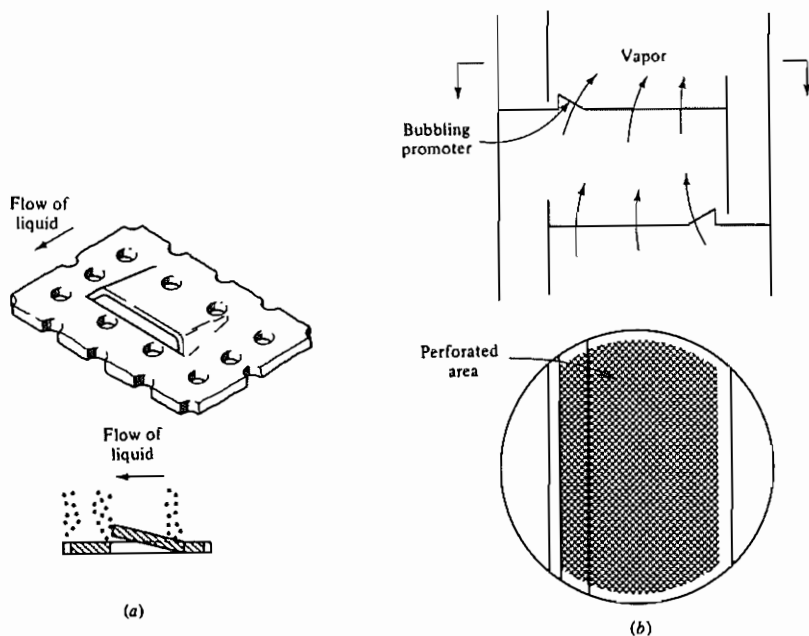


Figure 6.18 Linde sieve trays: (a) slotted sieve tray; (b) bubbling promoter, cross-flow tray. (From *Chemical Engineering Progress*, with permission.)

Valve trays These are sieve trays with large (roughly 35- to 40-mm-diameter) variable openings for gas flow. The perforations are covered with movable caps which rise as the flow rate of gas increases. At low gas rates and correspondingly small openings, the tendency to weep is reduced. At high gas rates the gas-pressure drop remains low but not as low as that for sieve trays. The Glitsch Ballast Tray valve design is shown in Fig. 6.20; other well-known designs are the Koch Flexitray, the Nutter Float-Valve [10], and the Wyatt valve [114]. Tray efficiencies seem to be roughly the same as those for sieve trays with 6-mm ($\frac{1}{4}$ -in) perforations [2, 13].

Counterflow trays These tray-resembling devices differ from conventional trays in that there are no ordinary downspouts: liquid and vapor flow countercurrently through the same openings. *Turbo-Grids* [109] are sheet metal stamped with slotted openings to form the tray, which is installed so alternate trays have the openings at right angles. *Kittel trays*, used extensively in Europe [60, 98, 110], are slotted double trays made of expanded metal arranged so that liquid movement on the tray is influenced by the passage of the gas. *Ripple trays* [65] are perforated trays bent into a sinusoidal wave, with alternate trays installed with the waves at right angles. *Leva trays* [73] are basically very short wetted-wall towers between close-spaced trays, with no weirs and hence no controlled

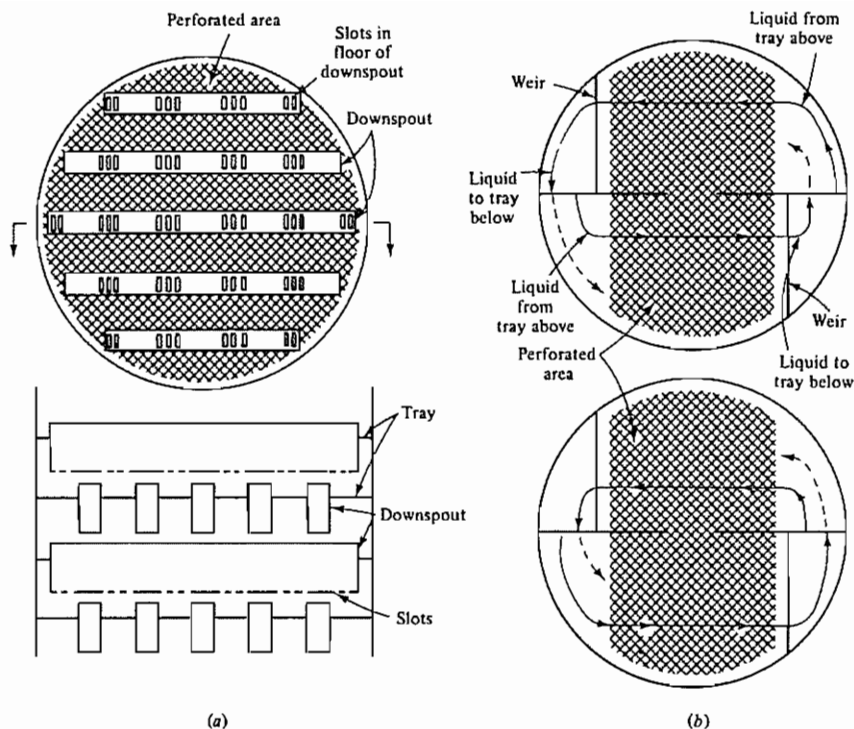


Figure 6.19 Linde tray arrangements: (a) multiple downspouts; (b) parallel-flow tray.

liquid depth on the trays. Since the gas-pressure drop is low, the trays are especially suited to vacuum distillation.

In most cases the design of proprietary trays is best left to the supplier.

Tray Efficiency

Tray efficiency is the fractional approach to an equilibrium stage (see Chap. 5) which is attained by a real tray. Ultimately, we require a measure of approach to equilibrium of all the vapor and liquid from the tray, but since the conditions at various locations on the tray may differ, we begin by considering the local, or *point*, efficiency of mass transfer at a particular place on the tray surface.

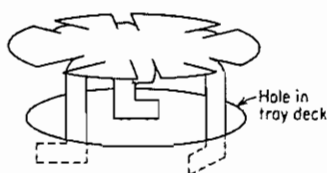


Figure 6.20 Glitsch Ballast Tray valve design (schematic). (Fritz W. Glitsch and Sons, Inc.)

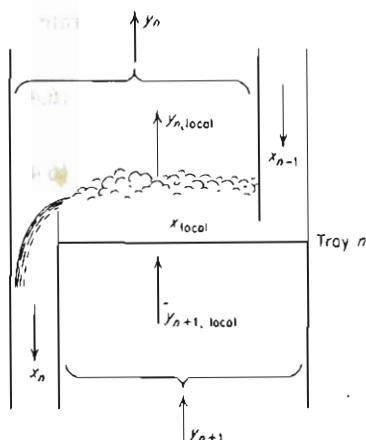


Figure 6.21 Tray efficiency.

Point efficiency Figure 6.21 is a schematic representation of one tray of a multitray tower. The tray n is fed from tray $n - 1$ above by liquid of average composition x_{n-1} mole fraction of transferred component, and it delivers liquid of average composition x_n to the tray below. At the place under consideration, a pencil of gas of composition $y_{n+1, \text{local}}$ rises from below, and as a result of mass transfer, leaves with a concentration $y_{n, \text{local}}$. At the place in question, it is assumed that the local liquid concentration x_{local} is constant in the vertical direction. The point efficiency is then defined by

$$E_{OG} = \frac{y_{n, \text{local}} - y_{n+1, \text{local}}}{y_{\text{local}}^* - y_{n+1, \text{local}}} \quad (6.47)$$

Here y_{local}^* is the concentration in equilibrium with x_{local} , and Eq. (6.47) then represents the change in gas concentration which actually occurs as a fraction of that which would occur if equilibrium were established. The subscript G signifies that gas concentrations are used, and the O emphasizes that E_{OG} is a measure of the overall resistance to mass transfer for both phases. As the gas passes through the openings of the tray and through the liquid and foam, it encounters several hydrodynamic regimes, each with different rates of mass transfer. The danger in attempting to describe the entire effect in terms of a single quantity, in cases of this sort, was pointed out in Chap. 5, but present information permits nothing better.

Consider that the gas rises at a rate G mol/(area)(time). Let the interfacial surface between gas and liquid be a area/volume of liquid-gas foam. As the gas rises a differential height dh_L , the area of contact is $a dh_L$ per unit area of tray. If, while of concentration y , it undergoes a concentration change dy in this

height, and if the total quantity of gas remains essentially constant, the rate of solute transfer is $G dy$:

$$G dy = K_y(a dh_L)(y_{\text{local}}^* - y) \quad (6.48)$$

Then

$$\int_{y_{n+1, \text{local}}}^{y_{n, \text{local}}} \frac{dy}{y_{\text{local}}^* - y} = \int_0^{h_L} \frac{K_y a dh_L}{G} \quad (6.49)$$

Since y_{local}^* is constant for constant x_{local} ,

$$\begin{aligned} -\ln \frac{y_{\text{local}}^* - y_{n, \text{local}}}{y_{\text{local}}^* - y_{n+1, \text{local}}} &= -\ln \left(1 - \frac{y_{n, \text{local}} - y_{n+1, \text{local}}}{y_{\text{local}}^* - y_{n+1, \text{local}}} \right) \\ &= -\ln(1 - E_{OG}) = \frac{K_y a h_L}{G} \end{aligned} \quad (6.50)$$

Therefore

$$E_{OG} = 1 - e^{-K_y a h_L / G} = 1 - e^{N_{iOG}} \quad (6.51)$$

The exponent on e is simplified to N_{iOG} , the number of overall gas-transfer units. Just as K_y contains both gas and liquid resistance to mass transfer, so also is N_{iOG} made up of the transfer units for the gas NT_G and those for the liquid NT_L . As will be shown in Chap. 8, these can be combined in the manner of Eq. (5.7):

$$\frac{1}{N_{iOG}} = \frac{1}{N_{iG}} + \frac{mG}{L} \frac{1}{N_{iL}} \quad (6.52)$$

The terms on the right represent, respectively, the gas and liquid mass-transfer resistances, which must be obtained experimentally. For example, by contacting a gas with a pure liquid on a tray, so that vaporization of the liquid occurs without mass-transfer resistance in the liquid, the vaporization efficiency provides N_{iG} , which can then be correlated in terms of fluid properties, tray design, and operating conditions. Values of N_{iL} obtained through absorption of relatively insoluble gases into liquids (see Chap. 5) can be similarly correlated.

The m of Eq. (6.52) is a local average value for cases where the equilibrium distribution curve is not straight. Refer to Fig. 6.22. It was shown in Chap. 5 that for a given x_{local} and y_{local} the correct slope for adding the resistances is that of a chord m' . For the situation shown in Fig. 6.21, the gas composition changes, leading to a change in m' from an initial value at $y_{n+1, \text{local}}$ to a final value at $y_{n, \text{local}}$. It has been shown [97] that the correct average m for Eq. (6.52) can be adequately approximated by

$$m = \frac{m'_{y_{n+1}} + m'_{y_n}}{2} \quad (6.53)$$

The location of a point of coordinates $(x_{\text{local}}, y_{\text{local}}^*)$ must be obtained by trial, to fit the definition of E_{OG} , Eq. (6.47). Equations (6.51) and (6.52) show that the lower the solubility of the vapor (the larger the m), the lower the tray efficiency will be.

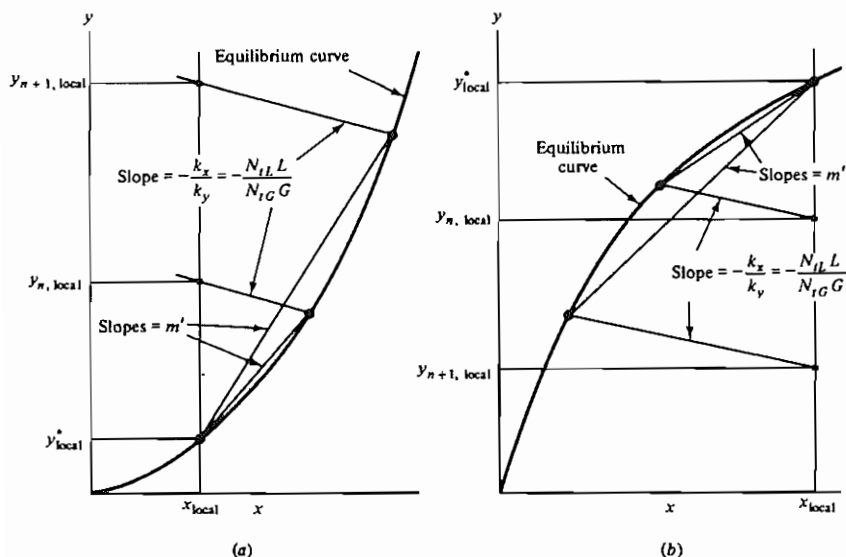


Figure 6.22 Values of m'_{local} : (a) for gas absorption and (b) for distillation and stripping.

Murphree tray efficiency The bulk-average concentrations of all the local pencils of gas of Fig. 6.21 are y_{n+1} and y_n . The Murphree efficiency of the entire tray is then [see Eq. (5.39)]

$$E_{MG} = \frac{y_n - y_{n+1}}{y_n^* - y_{n+1}} \quad (6.54)$$

where y_n^* is the value in equilibrium with the leaving liquid of concentration x_n .

The relationship between E_{MG} and E_{OG} can then be derived by integrating the local E_{OG} 's over the surface of the tray. Clearly, if all the gas entering were uniformly mixed and fed uniformly to the entire tray cross section, and if the mechanical contacting of gas and liquid were everywhere uniform, the uniformity of concentration of exit gas $y_{n+1, local}$ would then depend on the uniformity of liquid concentration on the tray. Liquid on the tray is splashed about by the action of the gas, some of it even being thrown backward in the direction from which it enters the tray (back mixing). The two extreme cases which might be visualized are:

1. Liquid completely back-mixed, everywhere of uniform concentration x_n . In this case, Eq. (6.54) reverts to Eq. (6.47), and

$$E_{MG} = E_{OG} \quad (6.55)$$

2. Liquid in plug flow, with no mixing, each particle remaining on the tray for the same length of time. In this case (Lewis' case I [75]), it has been shown that

$$E_{MG} = \frac{L}{mG} (e^{E_{OG}mG/L} - 1) \quad (6.56)$$

and $E_{MG} > E_{OG}$.

In the more likely intermediate case, the transport of solute by the mixing process can be described in terms of an eddy diffusivity D_E , whereupon [21]

$$\frac{E_{MG}}{E_{OG}} = \frac{1 - e^{-(\eta + Pe)}}{(\eta + Pe)[1 + (\eta + Pe)/\eta]} + \frac{e^\eta - 1}{\eta[1 + \eta/(\eta + Pe)]} \quad (6.57)$$

where
$$\eta = \frac{Pe}{2} \left[\left(1 + \frac{4mGE_{OG}}{LPe} \right)^{0.5} - 1 \right] \quad (6.58)$$

and
$$Pe = \frac{Z^2}{D_E \theta_L} \quad (6.59)$$

Here θ_L is the liquid residence time on tray and Z the length of liquid travel. Pe is a Péclet number (see Chap. 3), as can be seen better by writing it as $(Z/D_E)(Z/\theta_L)$, whence Z/θ_L becomes the average liquid velocity. $Pe = 0$ corresponds to complete back mixing ($D_E = \infty$), while $Pe = \infty$ corresponds to plug flow ($D_E = 0$). Large values of Pe result when mixing is not extensive and for large values of Z (large tower diameters). Although point efficiencies cannot exceed unity, it is possible for Murphree efficiencies.

Entrainment A further correction is required for the damage done by entrainment. Entrainment represents a form of back mixing, which acts to destroy the concentration changes produced by the trays. It can be shown [29] that the Murphree efficiency corrected for entrainment is

$$E_{MGE} = \frac{E_{MG}}{1 + E_{MG}[E/(1 - E)]} \quad (6.60)$$

Data Experimental information on N_{IG} , N_{IL} , and D_E is needed to use the relations developed above. Many data have been published, particularly for bubble-cap trays, and an excellent review is available [47]. Perhaps the best-organized information results from a research program sponsored by the American Institute of Chemical Engineers [21]. Most of the work was done with bubble-cap trays, but the relatively fewer data together with subsequent information from sieve trays [50] indicate that the empirical expressions below represent the performance of sieve trays reasonably well and even valve trays as a first approximation [47]. No attempt will be made here to outline in detail the range of conditions covered by these expressions, which should be used with caution, especially if the original report is not consulted.

Sieve trays†

$$N_{IG} = \frac{0.776 + 4.57h_w - 0.238V_a\rho_G^{0.5} + 104.6q/Z}{Sc_G^{0.5}} \quad (6.61)$$

$$N_{IL} = 40\,000D_L^{0.5}(0.213V_a\rho_G^{0.5} + 0.15)\theta_L \quad (6.62)$$

$$D_E = \left(3.93 \times 10^{-3} + 0.0171V_a + \frac{3.67q}{Z} + 0.1800h_w\right)^2 \quad (6.63)$$

$$\theta_L = \frac{\text{vol liquid on tray}}{\text{vol liquid rate}} = \frac{h_L z Z}{q} \quad (6.64)$$

There is good evidence [6, 7] that flow patterns on typical trays are not so well defined as Eq. (6.57) implies and that areas where the liquid is channeled and others where it is relatively stagnant may exist. These effects adversely alter the Murphree efficiency. There is further evidence that surface-tension changes with changes in liquid composition can strongly influence efficiencies [22, 127]. Particularly in large-diameter columns the vapor entering a tray may not be thoroughly mixed, and this requires separate treatment for converting E_{OG} to E_{MG} [35]. Multicomponent mixtures introduce special problems in computing tray efficiencies [58, 111, 125].

Illustration 6.4 Estimate the efficiency of the sieve tray of Illustration 6.3.

SOLUTION From the results of Illustration 6.3 we have

Vapor rate = 0.100 kmol/s	liquid rate = 0.25 kmol/s
$\rho_G = 0.679 \text{ kg/m}^3$	$h_L = 0.0106 \text{ m}$
$q = 5.00 \times 10^{-3} \text{ m}^3/\text{s}$	$h_w = 0.05 \text{ m}$
$V_a = 3.827 \text{ m/s}$	$Z = 0.824 \text{ m}$
$z = 1.063 \text{ m}$	$E = 0.05$
$y_{n+1, \text{local}} = 0.18 \text{ mole fraction methanol}$	

† Equations (6.61) to (6.63) are empirical and can be used only with SI units. For D_E and D_L in ft^2/h , h_L and h_w in inches, q in ft^3/s , V_a in ft/s , z and Z in feet, θ_L in hours, and ρ_G in lb/ft^3 , Eqs. (6.61) to (6.64) become

$$N_{IG} = \frac{0.776 + 0.116h_w - 0.290V_a\rho_G^{0.5} + 9.72q/Z}{Sc_G^{0.5}} \quad (6.61a)$$

$$N_{IL} = (7.31 \times 10^5)D_L^{0.5}(0.26V_a\rho_G^{0.5} + 0.15)\theta_L \quad (6.62a)$$

$$D_E = \left(0.774 + 1.026V_a + \frac{67.2q}{Z} + 0.900h_w\right)^2 \quad (6.63a)$$

$$\theta_L = \frac{2.31 \times 10^{-3}h_L z Z}{q} \quad (6.64a)$$

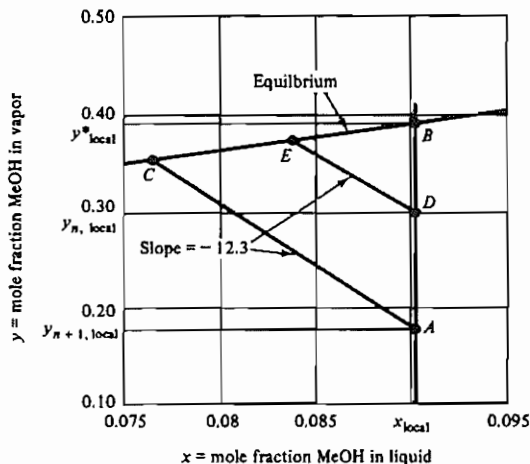


Figure 6.23 Illustration 6.4. Determination of m .

In addition, by the methods of Chap. 2, the following can be estimated: $Sc_G = 0.865$, $D_L = 5.94 \times 10^{-9} \text{ m}^2/\text{s}$. Substitution of these data in Eqs. (6.61) to (6.64) yields

$$\begin{aligned} \theta_L &= 1.857 \text{ s} & N_{IG} &= 0.956 \\ N_{IL} &= 4.70 & D_E &= 0.0101 \text{ m}^2/\text{s} \end{aligned}$$

For 15 mass % methanol, $x_{\text{local}} = (15/32)/(15/32 + 85/18) = 0.0903$ mole fraction methanol. The relevant portion of the equilibrium curve is plotted in Fig. 6.23 with data from "The Chemical Engineers' Handbook," 4th ed., p. 13-5. Point A is plotted with coordinates $(x_{\text{local}}, y_{n+1, \text{local}})$. At B, the ordinate is y_{local}^* . Line AC is drawn with slope

$$-\frac{N_{IL}L}{N_{IG}G} = -\frac{4.70}{0.956} \frac{0.25}{0.100} = -12.3$$

In this case it is evident that the equilibrium curve is so nearly straight that chord BC (not shown) will essentially coincide with it, as will all other relevant chords from B (as from B to E), and m equals the slope of the equilibrium curve, 2.50.

Equation (6.52) then yields $N_{IOG} = 0.794$, and Eq. (6.51) $E_{OG} = 0.548$. [This provides through Eq. (6.74), although unnecessary for what follows, $y_{n, \text{local}} = 0.297$, plotted at D, Fig. 6.23.] Equations (6.57) to (6.59) then give

$$Pe = 36.2 \quad \eta = 0.540 \quad E_{MG} = 0.716$$

and with $E = 0.05$, Eq. (6.60) yields $E_{MGE} = 0.70$. Ans.

Overall tray efficiency Another method of describing the performance of a tray tower is through the overall tray efficiency,

$$E_O = \frac{\text{number of ideal trays required}}{\text{number of real trays required}} \quad (6.65)$$

While reliable information on such an efficiency is most desirable and convenient to use, it must be obvious that so many variables enter into such a measure that really reliable values of E_O for design purposes would be most difficult to come by. As will be shown in Chap. 8, E_O can be derived from the individual tray efficiencies in certain simple situations.

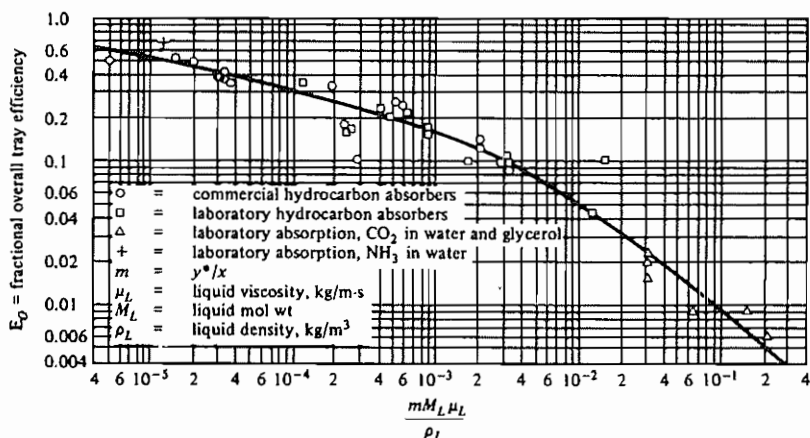


Figure 6.24 Overall tray efficiencies of bubble-cap tray absorbers. For μ_L in centipoises and ρ_L in lb/ft³, use as abscissa $6.243 \times 10^{-3} m M_L \mu_L / \rho_L$. (After O'Connell [90].)

Provided that only standard tray designs are used and operation is within the standard ranges of liquid and gas rates, some success in correlating E_O with conditions might be expected. O'Connell [90] was successful in doing this for absorption and distillation in bubble-cap-tray towers, and his correlations are shown in Figs. 6.24 and 6.25. They must be used with great caution, but for rough estimates they are most useful, even for sieve and valve trays.

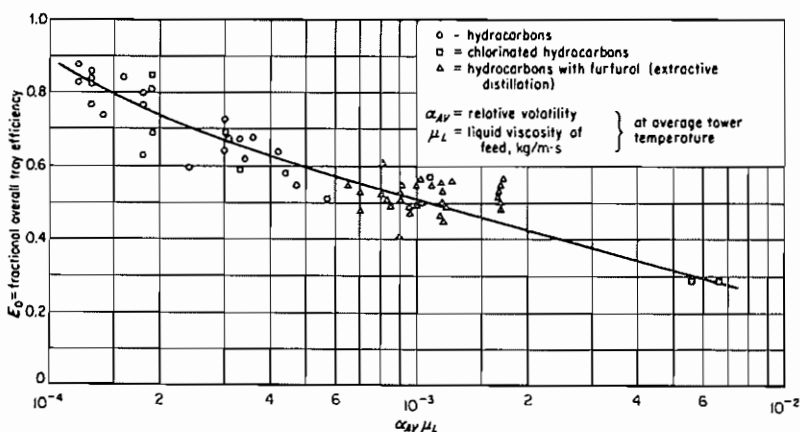


Figure 6.25 Overall tray efficiencies of bubble-cap tray distillation towers separating hydrocarbons and similar mixtures. For μ_L in centipoises, use as abscissa $\alpha_{AV} \mu_L \times 10^{-3}$. (After O'Connell [90].)

This group includes devices in which the liquid is dispersed into thin films or drops, such as wetted-wall towers, sprays and spray towers, the various packed towers, and the like. The packed towers are the most important of the group.

VENTURI SCRUBBERS

In these devices, which are similar to ejectors, the gas is drawn into the throat of a venturi by a stream of absorbing liquid sprayed into the convergent duct section, as shown in Fig. 6.26. The device is used especially where the liquid contains a suspended solid, which would plug the otherwise more commonly used tray and packed towers, and where low gas-pressure drop is required. These applications have become increasingly important in recent years, as in the absorption of sulfur dioxide from furnace gases with slurries of limestone, lime, or magnesia [51, 59, 68, 116]. Some very large installations (10 m diameter) are in service for electric utilities. The cocurrent flow produces only a single stage

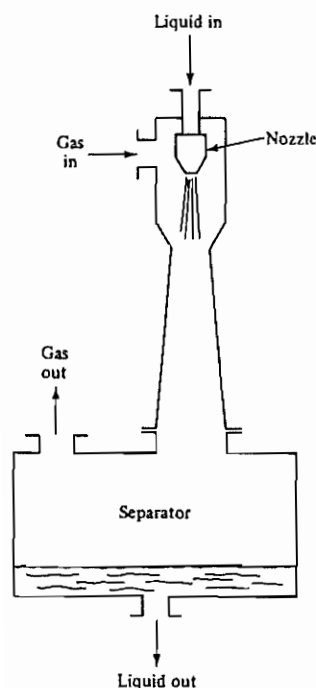


Figure 6.26 Venturi scrubber.